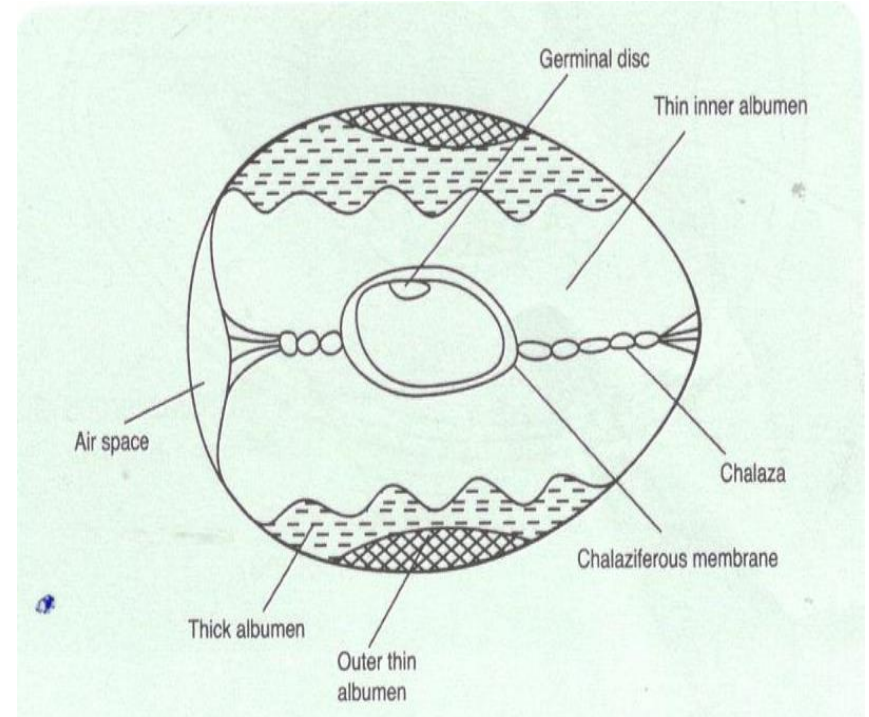
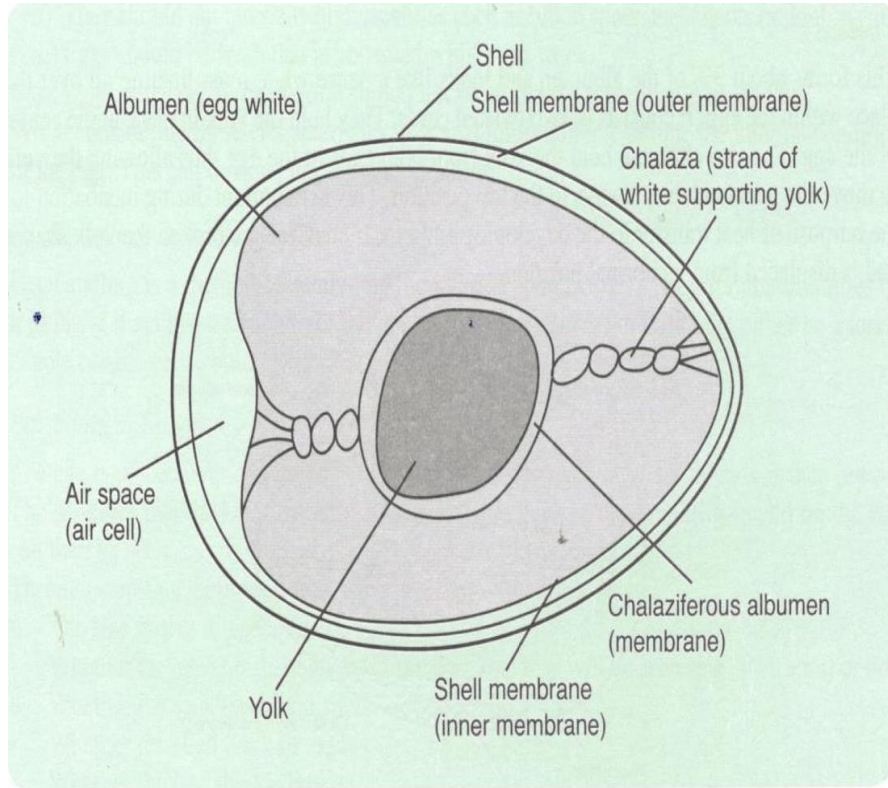


1. LIVESTOCK PRODUCTION 5 (POULTRY PDODUCTION)

12-May-21

Parts of an egg

2



1. **Shell-** It is the outer covering of the egg consisting of calcium and phosphorus compounds which make it hard.

Functions.

- i. It gives the egg its shape.
- ii. Protects the inner parts of the egg.
- iii. It allows gaseous exchange because it is porous.
- iv. Prevents entry of micro-organisms.

2. **Shell membrane-** It forms the inner lining of the shell.
 - It is made up of two membranes (inner and outer) whose contact is not easy to identify.
 - They separate at the broad end to form air space/air sac/air cell.

Functions.

- i. Outer and inner membranes separate to form air space. The air trapped is used by the developing embryo during incubation.
- ii. Determine the egg shape.
- iii. Protect inner parts of the egg.

3. **Air space/air sac/air cell-** It is formed at the broad end of the egg enclosed by two membranes.

- It is used for gaseous exchange.

Formation.

- It is formed immediately the egg is laid.
- The body temperature in a hen is higher than outside.
- Once the egg is laid the lower temperature on the outside causes the contraction of the egg contents creating the space which fills with air.

4. **The yolk-** It is yellow in colour and spherical.

- It contains nutrients (i.e. vitamins, fats, minerals and proteins).

Functions.

- i. Stores nutrients for the embryo.
- ii. Carries the germinal disc which develops into the embryo.

5. **Albumen/Egg white.**- it is jelly-like colourless fluid when fresh and turns white when cooked.

Functions

- a) Surrounds the yolk
- b) Stores food for developing embryo/ chick.
- c) Acts as a shock absorber.

6. **The chalaza-** It is a white dense mass of fluid.

- It consists of two twisted cords.

Functions.

- i. Hold the yolk in central position of the egg
- ii. The two chalazae hold the yolk from both ends of the egg ensuring the egg yolk/ germinal disc always move to the top position.
- This is important during incubation for the purpose of heat transfer to the developing embryo.

Incubation of eggs

6

- **Egg incubation** is the embryonic development of a fertilized egg into a chick under correct conditions which will ensure that a chick is finally hatched.

Selection of eggs for incubation.

- Eggs selected for incubation should have the following characteristics:
 1. They should be fertile.
 2. They should be of medium size (55-60 grams).

3. They should have smooth shells.
4. They should be oval in shape.
5. They should be free from any cracks in the shells.
6. They should be clean to ensure that the pores are open.
7. They should not have any abnormalities e.g. blood spots, meat spots, or double yolk.
8. They should be fresh i.e. collected within one week.

Study question.

List the qualities of shell considered when selecting eggs for incubation.

1. Texture/smoothness.
2. Absence of cracks.
3. Cleanliness.
4. Oval in shape.

Egg abnormalities

A. Internal abnormalities.

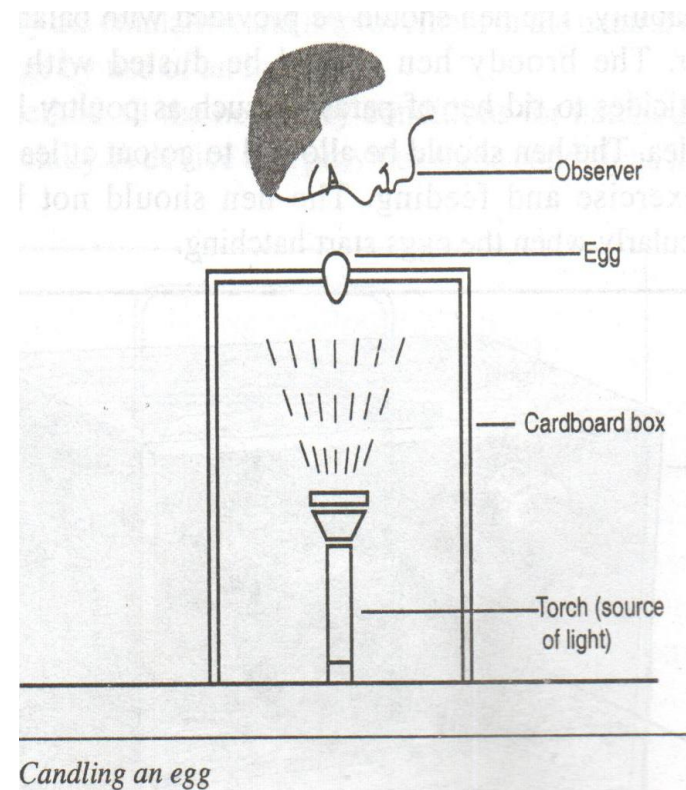
1. Blood/meat spots within the yolk.
2. **Double yolks**- two yolks surrounded by one membrane.
3. **Egg within egg**- two complete yolks each with its own membrane seen in one egg.
4. Yolkless eggs

External abnormalities.

- i. Thin shelled eggs.
- ii. Thin cracks on the shell.
- iii. Very porous shells.

EGG CANDLING.

- This is a method of examining the internal conditions of an egg for abnormalities by looking at it against strong light in a dark room.



Procedure of egg candling.

- The egg is placed over a hole made on a cardboard box.
- A light under the box is then put on.
- The observer then looks at the egg against the light below

Reasons for candling.

- i. To check egg abnormalities.
- ii. To monitor chick development during incubation.
- iii. To check whether the egg is fertile.

- It is done at least twice during incubation i.e. after 5-7 days/ 1 week to check for fertility (if fertile, blood veins will be seen), and on 18th day to confirm the presence of a chick.
- On 18th day the following are observed:
 - i. A large dark section of developing chick.
 - ii. A small clear section of air space.

What is observed through candling.

1. The size of air space/ cell.
2. Whether the egg is fertile or not. If it is fertile, the germinal disc will be seen as a black spot.
3. Whether the yolk has blood spots.
4. Whether the shell has hair cracks.
5. Whether the egg shell is broken.
6. Whether the shell is very porous

Conditions that reduce hatchability.

1. Presence of cracks.
2. Double yolks.
3. Having no yolk.
4. Meat/Blood spots.
5. Rough shell.
6. Unfertilized egg.
7. Very small/tiny eggs.
8. Eggs older than one week.

Causes of soft shelled eggs.

1. Lack of calcium in the feed.
2. Diseases e.g. Newcastle.

Methods of incubation.

11

A. NATURAL INCUBATION.

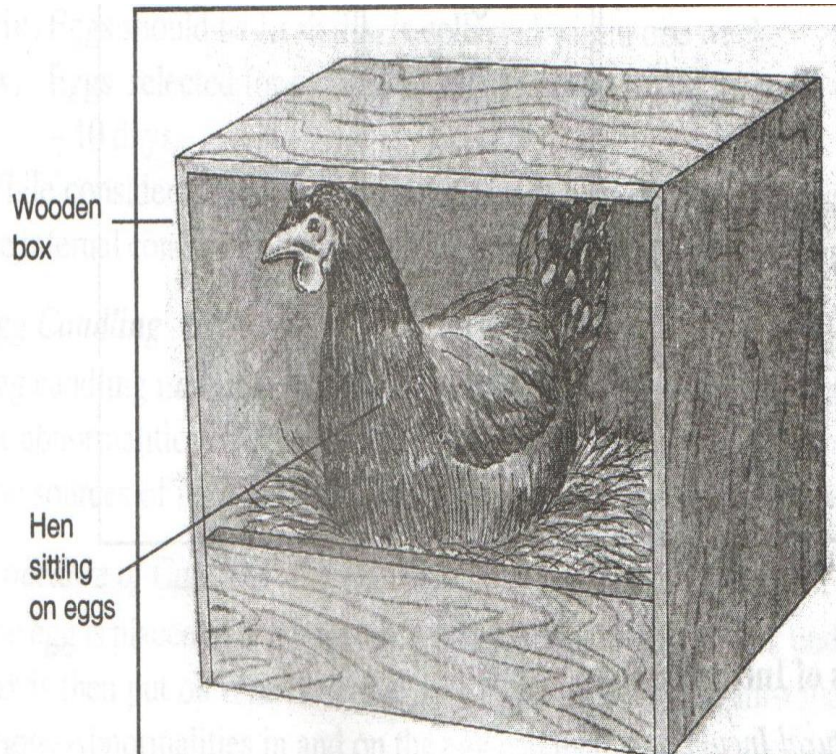
- This is where a hen is allowed to sit on eggs to provide optimum conditions for embryonic development/hatching.
- It is done through a broody hen.

Signs of broodiness in hen/poultry.

1. The hen stops laying eggs.
2. The hen continues to stay in the laying nest after laying
3. The hen plucks off her feathers which she uses in the incubation nest.
4. The hen becomes aggressive and produces a characteristic sound when approached in the nest.

Preparation and management of natural incubation.

- ✓ A nesting box (made from half debe, carton box, karai or wooden box) should be provided to a broody hen.
- ✓ The nesting box should be dry, spacious to allow movement of the hen and placed in the well ventilated place.
- ✓ The nesting materials e.g. dry grass, sawdust and wood shavings should be placed in the nest. This helps to maintain warmth in the nest.
- ✓ The hen is given an adequate number of eggs (about 10-15) to sit on.
- ✓ The broody hen should be provided with balanced feed and water.
- ✓ The broody hen should be dusted with appropriate insecticides to control parasites (e.g. louse or stick fast flea).
- ✓ The broody hen should be allowed to go out at least once a day for exercise and feeding.



Qualities of a good laying/ nesting box.

- i. Should be dimly lit/dark.
- ii. It should be spacious.
- iii. Should have dry beddings to keep it warm.
- iv. Kept in secluded parts of the house.

Advantages of natural incubation.

1. A small-scale farmer who cannot afford the incubator can multiply the flocks using this method.
2. It is less laborious and there is no need of turning the eggs and checking the temperatures.
3. Little skill is required.
4. The margin of risk is reduced.

Disadvantages of natural incubation.

1. Only few chicks can be hatched at a time by one hen.
2. The farmer cannot plan when to incubate.
3. Diseases can easily be transmitted from infected hen to chicks after hatching.
4. Hens can only be used when broody.

B. ARTIFICIAL INCUBATION

- ✓ In this method, all the conditions necessary for hatching are provided artificially in a device called **incubator**.

Conditions necessary for artificial incubation.

1. **Temperature-** it is necessary for embryonic development.
 - It should be maintained at 37.5-39.4 degrees celcius.
 - Lower or higher temperatures would kill the developing embryo.
2. **Fresh air/ ventilation-** there should be enough fresh air circulation because oxygen is necessary for embryonic development.
 - Ventilation facilitates gaseous exchange and helps to control humidity.
3. **Relative humidity-** it should be maintained at 60%.
 - Low humidity causes the embryo to lose moisture and die.
 - High humidity lowers hatchability and leads to production of bigger chicks which look marshy.
 - It is maintained by placing a damp cloth or water in a tray within an incubator.

Incubator

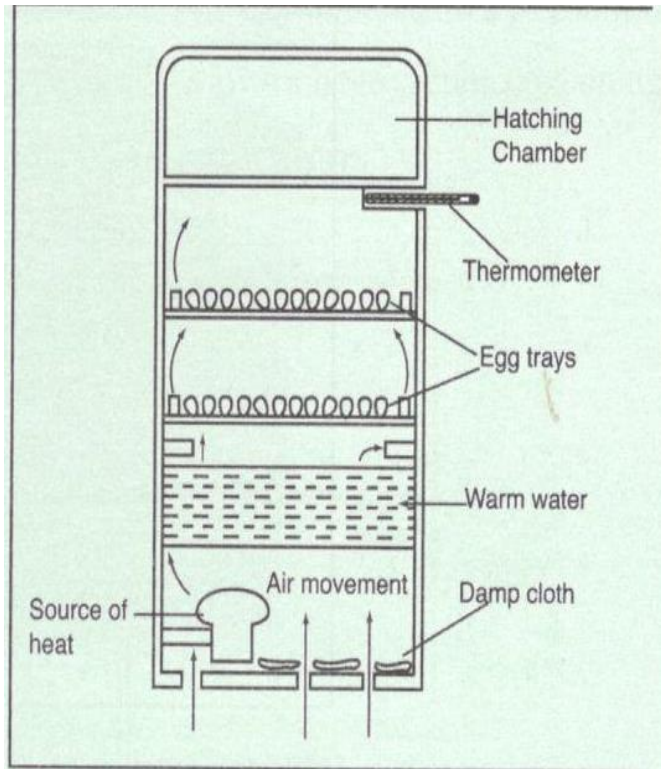


Fig. 1.5: Structure of a simple incubator #

Incubator

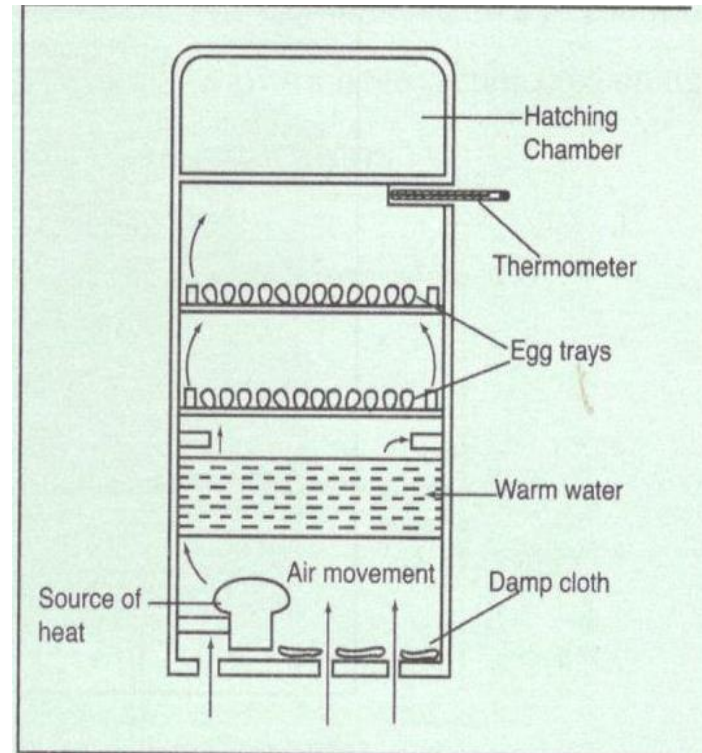


Fig. 1.5: Structure of a simple incubator #

4. **Egg turning-** this is done along the axis of the egg at 45 degrees from vertical and each time in different directions.
 - This is to avoid the germinal disc sticking onto the egg shell leading to lack of hatchability.
 - Wrong turning of the egg may result in breakage of blood vessels.

Preparation and management of artificial incubation/ incubator.

- ✓ Clean and fumigate/ disinfect the incubator before putting the eggs and put it in a clean room.
- ✓ Set well selected eggs.
- ✓ Turn the eggs after every 6-8 daily but in the first 24 hours and the last 3 days the eggs should not be turned.
- ✓ Remove any infertile egg on the 5th day and any broken egg.
- ✓ Add water/ place a damp cloth to maintain humidity.
- ✓ Maintain the temperature at the appropriate range.
- ✓ Ensure that there is proper ventilation.
- ✓ Transfer the eggs into the hatching compartment after 19th day.

Advantages of artificial incubation.

1. Many chicks can be hatched at a time.
2. It is possible to plan when to hatch.
3. It is easier to control diseases and parasites.
4. The incubator is usually ready when required.

Disadvantages of artificial incubation.

- i. The incubator is expensive to buy hence capital investment is high.
- ii. It requires more labour
- iii. It requires more skill than natural method.
- iv. It involves high risks of damaging all the eggs or embryos if management is poor.

Sources of chicks

- When sourcing from day-old chicks the following factors are considered:
 1. The reputation of the supplier.
 2. Time taken from the hatchery and the farm.
 3. Type of chicks available.

Examples of reputable hatchery.

1. Muguku poultry farm, Kikuyu.
2. Lake chicks hatchery, Kisumu.
3. Kenchick poultry hatcheries, Athi River.
4. Stockplan hatcheries, Mombasa.
5. Stigma suppliers, Nairobi.

BROODING

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- Brooding refers to rearing of chicks.

Methods of brooding

A. NATURAL BROODING.

- The hen is allowed to take care of the chicks.
- The hen provides warmth and protection to the chicks.
- Feeds and water are provided to the hen and chicks.
- The hen stays with the chicks until they are ready to take care of themselves.

Advantages of natural brooding.

- i. It is cheap.
- ii. It requires less labour.

Disadvantages of natural brooding.

- i. It is only possible when hens go broody.
- ii. It lowers egg production during brooding.

B. ARTIFICIAL BROODING.

- ✓ In this method, chicks are kept in a structure called artificial brooder for 6-8 weeks after hatching.

Preparation of the brooder before arrival of the chicks.

- ✓ The brooder should be made ready 2-3 days before the arrival of chicks.
- ✓ All equipment should be functioning.
- ✓ Newspapers should be spread on the floor of the brooder. This prevents the chicks from eating the litter.
- ✓ Some feeds should be spread on the newspapers and some placed in the feeders. This helps the chicks to know where the feeder is after they eat up all the feeds on the newspapers.
- ✓ After the chicks have learnt where to eat from the newspapers are removed.
- ✓ The brooder and the equipment should be thoroughly cleaned and disinfected before the arrival of chicks.

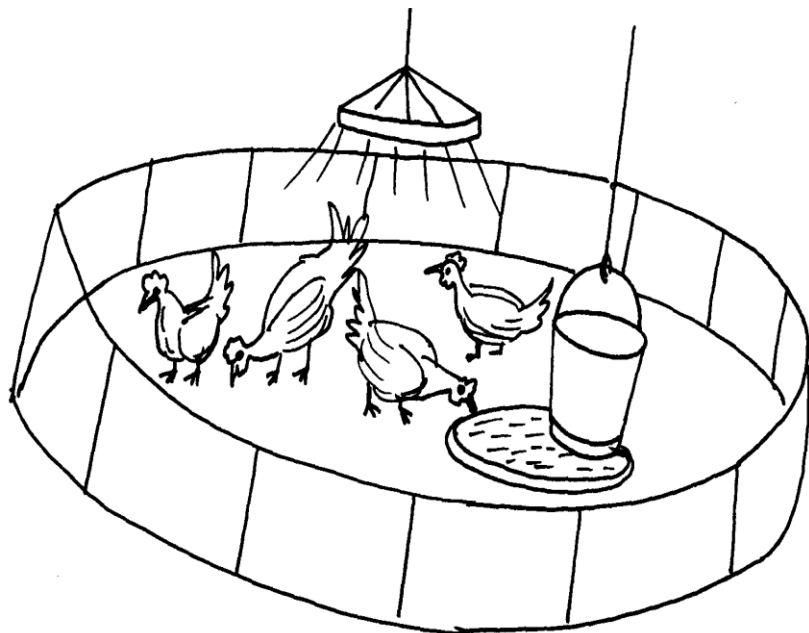
- ✓ When the chicks arrive and seem to be weak or suffering from stress, they should be persuaded to start drinking and glucose should be put in water.

Management after arrival of chicks.

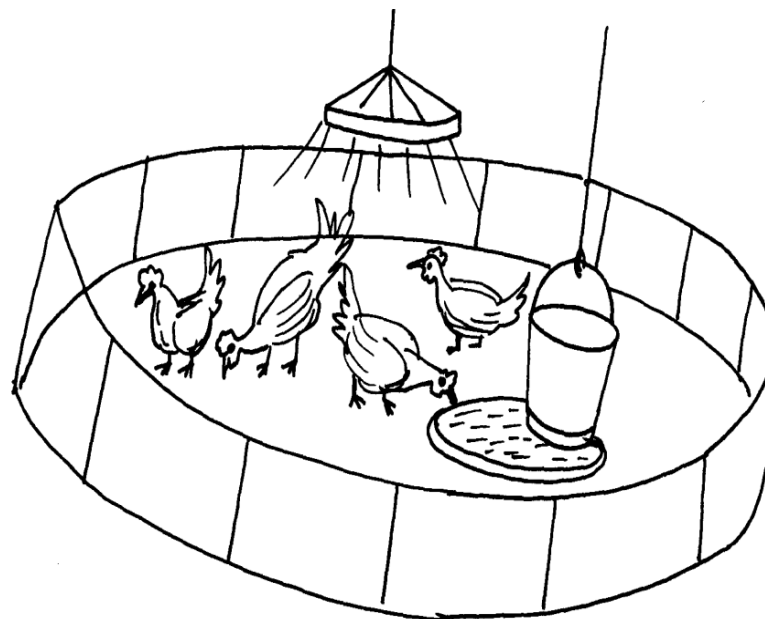
- ✓ When the chicks arrive they should be fed on chick mash for 8 weeks.
- ✓ The feed must be kept as clean as possible.
- ✓ The feeders should be kept clean everyday.
- ✓ Should be provided with plenty of clean water.
- ✓ Chicks should be vaccinated against gumboro disease after 2 weeks, newcastle at 3-4 weeks of age and fowl typhoid at 7 weeks of age.
- ✓ Coccidiosis disease controlled by giving coccidiostats to chicks through water or feed.
- ✓ The chicks should be dusted with appropriate chemicals to control external parasites.

- ✓ Roosts for the chicks to perch on are introduced in the 6th week.
- ✓ Grit/ sand is also given to help in digestion of the feed.
- ✓ In the 7th week growers mash is introduced gradually which is mixed with a ration of chick mash.
- ✓ The amount of chick mash is reduced gradually while increasing the amount of growers mash.
- ✓ When the chicks are 8 weeks old they are removed from the brooder.
- ✓ In the 9th week chicks are feed on growers mash only.

Brooder



Brooder



Requirements/ qualities of artificial brooder.

1. **Litter** – it is made from wood shavings/saw dust to maintain warmth, absorbs moisture and keep the chicks busy preventing cannibalism.
2. **Proper ventilation/ fresh air.-** should have holes on the walls for ventilation to allow proper gaseous exchange.
3. **Equipment-** they include:
 - a) **Feeders-** should have adequate feeders to allow for proper feeding without overcrowding.
 - They should be made in a such a way that the chicks do not step or defecate on the feeds.
 - To check whether the feeder are enough the farmer observes whether the chicks are feeding without overcrowding.

- b) **Waterers-** should have adequate waterers to allow for proper watering of chicks without overcrowding.
 - They should be kept clean and made in a such a way that the chicks do not step or defecate on them.
 - They should have the pointed top on which the chicks cannot perch.
- 5. **Well lit-** to allow chicks to see feeds and water.
- 6. **Shape of the brooder-** It should be round to avoid overcrowding at the corners which may cause suffocation.
- 7. **Heat source-** heat sources e.g. electric bulbs, lantern, kerosine and gas burners should be provided and controlled to maintain correct temperatures within the brooder.
 - There should be a wire guard round the heat source to prevent the chicks from being burnt when they crowd around it.
 - The temperature should be maintained at the optimum range.
- 8. **Dim light-** Dim or dull lights are recommended because too blight light has blinding effects on chicks and influence toe pecking.

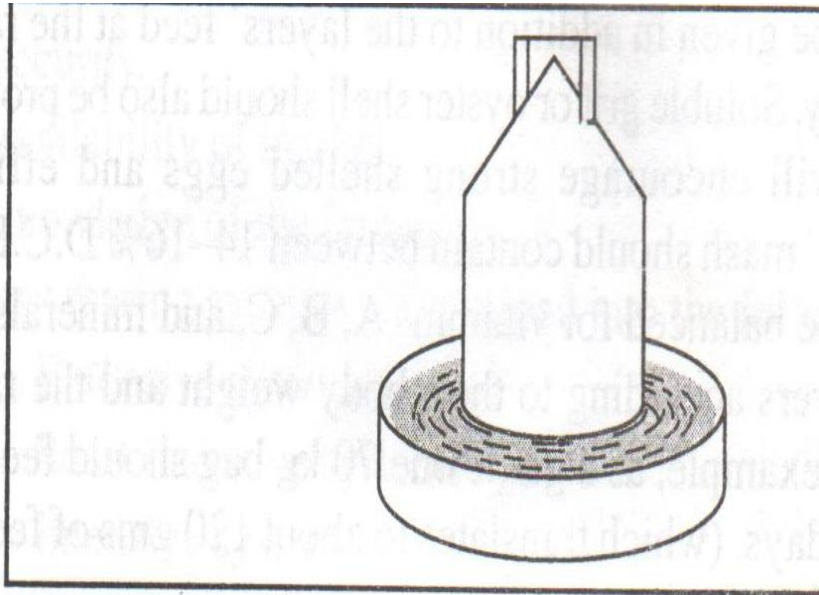


Fig. 1.8: Chicken Waterer

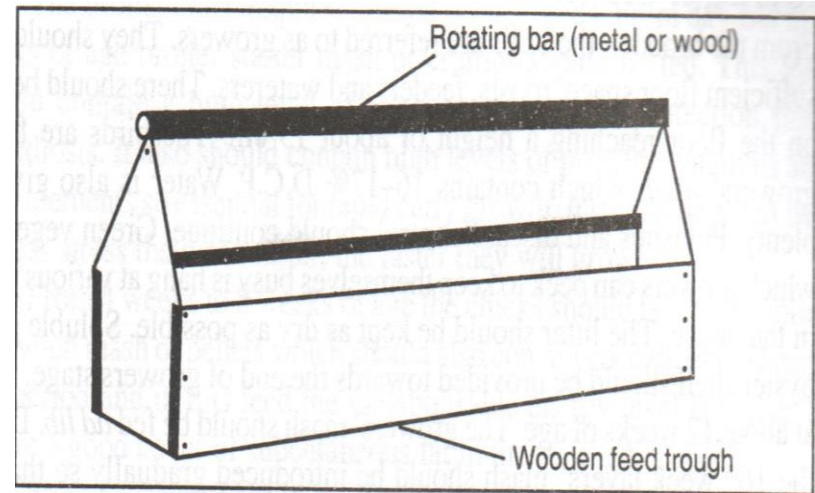


Fig. 1.7: Wooden feed trough

Methods of checking temperature in the brooder

1. Use of a thermometer.
2. Checking the behavior of chicks.
 - ✓ When the chicks move away from the heat source then the temperature is high.
 - ✓ If they overcrowd around the heat source then the temperature is low.
 - ✓ If they overcrowd at one side then there is draught from the opposite side.
 - ✓ When temperature is optimum then the chicks are evenly spread all over the brooder floor.

Behavior of chicks under heat stress.

- i. They move away from the heat source.
- ii. Opening/ spreading of wings.
- iii. They lie flat on their bellies.
- iv. Drinking a lot of water.
- v. Opening of beaks.

Causes of high mortality of one day old chicks.

- i. Coldness/chillness.
- ii. Too much heat.
- iii. Lack of feeds and water.
- iv. Parasites and diseases.

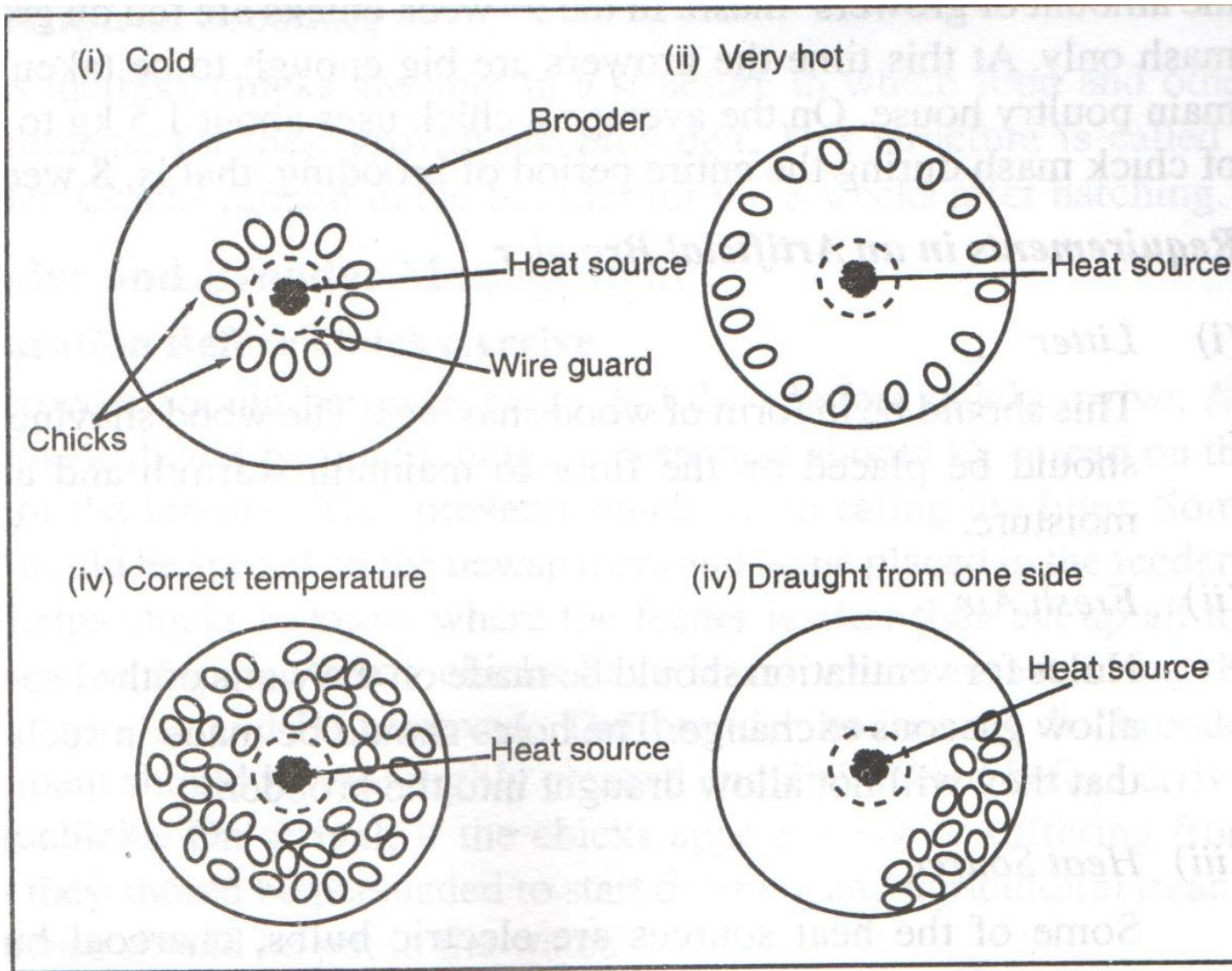


Fig. 1.6 Chick behaviour with heat variation

Study question

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1. Describe preparation of a brooder before the arrival of chicks.

- Prepare a brooder 2-3 days before the arrival of chicks.
- Wash and disinfect the brooder.
- Ensure that the brooder has round corners.
- Ensure that it is well ventilated but draught free.
- Ensure that the brooder is spacious.
- Spread litter on the floor.
- Spread newspapers on the litter.
- Provide enough clean feeders and waterers.
- Sprinkle some feeds on the newspapers when the chicks arrive.
- Provide dim light/right temperature.
- Provide a wire guard around the heat source.

2. Describe management of chicks in the brooder.

- On arrival supply water mixed with glucose.
 - Feed chicks on fresh chick mash.
 - Provide clean water.
 - Provide adequate feed.
 - Vaccinate the chicks if they were not vaccinated in the hatchery.
- Clean and disinfect the brooder occasionally to control parasites and diseases.
 - Remove any dead chicks and properly dispose them off.
 - Check and maintain the appropriate range of temperature in the brooder.

- Dust against ectoparasites.
- Deworm against endoparasites.
- Provide coccidiostats in water/feed to control coccidiosis.
- Provide dim light to avoid toe pecking.
- Introduce roosts from 6 week.
- Gradually introduce growers marsh from 7th week.
- Isolate and treat sick chicks
- Properly dispose off dead chicks.
- Provide grit to chicks.
- Keep proper records.
- Debeak at 8-10 weeks of age.
- Change the feeds gradually.

growers/pullets (between 9th week-point of lay) in a poultry house.

- Feed the growers on growers mash.
- Supplement with grains.
- Hang greens in the house to keep them busy and to provide vitamins.
- Introduce soluble grit during the last week to provide calcium important for shell formation.
- Provide feed and clean water *ad libitum*.
- Vaccinate against diseases.
- Dust against ecto -parasites.
- Drench against endo- parasites.
- Control predators.
- Provide enough waterers and feeders.
- Clean and disinfect the waterers and feeders regularly.
- Provide enough space/avoid overcrowding.
- Keep litter dry by regular turning/ scattering grains on the floor for the birds to scratch.
- Provide enough roosts and perches for birds to perch on.
- Introduce layers mash gradually from 18th week.

Management of layers in a poultry house.

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- ✓ Provide enough space.
- ✓ Keep the litter dry.
- ✓ Hang greens for exercise, provide vitamins and prevent cannibalism.
- ✓ Scatter some grains on the floor for exercise and turning of litter.
- ✓ Provide enough roosts and perches for birds to perch on.
- ✓ Provide enough feeders and waterers.
- ✓ Keep the equipment clean.
- ✓ Provide layers mash or pellets and clean water *ad libitum*.
- ✓ Supplement with grains, vitamins and minerals.
- ✓ Provide soluble grit or oyster shell which encourages strong shelled eggs and efficient digestion.
- ✓ Collect eggs twice a day (at noon and evening).
- ✓ Dispose off broken eggs.
- ✓ Debeak when necessary.
- ✓ Provide enough grit.
- ✓ Cull poor layers and cannibals.
- ✓ Isolate and treat sick birds.
- ✓ Vaccinate against newcastle and fowl typhoid diseases every 6 months.
- ✓ Avoid stress factors.
- ✓ Dust against ectoparasites.
- ✓ Deworm against endoparasites.
- ✓ Keep proper records.
- ✓ Provide enough laying nests.

Causes of low egg production in layers.

1. Overcrowded housing.
2. Fighting /pecking.
3. Lack of adequate clean water.
4. Parasite infestation.
5. Inadequate feeding.
6. Old age.
7. Broodiness.
8. Inadequate waterers/feeders.
9. Presence of predators /strangers.
10. Sudden changes of feeds.
11. Sudden noise.
12. Sudden change in weather.
13. Disease infestation.
14. Inferior/poor feeds.
15. Inadequate laying nests.

Management practices that ensure maximum egg production.

- ❑ Proper feeding
- ❑ Provision of enough laying nests
- ❑ Proper housing
- ❑ Parasite and disease control
- ❑ Regular collection of eggs
- ❑ Provision of correct light intensity
- ❑ Protection from predators
- ❑ Debeaking to stop cannibalism
- ❑ Provision of greens to keep them busy and for exercise

Categories of poultry feeds

- i. Chick marsh
- ii. Growers marsh
- iii. Layers marsh
- iv. Broiler starter marsh
- v. Broiler follow on
- vi. Broiler finisher

Management/rearing of broilers in a poultry house.

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- Feed the chicks on broiler starter mash.
- The feed should contain coccidiostat to control coccidiosis.
- Provide enough feed with high level of proteins, vitamins and mineral salts and clean water which are essential for rapid earlier growth..
- Gradually introduce broiler finisher mash.
- Have the correct stocking rate.
- Ensure proper ventilation
- Maintain hygiene in the house.
- Dust against ectoparasites.
- Deworm against endoparasites.
- Vaccinate against common diseases e.g. new castle.
- Dispose off dead chicken properly.
- Provide adequate waterers and feeders.
- Keep the litter dry and free from dust.
- Ensure adequate supply of greens and grains.
- Isolate and treat sick birds.
- Keep proper records.

Practices that help in control of parasites and diseases in a poultry house.

- ❑ Vaccination against diseases.
- ❑ Dusting against ecto parasites.
- ❑ Deworming against endo parasites.
- ❑ Treating sick birds.
- ❑ Isolating sick/dead birds to prevent the spread of diseases.
- ❑ Washing and disinfecting waterers and feeders.
- ❑ Keeping the litter dry.
- ❑ Providing a foot bath at the entrance to prevent infections.

Measures observed to prevent disease outbreak in poultry house/ Management practices that ensure proper hygiene in poultry house.

- ❑ Provide a foot bath at the entrance.
- ❑ Screen the poultry house against dust and wild birds.
- ❑ Remove any dead birds and dispose off properly.
- ❑ Control visitors into the poultry houses.
- ❑ Regularly clean and disinfect feeders and waterers.
- ❑ Ensure that the feed is free from pathogens.
- ❑ Isolate/cull sick birds.
- ❑ Treat sick birds.

POULTRY REARING SYSTEMS.

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Factors affecting the choice of a rearing system.

- i. Availability of land for rearing.
- ii. Topography of land to facilitate drainage.
- iii. Availability of labour.
- iv. Availability of appropriate equipment.
- v. Availability of capital.
- vi. Security.
- vii. Availability of market.
- viii. Knowledge of the farmer.

□ The rearing systems include:

- A. Extensive system.
- B. Semi-intensive system.
- C. Intensive system.

A. Free range system.

- ✓ This is an extensive system where the hens are allowed to move freely in a fenced ground with a simple house to provide shelter at night.
- ✓ Laying nests are also placed in this house.
- ✓ Feed and water troughs are placed outside but protected from rain by a simple shelter.

Requirements of free range system

- a) **Land-** it must be large, well drained and with trees for shelter. The external fence is required to prevent the birds from straying.
- b) **Runs-** these are partitioned areas within the land to facilitate rotation. This reduces disease and parasite build up.
- c) **House-** it should be constructed to provide shelter and an area for laying nests.
 - ✓ The house can be movable or constructed at the centre of the runs.

Advantages of free range/ extensive system.

1. Less feed is used since birds pick insects and vegetation from the ground.
2. Manure is evenly spread within the runs which facilitate vegetation to regenerate.
3. Cannibalism and egg eating are reduced because the birds are not crowded and are free within the runs.
4. It requires low capital investment.
5. Birds can pick grit from the soil hence there is no need of providing them with grit.
6. Birds have plenty of exercise thus helping to keep good health.

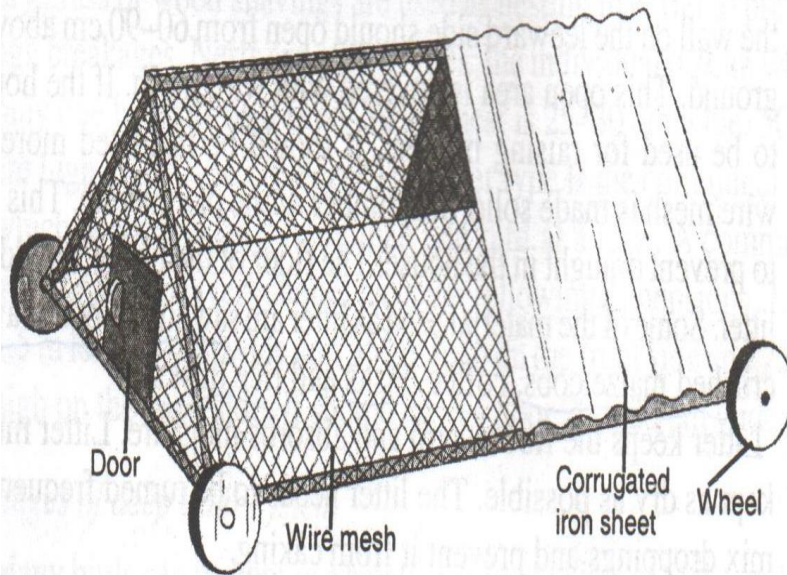
Disadvantages.

1. Requires a lot of land which may not be available if many birds have to be kept.
2. Birds can be stolen or eaten by predators.
3. Eggs can get lost in the runs.
4. Eggs can get dirty.
5. It is difficult to keep proper records/ close supervision for individual birds.
6. Breeding programme is not easy to follow.
7. The range can be easily infested with diseases and parasites/ may lead to parasite and disease spread.
8. Birds can cause damage to crops if the perimeter fence is not properly constructed.

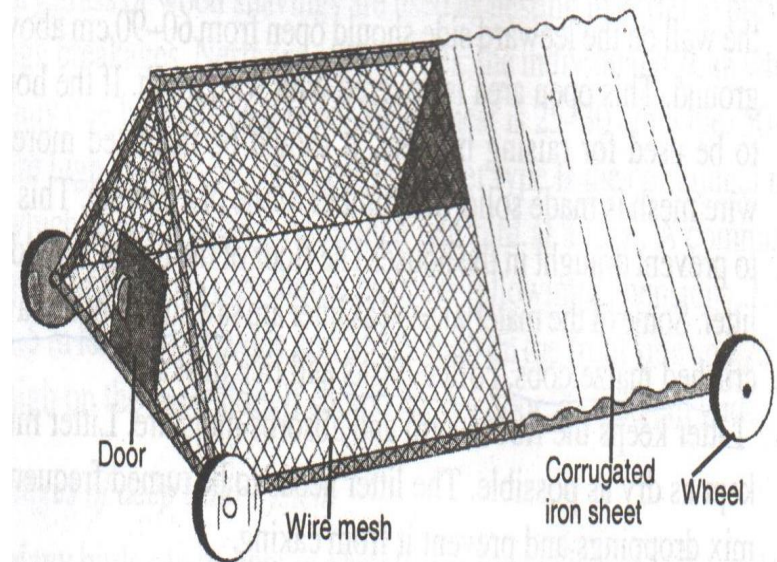
B. Semi-intensive system/fold system.

- ✓ In this system birds are confined in movable structures called **folds/arks.**
- Folds are moved daily to:
 - i. Reduce build up of diseases and parasites.
 - ii. provide fresh grass.
 - iii. Avoid accumulation of droppings.
 - iv. Allow even spread of manure which facilitate vegetation regeneration.
- For easy the movement the folds/arks are fitted with wheels.
- One part of the part of the fold is covered with corrugated iron sheet/ metal/ plastic/grass (thatch) for shelter and laying/for laying nests.
- The other part is covered with wire mesh for:
 - i. Sunning/basking and,
 - ii. feeding on vegetation.
- The feeders and waterers should be placed on the open side of the fold.

Fold system/ semi-intensive system



Fold system/ semi-intensive system



Advantages of fold/semi-intensive system.

1. Less feed is used as birds eat vegetation.
2. There is even distribution of manure/droppings.
3. Requires less feeding/Birds pick grit from the soil.
4. Reduces the build up of parasites and diseases.
5. Protects birds from predators.

Disadvantages of fold system.

1. Few birds are kept per fold/ accomodates few birds.
2. It requires a lot of labor to move folds from one place to another.
3. It is difficult to keep production records for individual birds.
4. Egg eating and cannibalism are very common.
5. Damage/breakage of the fold due to frequent movements.

c. INTENSIVE REARING SYSTEMS.

- In this system, birds are confined in houses or structures throughout the laying period.
- All the necessary requirements are provided in the structures.
- There are two intensive rearing systems, namely:
 - i. Deep litter system.
 - ii. Battery cage system.

i) Deep litter system.

- In this system, birds are totally confined in a house and stay indoors for the whole of their lives.

Requirements of a deep litter system

1. Site- the area where the house is built should be well drained.
2. House- it should have the following features:
 - a) A roof which is leak proof.
 - b) The wall on the leeward side should open from 60-90cm above the ground for ventilation. The open area is covered with wire mesh
- The wire mesh is made solid with timber, stone or concrete to prevent draught in the house.

- c) The floor of the house should have litter 15-30 cm deep made from sawdust, wood shavings, crushed maize cobs, coffee husks and rice husks.
 - Litter keeps the floor warm and absorbs moisture.
 - The litter should be kept as dry as possible, turned regularly to mix droppings and prevent it from caking.
 - Grains are thrown on the litter. As the birds scratch for
 - When new litter is introduced, it should be mixed with a little of the old so as to introduce bacteria which help the litter to rot
 - The floor space requirement should be 1 square metre per 2-3 birds.
3. **Feeders and waterers-** they should be enough and kept clean. They should be placed at various points in the chicken house so that chicken do not crowd and contaminate the feed and water.

4. **Roosts and perches-** they are timber frames on which birds perch. They should be moveable so that droppings do not accumulate too much at certain points causing dampness.
5. **Nests-** they are laying boxes.

- Laying nests should have the following requirements:
 - i. Large enough to accommodate the bird
 - ii. Kept in a secluded place
 - iii. Kept dark to avoid vices
 - iv. With dry nesting materials
 - v. With lockable doors

Advantages of deep litter system.

1. Many birds can be reared in a small area hence allowing high stocking rate.
2. There is no loss of eggs.
3. There is fast accumulation and collection of manure.
4. It requires less labour in collecting eggs, and turning the litter.
5. Birds are safe from predators and thieves.
6. The system can be used in rearing breeding stock.

Disadvantages of deep litter system.

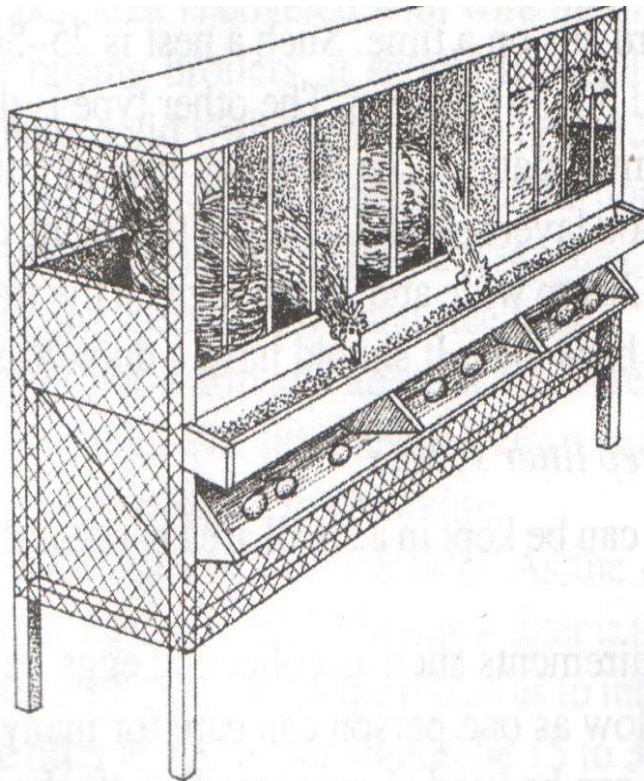
1. The incidence of cannibalism, egg eating, feather plucking and toe-pecking are common.
2. There is likelihood of parasite and disease accumulation in the litter.
3. It is difficult to keep record of individual birds.
4. Litter may be difficult to find in some areas.
5. Eggs may get dirty especially if laid on the floor or if the nesting boxes are not clean.
6. Litter can contaminate water and food.



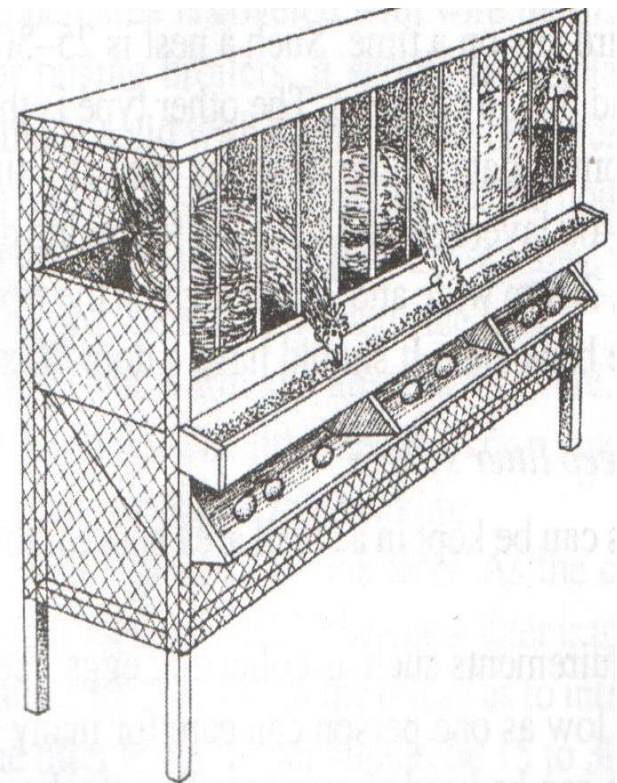
Fig 1.11: Deep litter system.

ii. **Battery cage system.**

- In this system birds are confined in wire mesh cages throughout the laying period.
- The cages are arranged in rows which are built over one another to form tiers.
- The cages have slanting floors to allow rolling of eggs to the tray. The eggs are collected from a tray behind the tiers.
- The lowest tier is normally 60-90cm above the ground to allow for easy cleaning.
- Feeders and water troughs are placed on the front side of the cages.



Battery cage system



Battery cage system

Advantages of battery cage system.

1. There is higher egg production due to less energy wastage by birds.
2. Accurate egg records can be kept because it is easy to know which bird has laid.
3. Cannibalism and egg eating are controlled.
4. Eggs are clean because birds do not step on them.
5. The system can easily be mechanized.
6. Birds do not contaminate food and water.
7. Handling is easy as hens are restricted to a small place.
8. Broodiness is discouraged as the birds do not reach the eggs.
9. A large number of birds can be kept in a small space hence higher stocking rate.
10. Sick birds can be detected easily and isolated for treatment.
11. Wire floors prevent re-infestation of parasitic worms and coccidia.
12. There is no bullying during feeding.
13. There is low labour requirement.

Disadvantages of battery cage system.

1. It requires high initial capital hence very expensive.
2. It requires high level of management or technical knowledge.
3. In case of disease outbreak, spreading is very fast.
4. Birds develop bruises on combs, breast and toes as they stick their necks out to eat and walk on the cages.
5. The birds do not have enough room for exercise.
6. The system is not good for brooding purposes.

STRESS IN POULTRY.

- Stress is any cause of discomfort in birds.

Causes of stress in poultry.

1. Any sudden change e.g. change of feed.
2. Presence of strangers and predators e.g. mongoose in the poultry house.
3. Handling of birds e.g. during vaccination, culling, debeaking etc.
4. Sudden noise e.g. passing of tractor or thunder.
5. Sudden change of weather e.g. from warm to cold weather.
6. Diseases and parasite infestation.
7. Lack of food and water.

Control measures of stress.

1. Keep the poultry house quiet by building it away from road where people and vehicles pass.
2. Insulate the poultry house to maintain uniform temperature.
3. Control diseases and parasites.
4. Change of routine program should be gradual.
5. Provide enough feed and water.

POULTRY VICES.

- A vice is abnormal behavior in birds and include egg eating and cannibalism.

a) Causes of egg eating

1. Presence of broken or soft-shelled eggs.
2. Bright light in the nests allowing the birds to see the eggs.
3. Idleness.
4. Inadequate nests forcing birds to lay eggs on the floor.
5. Lack of minerals e.g. calcium making the birds to look for them from elsewhere.
6. Inadequate feed.

Control of egg eating.

1. Collect the eggs regularly.
2. Making the laying nests dark.
3. Debeaking perpetual egg eaters.
4. Keeping the birds busy by supplying green leaves.
5. Feeding the birds on balanced diet/ ration
6. Scatter the grains on the litter to enable birds to scratch for them.
7. Provide enough feed.

- b) **Cannibalism-** this is a condition where birds peck each other.

Causes of cannibalism in poultry

1. **External parasites-** a bird may want to remove a parasite e.g. flea from combs of others.
2. **Overcrowding-** this makes the birds see others closely and it is easy for them to detect what to peck at.
3. **Bright light-** this makes the toe of chicks shine leading to toe pecking.

4. **Prolapse-** it occurs when the cloaca does not retract after laying egg hence the cloaca is pecked at by others.
5. **Mineral deficiency-** unbalanced feed makes the birds to try to get minerals elsewhere.
6. **Introduction of new bird in a flock-** this causes fighting leading to cannibalism after injury.
3. Provide birds with balanced feeds.
4. Birds should be kept according to age.
5. Control external parasites.
6. Keep the birds busy by hanging green leaves or vegetables in the house.
7. Debeak hens which peck others.
8. Cull perpetual cannibals.

Control of cannibalism

1. Avoid bright light in the house.

MARKETING OF POULTRY PRODUCTS.

1. Eggs

a) Sorting and grading of eggs.

Factors to consider when sorting eggs for market.

- i. Cleanliness-** eggs should be kept clean by wiping them using a damp clean piece of cloth.
- ii. The size of the egg-** large eggs have higher market price than small ones. Sorting should be done into 3 grades, small, medium and large before marketing.

- iii. Candling quality-** eggs of high candling quality are preferred.
- iv. Egg color-** some consumers prefer brown eggs to white eggs.
- b) Packaging of eggs-** they should be packed in trays with their broad end facing up to maintain gaseous exchange.
- c) Marketing of eggs-** they are sold directly to consumers or through cooperative societies.

2. Chicken meat.

a) Methods of killing birds-

1. Dislocating the head- the hen is starved for 12 hours to avoid the carcass from turning green.
 - The legs of the bird are held firmly with the left hand the head is twisted backwards and downward with a sharp thrust.
 - The bird is then hung by the feet so that blood may drain from the body.

2. Killing with a sharp knife- the legs and wings are held firmly and the neck is cut instantly.
 - The bird is then hung by the feet in order to allow blood drain.

Procedure of dressing chicken for sale.

- ✓ Lay the bird on its breast on the table.
- ✓ Pull and cut off the skin of the neck at the point where the neck enters the body.
- ✓ Cut the neck from the head at the point of dislocation/ the cut.
- ✓ Turn the carcass on its back.
- ✓ Grasp and gently pull the crop
- ✓ Insert the index finger as far as possible into the body cavity through the opening left after removing the neck and crop.
- ✓ Work around the finger close to the body to loosen the organs from the ribs and the back.
- ✓ Turn the bird and hold the abdomen with the left hand.

- ✓ Make a horizontal cut between the vent and the keel bone to expose the intestines.
- ✓ Insert a finger into the abdomen through the cut and loop around the vent to close the intestines.
- ✓ Hold the intestines away from the vent and then cut around with the bowels still attached.
- ✓ Insert the right hand into the abdominal opening gently grab the gizzard and pull it out accompanied with other internal organs.
- ✓ Cut the skin around the shank to remove the sinus just below the hocks and pull them out strongly from the body.
- ✓ Detach the shanks.
- ✓ Pack the carcass in polythene paper for sale.

Routine management practices in poultry.

1. Parasite control- Done:
 - i. By dusting, spraying or dusting the poultry house.
 - ii. Through deworming.
 - iii. By replacing the litter to prevent the parasite build up.
2. **Debeaking-** Done to prevent cannibalism and egg eating.

3. Feeding- Provide feed *ad libitum*.
4. Disease control- Done by use of antibiotics, vaccination and keeping the poultry house clean.
5. Egg collection- Done regularly to prevent egg eating in birds.

5. **Culling-** Refers to removal of unproductive birds from the flock.

Causes of culling in poultry/factors that necessitate culling.

- i. Poor growth.
- ii. Poor laying/production/yield.
- iii. Old age.
- iv. Vices.
- v. Poor health/Chronic diseases/parasite and disease attack.

Ways of differentiating between a poor and good layer.

- They include:
 - i. Observation of the physical characteristics.
 - ii. **Trap nesting-** a nest is built in such a way that when the hen enters to lay it does not come out.
- Those that enter to lay are marked thus those not marked are poor layers

Observing physical characteristics.

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Good layer.

1. Combs and wattles are large, warm, waxy and red.
2. Eyes are bright and alert.
3. The beak is pale.
4. The vent is oval, moist, reddish and active.
5. Abdomen/breast is soft.

Poor layer.

1. Combs and wattles are small, cold, dry and scaly.
2. Eyes are dull and pale yellow.
3. The beak is yellow in colour.
4. The vent is round, dry and less active.
5. The abdomen/breast is hard.

Good layer

6. The space between the keel and pelvic bone is wide i.e. can fit 3-4 fingers.
7. Temperament (It is alert and active).
8. Plumage/feathers are dry and rugged.
9. Moulting starts late.
10. Shanks are pale.
11. Broodiness is rare.

Poor layer

6. The space between the keel and pelvic bone is narrow i.e. can fit 1-2 fingers.
7. Temperament(It is lazy and dull).
8. Plumage/Feathers are beautiful/glossy.
9. Moulting starts early.
10. Shanks are yellowish.
11. Broodiness is common.

Marketing of eggs

Practices carried out on eggs in preparation for marketing.

- i. Checking out the abnormalities/candling.
- ii. Selecting eggs of the right size/weight.
- iii. Cleaning/wiping the dirt.
- iv. Sorting and grading.

Factors to consider when sorting and grading (selecting) eggs for marketing.

- i. Size of the egg.
- ii. Colour of the eggs.
- iii. Cleanliness of the egg.
- iv. Shell quality/ texture e.g. broken or rough.
- v. Candling qualities e.g. freshness
- vi. The shape of the eggs.

Desirable qualities of marketable eggs.

1. Large/heavy.
2. Brown.
3. Clean.
4. Smooth shelled.
5. Oval/normal shaped.
6. Fresh.

2.LIVESTOCK PRODUCTION VI

(CATTLE PRODUCTION)

RAISING OF THE YOUNG STOCK

a) Feeding of calves

- Ensure the calf suckles within the first 8 hours of birth.
- This is because within this time the digestive system is not fully developed hence colostrum is fully absorbed.
- Feed the calf on colostrum within the first 4-6 days.
- Introduce the feeding of whole milk after the fourth day.
- Feed the calf with warm milk at regular intervals.
- Introduce palatable dry feeds after the third week.
- Provide mineral supplements/licks.
- Any change in feed should be gradual.

Advantages/ reasons/ of feeding calves on colostrum.

1. It is highly digestible making it suitable for the digestive system of the calf which is not fully developed.
2. It is highly nutritious and contains vitamins for growth and disease resistance.
3. It has antibodies that enable the calf to resist early disease infection.
4. It has laxative effect i.e. it is good in cleaning the bowels of the calf.
5. It is highly palatable.

Preparation of artificial colostrum.

1. Whip a fresh egg in 0.86 litres of water.
2. Add the following:
 - i. $\frac{1}{2}$ litre of warm water.
 - ii. 1 teaspoonful of cod liver oil to provide vitamins.
 - iii. 1 tablespoonful of castor oil to provide laxative effect.
3. Mix the ingredients thoroughly and feed the calf three times a day.
- The castor oil is omitted in the mixture when the calf starts to pass dung normally.

Methods of calf rearing.

They include:

- A. Natural calf rearing method.
- B. Artificial calf rearing method.

A . NATURAL CALF REARING METHOD.

- In this method a calf is allowed to suckle the mother at will until weaning.
- The cow can suckle her own or 2-3 other calves hence called **foster mother**.
- Weaners are separated and fed on good quality pasture after weaning.
- Carry out all the management practices.
- Keep proper records.

Advantages of natural method.

- i. The calf takes milk at the right body temperature.
- ii. Milk is free from contamination.
- iii. Problems of scouring are minimized.

Disadvantages.

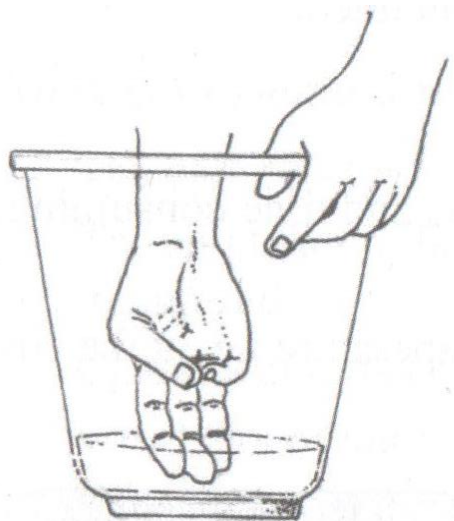
- i. The calf may be underfed (when the farmer removes all milk from the udder) or overfed leading to digestive disorders e.g. scours.
- ii. It is difficult to keep proper records of milk yield.
- iii. If the calf dies the cow dries up early.

B. ARTIFICIAL/ BUCKET FEEDING.

- In this method the calf is allowed to suckle colostrum for the first 8 hours then separated from the mother and reared in a calf pen.
- The calf is trained to drink milk from the bucket until weaning.
- All the management practices are carried out.

Procedure of training a calf for bucket feeding.

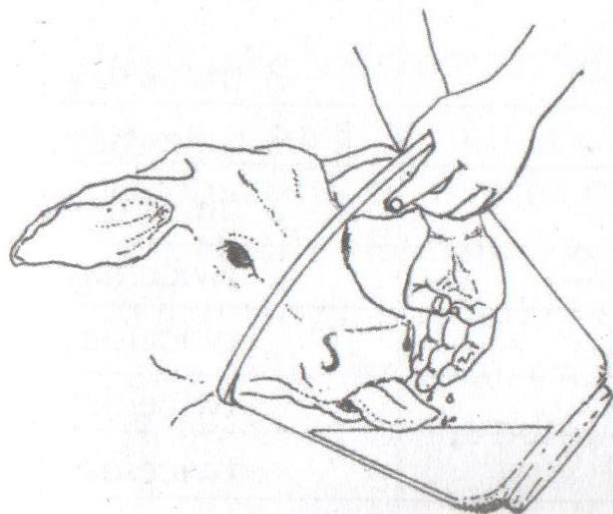
1. Put clean milk in a clean bucket.
2. Place the index finger into the calf's mouth. The calf starts sucking.
3. Lower the finger slowly until it is submerged in milk as the calf sucks. This allows the calf to drink milk.
4. Slowly withdraw the finger while the calf is sucking.
5. Repeat the steps until the calf starts to drink milk without any assistance.



(i)

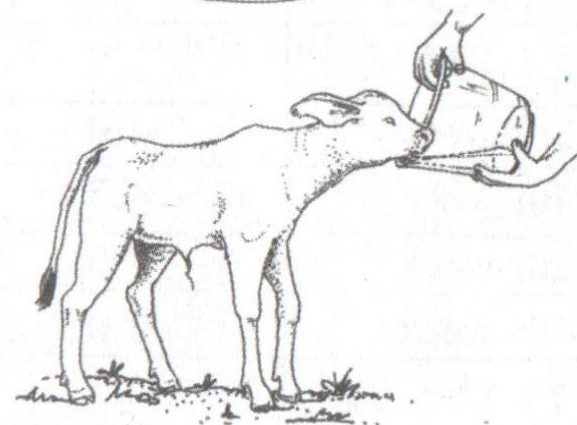


(ii)



(iii)

(iv)



Advantages of artificial/ bucket feeding.

1. It is easy to keep accurate records on milk production.
2. It is easy to control the amount of milk taken by the calf to avoid over/under feeding.
3. Cows can continue to produce milk even in the absence of the calf.
4. A farmer is likely to sell more milk thus maximizing profits.
5. It is easy to maintain high standards of sanitation.

Disadvantages

1. Training of a calf is tedious and time consuming.
2. Milk may not be fed at the right body temperature.

WEANING OF CALVES

- Refers to the practice of separating calves from the mothers (dams) and stopping them from suckling.

Weaning programmes.

A. Early weaning.

B. Late weaning.

A. EARLY WEANING PROGRAMME

- Done after 10 weeks.
- Feed the calf on colostrum during the first 4 days.
- The calf is fed on whole milk equal $\frac{1}{2}$ of its body weight up to eight week.
- Thereafter the amount of milk is reduced by 1kg until it is weaned.
- Introduce gradually early weaning pencils and young forage gradually after the 5th day along with whole milk.
- Introduce concentrates and clean water *ad libitum*.

B. LATE WEANING PROGRAMME.

- Done after 16 weeks.
- Feed the calf on colostrum in the 1st week.
- In the 2nd week provide 3.5. kg of whole milk per day.
- In the 3rd week provide 4.0.kg of whole milk per day.
- Provide milk at the right body temperature.
- From 4th week gradually replace whole milk with skim milk (without cream).
- From the 3rd week introduce calf pellets/pencils and green fodder gradually.
- Initially divide the milk into 2 equal parts and finally into 2 equal parts.
- Reduce the amount of milk gradually as the calf grows.
- Skim milk should be increased as whole milk reduces.
- From the 7th week the calf is not fed on whole milk.
- Introduce the concentrates as the calf copes with bulky solid feed.
- At the 16th week, introduce the calf fully to the forage crops.

CALF HOUSING.

- The calf should be protected from adverse environmental conditions e.g. wetness, high temperatures and chilling.

Types of calf pens.

- a) **Raised permanent pens with slatted floor-** it is raised 0.5 m above the ground with slatted floor to allow urine, dung and water to drain to keep it dry.
- b) **Permanent calf pen with concrete floor** the floor is slanted to allow drainage to take place.
- b) **Mobile calf pens-** are movable pens with slatted floor. They have roofs and are moved after every 2 days.

Features/Requirements of calf pens.

1. **Easy to clean-** they should have concrete floor to facilitate cleaning.
2. **Dry and warm-** dry litter should be placed on the floor to avoid dampness/ wetness.
3. **Adequate space-** it should be spacious for exercise, feeding and watering calves.
4. **Proper lighting-** it should be well lit because light is necessary for synthesis of vitamin D.
5. **Proper drainage-** should be constructed on a well drained place to avoid dampness.
6. **Proper ventilation-** to allow free air circulation.
7. **Draught free-** to prevent entry of cold winds and discourage infections e.g. pneumonia.
8. **Single housing-** to prevent disease and parasite spread and to prevent the formation of hair balls in the stomach.
9. **Leak proof-** the roof should not leak or avoid spilling water on the floor to discourage dampness and wetness of litter.

ROUTINE MANAGEMENT PRACTICES.

- 1) **Parasite control-** done through spraying and deworming/ drenching.
- 2) **Disease control-** done through vaccination against prevalent diseases.
- 3) **Identification-** done using appropriate methods e.g. ear tagging, tattooing, ear notching, branding and use of neck straps/ chains.

Reasons/ advantages of identification

- i) To facilitate selection and breeding.
- ii) To facilitate disease control and treatment.
- iii) To facilitate feeding, record keeping and culling,

- 4) **Castration-** done on male calves not required for breeding.

Reasons/ advantages of castration

- i. Castrated male calves are docile.
 - ii. For fattening.
 - iii. To control inbreeding.
 - iv. To control breeding.
 - v. To control breeding diseases e.g. brucellosis, vaginitis.
- 5) **Removal of extra teats-** done on females with extra (vestigial) teats using sterilized pair of scissors, sharp knife or teat clippers.
 - After cutting the wound is disinfected with antiseptics e.g. iodine to prevent infection.

Reasons for removal of extra teats.

- i) To prevent mastitis.
- ii) To prevent obstruction to machine milking
- iii) To avoid interference with development of normal teats.

- 6) **Disbudding/dehorning-**
Dehorning is the trimming of the horn to stop it from growing while **disbudding** is the removal of the horn bud before it develops into a horn.
- It is done using caustic potash stick (potassium hydroxide), dehorning wire/saw, disbudding iron, dehorning collodion, and use of rubber ring and elastrator.

Reasons for dehorning

- i) To make the animals docile and easy to handle.
- ii) To facilitate economic use of space when feeding, drinking or transportation of animals.
- iii) To reduce chances of causing injury to others and the farmer and avoid damaging the skin.

DISEASE CONTROL ROUTINE MANAGEMENT PRACTICES IN CALF REARING

1. Culling highly susceptible calves.
2. Spraying with appropriate acaricides to control ectoparasites/ ticks/ disease vectors.
3. Drenching/deworming with appropriate antihelminthes to control internal parasites.
4. Vaccination against diseases.
5. Castration of males not required for breeding to control venereal/breeding diseases.
6. Identification at the appropriate age to facilitate record keeping.
7. Removal of extra teats to control mastitis.
8. Proper feeding of the calf to prevent nutritional diseases.
9. Isolation and treatment of sick calves.
10. Maintenance of hygiene.

STUDY QUESTIONS.

- *Describe the management of a dairy calf using artificial rearing method from birth to weaning.*

Answer.

- ✓ Clean the mucous from the calf as soon as it is born.
- ✓ Ensure that the calf is breathing.
- ✓ Cut and disinfect the navel cord.
- ✓ Ensure that the calf sucks colostrum within the first 8 hours.
- ✓ Assist the calf to suckle.
- ✓ Introduce whole milk after 4th day.
- ✓ Ensure that the bucket is clean.
- ✓ Observe strict hygiene in the calf house.
- ✓ Provide adequate clean water.
- ✓ Ensure proper ventilation in calf house.
- ✓ Introduce green fodder in the 2nd or 3rd week.
- ✓ Give the calf early weaning pellets from the 3rd week.
- ✓ Wean at 8th week or late wean at 16th week.
- ✓ Take the calf outside on warm days for exercise and to get sunlight for manufacture vitamin D.
- ✓ Spray against ectoparasites.

- ✓ Drench/deworm against endoparasites.
- ✓ Vaccinate against infections.
- ✓ Dehorn/disbud the calf using the appropriate method.
- ✓ Put appropriate identification marks on the calf.
- ✓ Remove extra teats if any within the first 3 weeks after birth.
- ✓ Give the calf mineral supplements.
- ✓ Castrate male calves not intended for breeding.
- ✓ Treat sick calves.
- ✓ House the calves singly.
- ✓ Keep proper records.
- ✓ Reduce the amount of milk gradually towards weaning.
- ✓ Train the calf to feed from a bucket.

Describe the feeding practices in artificial rearing of a dairy calf.

- ✓ Train the calf to feed from a bucket (bucket feeding).
- ✓ Ensure the calf suckles the cow within the first eight hours to get colostrum.
- ✓ Feed the calf on colostrum for the first 4 days.
- ✓ Introduce the feeding of whole milk or milk substitutes after, the fourth day.
- ✓ Feed the calf 2 - 3 times per day for the-first-4 weeks.
- ✓ Feed the calf on the correct amount of milk up to weaning time.

- ✓ Provide adequate clean water from the third week.
- ✓ Feed the calf with warm milk at regular intervals.
- ✓ Introduce palatable, dry feeds such as concentrates (calf pellets) and good quality cut grass from the third week.
- ✓ Provide mineral supplements or licks.
- ✓ Any change in feeding should be done gradually to avoid nutritional disorders.
- ✓ Observe hygiene.

- **Describe the management practices that should be carried out to raise beef cattle using natural rearing methods.**

Answer

- ✓ Ensure that the calf is breathing.
- ✓ Apply artificial breathing if the calf is not breathing well.
- ✓ Cut and disinfect the navel cord to avoid infection.
- ✓ Ensure that the calf sucks colostrum.
- ✓ Leave the calf to stay with the mother to suckle at will.
- ✓ Wean the calf at 6-8 months.
- ✓ Separate the calves and allow them to graze on high quality pastures.
- ✓ Spray against ectoparasites.
- ✓ Dehorn the calves within the first 4 months.
- ✓ Castrate the bull calves not intended for breeding.

- ✓ Identify the calves as early as possible.
- ✓ Separate the uncastrated bulls from heifers at weaning time.
- ✓ Give mineral supplements where necessary.
- ✓ Deworm the calf regularly to control ectoparasites.
- ✓ Vaccinate the calves against prevalent diseases.
- ✓ Observe and treat sick animals.
- ✓ Separate sick animals from the healthy ones.
- ✓ Give supplementary feedstuff during dry periods.
- ✓ Provide adequate clean water all the time.
- ✓ Cull defective animals.
- ✓ Keep proper records.

Describe the feeding practices carried out on a calf from birth to weaning.

- ✓ Ensure the calf suckles the cow within the first 8 hrs to get colostrum.
- ✓ Feed the calf on colostrum for the first 4 days.
- ✓ Feed the calf 2-3 times per day for the first 4 weeks.
- ✓ Introduce whole milk/milk substitutes after the 4th day.
- ✓ Feed the calf on the correct amount of milk up to weaning.
- ✓ Feed the calf on warm milk / provide milk at the right body temperature to avoid scours.
- ✓ Provide adequate clean water for the calf from 3rd week.

- ✓ Introduce palatable dry feeds/ concentrates/ calf pellets/ pencils and good quality cut grass from the 3rd week.
- ✓ Any change in feed should be done gradually to avoid nutritional disorders.
- ✓ Clean the equipment used for feeding the calf.
- ✓ Train the calf to suck milk from the bucket.

Describe management practices that a farmer should carry out to improve milk production in a low yielding herd of dairy cattle.

Answer

- ✓ Select good animals on basis of high yielding cows.
- ✓ Select healthy animal.
- ✓ Select animals with high fertility
- ✓ Select animal with good dairy conformation.
- ✓ Cull poor producers.

- ✓ Use superior bulls/ semen from superior bulls to serve the cows.
- ✓ Breed heifers when fully mature, considering weight/age.
- ✓ Breed cows 60 to 90 days after calving to maintain a calving interval of one calf per year.
- ✓ Keep the animals healthy by vaccinating them regularly.
- ✓ Control external and internal parasites using appropriate drugs.
- ✓ Treat sick animals.
- ✓ Isolate the sick animals suffering from contagious diseases.
- ✓ Avoid physical injuries to the animals/predisposing disease factors.
- ✓ Improve sanitation/cleanliness in the farm.
- ✓ Feed cattle on a balanced ration/diet.
- ✓ Give adequate feed.

- ✓ Give clean and uncontaminated feed.
- ✓ Provide plenty of clean water.
- ✓ Provide minerals and vitamins supplements to the cow.
- ✓ Provide proper housing with good ventilation.
- ✓ Avoid overcrowding in the animal houses.
- ✓ Milk cows at regular intervals.
- ✓ Use proper milking technique.
- ✓ Handle the animals properly.
- ✓ Observe any signs of heat closely.
- ✓ Observe any signs of diseases and parasites closely.
- ✓ Keep proper records.

MILK.

- This is a white substance secreted in the mammary glands of all lactating mammals.

Milk composition/components.

- i. Milk sugar/lactose/milk carbohydrate.
- ii. Milk protein/casein.
- iii. Water.
- iv. Milk fat/butter fat.
- v. Minerals e.g. calcium and phosphorus.
- vi. Vitamins.

Factors affecting milk composition/quality of milk.

1. **Age of the animal-** butter fat becomes less as the animal grows old thus young animals produce milk with higher butter fat content than older animals.

2. **Physiological condition of the animal e.g. pregnancy, emaciation, sickness-** during late pregnancy and emaciation, cows produce milk with low butter fat content.
3. **Stage of lactation-** the butter fat content is highest at the middle phase of lactation period and lowers towards of end of lactation.
4. **Completeness of milking-** the last drawn drop from the udder contains high butter fat content.
5. **Breed differences-** different breeds of cattle produce milk of differing percentage composition e.g. jersey produces milk with the higher butter than Friesian.
6. **Season of the year-** butter fat content increases during cold season.
7. **Nutrition/type of feed eaten by the animal-** roughage feeds produce milk with higher fats, lactose and proteins compared to grains.
8. **Animal health/disease-** diseases like mastitis reduce lactose composition in milk because bacteria attack milk sugars. Animals under drugs produce milk of variable compositions.
9. **Time of milking-** milk produced in the morning has lower butter fat content than milk produced in the evening.

Factors affecting the quantity/amount of milk.

1. Stage of lactation- milk quantity increases for the first 30 days after which it slowly declines.
2. Length of lactation- amount/quantity of milk is higher if cows are milked for a long period of time.
3. Amount of water ingested- less amount of milk is produced if a lot of water is taken.
4. Handling of the animal during milking- rough handling/ beating leads to low amount of milk.
5. Prevailing weather conditions- stressing conditions leads to less amount of milk.
6. Amount and quality of feed taken by the animal- high amount of nutritious feed taken leads to high amount of milk.

STRUCTURE OF THE UDDER/ MAMMARY GLAND.

- It is composed of:
 1. Alveoli- they are milk secreting cells surrounded by capillaries.
 2. Lobule- consists of many alveoli. It receives milk from the alveoli.
 3. Lobe- consists of several lobules. It receives milk from the lobules through lactiferous ducts.
 4. Quarter- consists of many lobes. It has a gland cistern which receives milk from the lobes.
 5. Teat cistern- receives milk from the gland cistern.
 6. Teat canal/nipple/orifice- has sphincter muscle that allows milk out during milking.

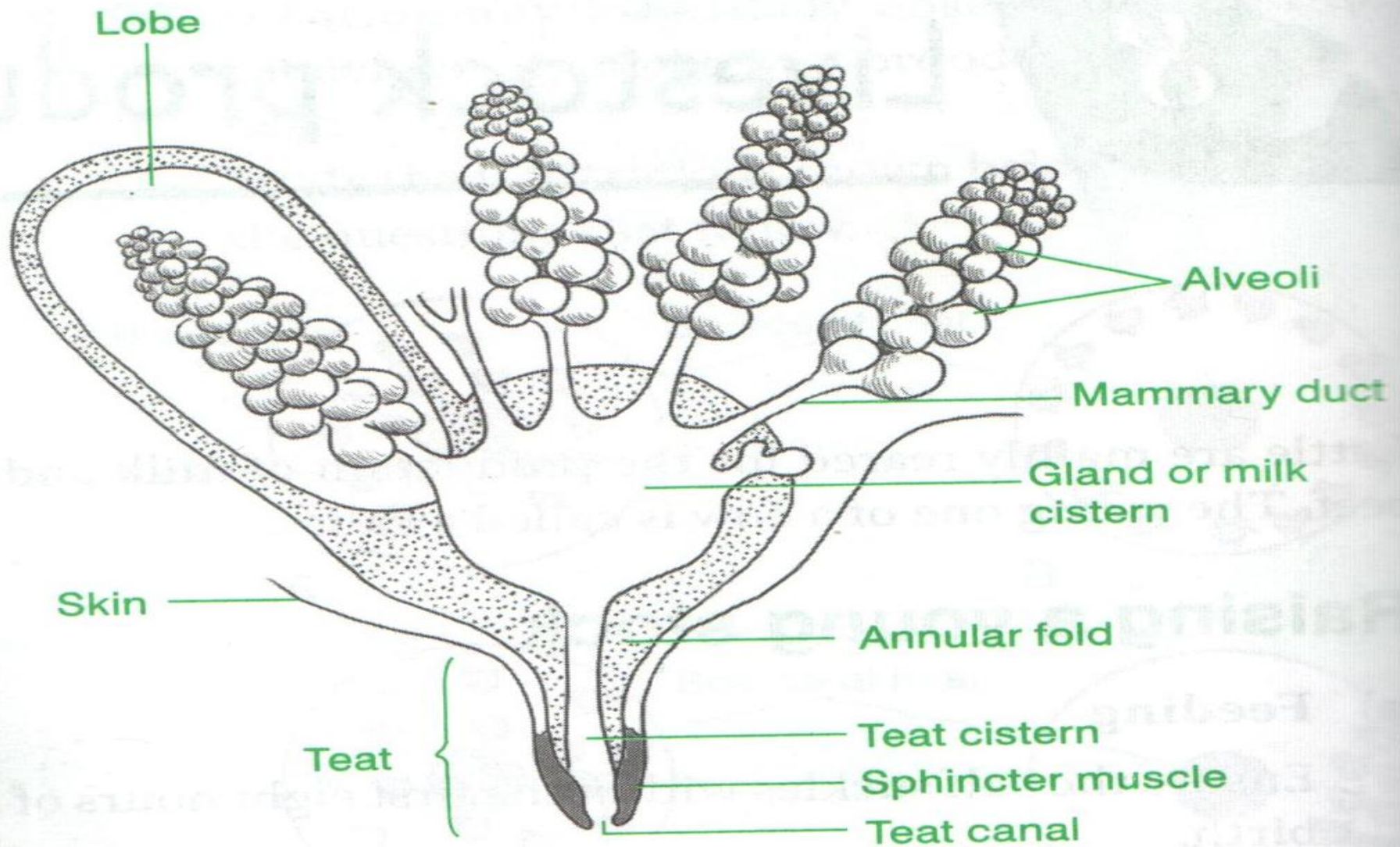


Fig. 8.1 The structure of the mammary gland of a cow

MILK SECRETION/

LACTOGENESIS.

- This is the formation of milk by secretory cells in alveoli. Milk is made from the final products of digestion.
- The alveoli are surrounded by a network of capillaries
- Blood sugar/glucose is converted into lactose/milk sugar, amino acids are converted into casein/milk proteins and fatty acids and glycerol are converted into butter fat/milk fat.
- Lactogenesis is controlled by the pituitary glands which secrete a hormone called **prolactin** during late gestation.
- Prolactin initiates lactogenesis in the alveoli which is stored in **the alveolar cavity, gland cistern and teat cistern.**

MILK LET-DOWN.

- This is the flow of milk from the upper region of the **udder/alveolar region to the gland and teat cisterns.**
- It is controlled by pituitary glands which secrete **oxytocin** hormone into the blood stream
- **Oxytocin** causes the muscles surrounding the alveoli to contract, squeezing the milk out.
- The milk then flows into the gland cistern through ducts
- Milking or suckling forces milk out.

Factors that stimulate milk let down.

- i. Sight of a calf.
- ii. Suckling of the calf.
- iii. Sight of milkman.
- iv. Washing/massaging the udder by warm water.
- v. Sight of feed/feeding.
- vi. Familiar sounds.
- vii. Regular milking routine.

Factors that hinder milk let-down.

- i. Pain.
 - ii. Fear.
 - iii. Anxiety.
 - iv. Sickness.
 - v. Change of milk man/milk routine.
 - vi. Strange surroundings.
 - vii. Poor milking techniques.
- These factors cause the **adrenal glands** to secrete **adrenalin hormone** which causes the relaxation of muscles hindering milk let-down.

CLEAN MILK PRODUCTION.

Qualities/characteristics of clean and high quality milk.

- i. Free from pathogens/ disease causing organisms.
- ii. Free from hair/dirt/foreign materials.
- iii. High storage/keeping quality.
- iv. Good flavor/taste.
- v. Its chemical composition is within the expected standards.

Essentials of clean milk production/ practices that ensure clean milk production.

1. Healthy milking herd/ cows- maintain healthy milking herd to prevent transmission of disease agents to milk and milkman.
 2. Clean milking cows- to prevent contamination of milk with dirt.
 3. Healthy milkman- to prevent transmission of zoonotic diseases.
 4. Clean milkman- to prevent contamination of milk.
 5. Clean milking shed- to prevent contamination of milk.
 6. Clean milking utensils/ equipment- to prevent contamination of milk.
- ✓ Utensils and equipment should be seamless, smooth with joints filled to facilitate easy and thorough cleaning.
 - ✓ They should be washed with hot water and sterilized by keeping them in the sun during the day.

6. **Milk filtration-** to remove solid impurities.
7. **Milk cooling and proper storage-** to give it long shelf life.
8. **Avoid flavours in milk-**
 - ✓ Strong flavored feeds (e.g. *Mexican marigold*, onion wastes, pineapple wastes e.t.c.) should be fed after milking to avoid flavors in milk.
 - ✓ Avoid direct exposure of milk to sunlight which causes oxidation of milk fats and avoid using utensils with traces of copper/iron.

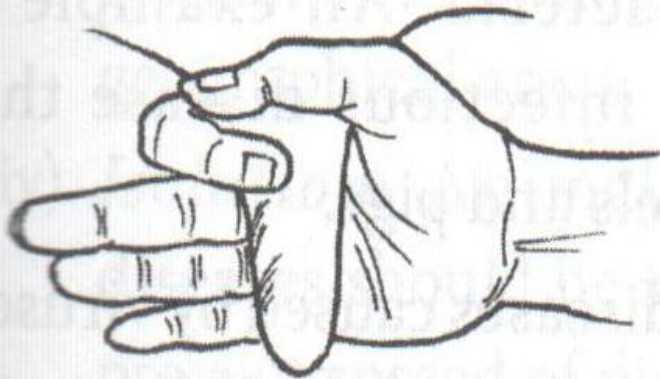
MILKING MATERIALS, EQUIPMENT AND UTENSILS.

1. **Udder clothes/towels-** for cleaning and drying the udder.
2. **Sieves/filters/Filtering pads-** filtering/ straining milk.
3. **Milking jelly/salve-** applied to teats after milking to prevent cracking of teats which predispose the cow to mastitis. It should not be applied before or during milking.
4. **Warm water-** washing the udder to remove dirt and stimulate milk letdown.
5. **Milking pails/buckets-** used for milking.

6. **Milking stool-** to sit on during milking.
7. **Strip cup-** for testing mastitis. The first strip of milk from each teat is put into the strip cup and checked whether it has clots/ flakes.
8. **Milk can/churn-** Hold milk during milking or transportation.
9. **Chain/rope-** Restraining the animal.
10. **Concentrates/feeds-** Stimulate milk let-down.
11. **Weighing scale-** for weighing milk.
12. **Cooler/ fridge-** for cooling milk.
13. **Milking machine-** for machine milking.

Milking procedure

- Milking is done by hand or machine.
- a) **Pre-milking procedure.**
 - Assemble all the equipment.
 - Put the cow in a crush/ milking parlour, restrain it and give it food.
 - Wash and dry the udder.
 - Check for the presence of mastitis.
- b) **Milking procedure**
 - Grasp the base of the teat tightly at the base between the thumb and the fore finger to prevent back flow of milk.
 - Sequentially close other three fingers tightly applying pressure from the top to the bottom forcing milk to drain into the bucket.
 - Relax the fingers simultaneously to allow the teat to be refilled and begin a new sequence.



The teat is grasped tightly at its base and the thumb is closed with the first finger.

The other three fingers are closed tightly to squeeze the milk out.

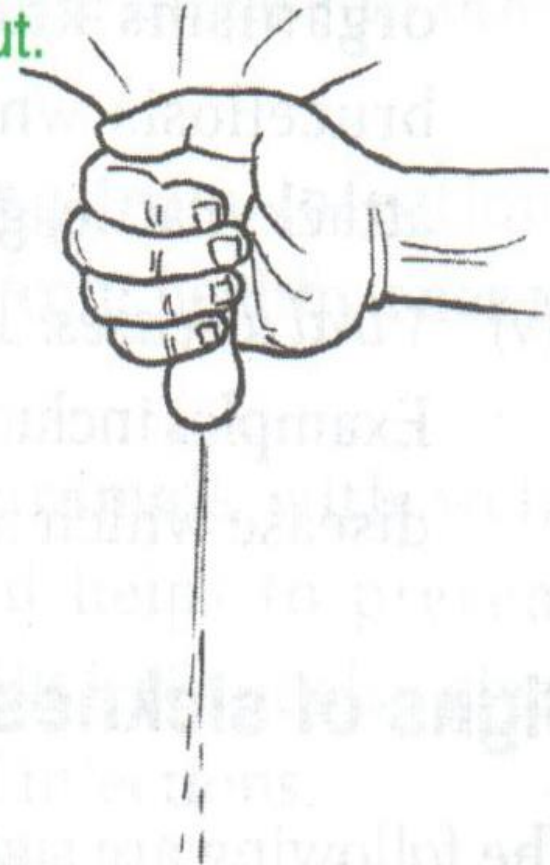


Fig. 8.2 Hand milking

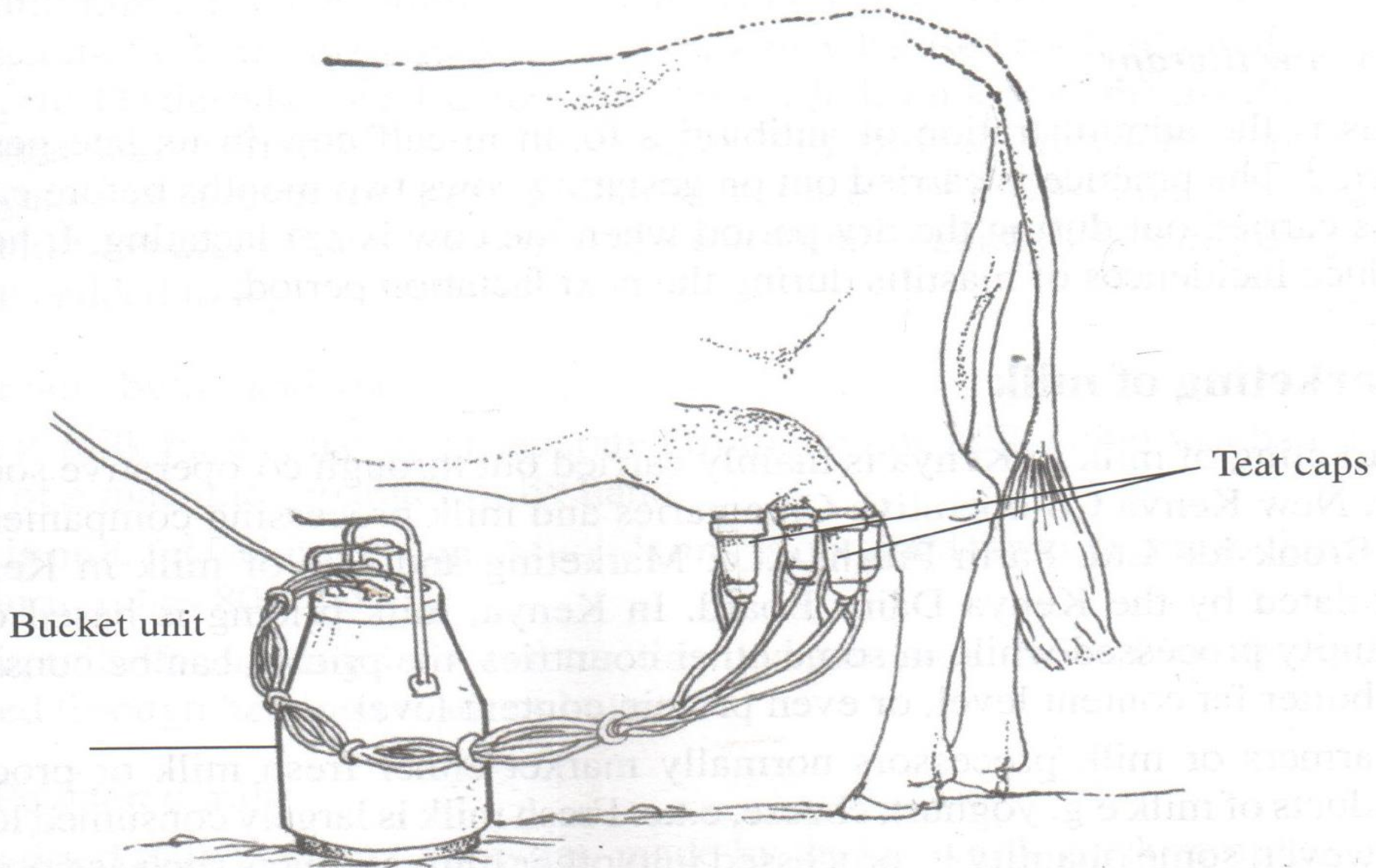


Fig. 2.4: Milking machine.

Machine milking.

- In this the machine is used to suck out milk from the teat cistern.
- When the teat cups are placed on the teat, the vacuum pump sucks and removes air from the sealed space creating a space.
- The vacuum created holds the teats, opens the teat canals and draws milk from the teat cistern into the milk tube.

Rules followed during milking

1. Avoid stripping the udder because:

- ✓ It causes physical injury on the teat causing scar.
- ✓ It tears the teat tissues making it pendulous/sag hence more prone to mastitis.
- ✓ It leads to high chances of milk contamination because milking salve will be necessary.

2. Milk quickly and evenly-
because oxytocin effect last for 5-10minutes.
3. Milking at regular times-
because milk let-down is initiated when milk time is reached.
4. Complete milking- to prevent drying off too soon, reduction in yield and prevent mastitis.
 - When using milking machine, the last drops of milk are stripped to avoid mastitis.
 - Hands should be dry as it is hygienic.

Post-milking practices./practices after complete milking.

- i. Teat dipping to control mastitis.
- ii. Application of milking jelly on the teats.
- iii. Weighing and recording of milk.
- iv. Sieving/ filtration/ straining of milk.
- v. Storage of milk in a cool place.
- vi. Cleaning the milking shed/parlor
- vii. Cleaning the milking equipment and utensils.

Disadvantages of hand milking over machine milking.

- i. It is slow.
- ii. Milk may be contaminated if the hands are dirty.
- iii. Partial milking may occur.
- iv. Poor milking technique can lead to mastitis.

DRY COW THERAPY.

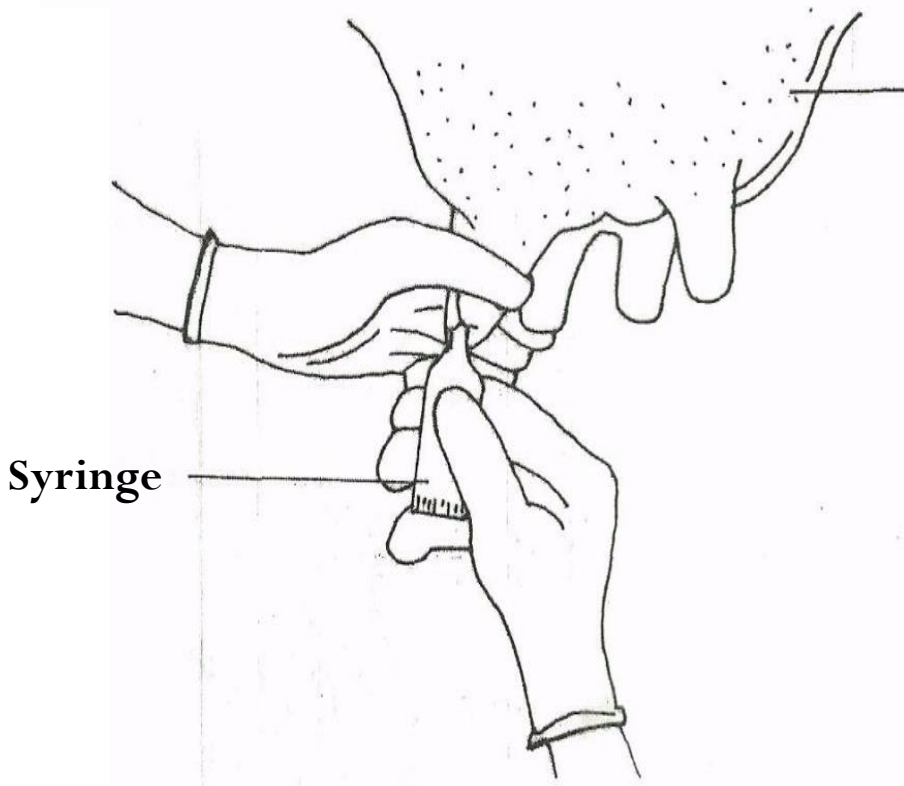
- ✓ It involves infusion of the infected teat/ quarter with antibiotics e.g. penicillin through the teat canal to kill bacteria/ prevent mastitis.
- ✓ It is also carried out during drying off period.
- ✓ This is because the little milk left at the teat canal acts as a culture medium for bacterial infection which cause mastitis.
- ✓ To confirm mastitis, stripping of milk into a strip cup is done before actual milking is done.
- ✓ Watery milk/ bloody milk/ clots/ pus on the strip cup indicate mastitis infection.

✓ To prevent spread of disease to other animals the following are done:

- i) Milking the infected animals last and discarding milk.
- ii) Using separate cloths for each animal.
- iii) Milking the infected quarter last.

✓ Other practices carried out on the udder to prevent mastitis infection include:

- i. Teat dipping.
- ii. Complete milking.
- iii. Proper milking technique.
- iv. Applying milking jelly after milking.



MILK PRODUCTS

1. Homogenised and pasteurised milk-

- **Homogenisation** involves breaking fat globules in milk to smaller particles which are evenly distributed in milk.
- **Pasteurisation-** involves heating the milk and then cooling immediately to destroy pathogens.

2. Ultra heat treated(UHT)milk- It is heat treated at a very high temperature for one second and then packed and cooled immediately.

3. Cream- A layer of fat that forms at the surface of milk when it is allowed to stand for some time.

4. Skim milk- This is milk left after the cream has been removed.

5. Butter - Obtained by churning the cream to separate the milk fat from non-fat solids.

6. Ghee - It is prepared by heating cream or butter in order to remove moisture.

7. Curd It is semi solid milk that forms after the whole/skim milk has been allowed to stand for sometime.

8. Cheese - Prepared by compressing milk curd until water has been drained completely.

9. Powdered milk- Done by drying the whole milk/skim milk and converting it into powder.

10. Yoghurt - It is thickened and flavored milk.

Marketing of milk

- Milk is sold through co-operatives, middle men or directly to consumers.

Market channels of meat /ways in which farmers sell meat/beef in Kenya.

- i. Kenya meat commission (KMC).
- ii. Livestock marketing division in the ministry of livestock development.
- iii. Local butcheries.
- iv. Licensed stock traders.

Qualities of good beef.

- i. Tender.
- ii. Good flavour.
- iii. Juicy.
- iv. Lean (soft with little fat).
- v. Cherry red colour.

Problems facing farmers in marketing of beef and milk.

- i. Lack of adequate storage facilities
- ii. Poor infrastructure
- iii. Lack of market information
- iv. Delayed payments of farmers
- v. Price fluctuations

3. FARM POWER AND MACHINERY

1

Advantages of farm mechanization.

1. Farm operations are carried out faster and on time.
2. Large areas can be covered within a short time.
3. It increases efficiency/better job is done.
4. There is high yield because operations are carried out on time.
5. It makes work easier and enjoyable.
6. Pest and diseases are controlled easily.
7. It encourages farmers to consolidate land.
8. Uses less labor.
9. Farmers benefit from economics of scale.

Limitations/disadvantages of farm mechanization.

- i. Machines are expensive.
- ii. Land fragmentation makes farm mechanization uneconomical.
- iii. Most machines are powered by a tractor that cannot be used on uneven topography.
- iv. Requires skilled labour.
- v. Unavailability of the spare parts to maintain the tractor.

SOURCES OF POWER IN THE FARM.

- They include:
 - i. Human.
 - ii. Animal.
 - iii. Wind.
 - iv. Water.
 - v. Biomass/bio-energy.
 - vi. Electricity.
 - vii. Solar energy.
 - viii. Tractor engine.

1. Human power

- Human beings work directly or utilize their capacity of organization.

Limitations.

1. It is slow.
2. Low work output.
3. Relies on the skill of the worker.

2. Animal power

- Draught animals are used to provide power.

Advantages of animal power.

- i. Require less skill as compared to tractor power.
- ii. Animals are cheaper to buy and maintain compared to tractor engine power.
- iii. Work output is higher than human power.
- iv. Can be work better in on small holdings than tractors.
- v. Animals can work in areas where it is impossible for tractors.

Disadvantages of animal power.

1. Require a big of land maintenance/grazing.
2. Are slower than tractors hence cannot cope with large amount of land.
3. They can damage crops when they are used for weeding.
4. They can fall sick hence reducing power output.
5. They require full maintenance/feeding when not in use.
6. Weather conditions may affect their efficiency.
7. They need training.
8. They easily get tired/fatigued.

Factors to consider in order to get optimum animal power.

- i. Training of animals.
- ii. Farmers should not be harsh to them.
- iii. Proper feeding.
- iv. Same sized animals should be harnessed in the same yoke.
- v. A yoke/cart of the right size should be used.
- vi. Proper timing to avoid harsh environmental conditions.
- vii. Proper harnessing of animals.

Instances where animal power is preferred to other sources of power

- i. Where land is not easily accessed by a tractor.
- ii. Uneven topography/uneven areas.
- iii. Where the cost of hiring a tractor is high.
- iv. Where land is small.

Factors that influence the power output of animals.

1. **Age-** mature animals produce more power than young animals.
2. **Breed and type-**indigenous animals are hardy hence have high power output than exotic breeds.
3. **Training-**properly trained animals have more power output.
4. **Proper harnessing-**properly harnessed animals have more power output.

3. **Wind power-** It is obtained from wind and is used for winnowing, turning turbines and pumping water.

Advantages.

- i. It is free and inexhaustible.
- ii. It requires minimal maintenance after installation.
- iii. Installation can be done even in remote areas.

Disadvantages/limitations.

- i. It is unreliable because of the flow and strength of wind.
- ii. Initial installation is costly.
- iii. It is limited to areas where wind is prevalent.
- iv. The direction of wind cannot be controlled.

4. **Water power-** Energy is obtained from running water. It is used for driving turbines to produce hydroelectric power, drive maize grinding mills and pumping of water.

Advantages.

- i. Once installation is complete production of power is continuous.
- ii. High power is produced.
- iii. After installation it can function for a long time.

Disadvantages/limitations.

- i. Most rivers are small and seasonal.
- ii. Very high initial capital required for installation.
- iii. Most farms do not have access to large quantities of moving water.

5. **Biomass/Bio-energy.**

- It is obtained from organic matter.
- It include:
 - i. Biogas.
 - ii. Wood fuel/charcoal.
- i. **Biogas.**
 - This is a combustible gas that is produced after fermentation of organic matter producing methane gas which is highly inflammable.
 - It is produced in the a structure called **digester.**
 - The main raw material is **animal waste.**

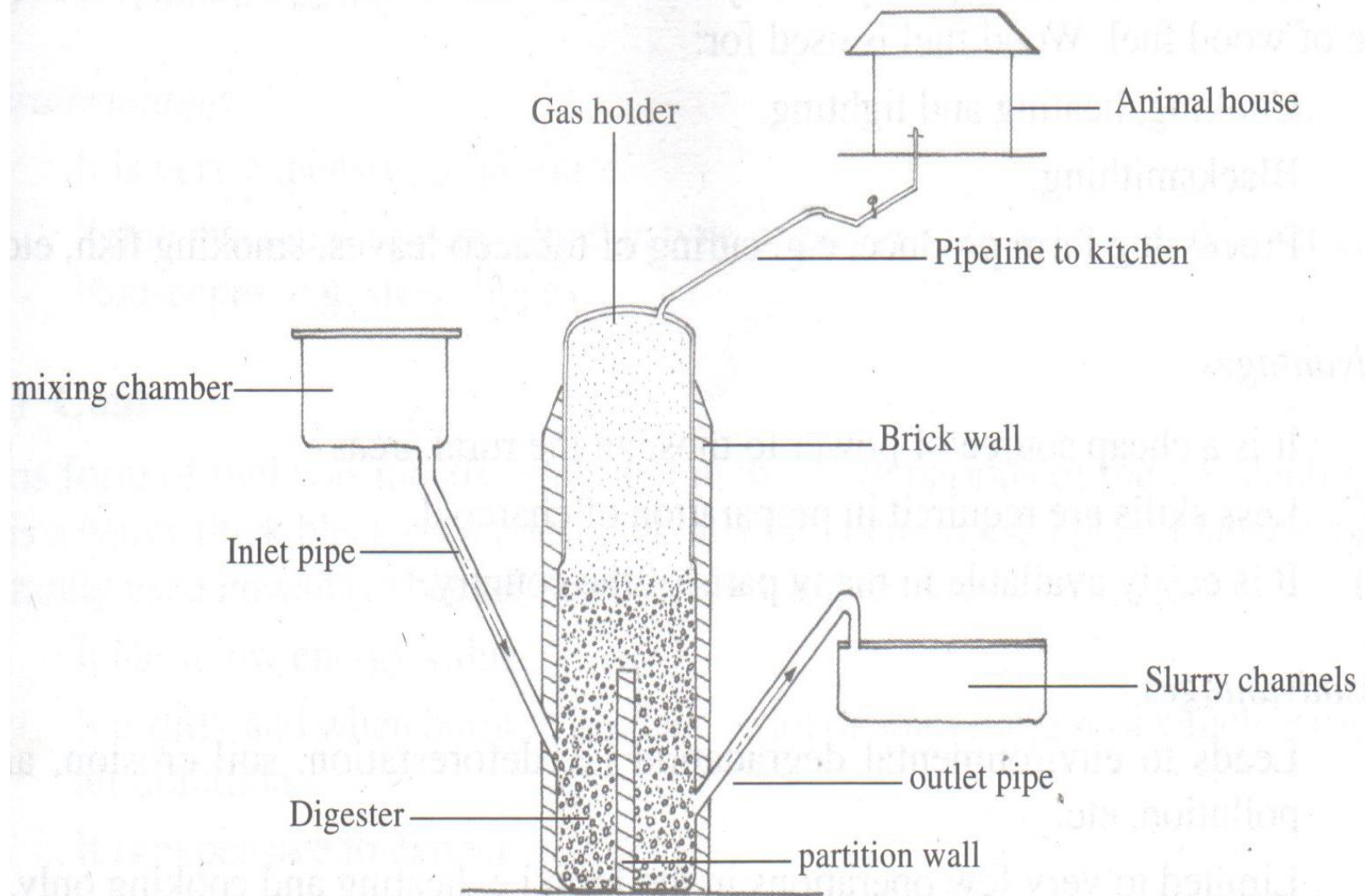


Fig.3.6: A Biogas plant.

Uses of biogas.

- i. Lighting and cooking.
- ii. As a source of power in internal combustion engines.

Biogas production.

- i. Organic materials are mixed with water in the mixing chamber and pushed into the digester by the inlet pipe.
- ii. After fermentation, gas bubbles/methane is produced which is directed into the kitchen for use.
- iii. **Slurry** is pushed out into the slurry pans which is collected and used for making manure.

Advantages /benefits of using biogas.

- i. It is economical for farmers with zero grazing cows and resources to construct a biogas plant.
- ii. It is put to many uses e.g. lighting, cooking etc.
- iii. The by-product is a better source of fertilizer.
- iv. Requires low maintenance costs.
- v. Potentially harmful wastes of cows and pigs are removed hence does not pollute the environment.
- vi. It is a renewable/sustainable source of energy.

Disadvantages.

- Requires high initial investment.
- It requires high management skills to produce the gas.
- Requires a large number of animals to sustain the production of gas/requires large amount of raw materials.

- ii. **Wood fuel and charcoal-** it is obtained from charcoal and firewood.

Advantages.

- i. It is cheap.
- ii. Requires less skill.
- iii. It is readily available.

Disadvantages.

- It leads to destruction of vegetation hence soil erosion, destruction of water catchment areas etc.

6. **Fossil fuel.**

- It includes:

A. Coal.

B. Petroleum.

C. Natural gas.

A. **Coal-** Obtained from compressed vegetative remains.

Uses.

- Production of steam power.
- Production of electric power.
- Used for domestic purposes e.g. lighting.

Limitations of using coal.

- i. It is bulky to transport.
- ii. Has low energy comparatively.
- iii. It has health hazard as it is black and dusty.
- iv. It can cause pollution.

B. **Petroleum-** It is an inflammable mixture of hydro- carbons and includes petrol, kerosene and oil.

Uses.

- i. Used to power engines.
- ii. Production of power in industries.
- iii. Heating and lighting.
- iv. Used as lubricants and greases.
- v. Heavier oil is used in road construction.

Limitations of petroleum.

- Not readily available.
- It is expensive to mine, refine and transport.
- It is non renewable.
- It can cause environmental pollution.

C. **Natural gas-** It consists of liquid carbons obtained from animal and plant matter laid in sedimentary rocks.

- It is used in polymerization.

7. **Solar radiation-** it is obtained from the sun.

Uses of solar energy

1. Drying farm produce.
2. Heating water.
3. Distillation of clean drinking water.
4. Cooking.
5. Generating electricity.

Advantages.

- i. It is free.
- ii. It is inexhaustible.
- iii. It is readily available.
- iv. It does not cause pollution.
- v. Its use leads to minimal destruction of forests.

Disadvantages.

- i. Solar panels are expensive.
- ii. Power supply/trapping changes/fluctuates depending on weather conditions.
- iii. Solar trapping is limited to day light.
- iv. It requires skill to install and maintain/handle the devices.

8. Electrical power.

- It includes:
 - A. Hydro-electric power.
 - B. Geothermal power.
 - C. Nuclear power.
 - D. Storage battery power.

A. Hydro-electric power

- Obtained from running water that turn the turbines that generate power.

Advantages.

- i. Does not pollute the environment.
- ii. Can generate a lot of power.
- iii. It is inexhaustible.
- iv. It is cheap to maintain after installation.
- v. Power produced can be converted into so many other uses.

Disadvantages.

- Requires high initial capital investment.
- Water supply can be unreliable.

B. Geothermal power.

- It is obtained from the underground in form of steam.
- The steam is used to turn the turbines that is used to generate power.

Advantages.

- i. It is free hence cheap.
- ii. Steam can be cooled into water and put into other uses.
- iii. Does not contribute to green house effect/does not pollute the environment.

Disadvantages.

- i. May contain toxic gases.
- ii. Hot steam can burn the vegetation.
- iii. Construction of a plant is expensive.
- iv. It can be exhausted with time.

C. Nuclear power- Obtained from the nucleus of the atom.

Advantages.

- i. A lot of power is produced.
- ii. Requires little repair.

Disadvantages.

- i. Production of rays with health effects.
- ii. Requires high level of skill and technology.

D. Storage battery power- Power is stored in the battery.

Advantages.

- i. It is cheap to buy and maintain.
- ii. Can be converted into other forms of energy.

Disadvantages.

- i. Requires constant recharging.
- ii. It is not long lasting.
- iii. Requires skills to maintain.

9. Tractor power.

- It is obtained from the tractor engine(internal combustion engine).

Advantages of using a tractor in farm mechanization.

1. It is efficient and faster.
2. Enables farm planning and budgeting on farm operations to be carried out in time.
3. It can be hired out to earn income.
4. It can be used as security to get loan.
5. It can be used in many farm operations.
6. It requires less labour.
7. Does not require maintenance when not in use as in animal power.

Disadvantages/limitations of using a tractor.

- i. It is expensive to buy and maintain.
- ii. Cannot be used in uneven topography.
- iii. Requires skilled labour.
- iv. Depreciation due to tear and wear.
- v. It is uneconomical in small fields.
- vi. It is not available to all farmers at the same time.
- vii. It can break down hence interfere with farm operations.
- viii. Emits exhaust gases that cause pollution.

Farm operations carried out using a tractor.

- Transportation of farm produce and workers.
- Pumping water, digging of post holes, shelling, threshing, baling of hay, mowing through P.T.O (power take off).

Factors to consider before buying a tractor.

- i. Size of land.
- ii. Cost of tractor.
- iii. Availability of service facilities e.g. mechanics, drivers and spare parts.
- iv. Topography of land.
- v. Time available for operation.
- vi. Availability of capital.
- vii. Number and type of enterprise.
- viii. Availability of tractor hire services.

Advantages of using tractor hire services instead of owning a tractor.

- i. It is cheaper.
- ii. Farmer does not incur maintenance costs of the tractor.
- iii. There is no risk of owning the tractor.
- iv. Farm operations are carried out faster.

Disadvantages of using tractor hire services instead of owning a tractor.

- i. Tractor may not be available when required.
- ii. Some operators may charge high fees for the service.
- iii. Some operators can carry out poor quality work

External parts of internal combustible engine.

- A. Cylinder head-** upper part of the engine.
- B. Cylinder block-** middle part with the cylinders.
- C. The sump-** lower part that houses the connecting rod and crankshaft. Oil is stored in this part.

Internal parts of combustion engine.

A. Cylinder/combustion chamber.

B. Piston.

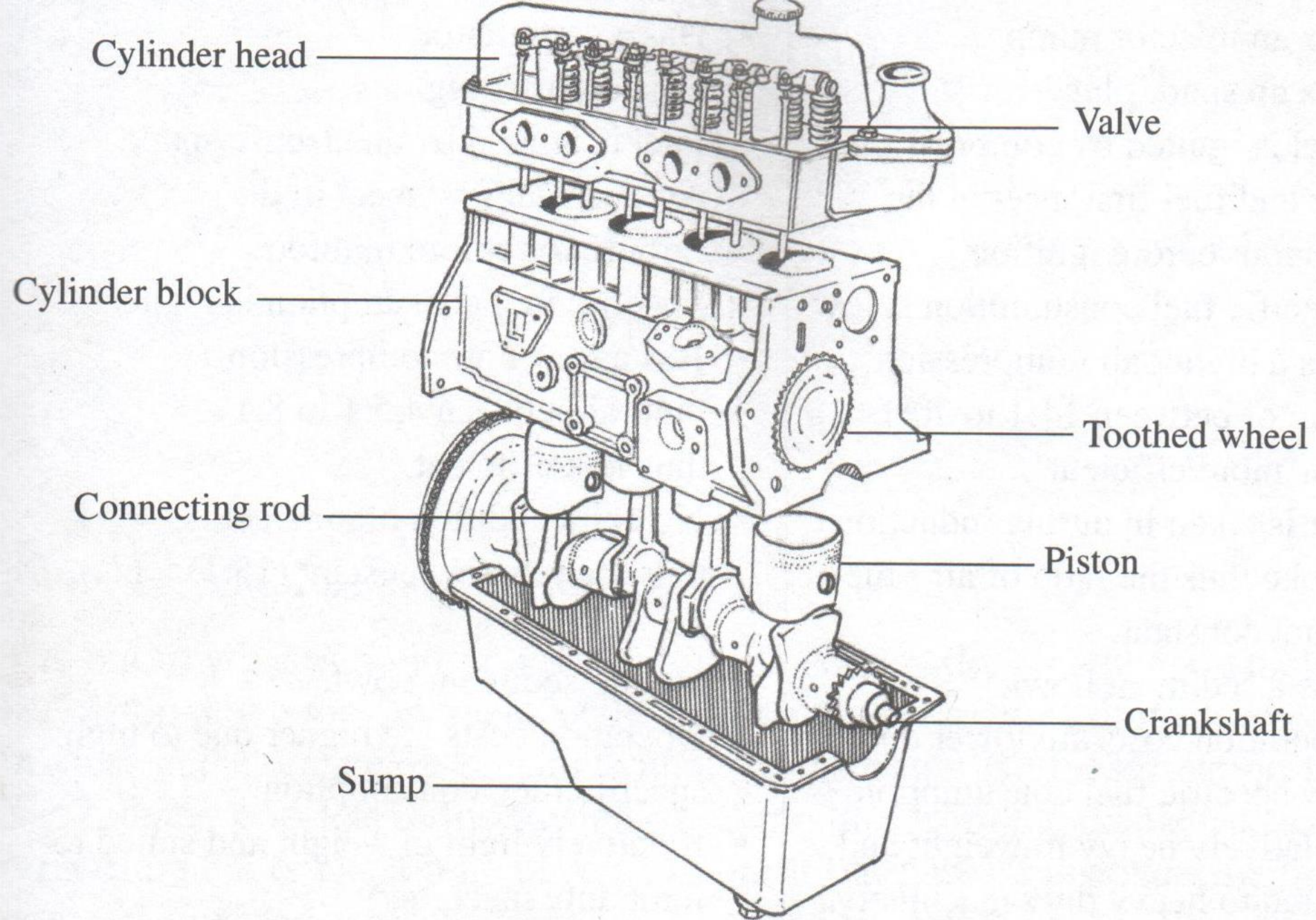
C. Connecting rod.

D. Crankshaft.

E. Cam shaft.

F. Valves.

G. Flywheel.



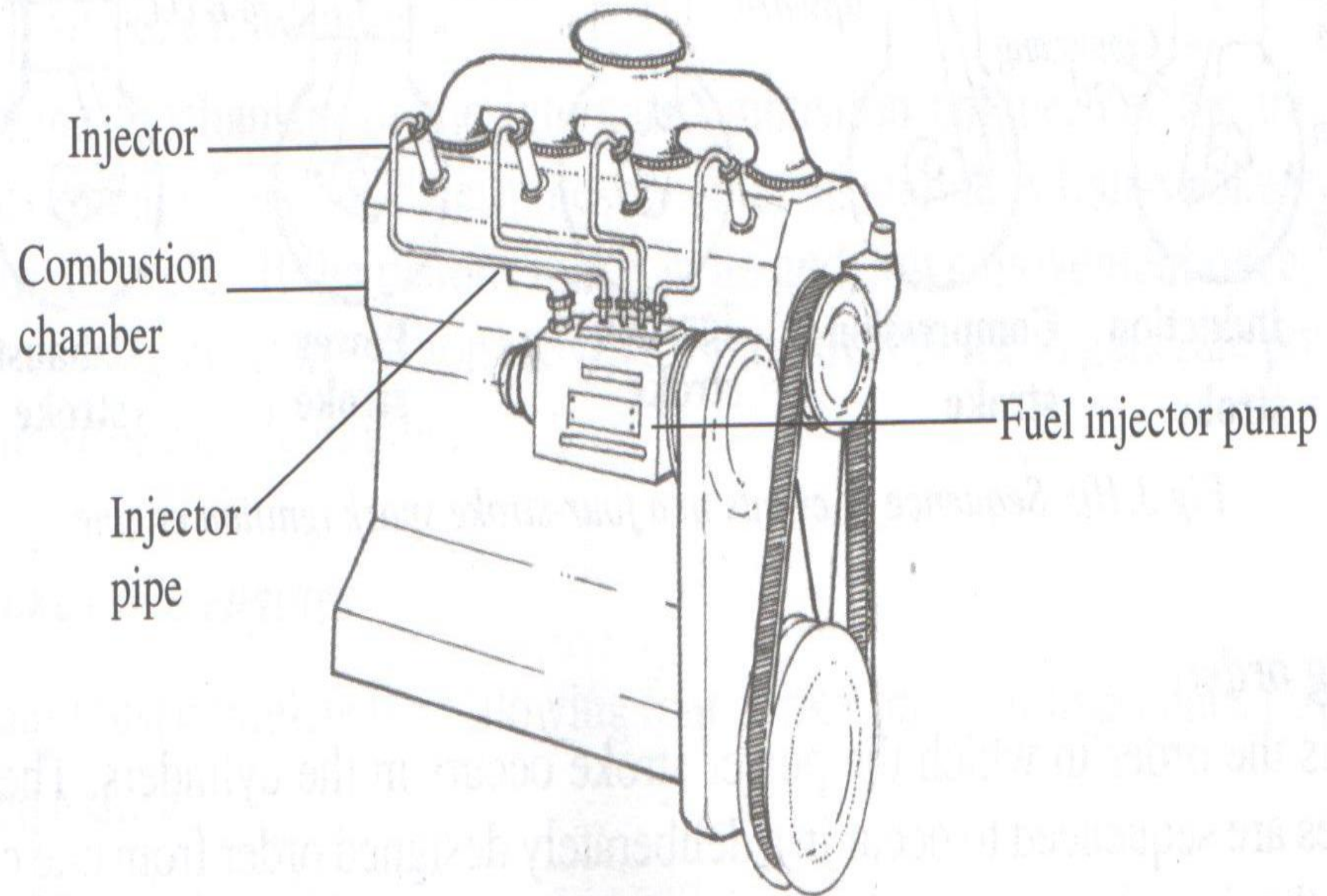


Fig.3.12: Diesel engine.

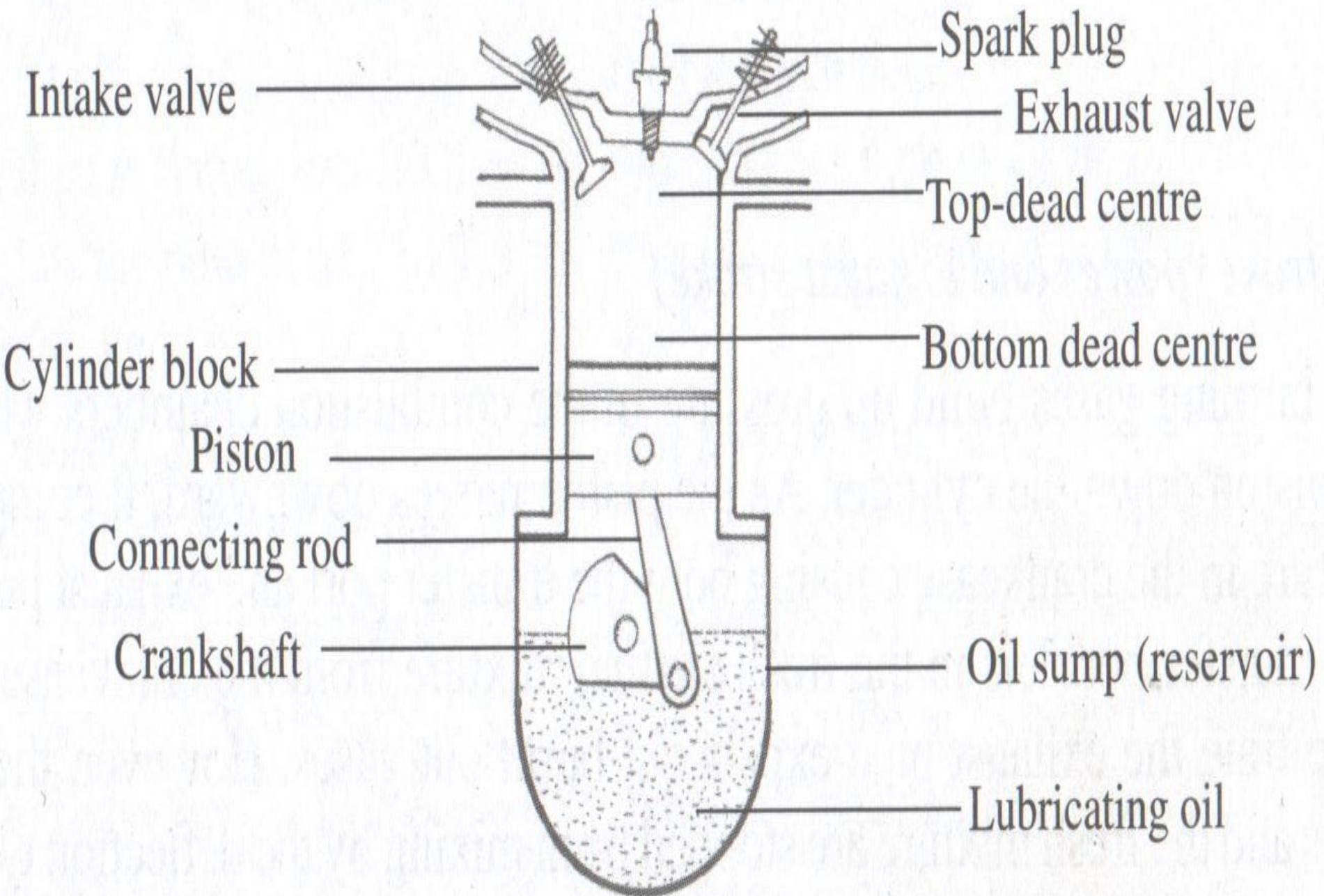


Fig.3.16: Basic parts of an internal combustion petrol engine.

A. Cylinder/Combustion chamber.

- This is air tight chamber whose top is covered with a cylinder head.
- Fuel and air mixture is introduced and burned inside this chamber/combustion takes place in this chamber.
- The top part of the cylinder is called **Top Dead Centre (TDC)** and the bottom part of the cylinder is called **Bottom Dead Centre (BDC)**.
- When the piston moves up and down to cover the distance between the BDC and TDC then it has completed **one cycle**.

B. Piston.

- It moves up and down within the cylinder compressing air and fuel to produce ignition.
- It also conducts away excess heat from the cylinder.
- Piston head prevents the leakage of fuel.

C. Connecting rod/piston rod.

- It connects the piston to the crankshaft transmitting power from the cylinder to the crankshaft.

D. Crankshaft.

- It is irregularly shaped rod onto which the pistons are fitted.
- It rotates causing the pistons to move alternately up and down to provide power.
- It is covered by a **crankcase** whose bottom is sealed with a **sump** with oil.

E. Camshaft.

- It is a rod fitted with cams that operate the opening and closing of **valves**.

F. Valves.

- They are openings in the cylinder
- They include:
 - i. **Inlet valve-** allows in fuel and air mixture.
 - ii. **Outlet/exhaust valve-** allows exit of burnt gases.

G. Fly wheel.

- It is fitted at one end of the crankshaft to assist in turning.
- It transfers power to the **clutch**.

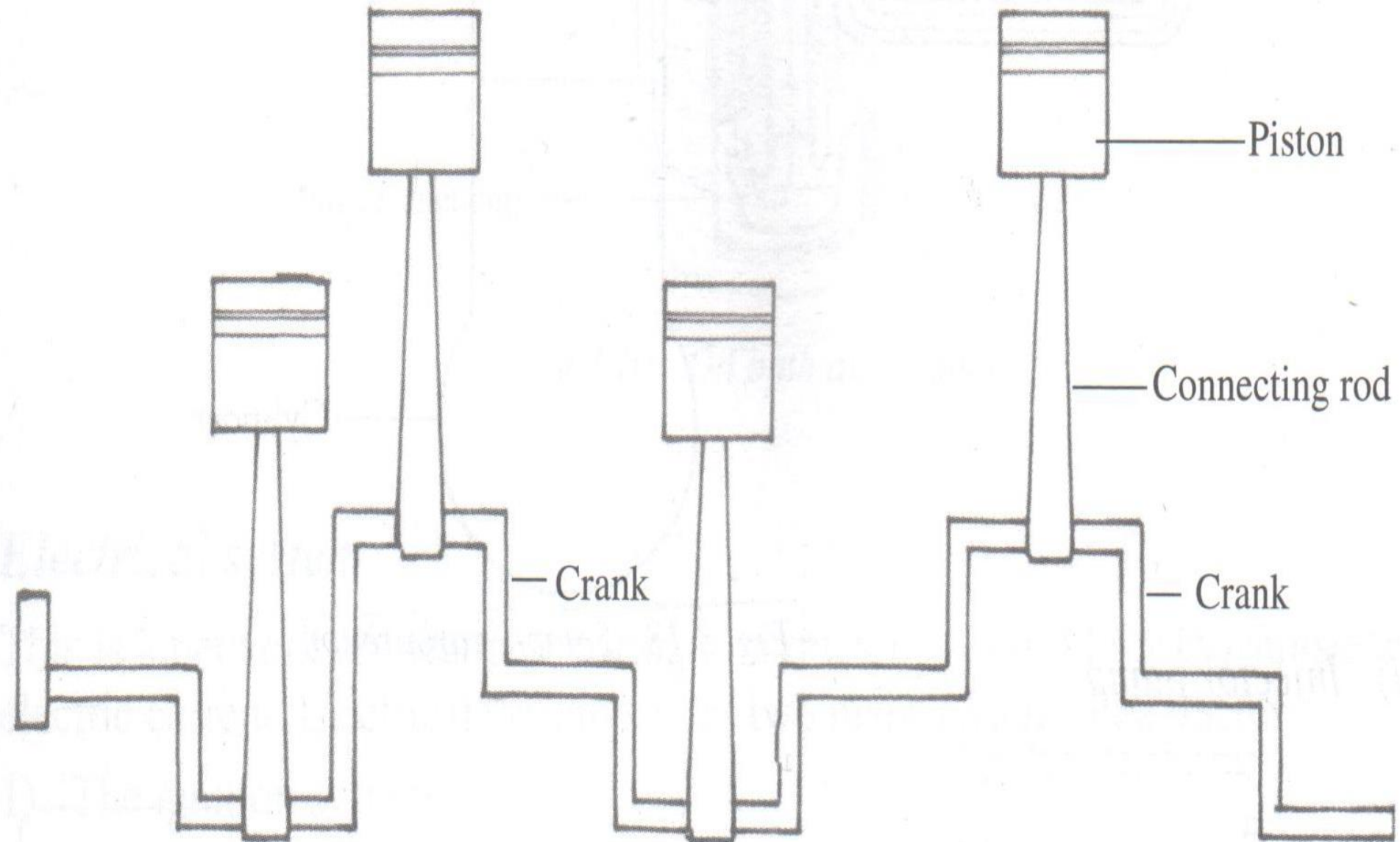


Fig.3.17: Cranks and pistons.

CATEGORIES OF INTERNAL COMBUSTION ENGINES.

A. Diesel/compression ignition engine.

B. Petrol/spark ignition engine.

Terms used in the engine.

i. **Compression ratio-** This is the ratio of the total amount of air in the cylinder to the amount of compressed air.

ii. **Firing order-** This is the order in which the power stroke occurs in the cylinder.

A. Diesel/compression ignition engine- It uses diesel as a source of power.

- The diesel fuel enters the cylinder through an injector by use of injector pump.
- The fuel/diesel mixes with air in the cylinder and the mixture is compressed leading to ignition.

B. Petrol/spark ignition engine- It uses petrol as a source of power.

- It has an electric spark which is produced by the spark plug which ignites the compressed petrol-air mixture in the cylinder.
- It has a **carburetor** which mixes petrol and air before injecting it into the cylinder.

DIFFERENCES BETWEEN PETROL AND DIESEL TRACTOR ENGINES

Petrol engine.

1. Uses petrol as fuel which is highly flammable.
2. Has a carburetor for mixing petrol and air.
3. Has a spark plug.
4. Air and fuel/petrol are first mixed in the carburetor before injected into the cylinder.
5. Fuel is ignited by electric spark.

Diesel engine

1. Uses diesel as fuel which is a low inflammable fuel.
2. Has injector pump to inject fuel into the cylinder.
3. Does not have a spark plug.
4. Air and fuel/diesel are mixed within the cylinder.
5. Fuel is ignited by compression.

- 6. Uses a battery to provide electric spark
- 7. Does not have sedimental bowl
- 8. Produces little smoke
- 9. Produces less power
- 10. It is light in weight
- 11. Uses more fuel hence expensive
- 12. Produces less noise
- 13. Has low compression ratio

- 6. It operates without a battery
- 7. Has a sedimental bowl to trap sediments in fuel
- 8. Produces a lot of smoke
- 9. Produces more power
- 10. It is heavier in weight
- 11. Uses less fuel hence more economical
- 12. Produces a lot of noise
- 13. Has high compression ratio

TYPES OF ENGINES.

A. Four-stroke cycle engine.

B. Two-stroke cycle engine.

A. 4-stroke cycle engine.

- In this type a piston goes through four strokes to make a complete cycle i.e. moves up and down twice to generate power.
- Power in a tractor engine is the **rotary motion** of the **crankshaft**.
- It is found in heavy machines e.g. tractor.

Four strokes.

1. Induction/intake stroke (I)
2. Compression stroke (C)
3. Power stroke (P)
4. Exhaust stroke (E)

1. INDUCTION/INTAKE STROKE.

- The piston moves downwards creating a space/partial vacuum in the upper part of the cylinder.
- The inlet valve open while the exhaust valve remains closed.
- Air and fuel enters/is sucked into the cylinder (through the inlet valve).

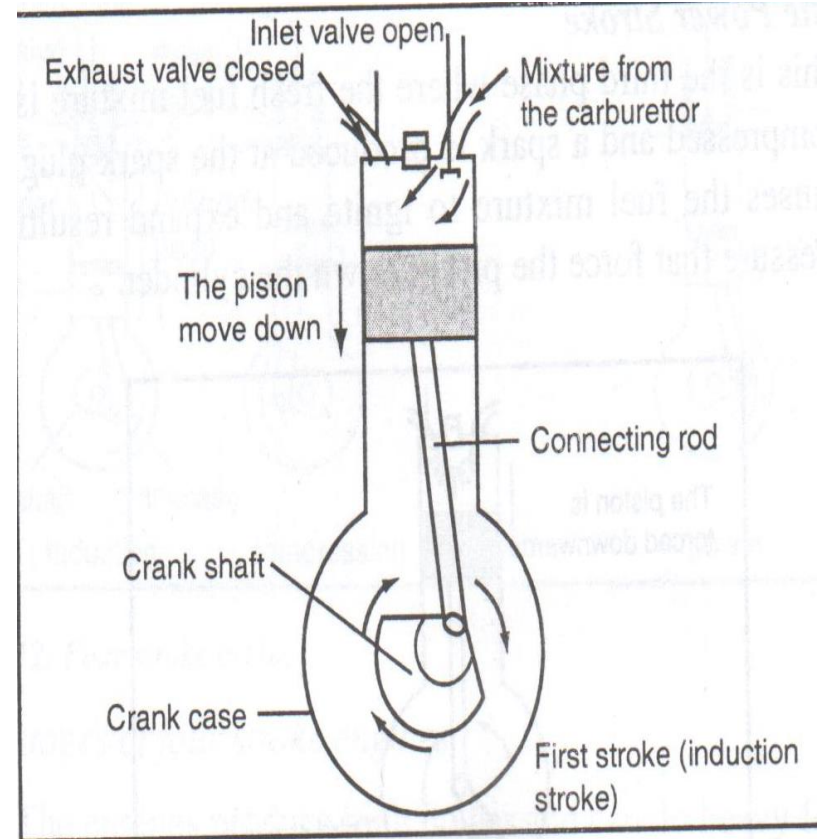


Fig. 3.8: Induction stroke

2. COMPRESSION STROKE.

- The inlet and exhaust/outlet valves remain closed.
- The piston moves upwards to compress the air fuel mixture.
- This causes ignition due to high temperature.

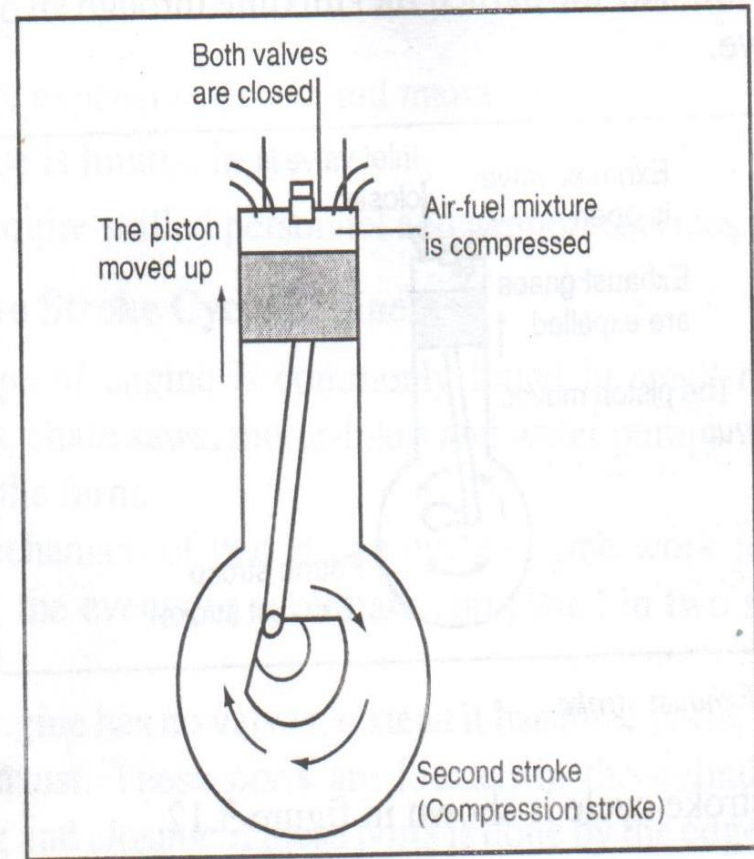
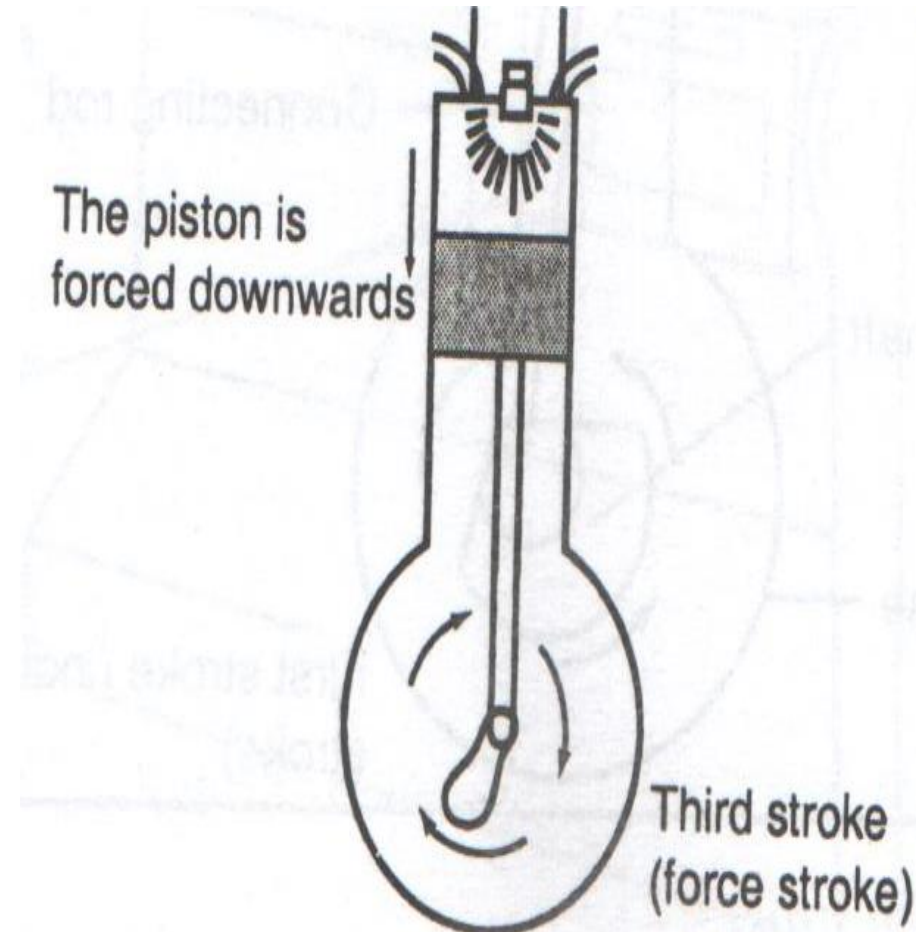


Fig. 3.9: Compression stroke

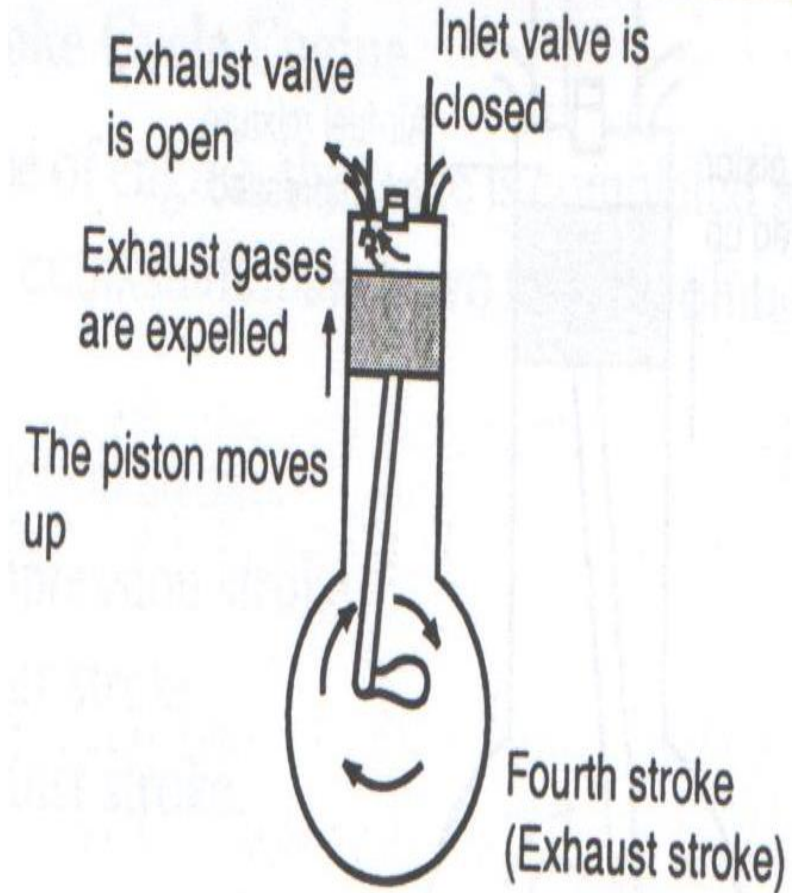
3. POWER/IGNITION STROKE.

- The inlet and exhaust/outlet valves remain closed.
- The ignited fuel expands producing pressure.
- The pressure forces the piston to move downwards generating power.



4. EXHAUST STROKE.

- The exhaust valve opens while the inlet valve remain closed.
- The piston moves up and forces out exhaust gases.



Advantages of 4-stroke engines.

- i. They produce a lot of power hence can do heavy work.
- ii. They are efficient in fuel and oil utilization.
- iii. They can perform a wide range of farm operations.
- iv. They are efficiently cooled by water.
- v. Exhaust gases are efficiently expelled from the cylinders.
- vi. They have heavy crankcase which enables them to withstand vibrations.

Disadvantages.

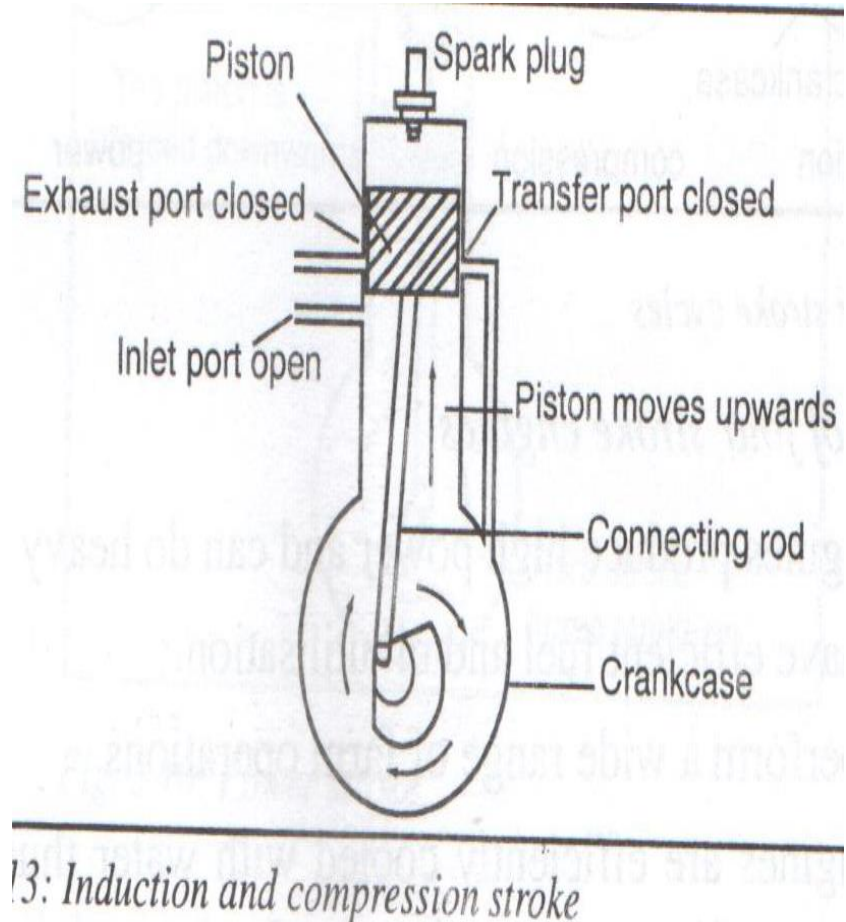
- They are expensive to buy and maintain.
- They require skilled labour.
- Their use is limited to some areas.

B. TWO-STROKE ENGINES

- In this the cycle is completed with two strokes of the pistons.
- They are found in small machines e.g. mowers, chain saws, motorbikes and water pumps.
- The engine has 3 ports instead of 2 valves.
- The 3 ports are located on the cylinder wall and their opening and closing is controlled by the edge of the piston as it moves within the cylinder.
- The strokes include:
 1. Induction and compression stroke.
 2. Power and exhaust stroke.

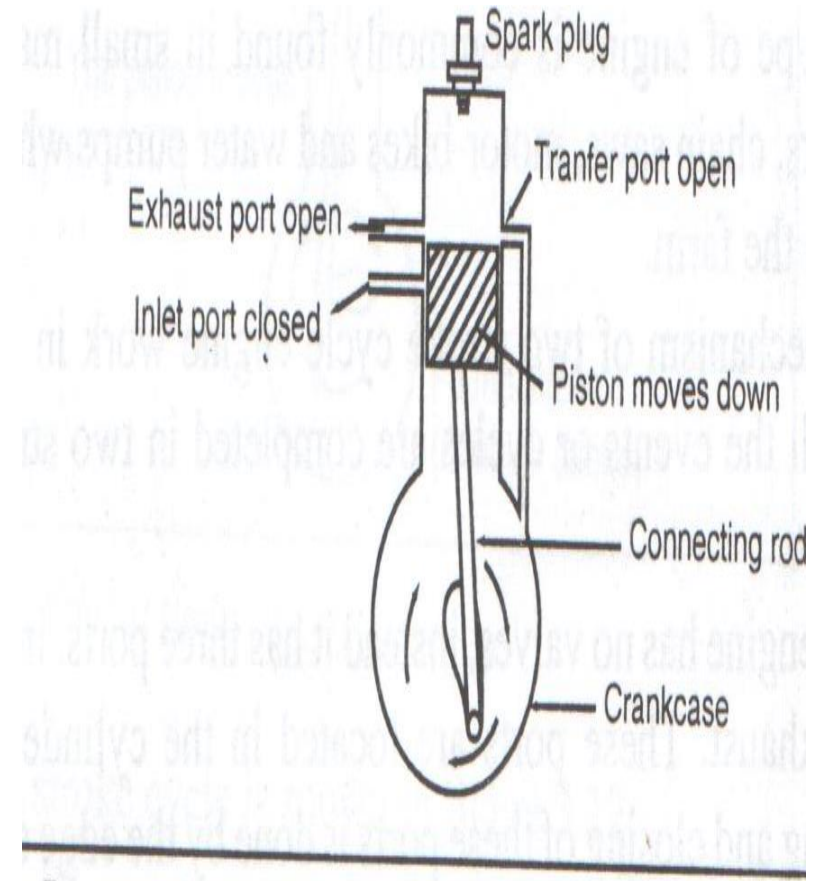
1. INDUCTION AND COMPRESSION STROKE.

- The piston moves up.
- This uncovers the **inlet port** but covers the outlet/exhaust and transfer ports.
- The piston compresses the air-fuel mixture in the cylinder/combustion chamber.
- The new air-fuel mixture is drawn into the crankcase through the inlet port.



2. POWER AND EXHAUST STROKE.

- The compressed air-fuel mixture ignites/burns creating pressure.
- The pressure pushes the piston downwards to compress the air-fuel mixture in the crankcase.
- The downward movement of the piston uncovers the transfer port and **exhaust port** but closes the **inlet port**.
- The air-fuel mixture in the crankcase is transferred to the cylinder/combustion chamber through the **transfer port**.
- The burnt/exhaust gases are expelled from the combustion chamber through the **open exhaust port**



: Power and exhaust stroke

Advantages of two stroke engines.

- i. They are cheap to buy and maintain.
- ii. They use little fuel/economical.
- iii. They can be used in a wide range of farm lands including hilly areas.
- iv. They can be used for small tasks in the farm which cannot be economical to use a four stroke engine.

Disadvantages.

- i. They produce less power hence cannot be used for heavy duties.
- ii. They are inefficient in burning of fuel.
- iii. They are air cooled thus limiting the size of engine.
- iv. They are slow in operation.

DIFFERENCES BETWEEN A FOUR STROKE AND TWO STROKE ENGINE

Two stroke engine.

1. Produces less power
2. Produces less power/does less work.
3. It is air cooled.
4. Crankcase is involved in the cycle
5. It is less efficient in oil and fuel utilization.
6. It is easy to transport to different places.
7. It requires 2 complete upward and downward movement of piston and one revolution of crankshaft.
8. Has no oil in the sump.
9. It is simple in construction with no valves.
10. Uses less fuel
11. It is cheap to buy and maintain.
12. Has ports
13. Require less skill to operate

Four stroke engine.

1. Produces more power
2. Produces more power/does heavy job.
3. It is water cooled.
4. Crankcase is not involved in the cycle
5. It is more efficient in oil and fuel utilization.
6. It is difficult to transport due to heavy weight.
7. It requires 4 complete upward and downward movement of the piston and two revolutions of the crankshaft.
8. Has oil in the sump for lubrication.
9. It is complex in construction with 2 valves.
10. Uses more fuel
11. It is expensive to buy and maintain.
12. Has no ports.
13. Requires skills to operate

SYSTEMS OF A TRACTOR.

1. Fuel system
2. Cooling system
3. Lubrication system
4. Ignition system
5. Electrical system
6. Power transmission system

A. FUEL SYSTEM.

- There are two types of fuel system:

A. Petrol fuel system.

B. Diesel fuel system.

Working mechanism in petrol fuel system.

- In this system, petrol is put into a tank which has a filter to filter fuel and trap sediments.
- The fuel is then passed to the carburetor which mixes it with air then the mixture is passed to the engine cylinder.

Maintenance practices of petrol fuel system.

1. The carburettor jets should be cleaned regularly to avoid blockage.
2. The fuel filter should be cleaned by washing in petrol.
3. The hole in the fuel tank should always be kept clean.

Working mechanism in diesel fuel system.

- In this system, diesel is put into the tank then delivered into the injector pump.
- The injector pump then delivers the fuel to the engine cylinder.

Maintenance of diesel fuel system

1. Filters should be replaced as recommended by the manufacturers.
2. Bleeding should be done incase air is trapped in the system.
3. The sediment bowl should be cleaned regularly.

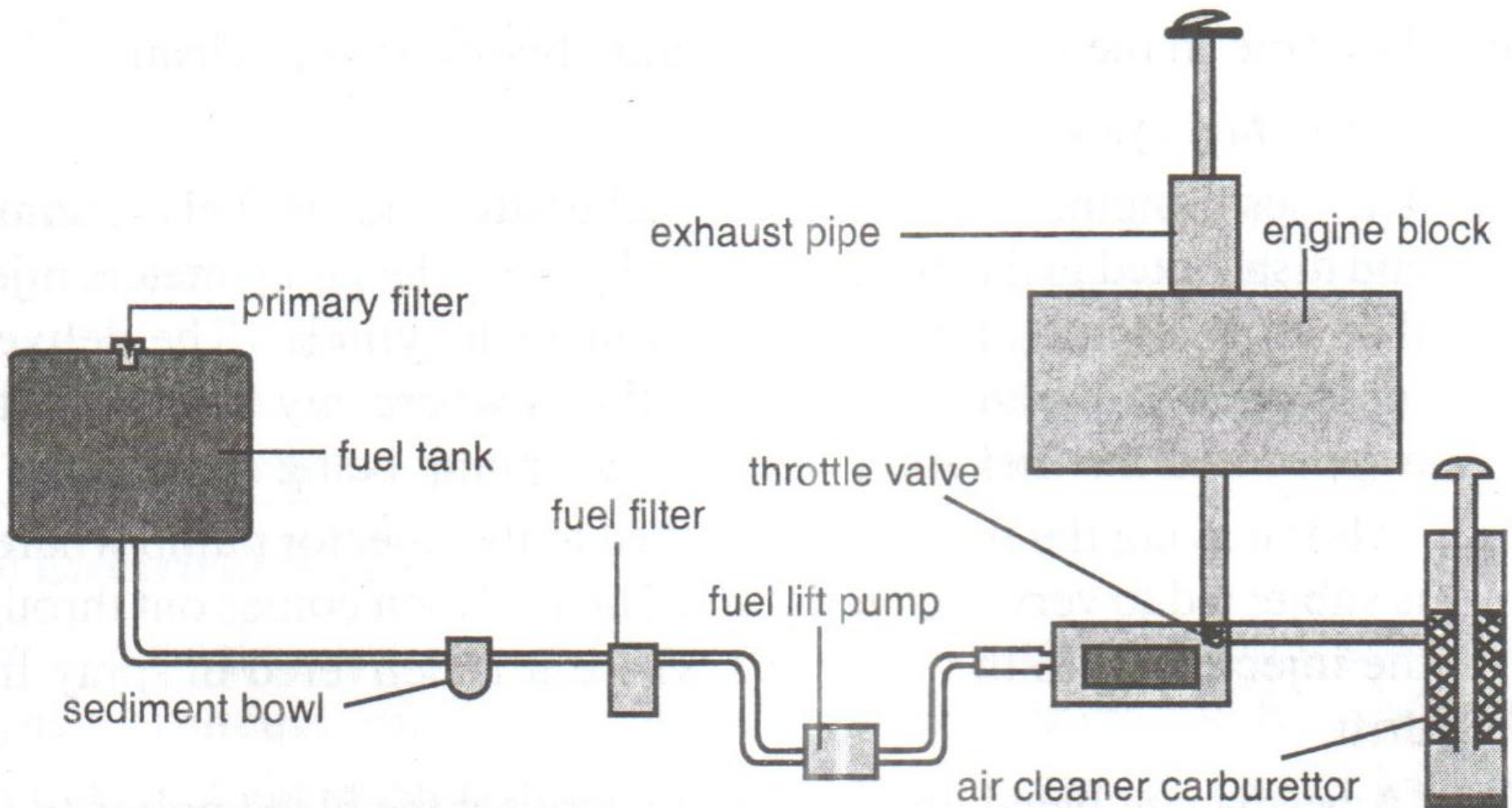
Bleeding a tractor.

- It refers to the removal of air bubbles trapped in the fuel system which may cause the engine not to start.
- Bleeding/bleed screws are opened and fuel pumped to the injector pump. Air bubbles start to move out through the bleeding screws.
- As fuel starts to escape through the bleeding screws they are the screws are tightened as all the air has escaped.

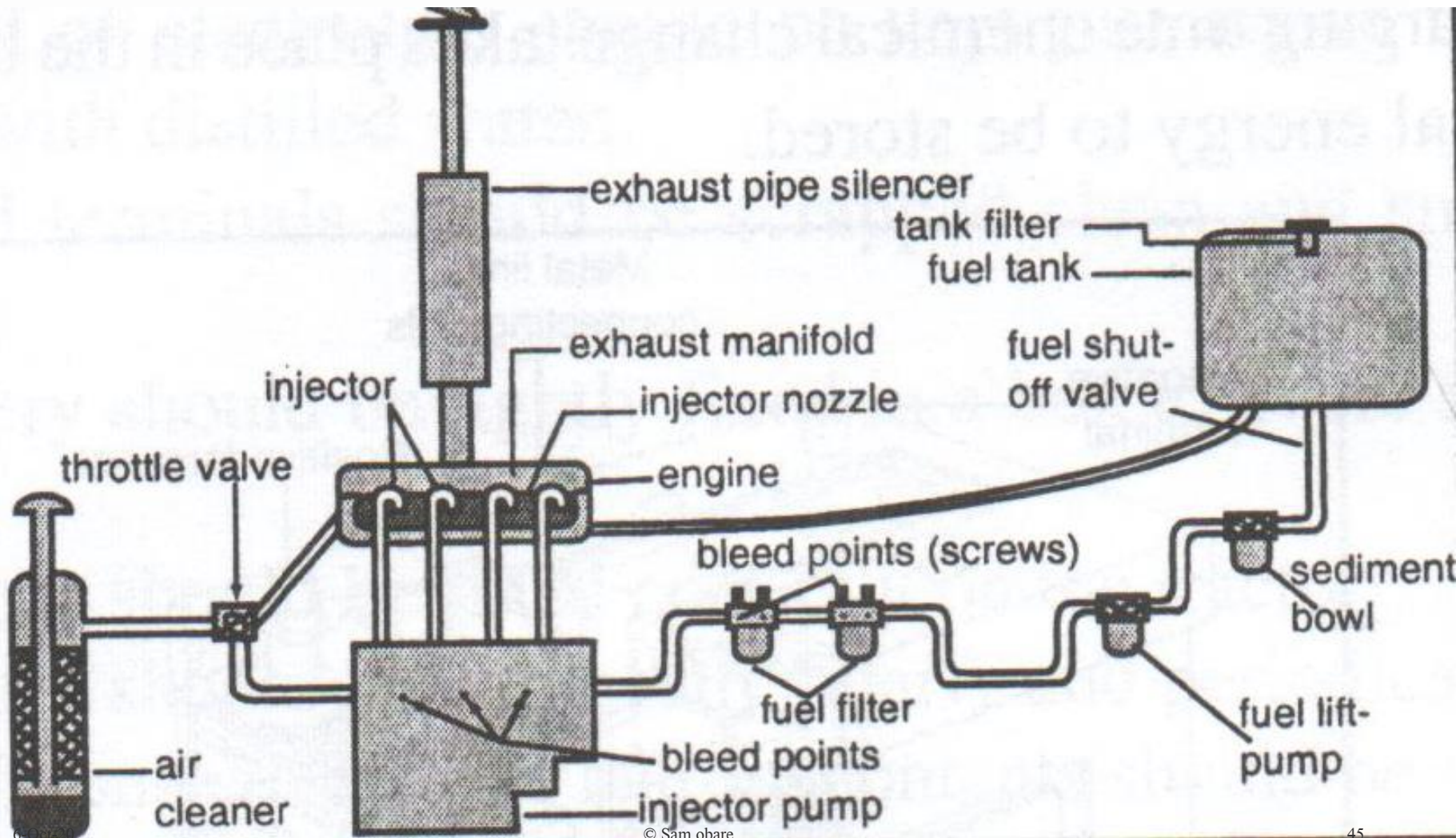
PARTS OF A FUEL SYSTEM

1. **Tank**- Stores fuel.
2. **Sediment bowl**- Deposits /traps any dirt and sediments in the fuel.
3. **Lift pump**- Forces /pumps the fuel into the injector pump.
4. **Fuel filters**- Removes/removes any dirt from the fuel.
5. **Fuel return pipe**- Carries/transfers the excess fuel from the injector pump back to the fuel tank.
6. **Air cleaner**- Removes dust particles from air before it enters into the carburetor or injectors.
6. **Injector pump**- It injects diesel into diesel engines/cylinders through injector nozzles.
 - It regulates the amount of fuel discharged hence controlling the engine speed.
7. **Carburetor**- it turns liquid petrol into vapour and mixes it with air.
 - Controls the ratio of air and fuel
 - Introduces the air-fuel mixture into the petrol engine

A. PETROL FUEL SYSTEM



B. DIESEL FUEL SYSTEM



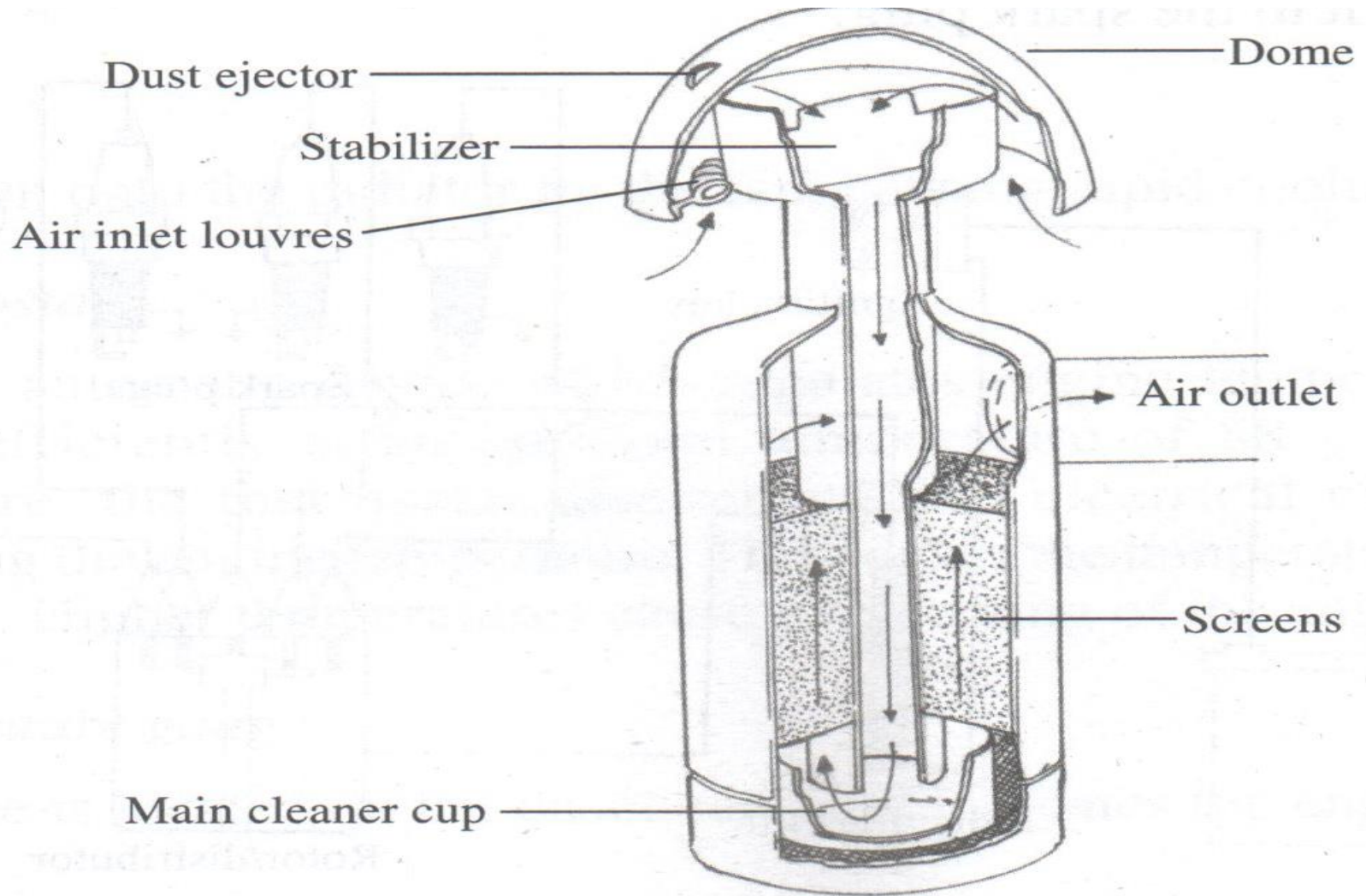


Fig.3.20: Oil bath air cleaner.

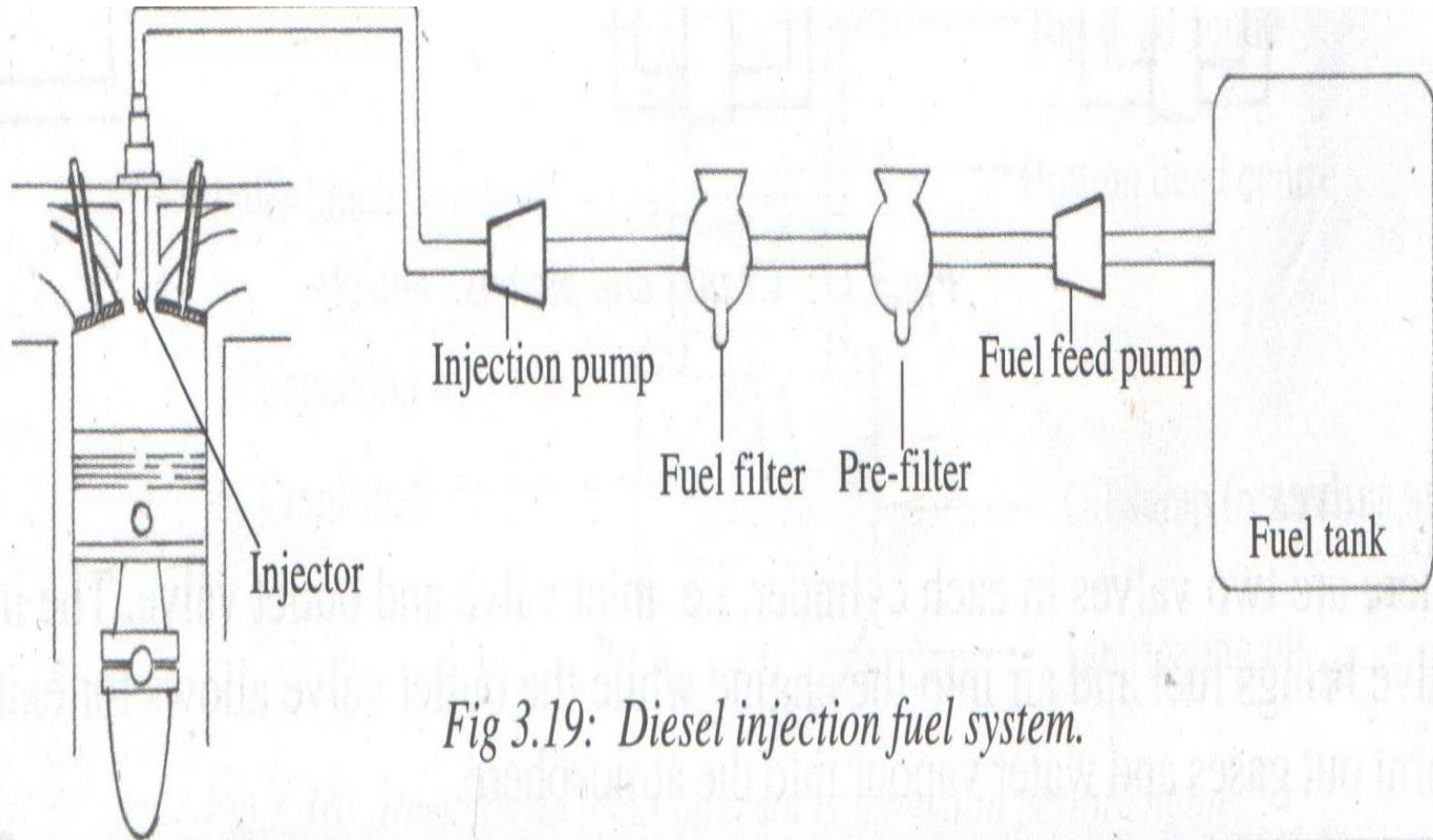
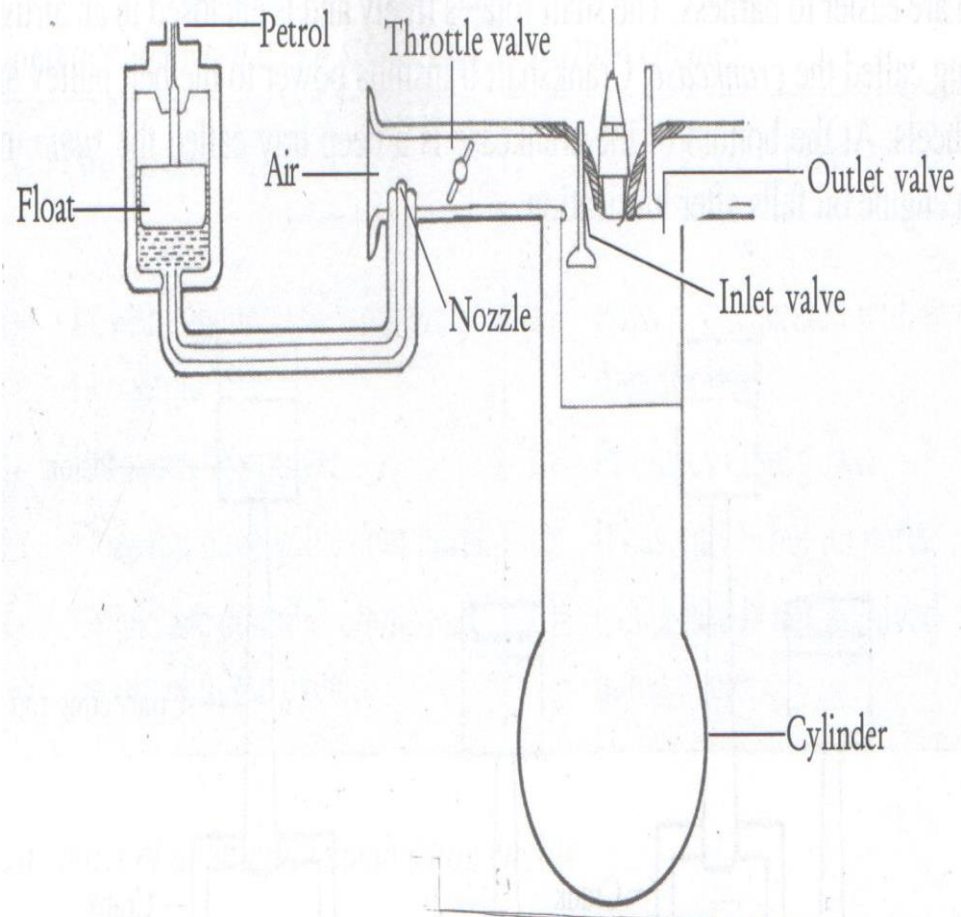


Fig 3.19: Diesel injection fuel system.

Carburettor.



2. COOLING SYSTEM.

- It helps to cool the engine to avoid overheating.
- Overheating causes expansion of engine components which distorts its shape causing:
 - i) Fuel leakage.
 - ii) Valve burning.
 - iii) Loss of engine power.
 - iv) Cracking of the cylinder or its head.
 - v) Dilution of lubricating oil.
- Engines are cooled by air or water.

A. Air cooled system.

- It is used in light weight farm machines engines e.g. mowers, motor-bikes.

Characteristics of air cooled engines.

1. They are simple in construction.
2. They have fins and a fan blade which assist the circulation.
3. They are light in weight for they do not have radiators and water jackets.

Types of air cleaners.

1. The oil bath air cleaner/wet air cleaner.
2. The dry type air cleaner.

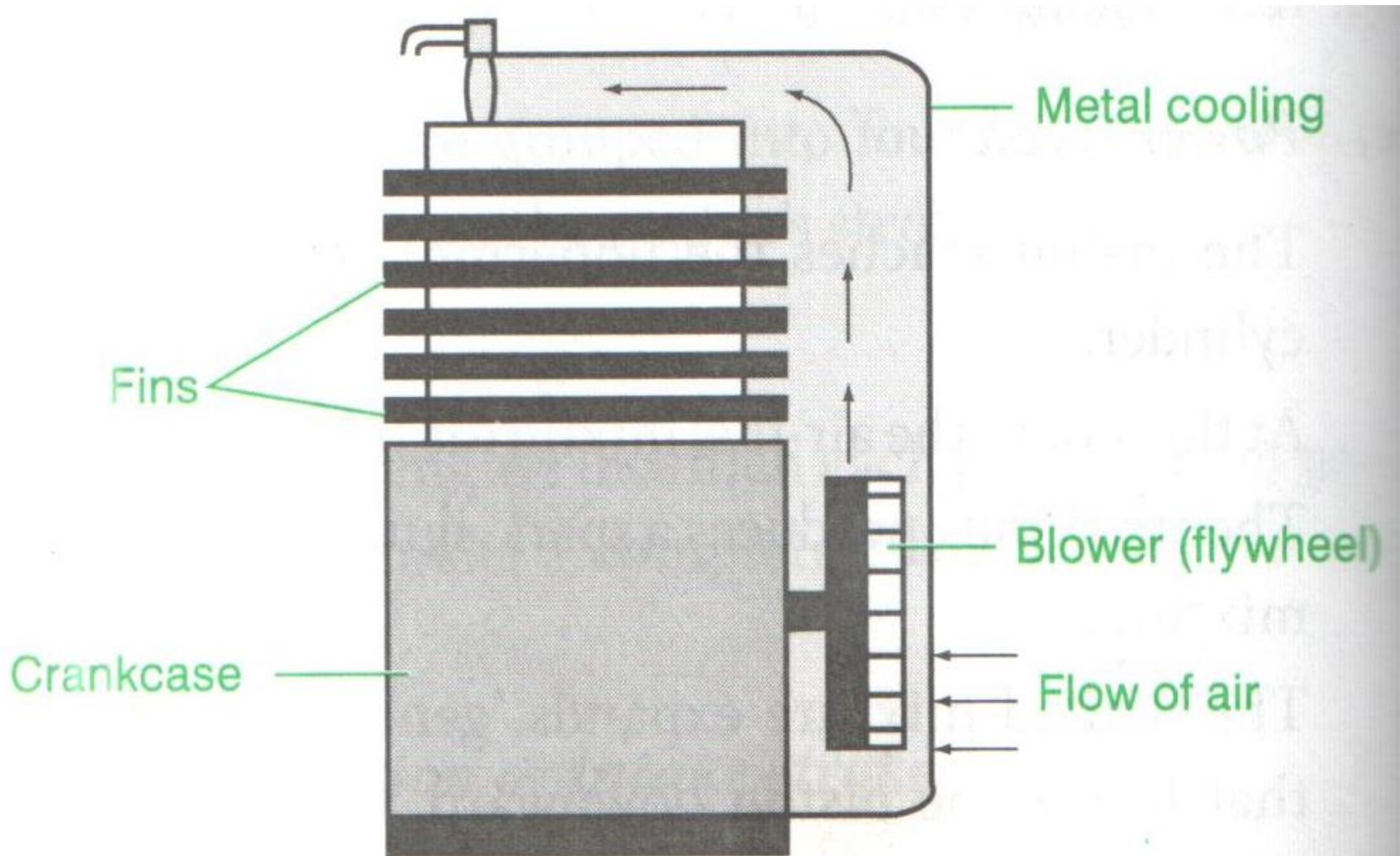
Problems of air cooled engines.

1. They get hot quickly and use heavy lubricating oils.
2. The cooling is not adequate under all conditions especially when carrying heavy loads.

B. WATER COOLED SYSTEM.

- Water is used to absorb heat from the engine

AIR COOLED SYSTEM



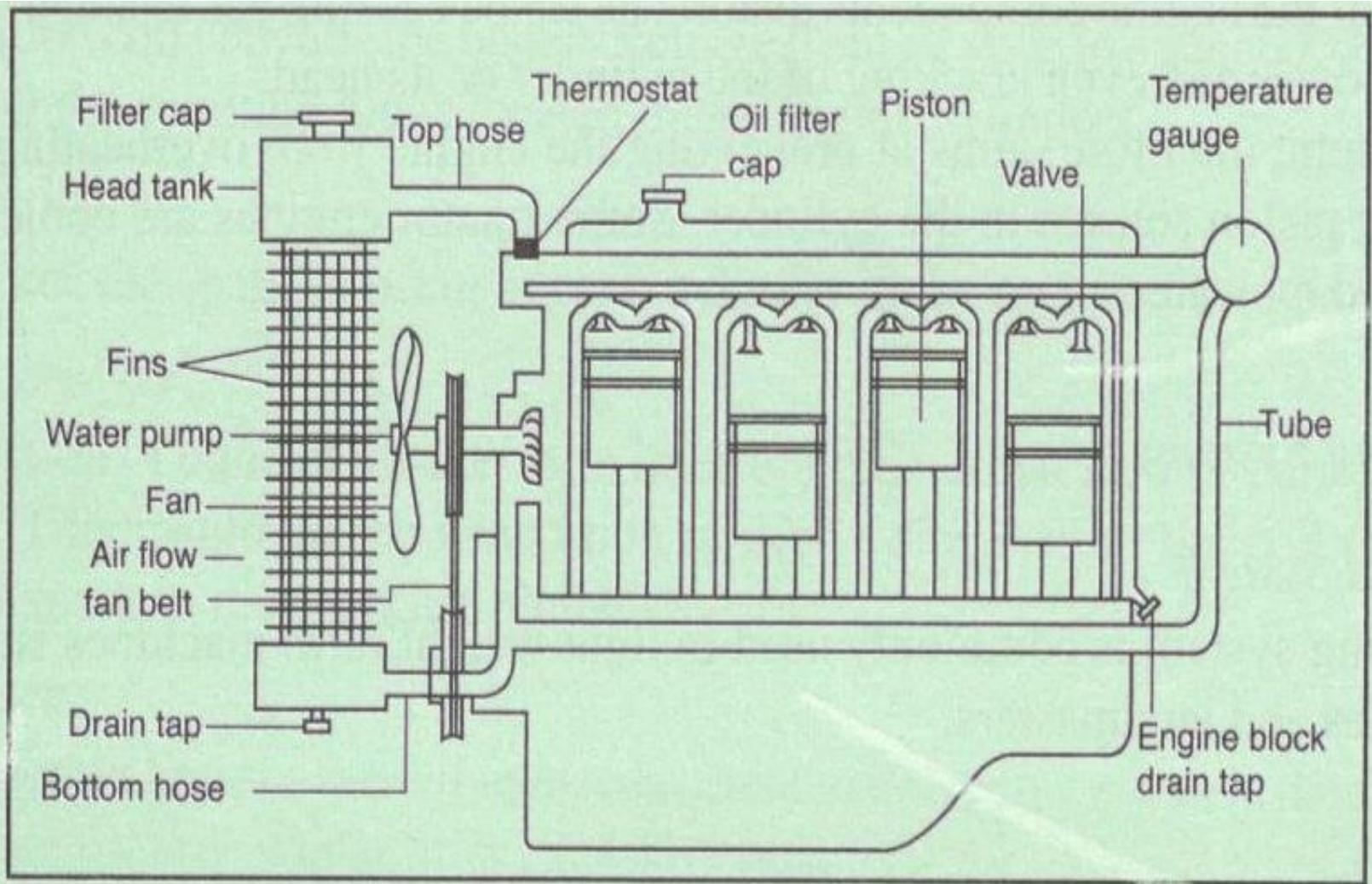
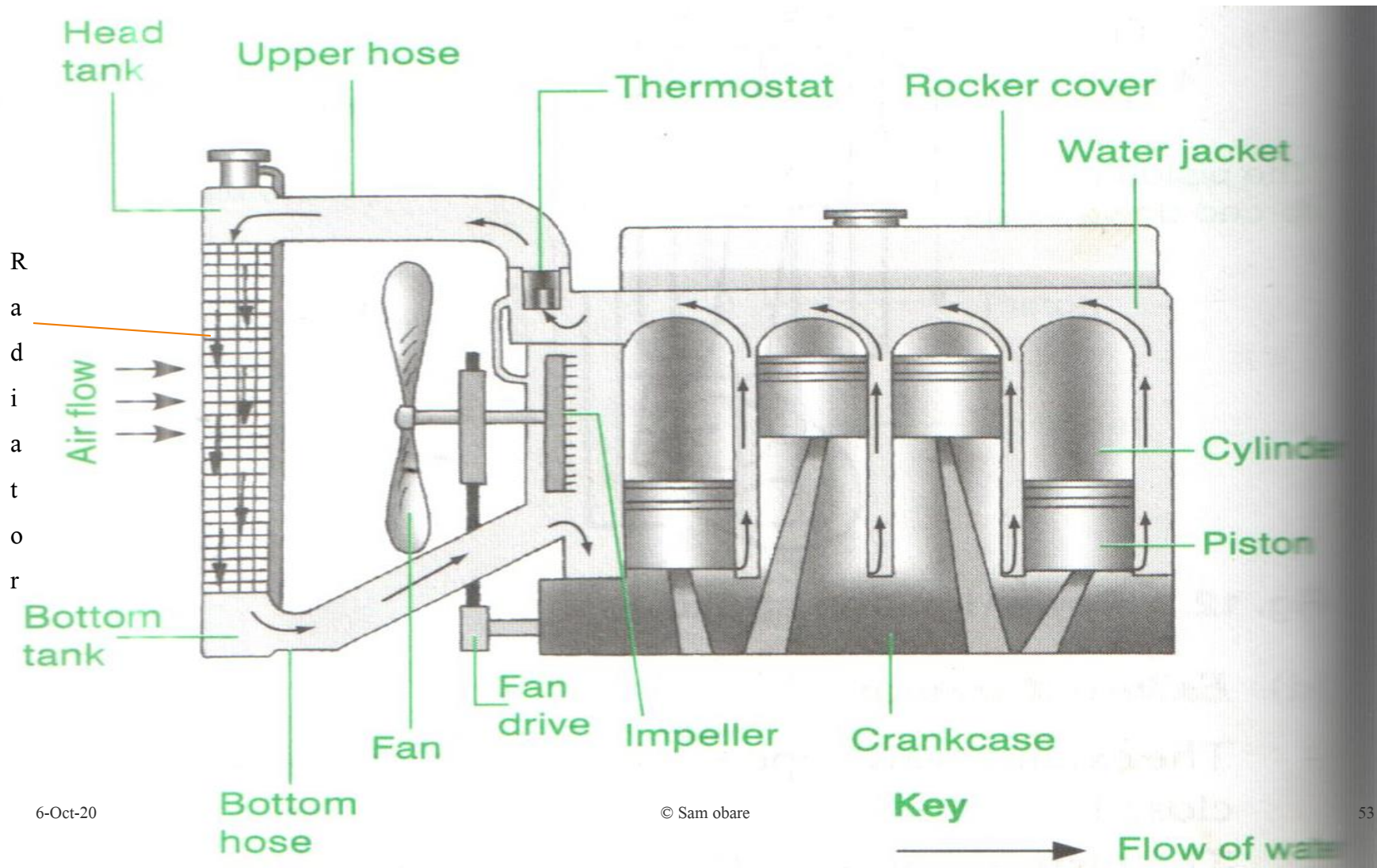


Fig. 3.21: Water cooling system

WATER COOLING SYSTEM



PARTS OF WATER COOLING SYSTEM.

<u>Part</u>	<u>Function.</u>
1. Head tank-	• Holds water used for cooling.
2. Radiator-	• Cools hot water from the engine.
3. Water pump-	• Pumps water to the engine.
4. Water jackets-	• Allow water to flow around the engine to cool it.
5. Fan-	• Blows air to the radiator to cool water.
6. Upper hose-	• Carries hot water from the engine to the head tank
7. Lower hose-	• Carries cool water from the radiator to the water jacket for circulation around the engine
8. Filter cap-	• It seals/closes the system to prevent entry of air
9. Thermostat-	• Controls engine temperature
10. Temperature gauge-	• Indicates the engine temperature to the driver

CARE AND MAINTENANCE OF WATER

COOLING SYSTEM

- i. Lubricate the water pump regularly.
- ii. Use clean water only in the radiator.
- iii. Remove dirt and rubbish from the radiator fins.
- iv. Fit the pipes properly to avoid leakages.
- v. Repair the leakages.
- vi. Top up the level of water in the radiator before using the tractor.
- vii. Ensure the fan belt is tightly fitted.

3. LUBRIICATION SYSTEM

Functions/ importance of lubrication system

- i. It increases the efficiency of the machine.
- ii. Reduces the rate of wear and tear of the moving parts.
- iii. It reduces the heat created by the rubbing surfaces and acts as a seal between them.
- iv. It removes dust, dirt, soot and metal chippings from the oil paths to the sump.
- v. Oiling prevents rusting of stationary machines.

Types of lubrication system.

1. **Splash feed lubrication system-**
Crank of crankshaft are dipped into the oil and splashed as they move up.
 2. **Oil mist type-** Oil is mixed with fuel/petrol in the tank.
 3. **Force feed-** Oil circulates through the whole engine and drips back into the sump.
- It is the common and modern method in the 4-stroke engines.

Types of lubricants

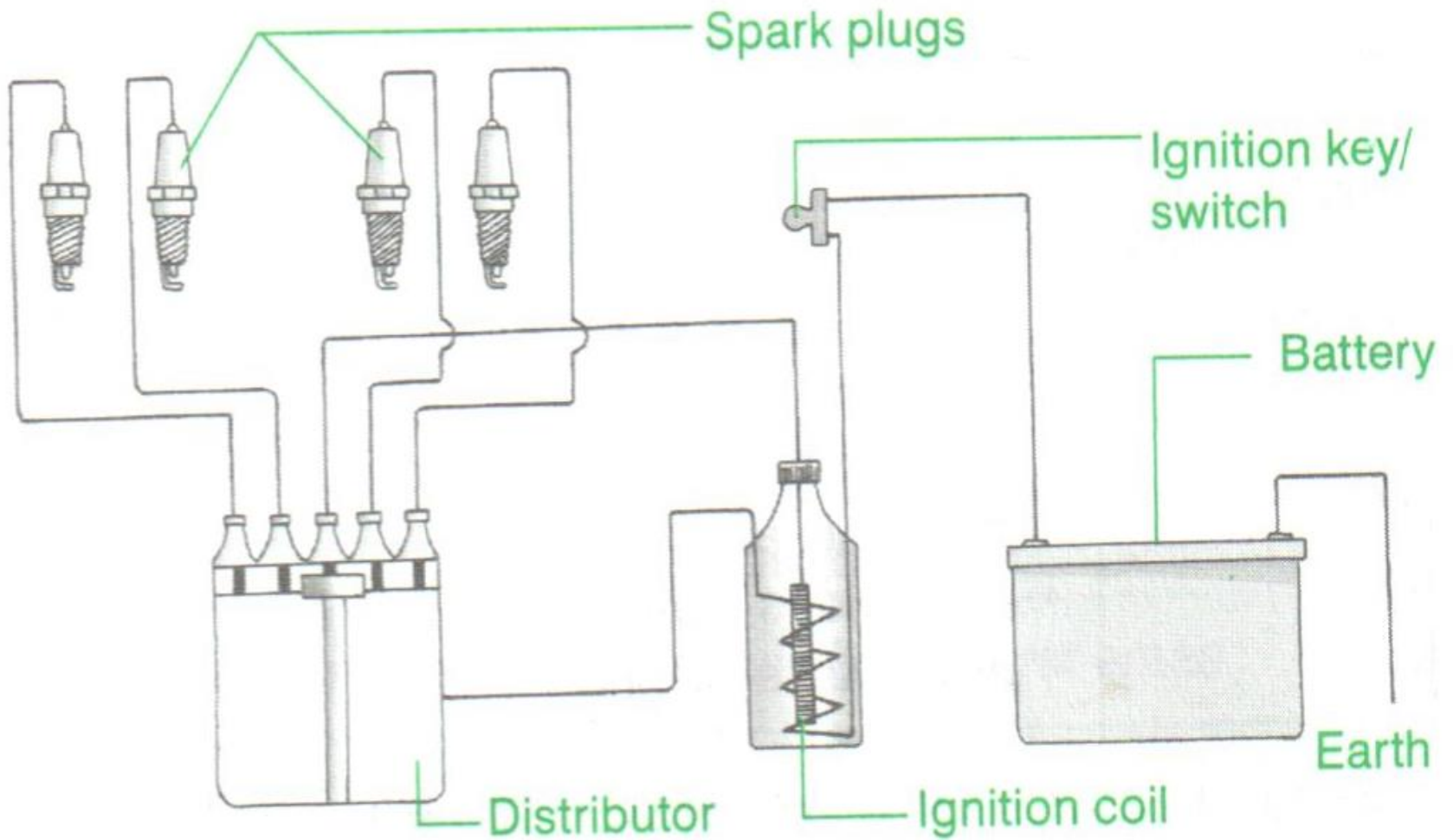
- They are identified by thickness (viscosity index) which is indicated by the S.A.E (Society of Automobile Engineers).
- They include;
 1. SAE 10- thin engine oil which gives little protection when heated.
 2. SAE 50- thicker engine oil recommended for protecting bearings.
 3. SAE 90-150- transmission oils recommended for clutch, gear box, wheel lubrications and ball bearings.
- The lower the SAE number the thinner the oil and the higher the SAE number the thicker the oil hence better in protection.

CARE AND MAINTENANCE OF LUBRICATION SYSTEM.

1. Replace oil filters when necessary.
2. Avoid using contaminated/old oil.
3. Dirty oil should be drained while still hot to avoid sticking to the wall of the sump.
4. Use the correct type of lubricants as per the instructions of the manufacturer.

4. IGNITION SYSTEM.

- It produces the spark which is required to ignite the fuel-air mixture.



PARTS OF IGNITION SYSTEM

Part.

Use.

- | | |
|--------------------|--|
| 1. Battery | 1. Acts as the source of electrical energy. |
| 2. Ignition coil | 2. Increases voltage of battery to generate a spark at the spark plugs. |
| 3. Condenser | 3. It absorbs the self-induced current in the primary circuit, therefore preventing the contact breaker points from excessive heating. |
| 4. Distributor | 4. It causes the spark to occur at each cylinder in the required firing order or sequence. |
| 5. Spark plug | 5. It produces spark which ignites the air-fuel mixture. |
| 6. Ignition switch | 6. It completes the circuit. |

WORKING MECHANISM OF IGNITION SYSTEM.

- The **battery** /generator supplies sparks which are required for ignition to take place.
- The **ignition coil** changes the low voltage from the battery to a high voltage current required in the **spark plug** in the petrol engine.
- The **condenser** absorbs the self induced current in the primary circuit therefore preventing the contact breaker points from excessive heating.
- The **condenser** passes on the electric current to the **distributor** which distributes the current to the spark plug of each cylinder.
- This causes the spark to occur on each cylinder igniting the air-fuel mixture then the tractor engine starts.

Common faults of ignition system

1. Sudden stopping- Caused by:

- a) poor terminal connections and
- b) faulty ignition system.
- It is corrected by:
 - a) Proper tightening of the terminals.
 - b) Proper cleaning and readjustment of the terminals.

2. Continuous engine running- it is caused by:

- a) Broken leads.
- b) Poor terminal connections.
- c) Faulty contact breaker.
- It is corrected by:
 - a) Replacing broken leads.
 - b) Cleaning and tightening

Maintenance practices of ignition system.

1. The carbon deposits on the spark plug electrodes should be removed.
2. The spark plugs whose electrodes are worn out should be replaced.
3. The contact breaker points should be cleaned.
4. The breaker points should be adjusted so that they lie between 0.30 mm and 0.50 mm.
5. The condenser should be replaced regularly.
6. The ignition system should always be kept dry
7. Ignition wires with poor insulation should be replaced.

5. ELECTRICAL SYSTEM.

- It consists of four circuit units:

- i. Ignition circuit.
- ii. Starter motor circuit.
- iii. Generator circuit.
- iv. Lighting circuit.

Uses of electrical system.

1. It ignites/starts the engine at the turn of key.
2. Provides the electric spark for igniting the air-fuel mixture in the cylinders.
3. It powers the lights at the bulbs.

Working mechanism

- The source of power is the **battery** that stores electrical energy.
- The **starter motor circuit** converts electrical energy to mechanical energy that is used to start the engine.
- The **generator** circuit supplies energy that recharges the battery.
- The **ignition circuit** provides a spark that ignites air-fuel mixture.
- The **lighting circuit** lights the bulbs.

Care and maintenance practices on tractor battery.

1. The level of electrolyte should be kept just above the plate by topping with distilled water.
2. Corroded terminals should be scrapped clean and smeared with grease.
3. The battery should be tightly in a box to avoid spillage and damage.
4. The battery should be fitted correctly on the tractor.
5. The battery should be charged regularly and periodically.
6. In case of long storage the battery contents should be emptied and battery kept upside down.
7. The generator fan belt should always be functional to ensure the battery is always charged.

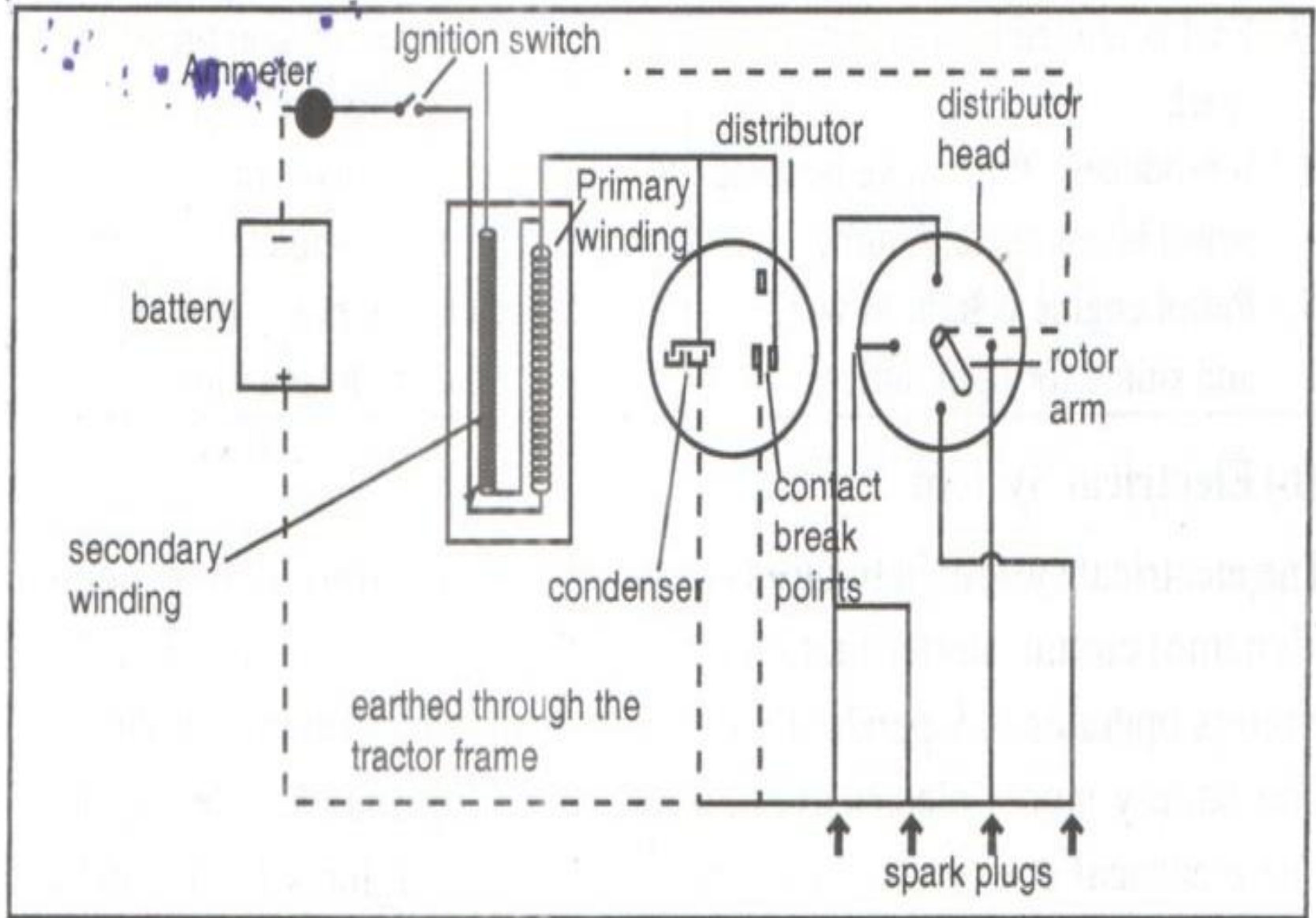


Fig. 3.18: Electrical system of a tractor

6. POWER TRANSMISSION SYSTEM

- It is used to transmit power from the tractor engine to the various parts of the tractor.

Parts of power transmission system.

1. **Clutch-** It consists of crankshaft, friction disk and pressure plate.
 - i. Connects or disconnects the drive shaft from the engine.
 - ii. It enables the tractor to take off gradually and smoothly.
 - iii. Provides power from the engine to the power take off (P.T.O.).

2. Gear box.

Functions.

- i. Helps the driver to select any forward or reverse gear.
- ii. It allows change in the speed ratio by altering gears conveniently in the gear box.
- iii. Enables the driver to stop the tractor without suddenly stopping the engine/ without keeping his foot pressed on the clutch..

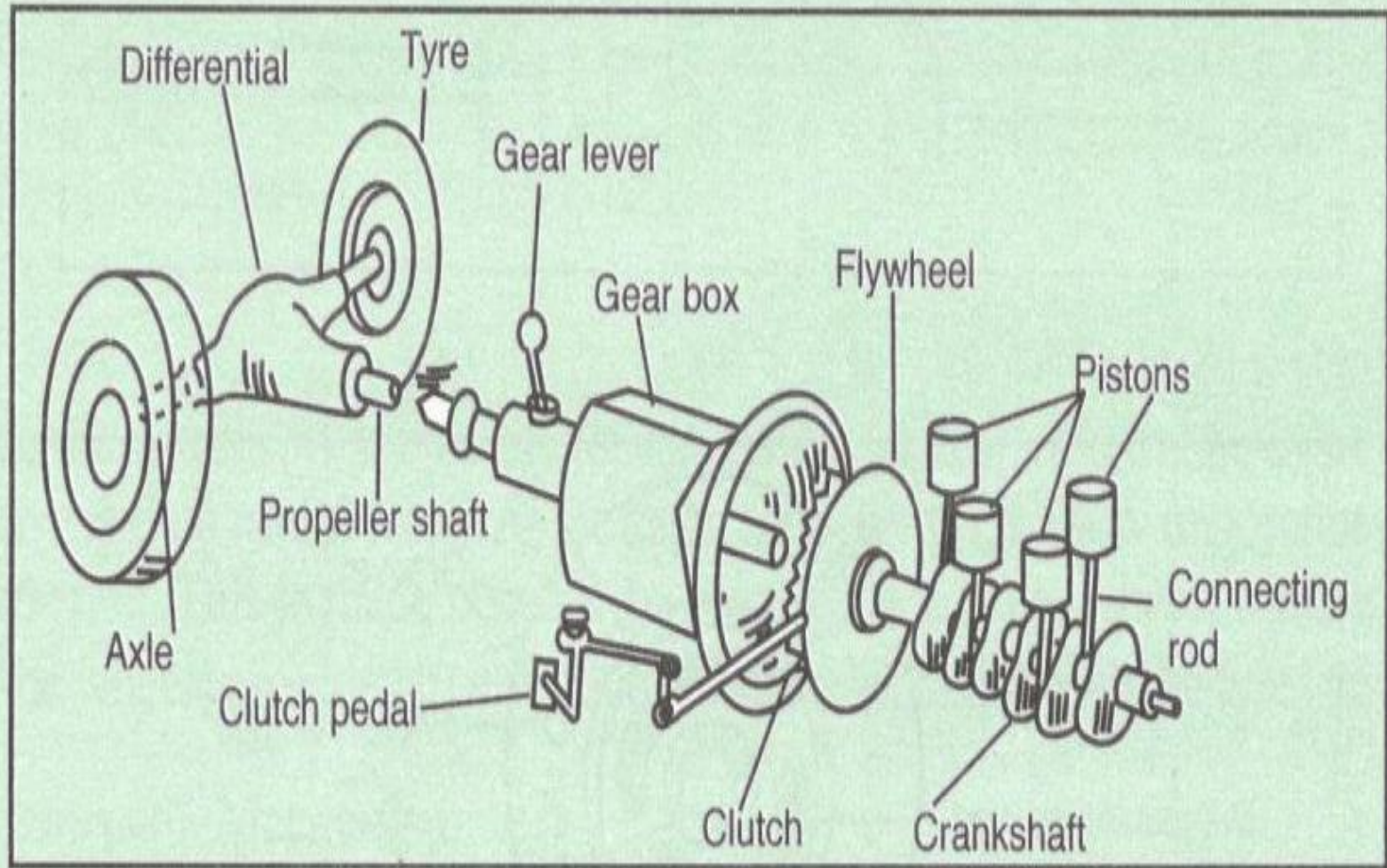


Fig. 3.22: Tractor transmission systems

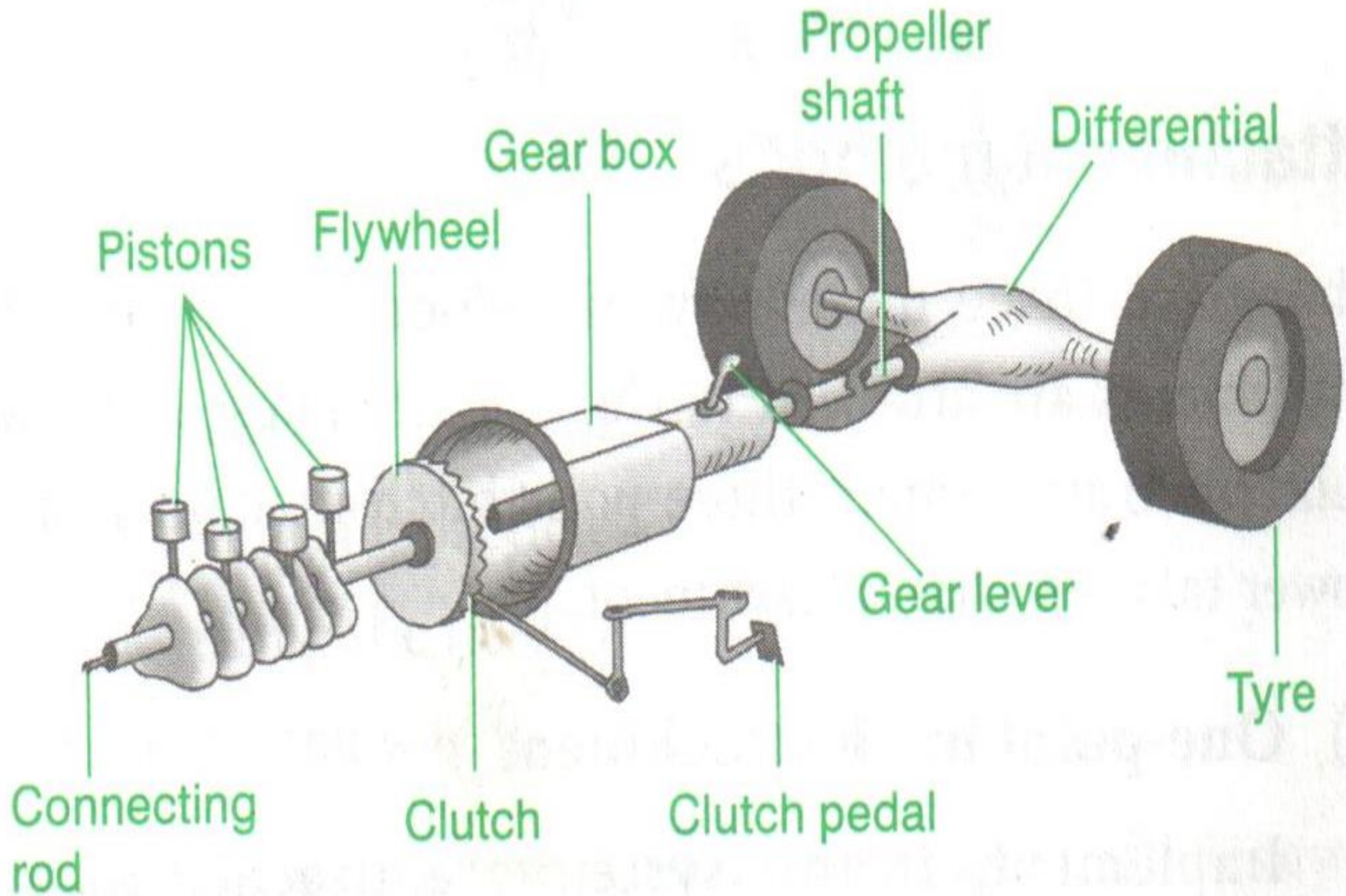


Fig. 12.11 Tractor power-transmission system

3. Differential.

- i. It allows changes in the direction of the drive to the right angles so that power is transmitted to the rear wheels.
- ii. It enables the rear wheels to travel faster or slower than the other especially when negotiating a corner.

4. Final drive.

- It includes the wheels and tyres.
- It is used for forward and backward movement of the vehicle/tractor.
- The tyres allow maximum grip to the ground (traction). The larger wheels provide a large surface area of contact between the tyres and ground.

POWER TRANSMISSION MECHANISM/ **WAYS IN WHICH POWER FROM THE** **ENGINE IS MADE AVAILABLE FOR** **USE.**

- Power transmitted from the engine is made available for use through the following:
 - A. The propeller shaft.
 - B. The power take off (PTO) shaft.
 - C. The hydraulic system.
 - D. The draw bar.
- **A. The propeller shaft.**
 - This connects the gear box to the differential that has axles which drive the wheels making the tractor move.
 - During the forward movement the tractor either pulls or pushes the implements attached onto it.

B. The power take off (PTO) shaft.

- It is located at the rear of the tractor and rotates at the same speed as the crankshaft.
- It can be connected to the machines that work by use of rotary motion **e.g. mowers, rotavators, sprayers, fertilizer sprayers.**

C. Hydraulic system.

- It is operated by use of the lever that is near the driver's seat.
- It is attached to three point linkage which raises or lowers the implements mounted to the tractor e.g. **ploughs, mowers, planters and sprayers.**
- The three point linkage raises/lowers the mounted implements during farm operation.

D. Draw bar.

- This is a bar at the rear part of the tractor.
- It does not get power directly from the engine instead it serves as a point of attachment for the trailed implements which can be used for harrowing, transportation or rolling.

MAINTENANCE PRACTICES/SERVICE **CARRIED OUT ON A TRACTOR.**

SHORT TERM SERVICE/ PRACTICES

1. The engine oil should be checked daily by use of a dip stick. If the oil level is low it should be added.
2. The fuel level should be checked at the start of every day's work and added if necessary.
3. Water level in the radiator should be inspected and if possible topped up.
4. The level of electrolyte in the battery should be checked daily. If the electrolyte level drops below the recommended, top up with distilled water
5. The nuts and bolts should be tightened every day. The lost nuts and bolts should be replaced before the day's work.
6. Grease should be applied by use of grease gun through the nipples.
7. Large sediments from the sediment bowl should be removed.
8. The tyre pressure should be checked every morning before the day's work using pressure gauge. It should be inflated/ deflated as required by the manufacturer.

9. The fan belt tension should be checked to ensure that it is correct.
10. The brake shaft bearing should be greased. Ensure that the brake fluid level is maintained at the recommended level.

LONG TERM SERVICE/ PRACTICES

1. The engine oil should be drained completely from the sump and replaced with a new oil.
2. The steering gear box should be inspected and be refilled if the level goes below the recommended level.

3. The oil in the differential should be replaced as recommended.
4. The linkage and the pulley attachment should be greased. The pulley oil level should also be checked and added if need be.
5. The dirty oil should be removed and replaced with clean one.

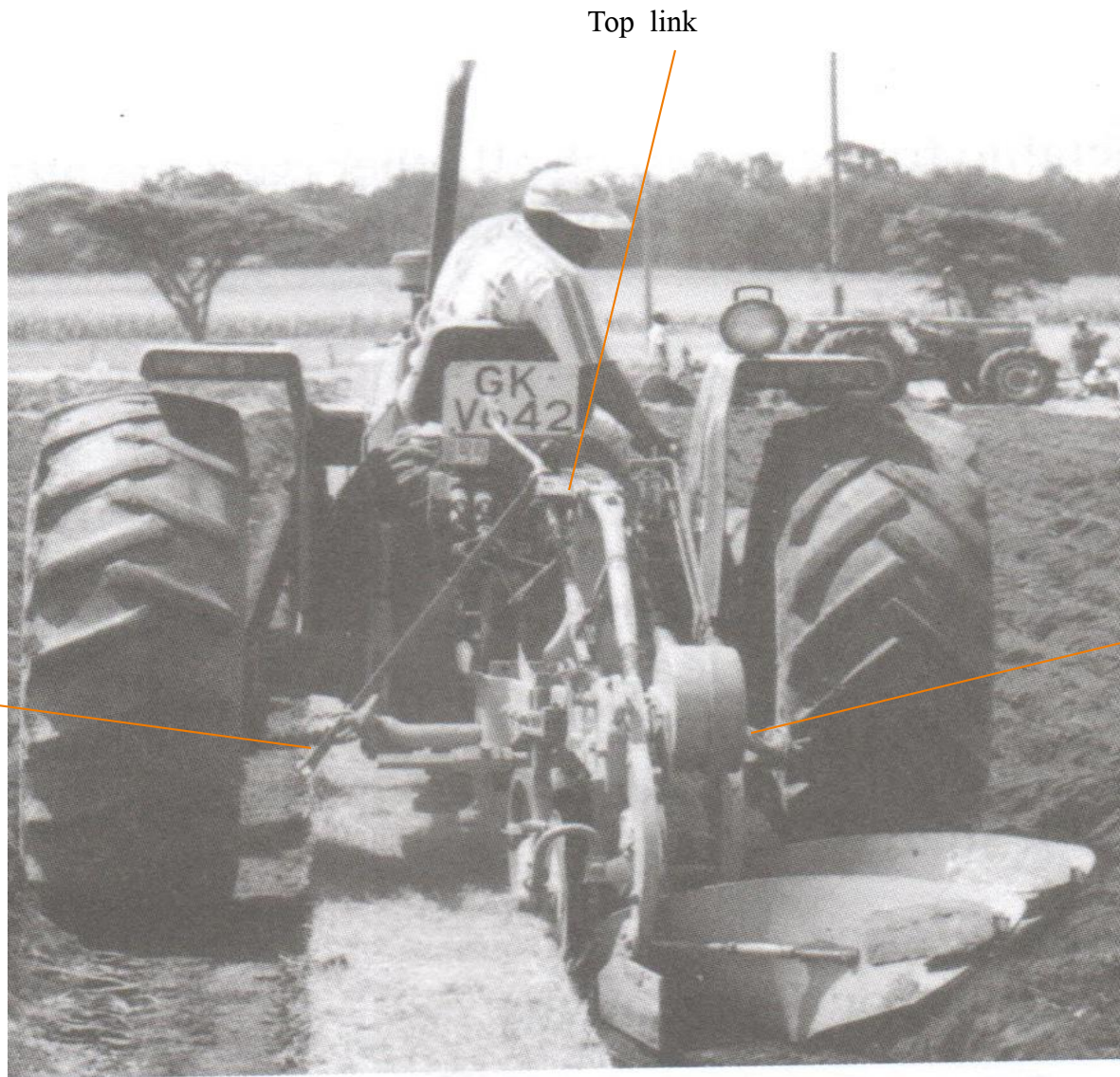
METHODS OF ATTACHMENT TO THE TRACTOR

A. One-point hitch attachment.

- In this the implements are attached to the tractor at one point/ draw bar.
- The implements attached include:
 - i. Trailers.
 - ii. Heavy harrows.
 - iii. Planters.
 - iv. Rollers.
 - v. Manure spreaders.

B. Three-point hitch attachment.

- The implements are attached to the tractor at 3 points (top linkage point and two lower linkage attachment points)
- They are operated by hydraulic system of the tractor.
- The implements attached include ploughs, harrows, sub-soilers, planters, mowers, ridgers, sprayers, rotavators.



Lower right link

Lower left link

Tractor drawn implements.

- 1) Trailers.
- 2) Ploughs.
- 3) Harrows.
- 4) Sub-soilers.
- 5) Ridgers.
- 6) Rotary tillers/rotavators.
- 7) Mowers.
- 8) Planters and seeders.
- 9) Cultivators and weeders.

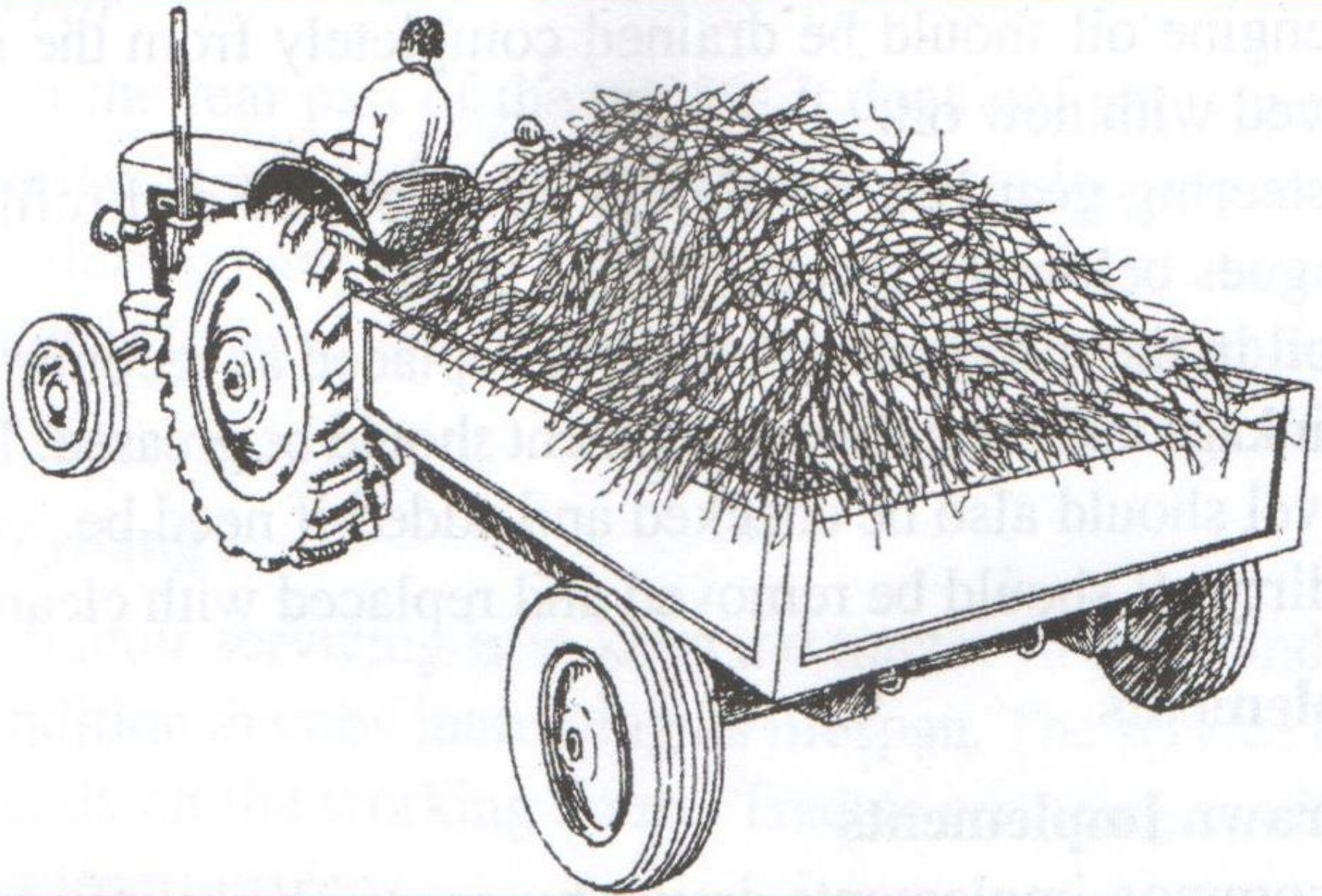
Factors that lead to depreciation of farm equipment.

1. Age of equipment.
2. Change of technology/obsolescence.
3. Wear and tear.
4. Lack of maintenance practice.
5. Improper storage/exposure to weather.
6. Wrong use of the equipment.

1. TRAILERS.

Maintenance practices.

- A trailer is a container drawn by the tractor.
 - It is attached to the draw bar of the tractor.
 - It is used to transport the agricultural produce, implements and input.
1. All moving parts should be lubricated.
 2. The tyre pressure should be checked regularly and adjusted accordingly.
 3. Worn out tyres should be replaced.
 4. Loose nuts and bolts should be tightened and worn out nuts and bolts should be replaced.
 5. The implement should be kept under a shed if not in use.



Trailer



2. PLOUGHS.

- They are used for primary cultivation and include:

A. Disc plough.

B. Mould board plough.

A. **DISC PLOUGH-** It has discs that roll and ride over the obstacles e.g. stones and stumps.

Parts of a disc plough

a) The beam

i) it acts as a point of attachment for other parts of the plough.

ii) Adds weight to the plough to increase the depth of cultivation.

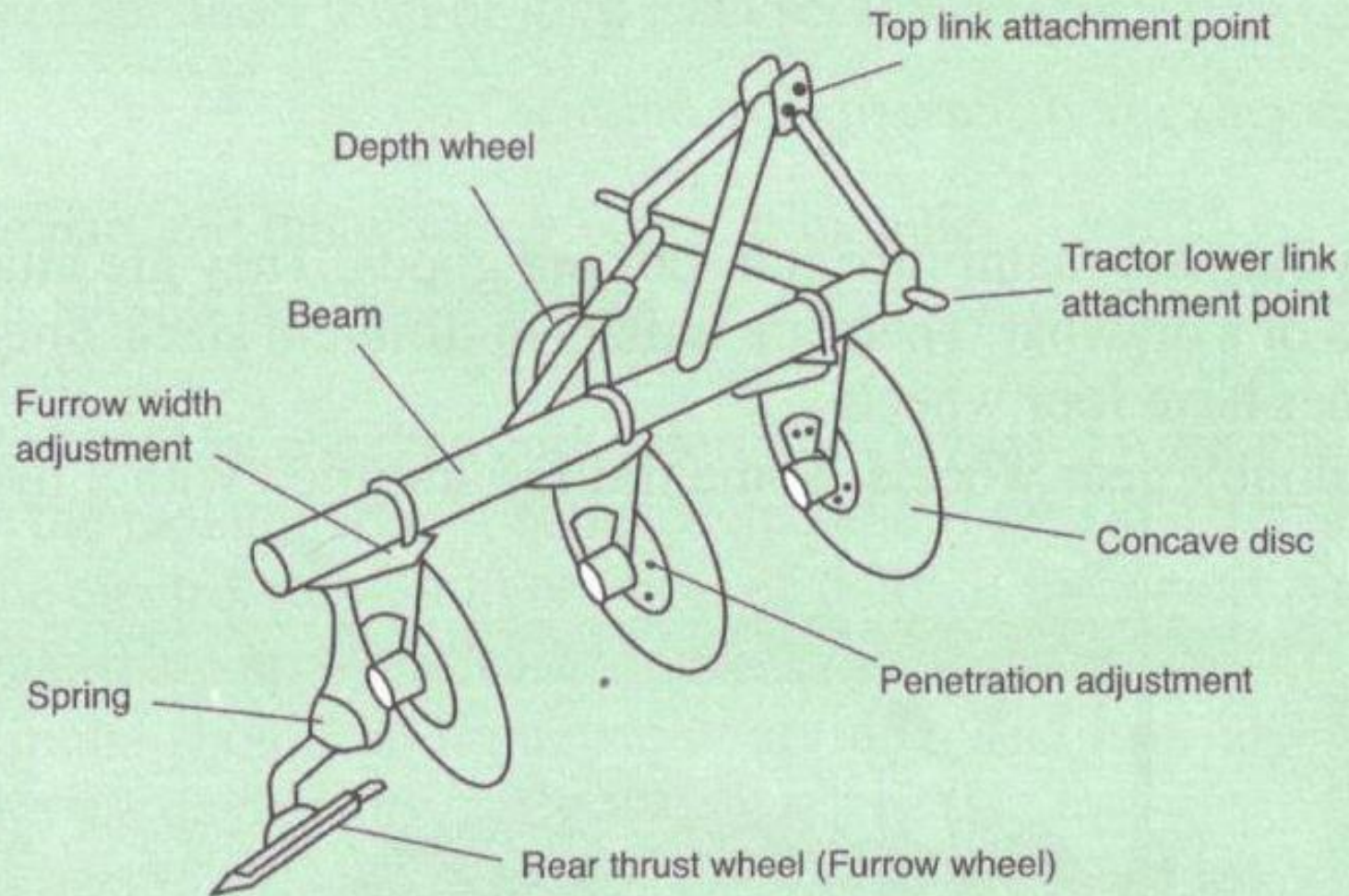


Fig. 3.26: Disc plough

b) Discs- they are concave in shape.

They cut, turn and invert furrow slices.

c) Scrapers- they are attached to each disc. They remove soil from the disc and help to turn and invert furrow slices.

d) Standards/ hangers- they connects discs to the beam.

e) Furrow wheel- it balances the whole implement.

Adjustments done on the disc plough.

1. Changing the cutting angle by adjusting the beam at some degree.
2. Adjusting the depth of cultivation by:
 - i. Changing the height of the depth wheel.
 - ii. Adjusting the hydraulic system.
 - iii. Adding/removing weight from the beam.

Conditions that require the use of disc plough.

1. A field with stones, stumps or roots.
2. Wet, heavy and sticky soil.
3. Hard and dry soil.
4. Soils prone to erosion.
5. Soil covered with thick vegetation and trash.

Advantages of using a disc plough.

1. It requires less power to pull as the discs are able to roll.
2. It can be used on fields with obstacles e.g. stones, stumps.
3. It has relatively low maintenance costs because discs last long.
4. Requires less skill to operate than mould board plough.

Disadvantages of using a disc plough.

- i. It leaves a rough field.
- ii. It does not give uniform depth of ploughing.

Maintenance practices of disc plough.

1. Moving parts (hubs and wheel bearings) should be lubricated.
2. The plough should be cleaned after work.
3. Broken discs should be repaired.
4. Old engine oil should be applied during long storage.
5. Loose nuts should be tightened. Lost/ worn out nuts and bolts should be replaced.
6. Blunt discs should be hammered.
7. Store it under a shed if not in use.

B. MOULDBOARD PLOUGH.

- It has mould boards that completely turn the furrow slices.
- It is hitched by three point linkage.

Parts of a mouldboard plough.

1. **Share** - Makes horizontal cut of the furrow slices and starts turning it over.
2. **Mouldboard-** completes the turning of furrow slices.
3. **Disc coulter**—it is located above the share. It cuts the furrow slices separating them from the unploughed land.

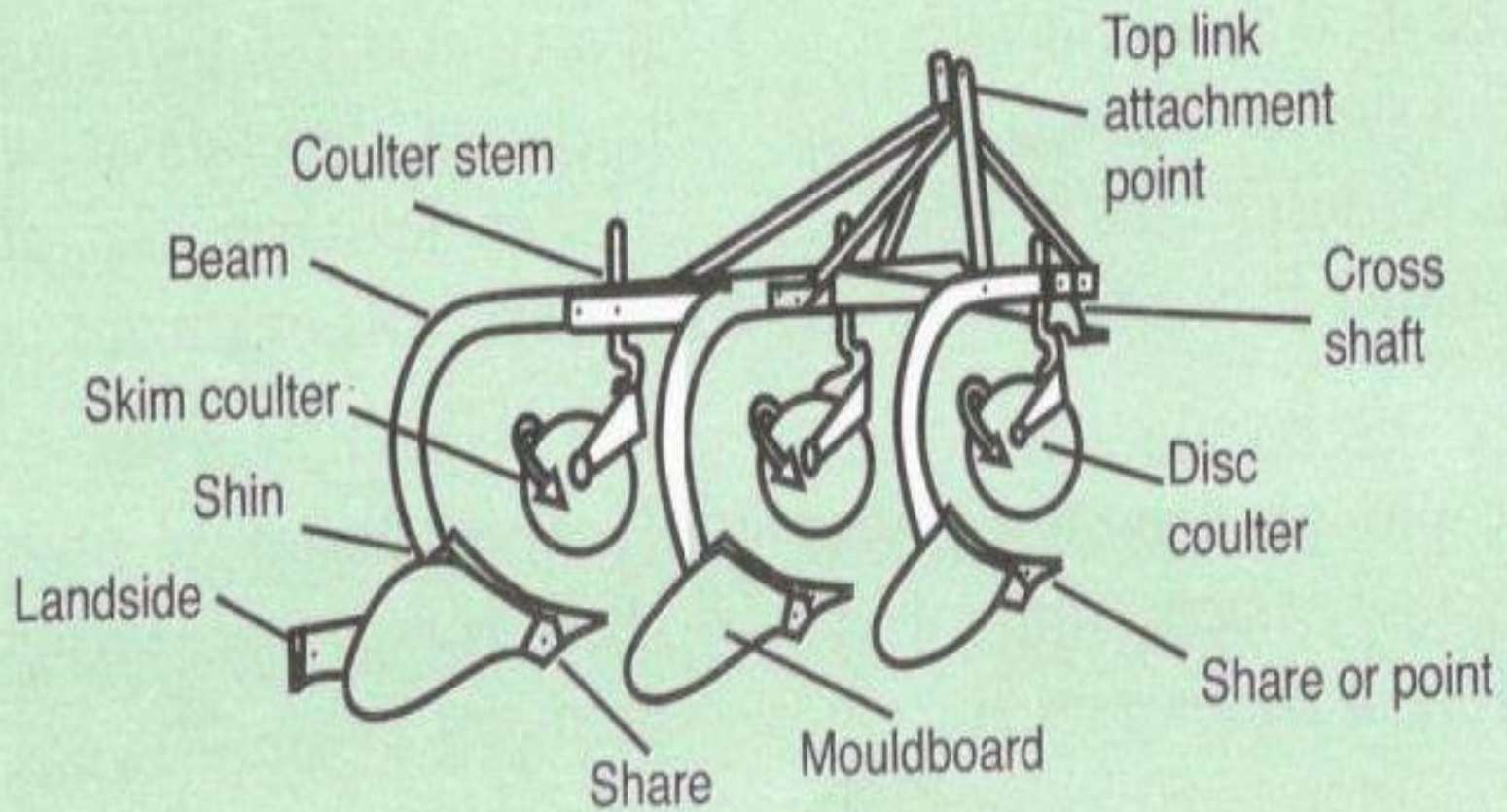


Fig. 3.27: A mouldboard plough



4. **Landside-** Stabilizes the plough by pressing against the wall of the land being ploughed.
5. **Beam/frame-** Provide a point of attachment for other parts of the plough.

Adjustments done on mould board plough.

1. Adjustment of the hydraulic system.
2. For shallow cultivation shortening of the top link.
3. For deep cultivation lengthening of the top link.

4. Controlling the depth of ploughing by lowering the depth wheel.
5. Controlling the furrow width by cross shaft adjustment.

Advantages of using mould board plough.

- i. There is uniform depth.
- ii. Produces a clean and good seed bed.

Disadvantages of using mouldboard plough

1. It can be damaged by obstacles.
2. It cannot plough effectively on dry and hard soils.
3. It can easily cause hard pans because of constant depth of ploughing.
4. Has higher maintenance costs due to frequent replacement of shares.

Maintenance practices of mouldboard ploughs

1. The moving parts should be lubricated.
2. Shares should be kept tight and sharp.
3. Loose nuts and bolts should be tightened.
4. Worn out parts should be replaced.
5. The plough should be cleaned after day's work.
6. Apply old engine oil for long storage.
7. Store it under a shed if not in use.

OPERATIONAL/FUNCTIONAL

DIFFERENCES BETWEEN DISC AND MOULD BOARD PLOUGHS

Disc plough.

1. It does not fully invert soil slices leaving the land rough
2. It requires more secondary operations because slices are not completely inverted leaving land rough.
3. It can be used on a field with obstacles e.g. stones, roots, stumps because of the rolling action of the discs.

Mould board plough.

1. It completely inverts soil slices facilitating decomposition leaving the land fine.
2. It requires fewer secondary operations because slices are fully inverted.
3. It cannot be used on a field with obstacles e.g. stones, roots and stumps because it cannot ride over them.

Disc plough.

- 4. It cuts/ploughs at different levels because of rolling ability.
- 5. It is not easily broken by obstacles because it rides over them.
- 6. It requires less power to pull when operating.

Mould board plough.

- 4. It cuts/ploughs at the same depth.
- 5. It is rigid and therefore can be easily broken by obstacles it comes across.
- 6. It requires more tractor power to pull when operating.

3. HARROWS.

- They are used for secondary cultivation/ harrowing and carry out the following functions:

1. Leveling the seed bed.
2. Breaking large soil clods.
3. Mixing up soil.
4. Destroying weeds.
5. Covering broadcasted seeds.

Types of harrows.

- A. Disc harrows.
- B. Spring-tined harrow.
- C. Spike toothed harrow.
- D. Zigzag harrow.

A. DISC HARROWS.

- It has concave discs which are either plain-edged or notched.

Uses.

1. Mixing soil particles.
2. Levelling the field.
3. Destroying weeds.

Advantages of using disc harrows.

1. They can work the top soil without bringing up the buried weeds.
2. They chop the green manure prior to primary ploughing.
3. There are few operations done hence less moisture is lost.
4. They work well on heavy soils.

Disadvantages of using disc harrows.

1. They can help in propagation of weeds with underground structures.
2. They are expensive to buy and maintain.

Maintenance practices on disc harrows.

1. Lubricate moving parts.
2. Replace/repair broken parts/ discs.
3. Tighten loose nuts and bolts.
4. Remove mud/trash/ soil after use.
5. Apply old engine oil on all parts for storage for long periods.

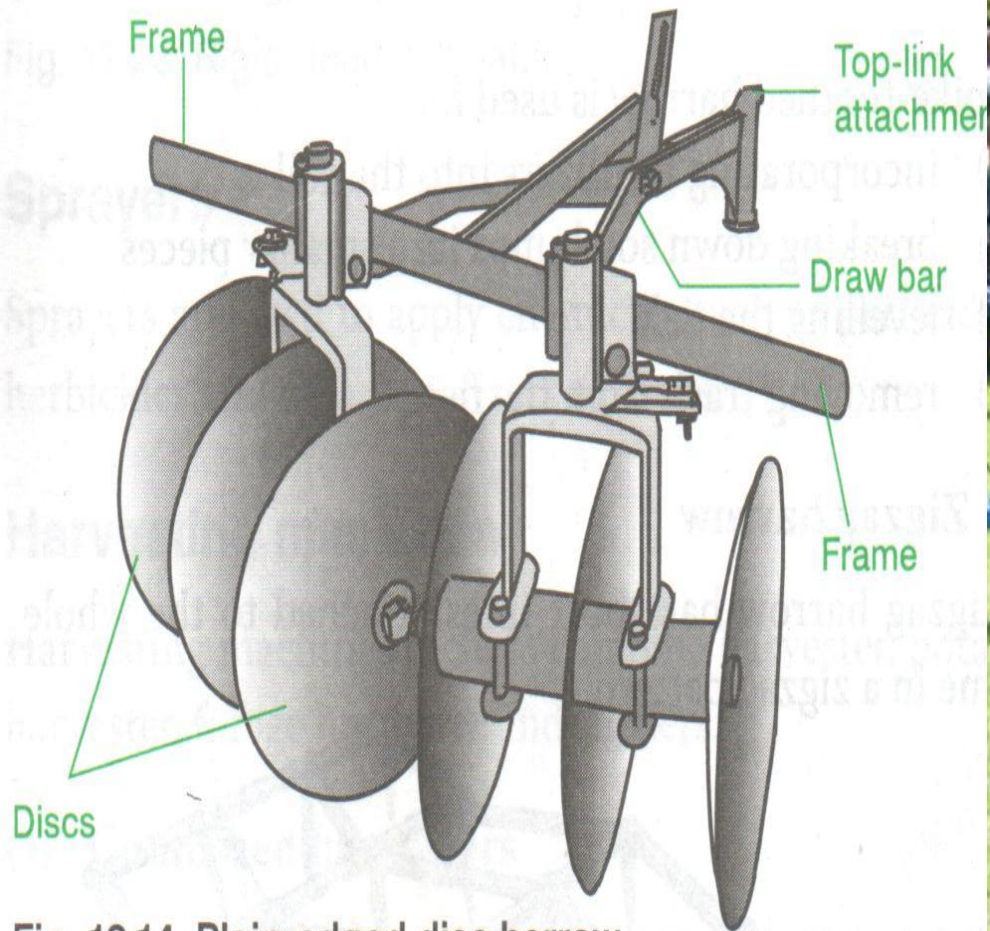


Fig. 12.14 Plain-edged disc harrow



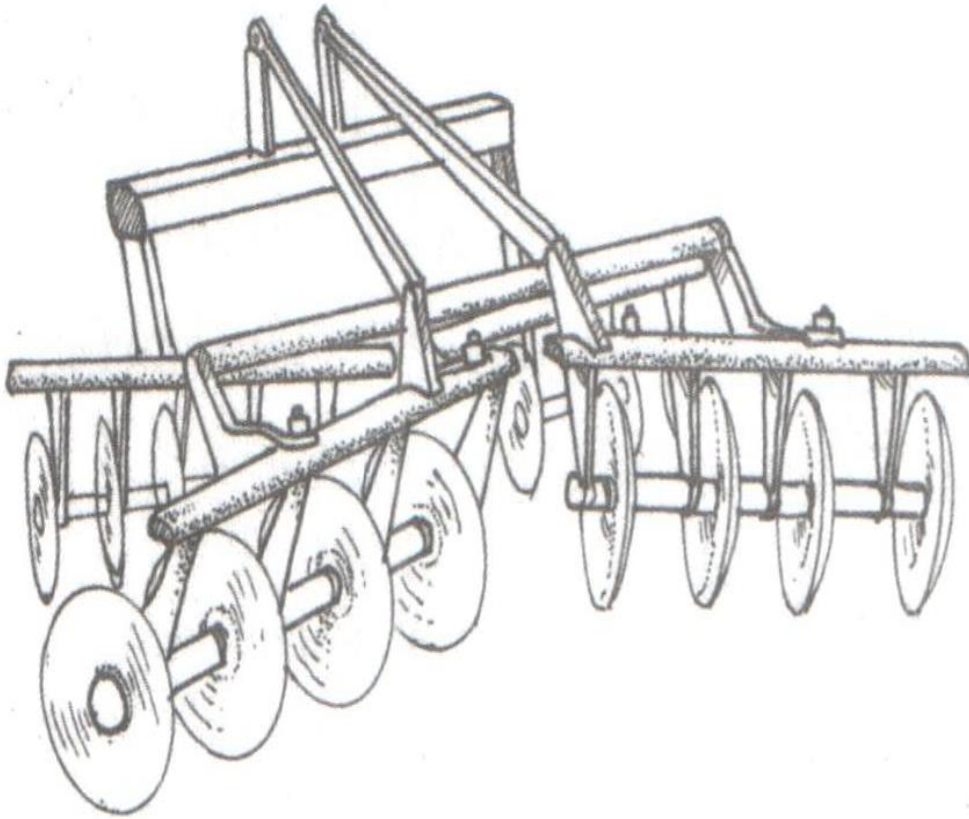


Fig.3.29: Four-gang harrow in a tandem.

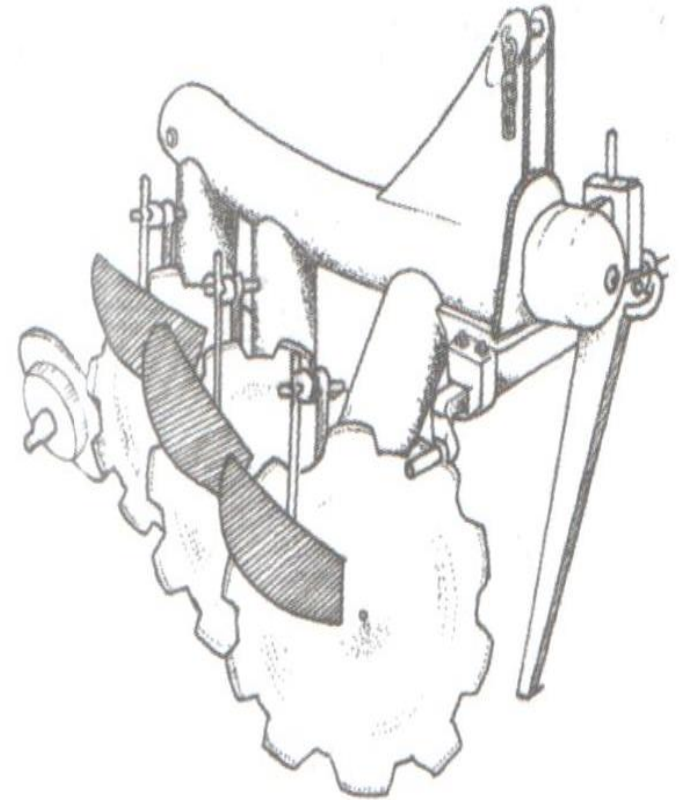


Fig.3.30: Notched disc harrow.

B. SPRING TINE HARROW.

- It has coiled spring-like tines that break the soil clods by means of their vibrating effect.

Uses.

1. Levelling the field.
2. Breaking soil clods.
3. Burying the trash.
4. Aerating the soil.
5. Removing weeds.

Maintenance practices.

1. Tighten loose nuts and bolts and replace lost nuts and bolts.
2. Replace worn out tines.
3. Apply old engine oil for long storage.
4. Clean after use.
5. Straighten the spikes when crooked.

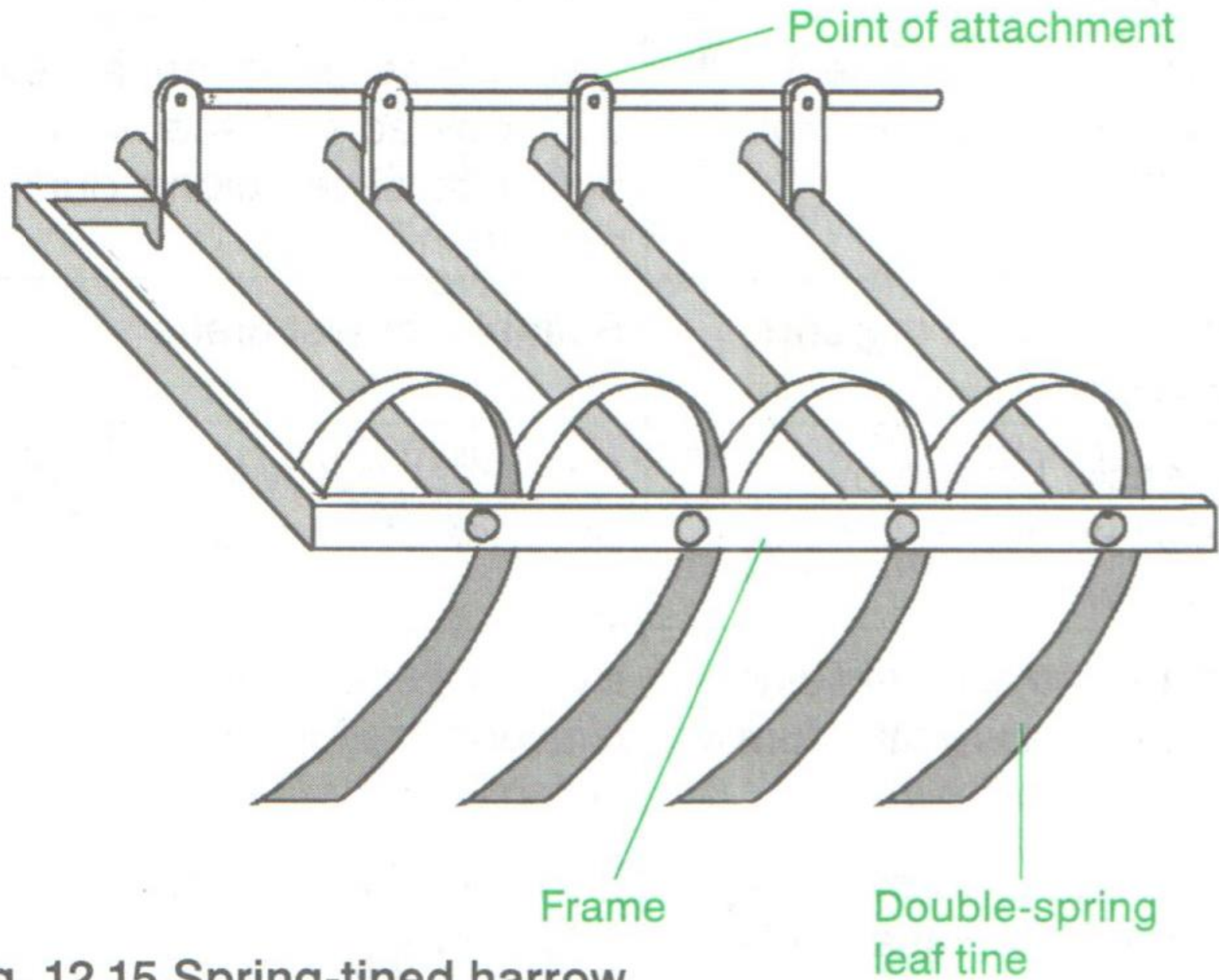


Fig. 12.15 Spring-tined harrow

C. SPIKE-TOOTHED HARROW.

- It consists of many strong and pointed spikes or tines.

Uses.

1. Breaking soil clods.
2. Levelling the soil surface.
3. Removing trash from the field.
4. Incorporating fertilizers into the soil.

Maintenance practices.

1. Replace/repair worn out parts.
2. Clean after work.
3. Tighten loose nuts and bolts.
4. Oil for long storage.

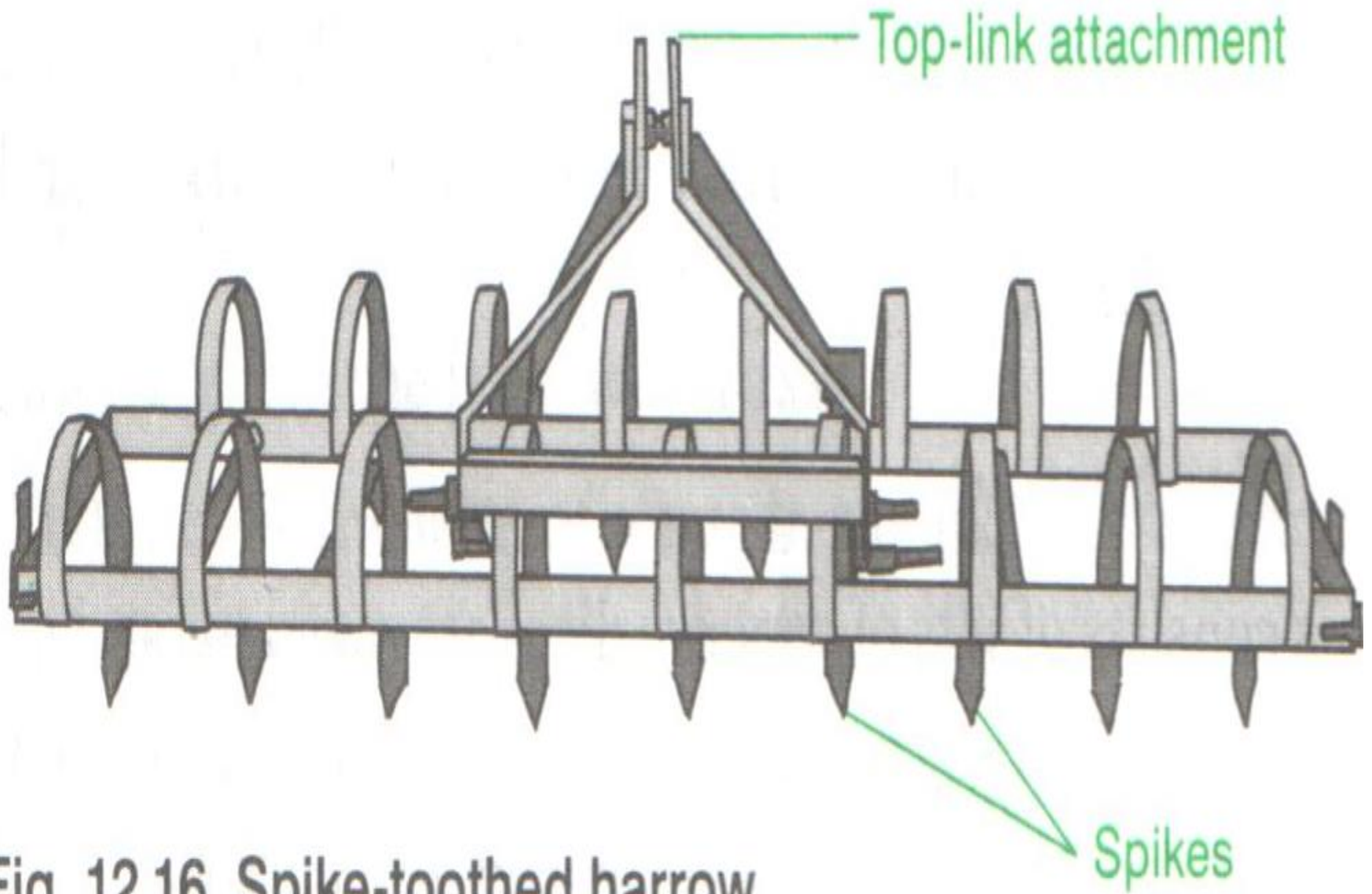


Fig. 12.16 Spike-toothed harrow

D. ZIGZAG HARROW.

- The tines are attached to the frame in a zigzag manner.

Uses.

1. Incorporating fertilizers into the soil.
2. Breaking up of soil clods.
3. Levelling the field.
4. Removing trash after harvesting crop.

Maintenance practices.

1. Replace/repair worn out parts.
2. Clean after work.
3. Tighten loose nuts and bolts.
4. Oil for long storage.

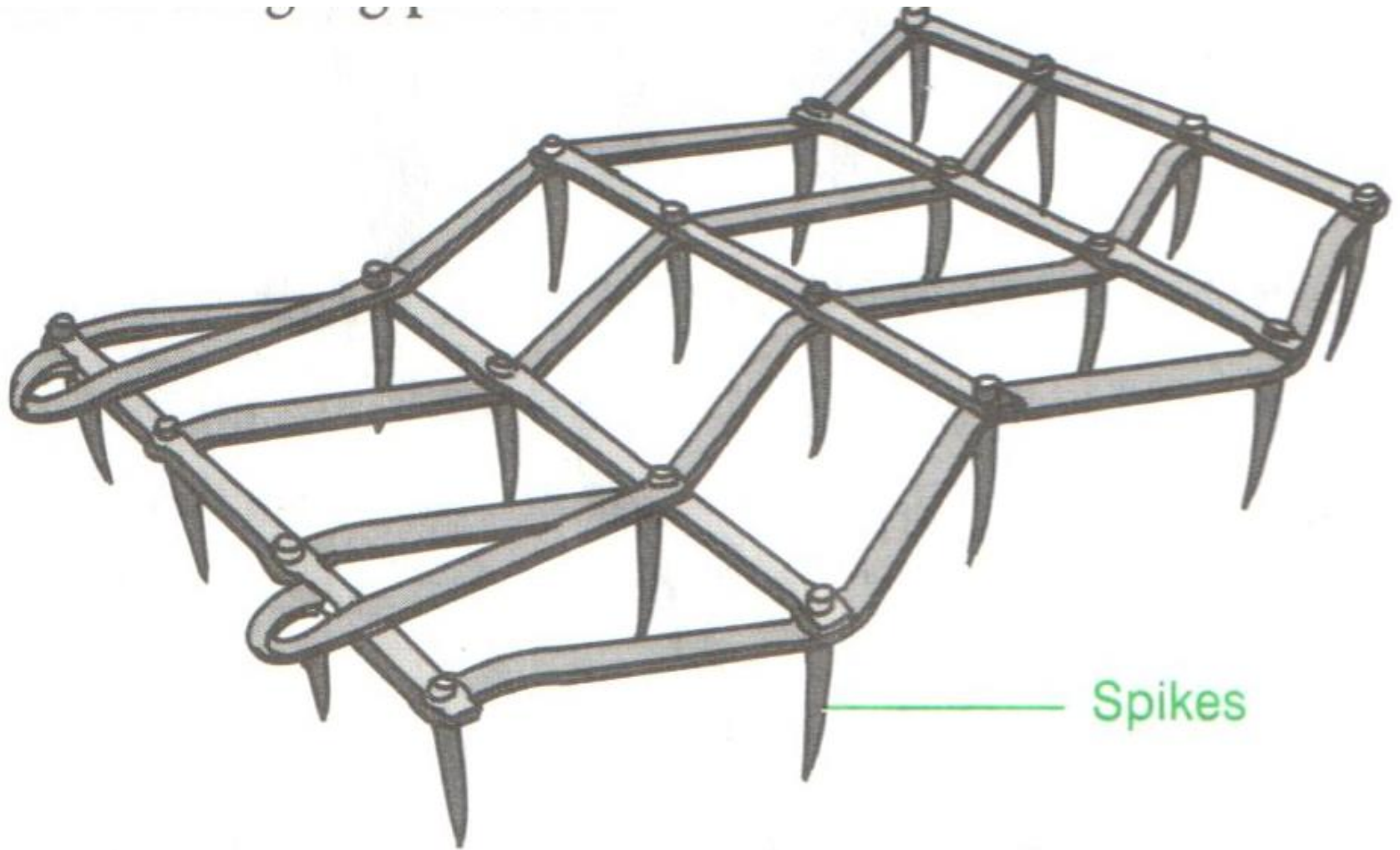


Fig. 12.17 Zigzag harrow

4. SUB-SOILERS.

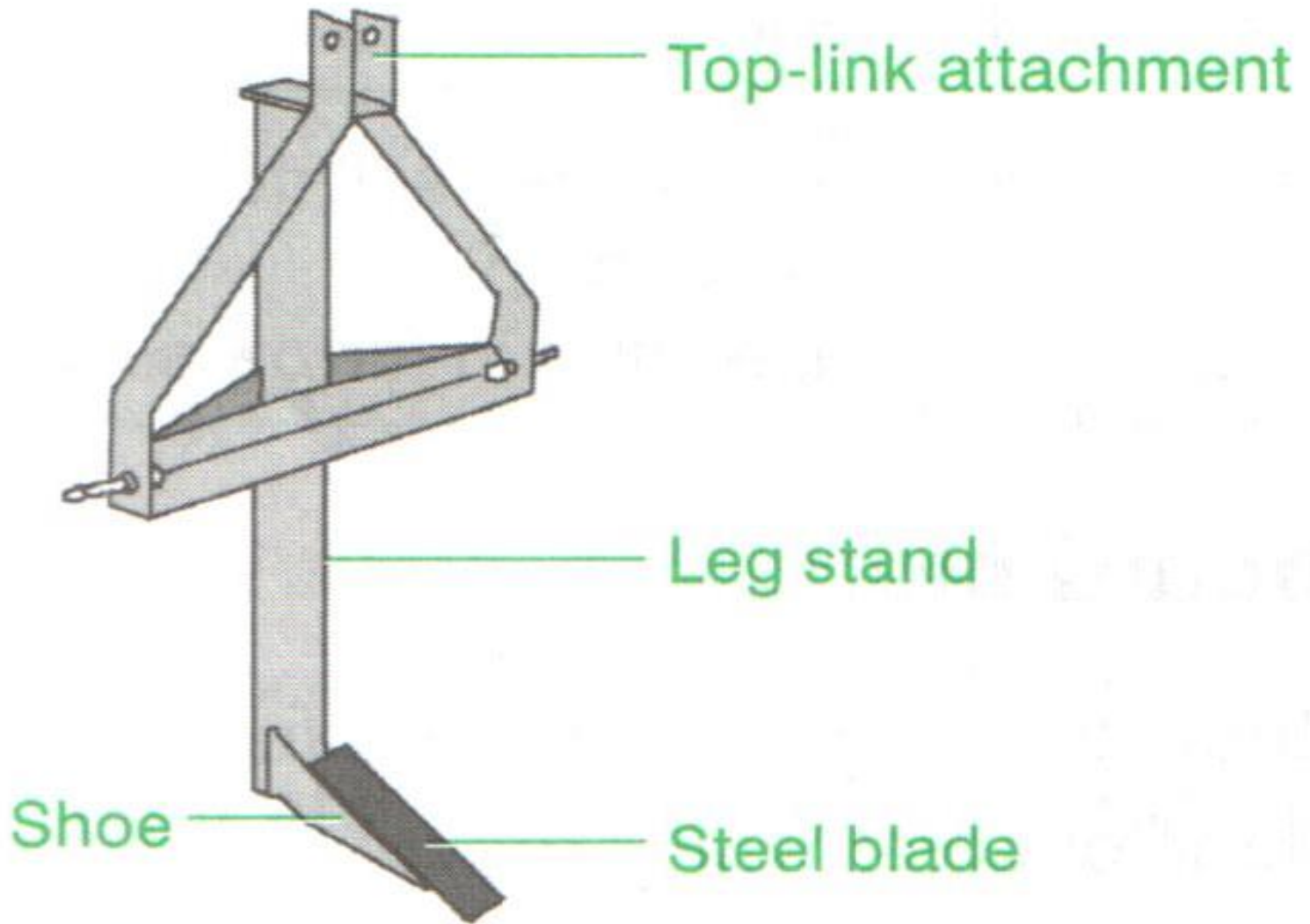
- They are heavy implements mainly used to break hard pans.

Uses.

1. Breaks hard pans.
2. Improves drainage/water infiltration.
3. Improves soil aeration.
4. Destroys/uproots deep rooted weeds.
5. Facilitates development of root crops.
6. It loosens top soil without bringing the sub soil to the surface to ensure minimum tillage.

Maintenance practices.

1. Tighten loose nuts and bolts.
2. Replace/repair broken parts.
3. Oil for long storage to prevent rusting.
4. Clean after use.
5. Store under a shed when not in use.



Subsoiler

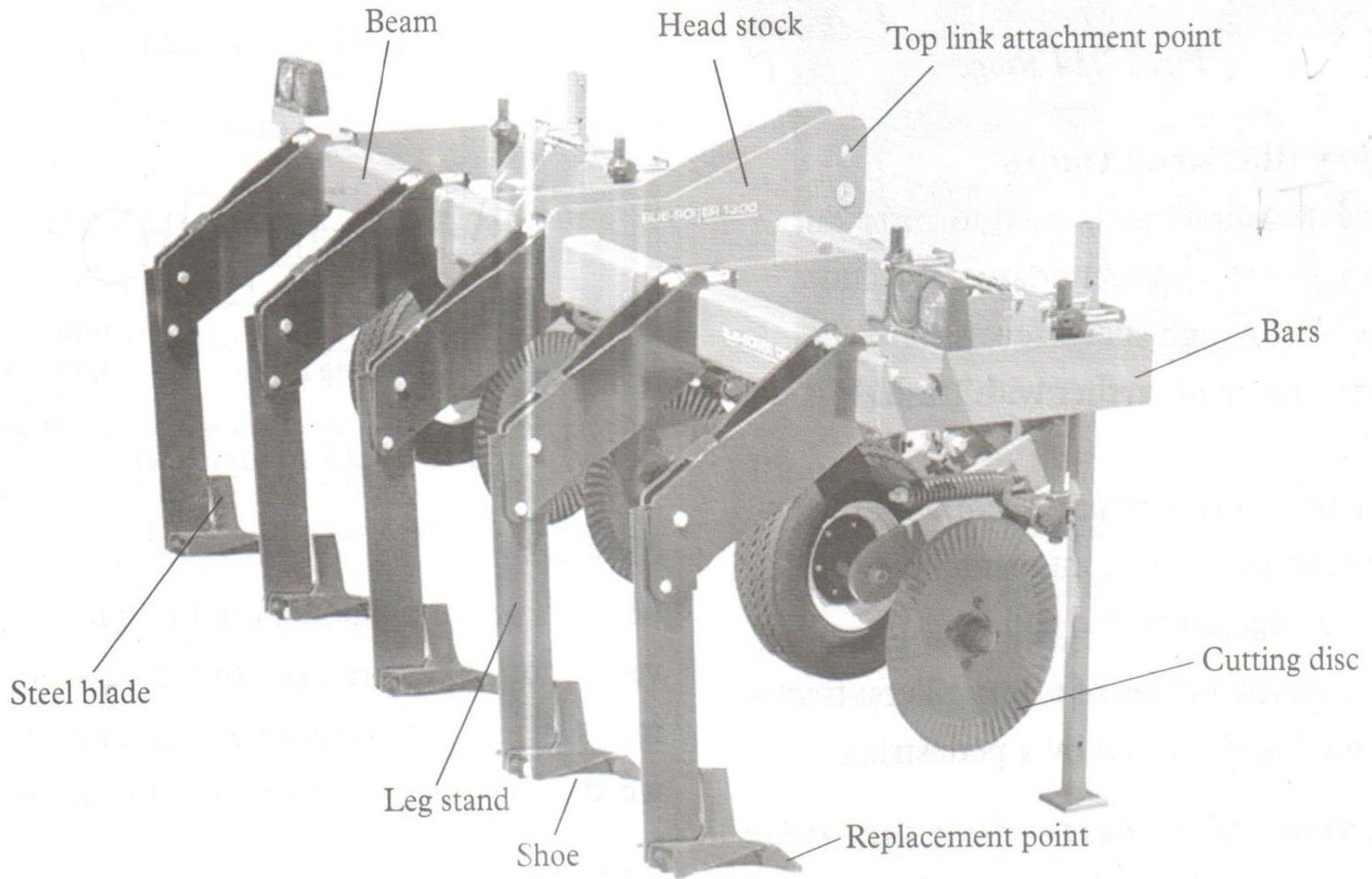


Fig. 12.13 Sub-soiler

5.RIDGERS

- A ridger is a double mouldboard plough that creates a ridge and a furrow when pulled.

Maintenance practices.

1. Clean the mouldboards after a day's work.
2. Replace worn-out parts/ shares.
3. Repair broken parts.
4. Store properly under a shed when it is not in use.
5. Tighten loose nuts and bolts.
6. Oil the unpainted parts to prevent rusting.

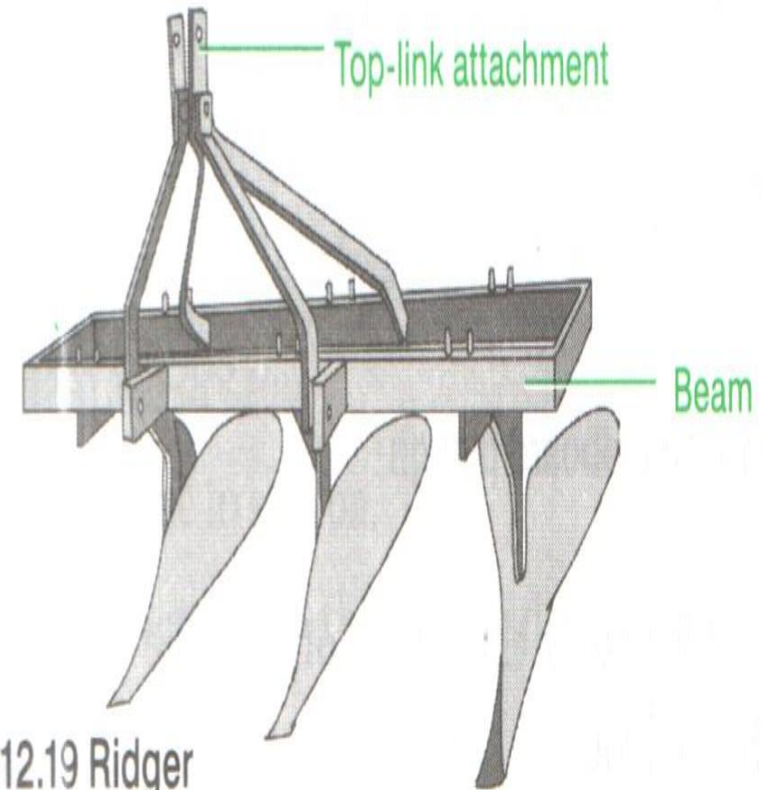


Fig. 12.19 Ridger

6. ROTARY TILLERS (ROTAVATORS)

- Are used in soft soil conditions e.g. in rice paddy fields to produce a fine mud in preparation for planting.

Functions of parts of a rotavator.

- 1) **Rotor**- provides attachment for the blades/flails.
- 2) **Blades**- cut out slices.
- 3) **Shield**- prevents the flying clods and stones from injuring the operator.
- 4) **Teeth adjustment**- changing the forward speed.
- 5) **Depth adjustment**- controls the depth wheel.

Maintenance practices on the rotavators.

1. Clean after a day's work.
2. Worn-out parts should be replaced.
3. The moving parts should be lubricated.
4. Unpainted parts should be oiled during long storage.
5. Nuts and bolts should be tightened.

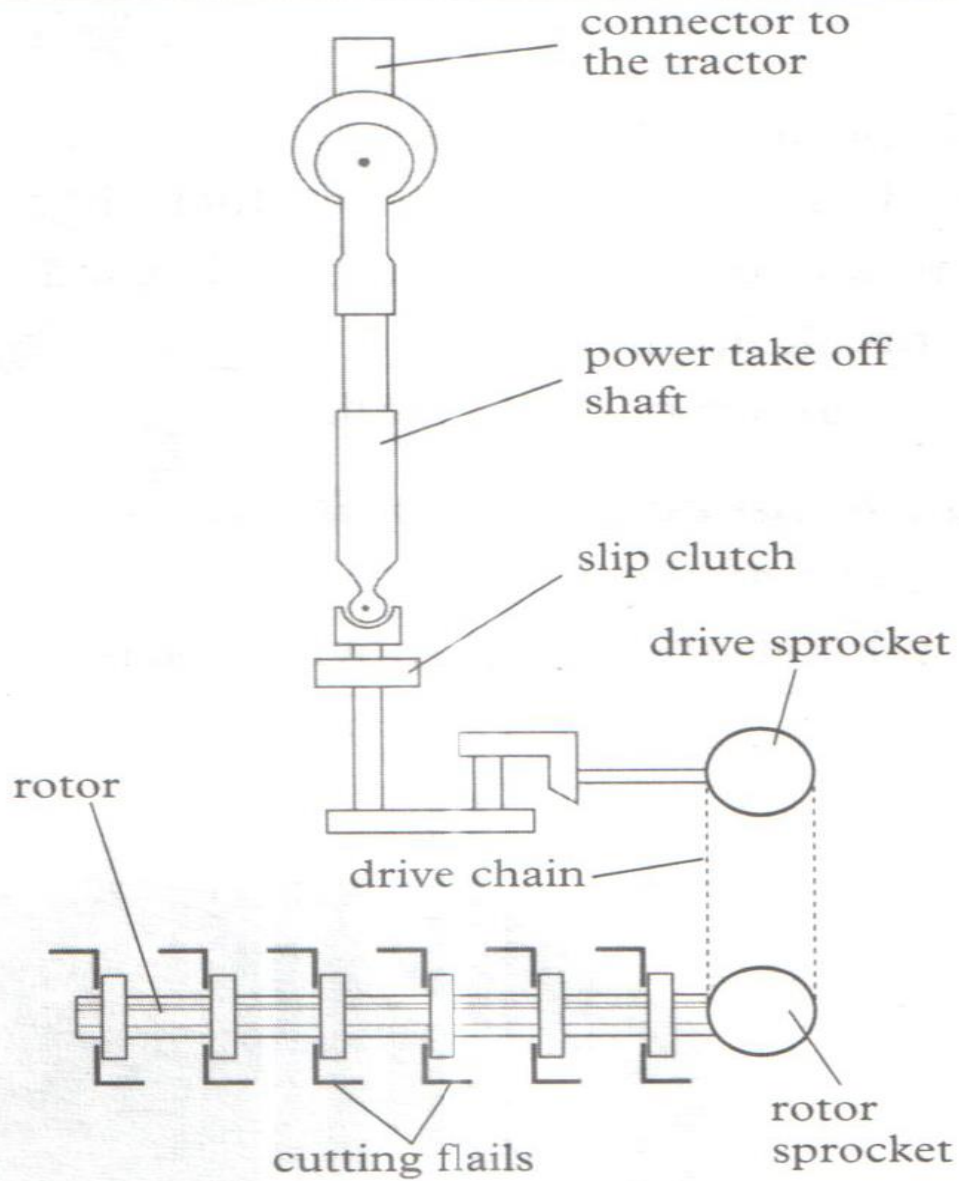


Fig. 12.15 Rotary tiller

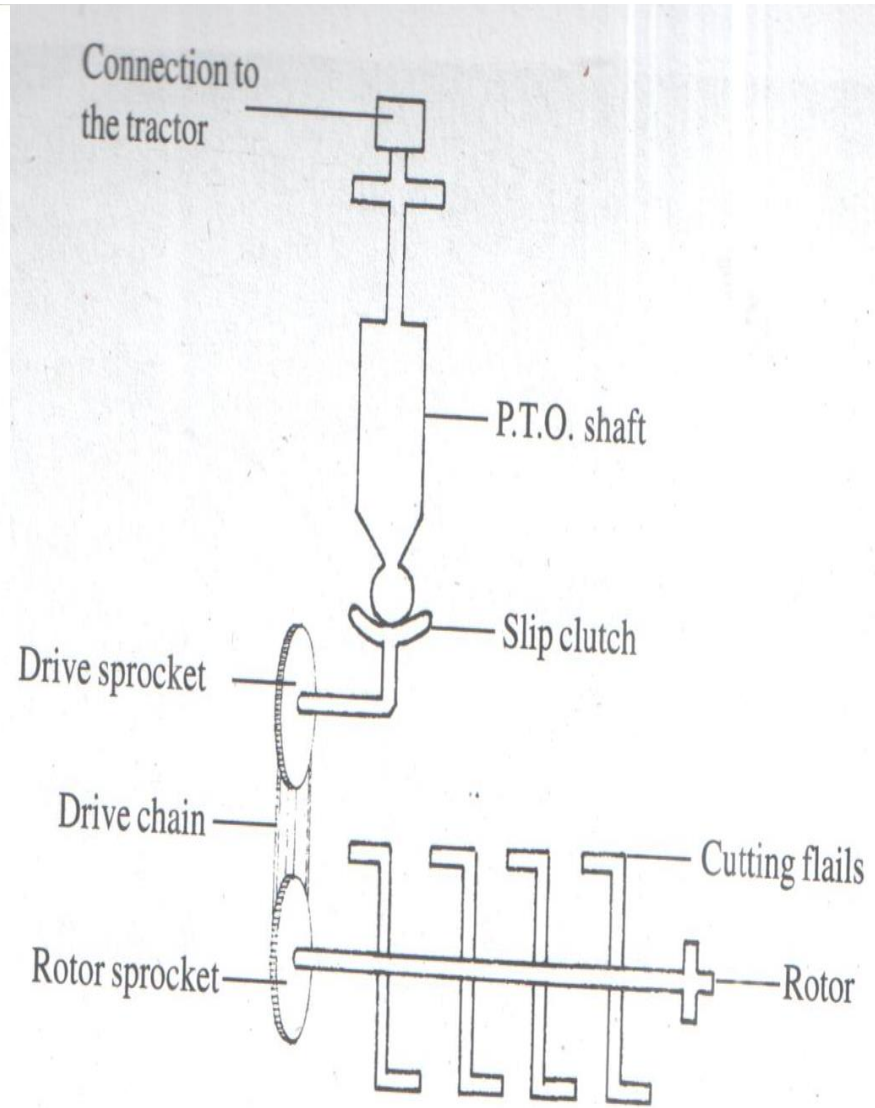


Fig.3.37: Parts of a rotavator.

7. MOWERS.

- They are used for:

1. Cutting/ clearing grass (mowing) for hay making and cutting grass.
2. Clearing vegetation before ploughing.
3. Cutting grass in lawn.

Types of mowers.

- A. Rotary/gyro mower.
- B. Reciprocating mower.

- A. **Rotary mower-** it is driven by PTO shaft. Cutting of vegetation is done by rotary motion of the blades.
 - The blades are enclosed by a shield to prevents flying objects thrown by blades from causing damage to the people/ property.

Maintenance practices of a rotary mower

1. Blunt blades should be sharpened.
2. Worn-out blades and guard should be replaced.
3. Clean the mower after a day's work.
4. Tighten loose nuts and bolts.
5. Rotating parts should be lubricated.
6. Unpainted parts should be oiled before long storage.

B. Reciprocating mower-

- It has fingers fixed on the cutter bar which is held horizontally during the operation.
- It is connected to PTO shaft and the reciprocating movement moves the cutting blades of the cutting knife between the fingers on the cutting bar.
- This movement of the blades between the fingers cuts the vegetation.

Parts of a reciprocating mower

1. **Swath stick-** directs cut grass to fall in the right direction.
2. **Swath board-** separates cut grass from the uncut.
3. **Shoe-** separates the grass to be cut from the rest.
4. **Cutter bar-** it forms the point where other parts of the mower are attached.
 - It also holds the grass being cut.
5. **Wear plate-** this is the part onto which knife bars move to and fro.
6. **Knife clips-** they hold the knives onto the ledger plate.

7. **Pitman-** converts vertical rotary movement to horizontal reciprocating movement.
8. **Ledger plate-** this is the part onto which knives are fixed.

Maintenance practices on reciprocating mower

1. Sharpen the cutting blades.
2. Tighten loose nuts and bolts.
3. Lubricate moving parts.
4. Repair broken parts.
5. Replace worn- out parts.
6. Clean after a days' work.
7. Oil the unpainted parts before long storage.
8. Store properly under a shed.

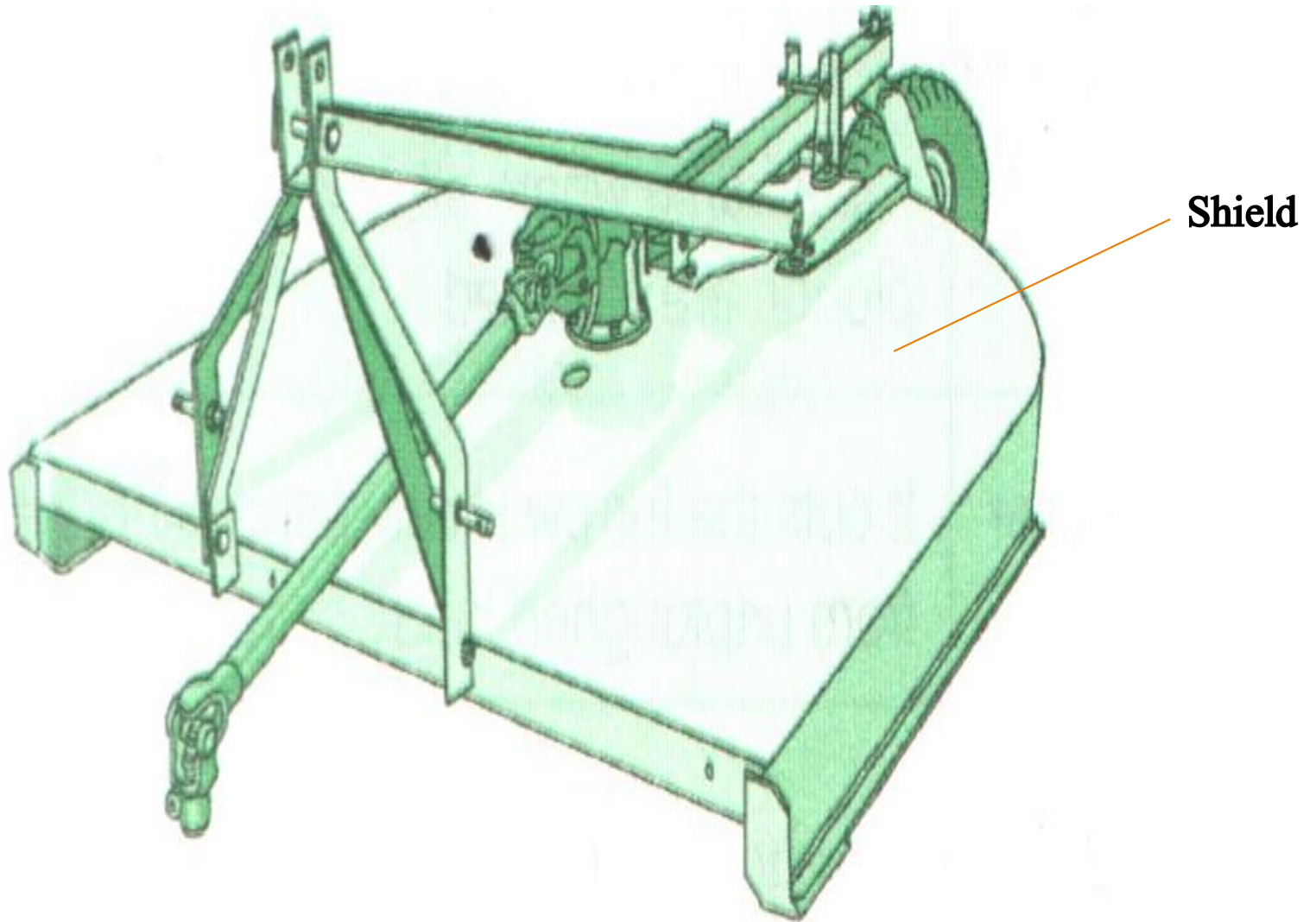


Fig. 12.20 Rotary mower



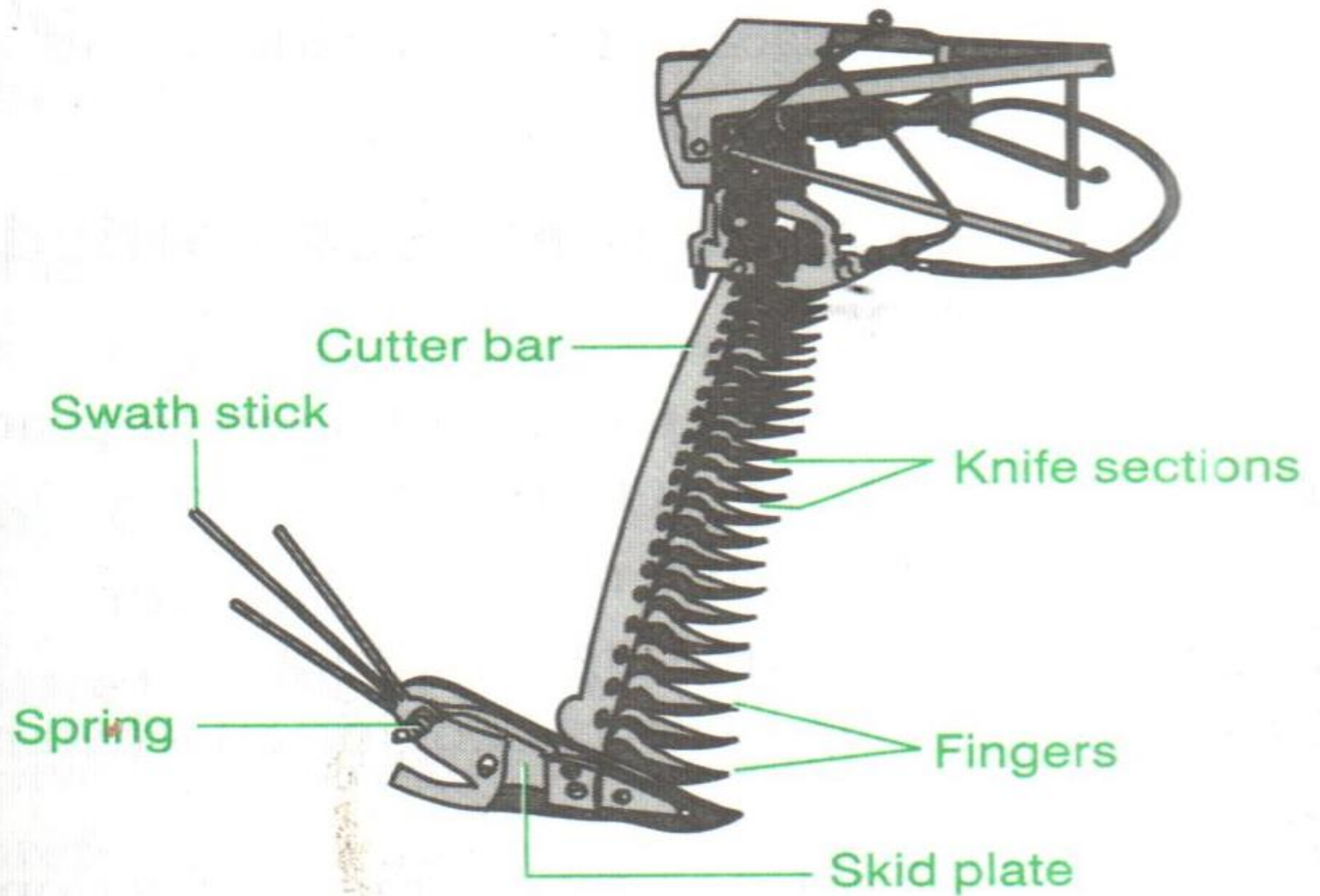


Fig. 12.21 Reciprocating mower

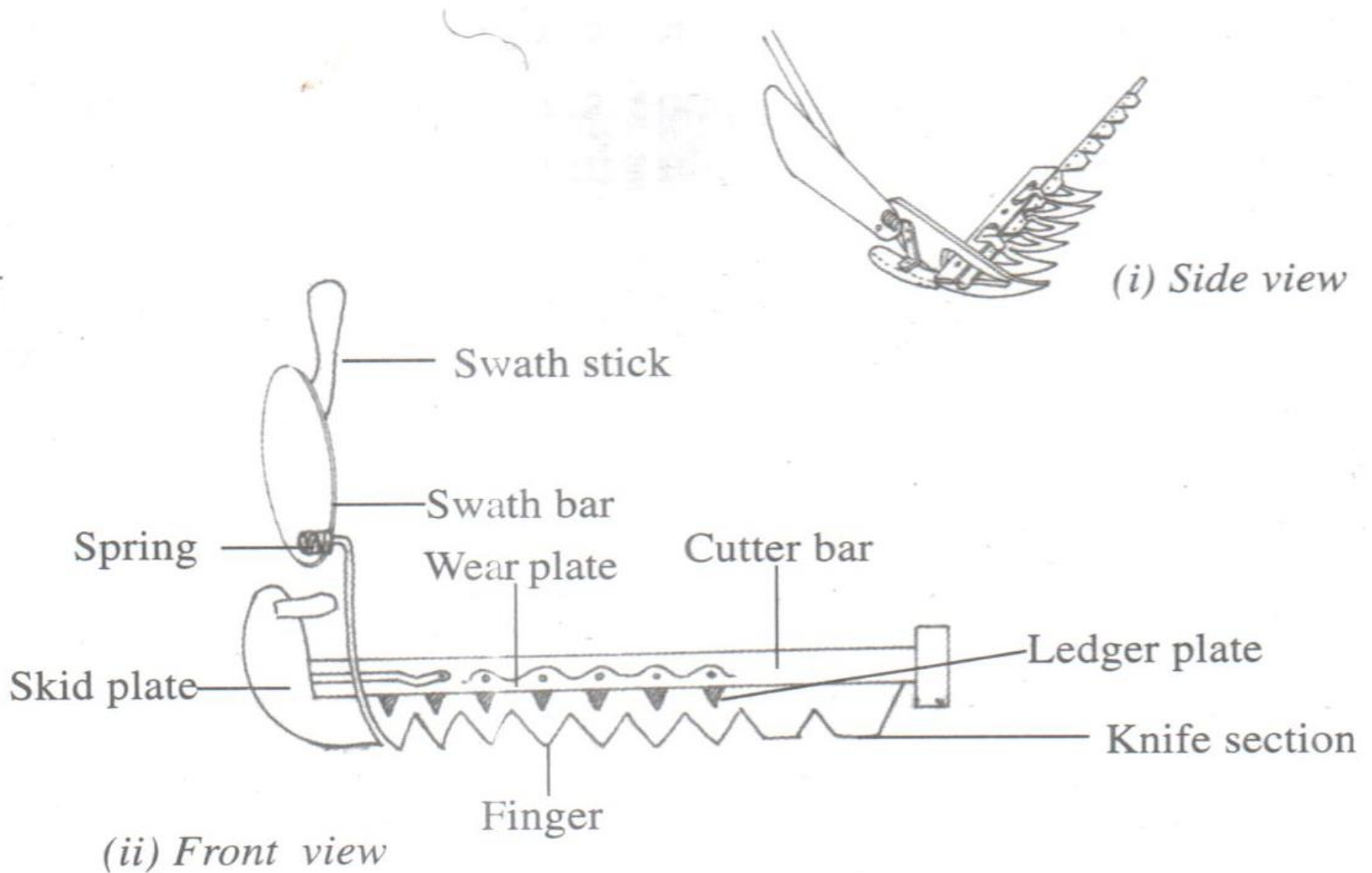


Fig. 3.38: Parts of a reciprocating mower.

8. PLANTERS AND SEEDERS

- They are used for planting seeds at the correct spacing, depth and rate to ensure proper plant population.

Parts of planters and seeders.

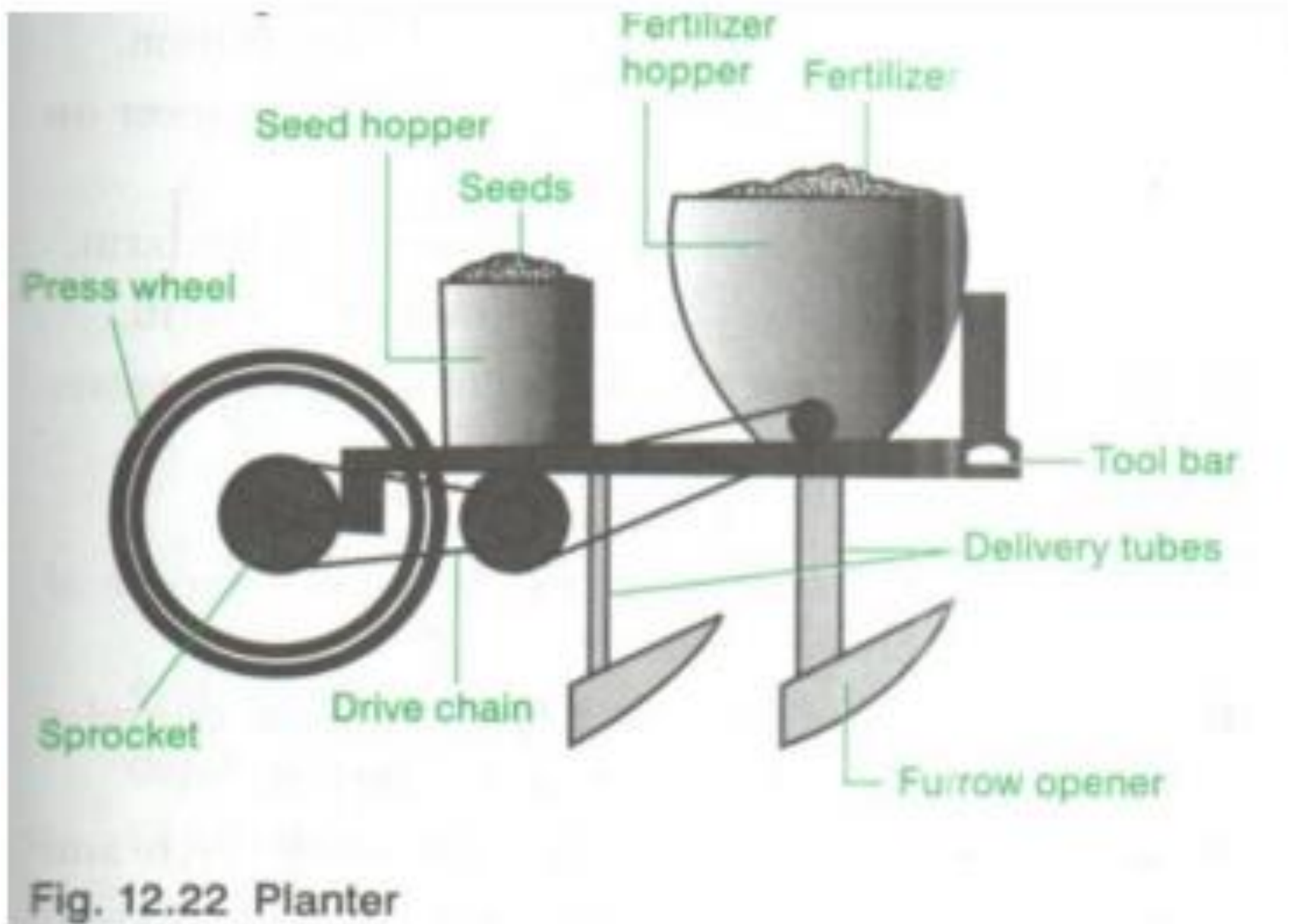
1. **Seed hopper-** this is where seeds are placed.
2. **Fertilizer hopper-** this is used for distributing fertilizers
3. **Tool bar-** it is used to connect the tool onto the tractor.
4. **Seed plates-** they are plates that pick up the seeds and carry them around until they drop into the delivery tubes.
5. **Delivery tubes-** they convey the seed and fertilizer from the seed and fertilizer hoppers/ containers into the soil.
6. **Press wheel-** it covers the seeds.

Maintenance practices on the planters and seeders

1. Lubricate the moving parts.
2. Clean the seed hoppers and fertilizer containers after use.
3. Loose nuts and bolts should be tightened.
4. Broken parts should be repaired.
5. Worn-out parts should be replaced.
6. The coulter tubes should be free from any blockage for efficient seed passage.

Adjustments on the planter.

1. Each type of seed should have its own plate.
2. The depth of cutting/ planting should be adjusted according to the size of the seed.



9. CULTIVATORS/ WEEDERS

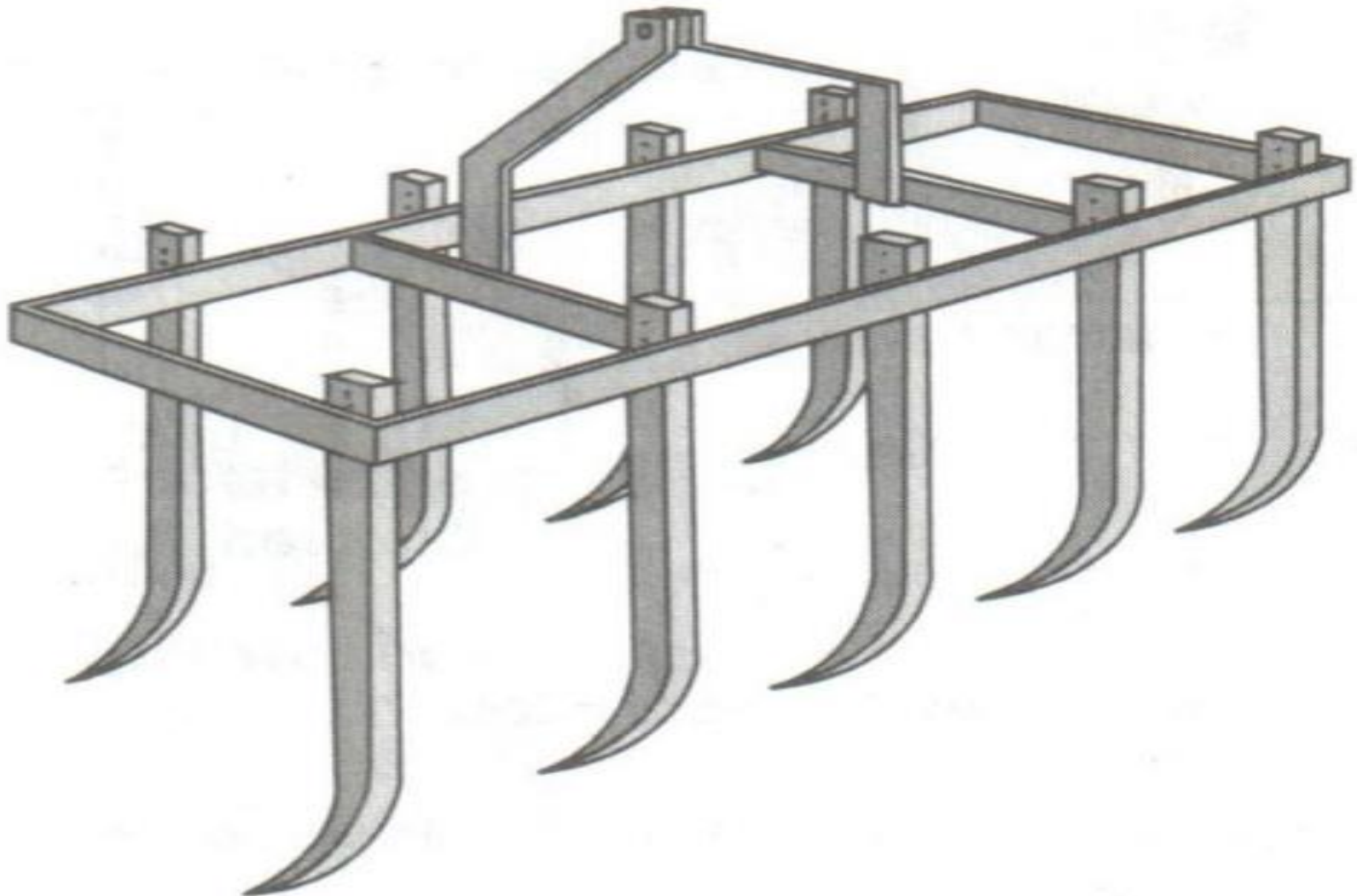
- They are used for shallow cultivation and weeding.

Types of cultivators.

1. Rigid-tined cultivator.
2. Spring-tined cultivator.
3. Spring-mounted cultivator.

Maintenance practices.

1. Loose nuts and bolts should be tightened.
2. Clean/ remove soil/ trash after use.
3. Broken tines should be repaired.
4. Worn-out tines should be replaced.
5. Proper storage under a shed during long storage.



23 Rigid-tined cultivator

10. SPRAYERS/BOOM SPRAYERS

- They are used for application of pesticides, herbicides or foliar fertilizers.

Parts of a sprayer.

1. **Tank-** stores the liquid chemicals.
2. **Pump** – creates pressure in the tank.
3. **Boom/Spray bar-** this is a metal bar with nozzles that break the liquid into very fine droplets.

Maintenance practices of sprayers.

1. The tank should be drained before and after use.
2. All the nozzles should be removed and cleaned when blocked.
3. All parts should be washed thoroughly with clean water and dried.
4. Metallic parts should be cleaned and painted to prevent rusting.

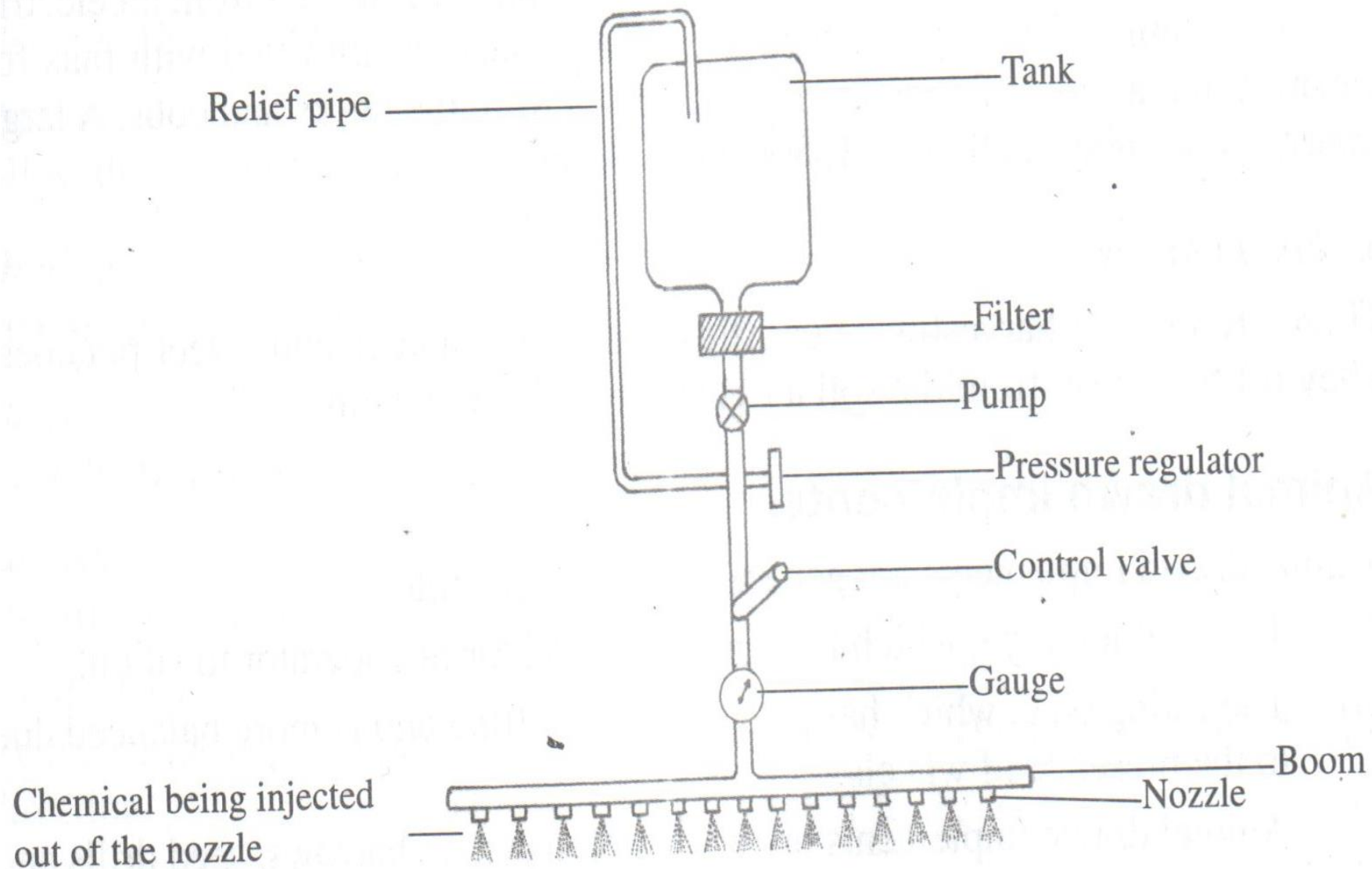


Fig.3.41: Parts of a Boom sprayer.

11. HARVESTING MACHINES.

- They are used for harvesting and include:

- a) Combine harvester- for harvesting grain crops.
- b) Root crop harvesters e.g. ridgers.
- c) Forage harvesters.

a) Combine harvester.

- It is used to harvest cereals e.g. wheat.
- It is designed to perform various tasks in one operation e.g. cutting grains, threshing, winnowing and even bagging.
- It is not tractor-drawn implement but it is self-propelled machine.

Parts of a combine harvester

1. **Cutter bar-** has knives that cut the heads of the grain.
2. **Pick up reel-** rotates as it pushes the grain heads upward for cutting.
3. **Elevator-** directs the cut grain heads to the threshing drum.
4. **Threshing drum-** threshing of grains.
5. **Winnowing fan-** used for winnowing.
6. **Straw walker-** receives the trash from the fan and directs it out of the machine.

7. **Grain delivery tube/grain elevator -** directs the clean grains into the collecting tank.
8. **Grain tank-** holds clean grains.
9. **Stone trap-** traps stones picked up together with the grain heads.

Maintenance practices.

1. Tighten loose nuts and bolts.
2. Clean after work.
3. Lubricate moving parts.
4. Replace worn out parts.
5. Keep it under a shed if not in use.
6. Oil for long storage.

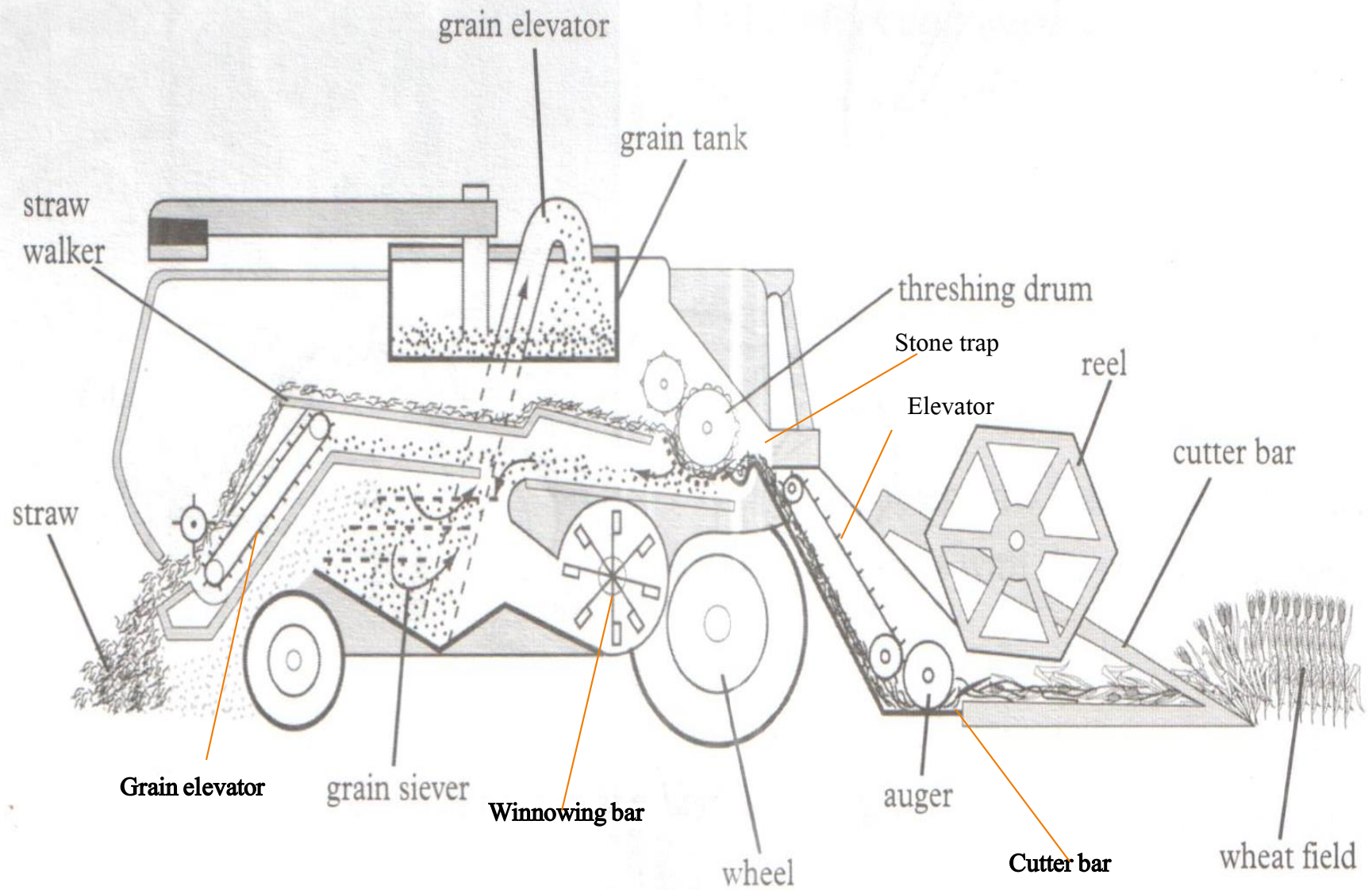


Fig. 12.22 Combine harvester

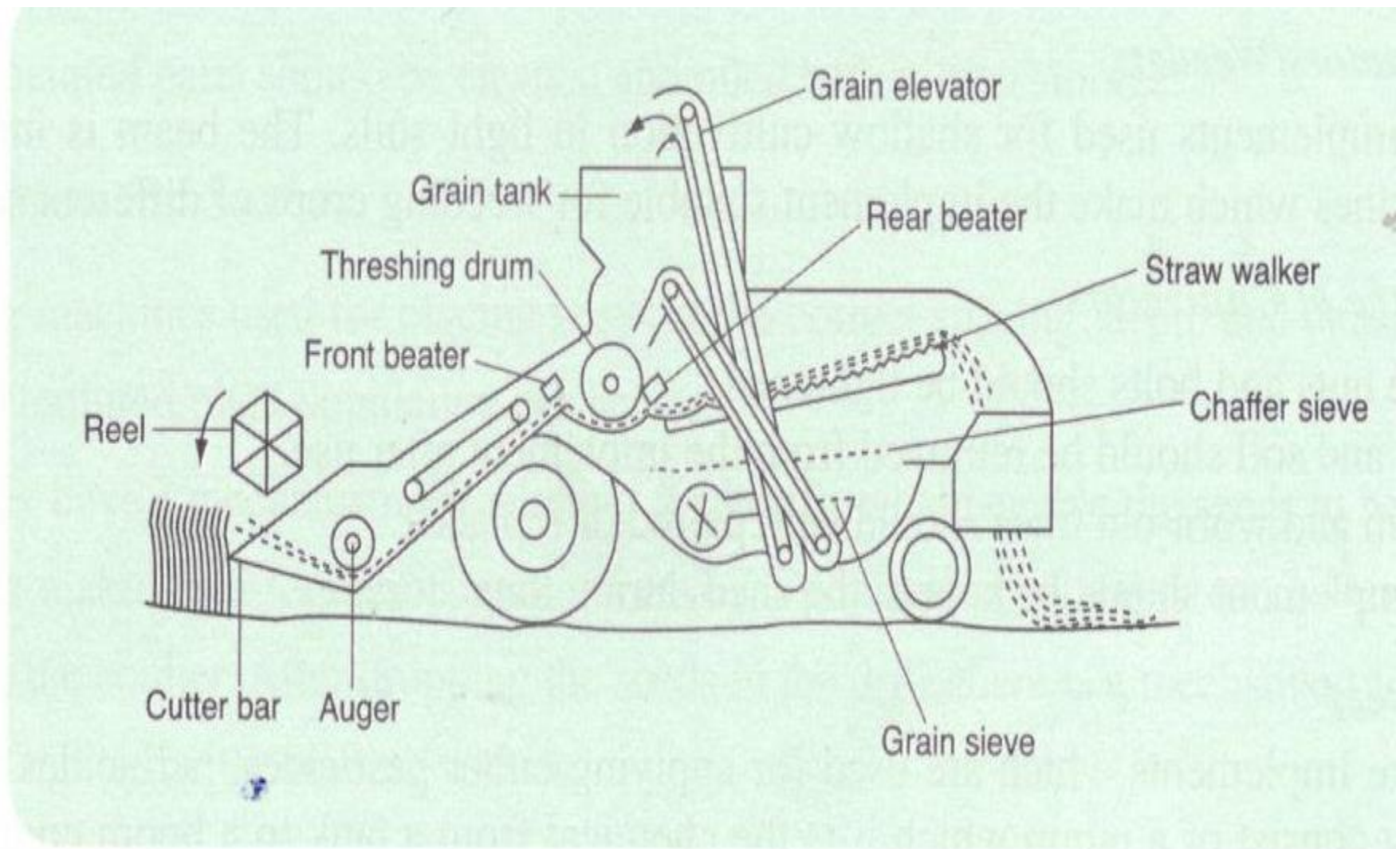


Fig. 3.34: A combine harvester



b) Root crop harvester/ ridger/ lifting plough.

- It is used to lift various root crops from the soil.
- The root crops should be planted on ridges for easy removal and minimize damage during harvesting.
- It is used to harvest irish potatoes, sweet potatoes, cassava, yams and carrots.

c) Forage harvester.

- It is used to harvest forage crops e.g. maize, sorghum and napier grass.

Parts of a forage harvester.

1. **Rotor-** has flails which cut the crop as it moves.
2. **Chute-** collects and delivers the cut material.
3. **Flap-** controls the delivery of forage.

Maintenance practices on forage harvester.

1. Lubricate moving parts.
2. Sharpen the cutting edges.
3. Tighten loose nuts and bolts.
4. Remove trash and dirt after use.
5. Store the implement under a shed when not in use.
6. For long storage oil to prevent rusting.

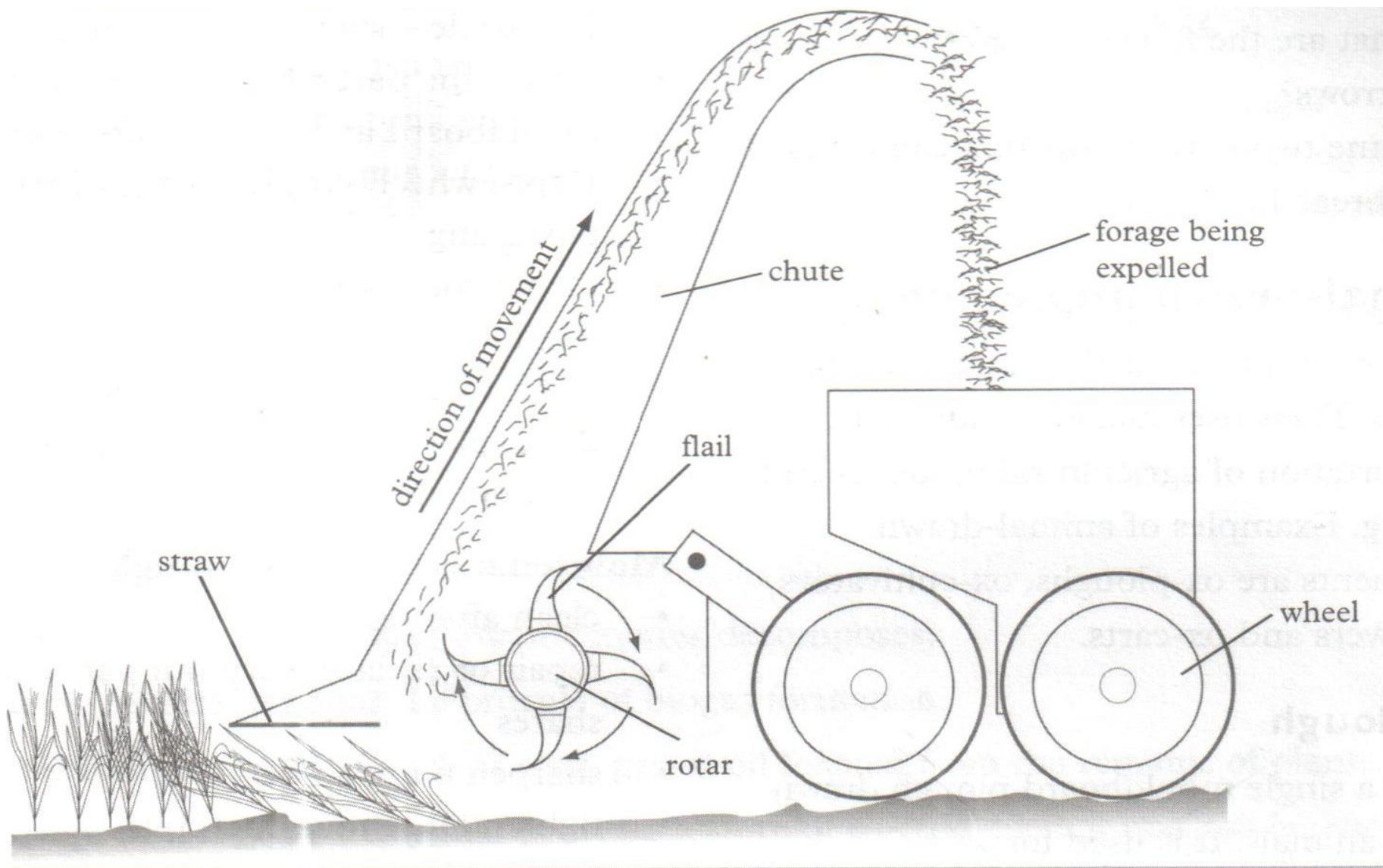


Fig. 12.23 Forage harvester



12. SHELLERS

- They are used for shelling maize i.e. to remove the maize grains from the cobs
- It is fitted with fans and screens to separate grains from the husks/cobs

Parts of shellers.

- Hopper-** holds the cobs.
- Screen-** screens the grains from the cobs.
- Fan-** separates seeds from broken cobs by blowing air onto them.

Maintenance practices.

1. Remove any stuck maize cobs.
2. Replace/repair any broken parts.
3. Tighten the belt tension.
4. Store under a shed when not in use.

SHELLER

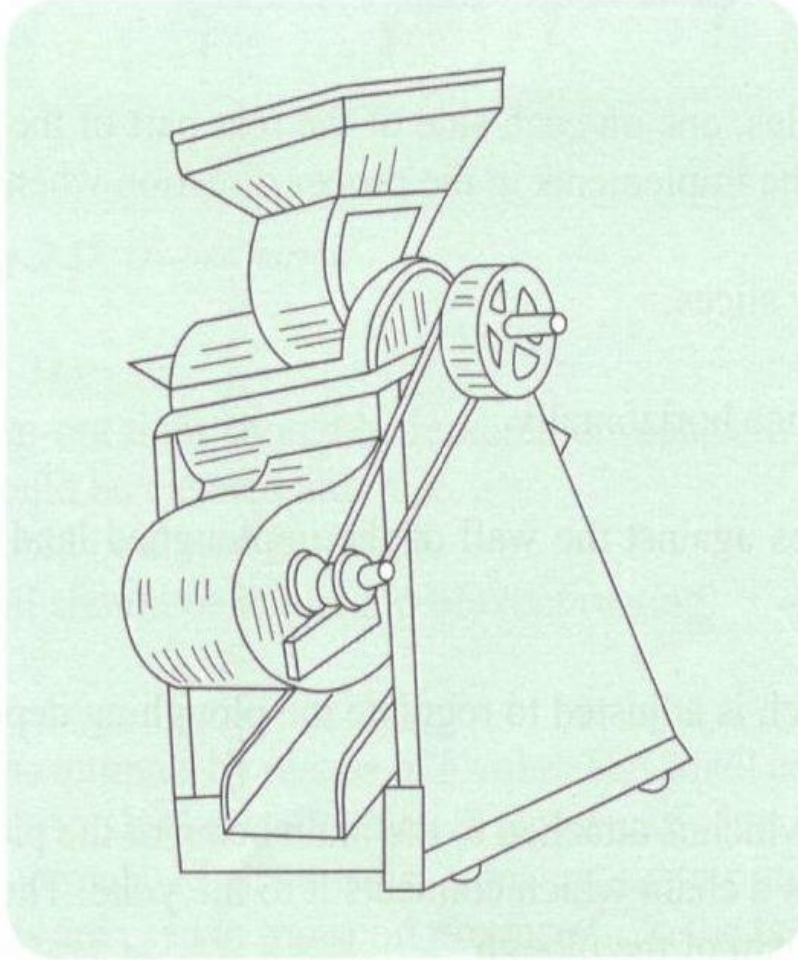


Fig. 3.35: A power maize sheller

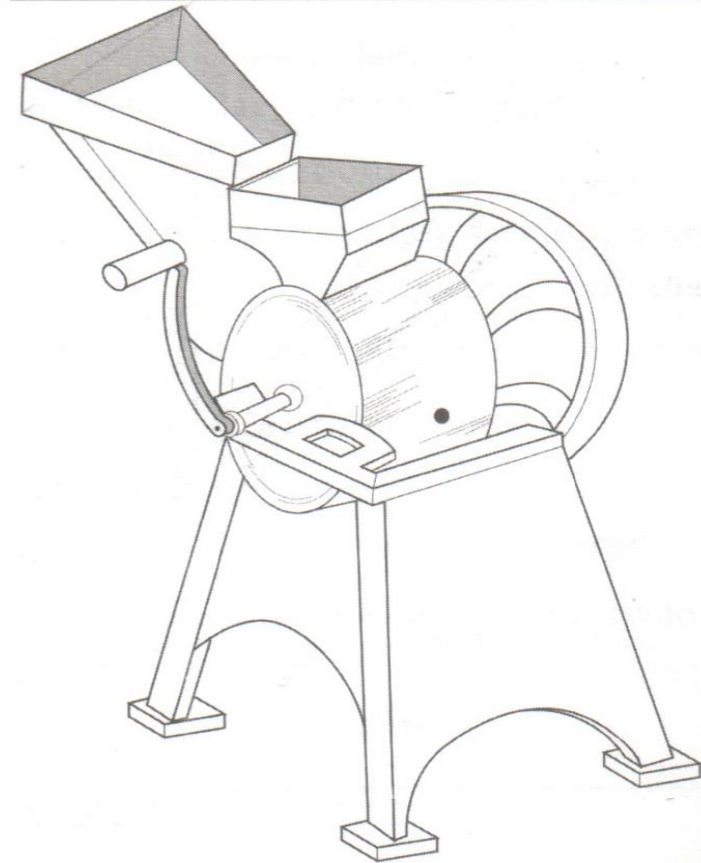


Fig. 12.24 Decorticator

ANIMAL-DRAWN IMPLEMENTS

- They are pulled by animals and include:

- A. Ox-plough.
- B. Ox-tine harrow.
- C. Ox-carts.
- D. Animal-drawn ridger

A. OX-DRAWN MOULDBOARD PLOUGH

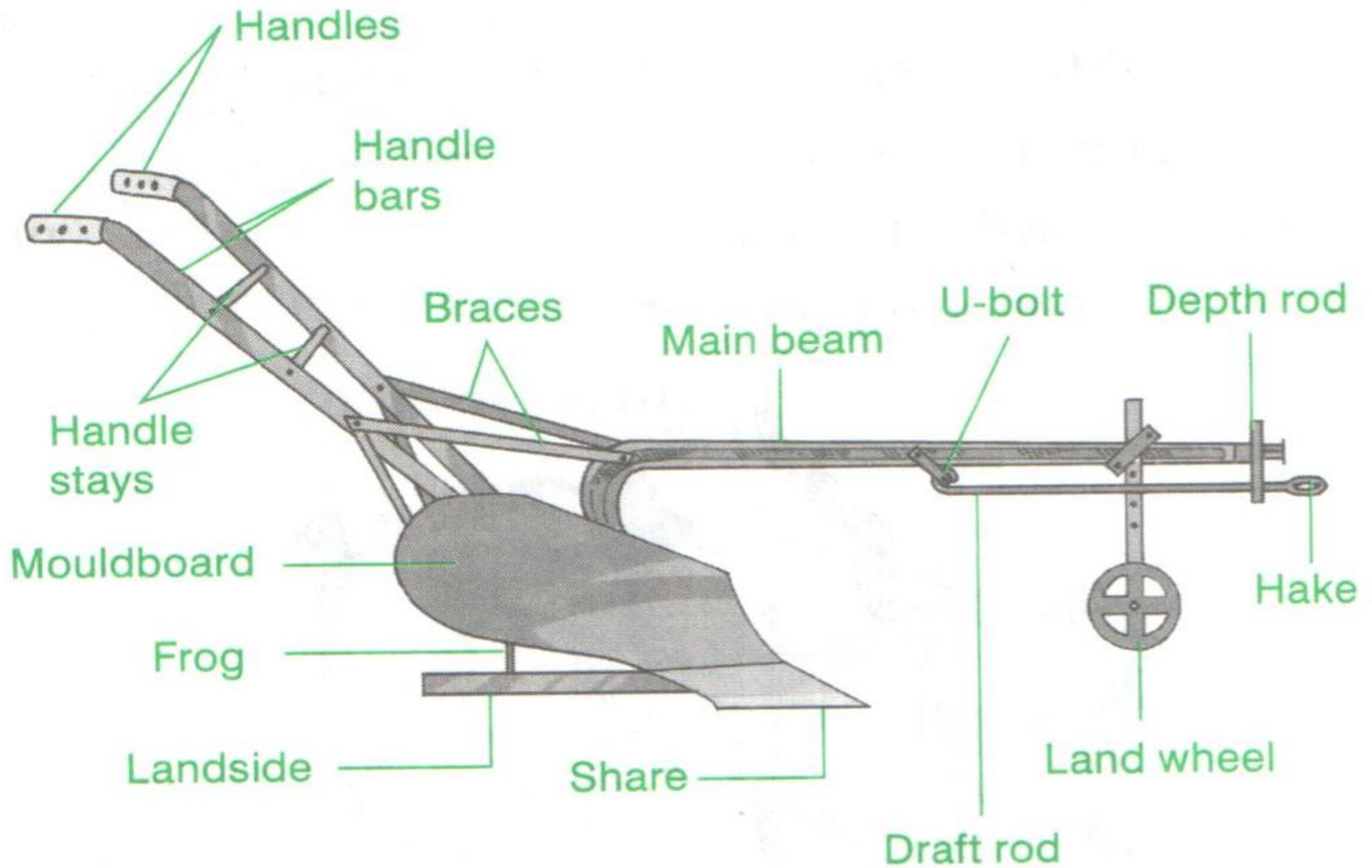
- It is used for ploughing, weeding, opening a furrow for planting seeds, harvesting crops e.g. groundnuts.

Parts of ox-drawn mouldboard plough

1. **Beam-** it provide a point of attachment for all parts of the plough.
 - It also adds weight to the plough for better penetration.
2. **Handles-** they enable the operator to set the implement in the proper direction when ploughing.
3. **Mouldboard-** it inverts the furrow slices.
4. **Share-** it cuts the furrow slice horizontally.
5. **Landside-** it presses against the wall of the uncultivated/ unploughed land hence stabilizing the plough.
6. **Land wheel-** it is adjusted to regulate the depth of ploughing.
7. **Draft rod-** it provides a point of connection of the implement to the chain used to pull the implement.
8. **Depth rod-** it adjusts the depth and width of ploughing.

Maintenance practices of ox-drawn mouldboard plough

1. The land wheel bearing should be lubricated.
2. Worn-out shares should be replaced.
3. Blunt shares should be sharpened by hammering.
4. Loose nuts and bolts should be tightened.
5. The implement should be kept in a shed.
6. Apply oil engine for long storage.



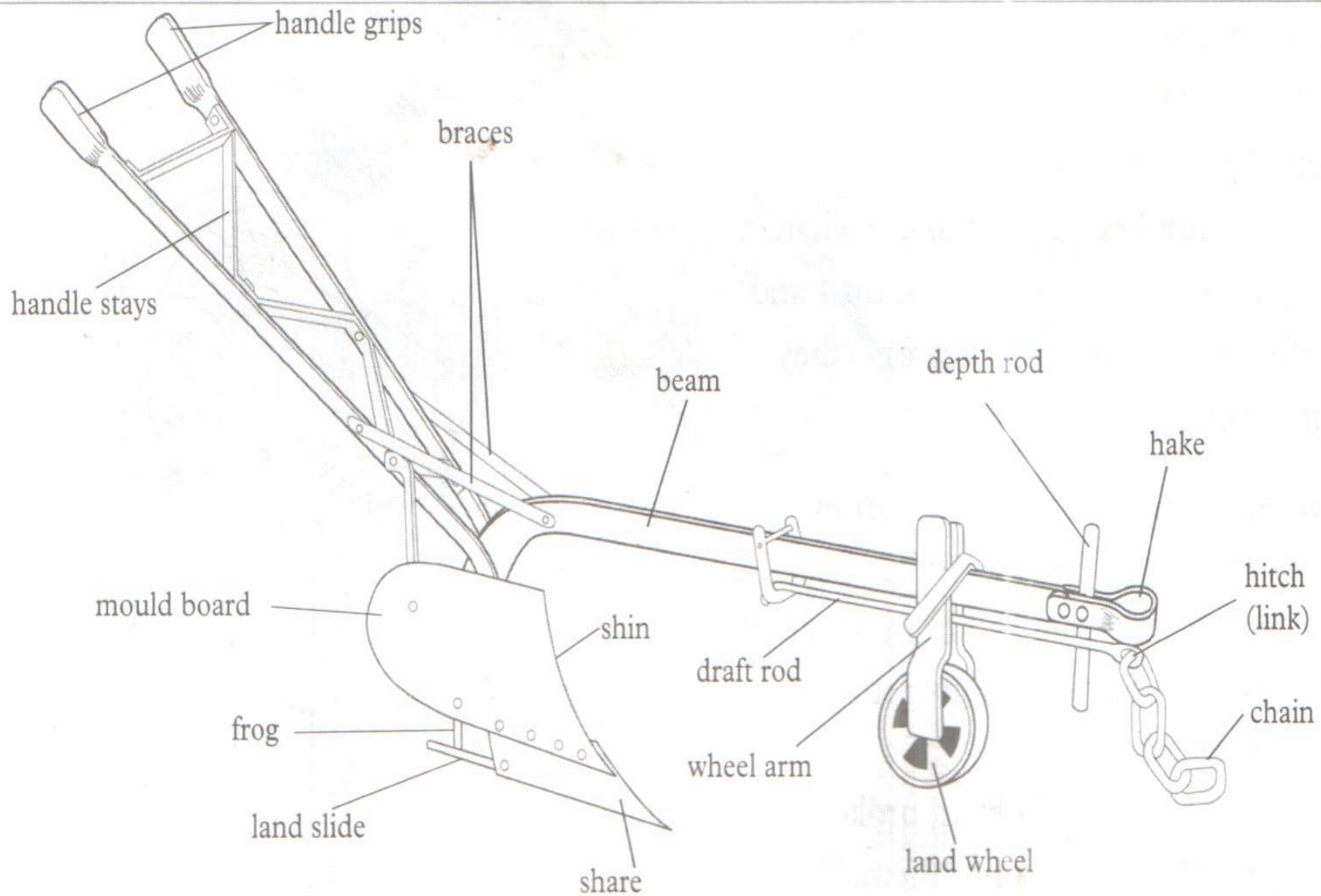


Fig. 12.25 Ox-plough

DIFFERENCES BETWEEN TRACTOR DRAWN AND ANIMAL DRAWN MOULD BOARD PLOUGH

Tractor drawn.

1. Require more skill to operate.
2. Work is done faster.
3. Cannot cultivate slopy areas.
4. There is more soil compaction.
5. It can cultivate very hard soil.
6. Used for fewer operations.
7. It ploughs deeply.
8. Requires one person to operate
9. It is expensive to buy.
10. It is expensive to maintain.

Animal/ox-drawn.

1. Require less skill to operate.
2. Work is done slowly.
3. It can be used to cultivate slopy areas.
4. There is less soil compaction.
5. It cannot cultivate very hard soil.
6. Used on more operations.
7. Ploughs shallowly.
8. Requires more people to operate
9. It is cheaper to buy.
10. It is cheaper to maintain.

B. OX-TINE HARROW

- Used to:

1. Break soil clods.
2. Collect trash.
3. Level the seed bed.
4. Cover planted seeds.

Maintenance practices on ox-tine harrow.

1. Any broken or worn-out tines should be repaired or replaced.
2. Clean the implement after use.
3. Store the implement under a shed.
4. For long storage, oil the implement to prevent rusting.

C. OX-DRAWN CARTS.

- They are harnessed to the animals by means of a yoke and are used to transport farm produce and goods.

Maintenance practices carried out on ox-drawn carts

1. Lubricate/grease moving parts to reduce friction and wearing.
2. The tyre pressure should be checked before the start of the work and adjusted accordingly.
3. Replace worn-out parts.
4. Repair broken parts.
5. Tighten loose nuts and bolts.
6. Replace lost nuts and bolts.
7. Paint for long storage to avoid rusting.
8. Clean after use.
9. Store under shed.

Advantages of animal-drawn implements compared to tractor-drawn implements.

1. They are cheaper to buy and maintain than tractor-drawn implements.
2. A person who operates the animal-drawn implement require less training than the one who operates the tractor.
3. They are cheap to maintain than a tractor-drawn implements.
4. They can be used in areas where the land is steep which are unsuitable for tractor-drawn implements.

Disadvantages of animal-drawn implements compared to tractor-drawn implements.

1. They are more tedious in guiding the implement and animals.
2. Animals may destroy crops on the farm.
3. The output of animals is less.
4. They are slower than tractors.
5. Extra costs are required for rearing animals.
6. Animals get tired at times and this slows down the work.
7. Some areas are prone to diseases hence difficult to use some animals.
8. More land is required for growth of animal feeds.
9. It is more tedious i.e. operators get tired easily

4.AGRICULTURAL ECONOMICS 3 (PRODUCTION ECONOMICS)

1

1. **Agricultural economics-** is the study of how man chooses to allocate scarce production resources to produce goods and services.
2. **Production economics-** is the branch of economics which deals with the study of how scarce resources/ factors of production (land, capital, labor and management) are combined in production process to produce goods and services in the most cost-effective and profitable way possible.

Reasons for production.

- i. To earn a living.
- ii. To satisfy basic human wants.
- iii. To improve the standard of living.

National income.

- This is the total value of goods and services produced by citizens of a country in a given year.
- It helps to measure the economic development/ economic growth of a country.

Factors that determine national income

1. **Household-firm relationship-** **Household** is a unit comprising the farmer and members of the family while a **firm** is a business unit involved in production.
 - The two are related closely and are important in an economy in the following ways:

a) **Household as a producer and consumer.**

- As a producer, the household produces raw materials e.g. sisal, tea, coffee, cotton, sugarcane, milk, wool to be used in industries/ firms.
- As the consumer, the household uses income earned by the sale of these products to buy household goods and services e.g. farm inputs and domestic appliances.
- The greater the household revenue, the greater the consumption of industrial goods hence more industrial development.

b) **The firm as a consumer and producer.**

- The firm as a consumer buys raw materials e.g. cotton, pyrethrum, wool, milk, hides and skins e.t.c. from the household.

- As the producer, the firm processes these materials into finished goods e.g. butter, cloth, pesticides and shoes and also manufactures goods and provides services used by household.

Role of households and firms in the country's economic growth.

- The interaction between the households and firms leads to income generation which will be used to expand the firms. This creates more employment and revenue.
- The income of households and firms is taxed by the government to earn the government some revenue which is used to finance national development programmes and other public services e.g. health and education.

2. **Gross Domestic Product (GDP)**- This is the total sum of goods and services produced by a country within a period of one year.

- It includes goods and services produced by foreigners and foreign companies within the country.
- It is not the best indicator of economic development because foreigners and foreign investments remit their profits, incomes and interests to their countries and Kenyan firms have invested and some Kenyans are working in those countries.

Role of GDP.

- i. Used to calculate economic growth/development.
- ii. Show and compare the standards of living in different countries.
- iii. Used to plan the economy.

3. **Gross National Product (GNP)**- This is the value of goods and services produced by nationals of a country regardless of where they are within a year.

- When GNP is expressed in monetary form it is called Gross National Income (GNI).
- 4. **Per capita income**- This is the average income of citizens of a country per year. It is arrived at by dividing the gross national income by the population of a country.

Per capita income= Gross national income (GNI)

Population

- The more developed countries have higher per capita income than the less developed countries.
- It is not the best indicator of economic development because of uneven distribution of income.

Observable indicators of economic development or growth of a nation.

1. Development of infrastructure.
2. Increase in recreation activities.
3. Improvement in the level of technology/ more industrialization.
4. Housing status of citizens.
5. Number of pupils/ students per teacher.
6. Per capita income.
7. Gross domestic product.
8. Gross national product.

ROLE OF AGRICULTURE TO THE ECONOMIC DEVELOPMENT /KENYAN ECONOMY.

1. **Source of food-/ food supply** –it supplies food to both urban and rural communities to ensure a healthy population which spends more energy in other aspects of economic development leading to a wealthy nation.
2. **Source of employment-** it provides direct employment to Kenyans as full time farmers, farm employees and those based in agro-based industries.
 - It also provides indirect employment to Kenyans employed in extension services, research stations, transport and industries.
 - The government taxes the income to finance national development programmes e.g. health, education, communication, water etc.

3. **Provision of foreign exchange-** when agricultural products are exported they earn the country foreign exchange which is used for economic development activities.

4. **Source of raw material for industry-** it supplies raw materials to processing industries e.g. tea, maize, leather, milk etc. to create more employment and revenue which is taxed for economic development.

5. **Source of market for industrial goods-** the final products of agro- industries e.g. jembes, pangas, wheelbarrow find their market in the agricultural sector.

- This improves production and revenue which is taxed to improve economy.

6. **Source of money/ capital/ income-** when agricultural products are sold money is earned and used in purchasing farm inputs e.g. chemicals, tools and improves the living standards of farming community.

- The government taxes the income and revenue to finance development projects.

Study question

6

- The table below shows the population and gross domestic products of countries A and B.

Country	Gross domestic product (in millions)	Population (in millions)
A	1800	36
B	1200	15

- a) Calculate the per capita income for each country.

$$A = 1800/36 = 50$$

$$B = 1200/15 = 80$$

- b) Which of the countries is more developed economically?

B

- c) Give a reason for your answer in b) above.

Has a higher per capita income.

- d) How can agriculture increase the gross domestic product of a country?

By creating employment/developing industries/increasing production.

FACTORS OF PRODUCTION/ PRODUCTION RESOURCES.

1. **Land-** This is the solid part of the earth where capital is placed.
- It cannot be increased but it is only the quality that is increased.

Ways of making land more productive

- i. Application of organic matter/ manure.
- ii. Application of fertilizers.
- iii. Irrigation.
- iv. Drainage.
- v. Pest, weed and disease control.
- vi. Soil and water conservation.

Characteristics of land.

- i. It is a basic factor of production.
- ii. It is fixed/ not moved and scarce.
- iii. Productivity can be increased.
- iv. It is subject to law of diminishing returns.
- v. It is a natural resource.

Methods of acquiring land.

- i. Purchasing/buying.
- ii. Inheritance.
- iii. Settlement and resettlement by the government.
- iv. Compensation due to death etc.

2. **Labour-** It is human effort employed in production process for a certain period of time.

- It is measured in terms of amount of work done within a specific period of time e.g. man-hours, man-months or man-years.

Ways of improving labour efficiency/productivity.

- Training workers.
- Farm mechanization.
- Giving of incentives to workers e.g. rewarding good workers.
- Supervision of work done.
- Assignment of specific duties.
- Provision of guidance and counseling to workers.
- Rewarding workers with exemplary qualities e.g. honesty.
- Improving terms and conditions of service e.g. good housing and remuneration.
- Provision of transport within the farm.

Changes that indicate improvement in efficiency of labour.

- Improvement in quality of production.
- Shortened/ reduced time of performing some tasks.
- Increase in returns per unit labour.

Characteristics of labour.

- It is a basic factor.
- Cannot be stored hence perishable.
- It is sold in exchange for money.
- It is human.
- It has a weak bargaining power.
- If it is skilled, one cannot adjust its supply rapidly.
- It is mobile.

Types of labour.

- i. Family labour- consisting of members of the family.
- ii. Hired labour- it includes:
 - a) Casual labour-it is hired on agreeable terms in relation to amount of work done.
 - b) Permanent labour- hired on monthly basis.

Factors that influence the supply of casual labour.

- i. Amount of money paid.
- ii. Number of people in the market.
- iii. Health of the worker.
- iv. Ability/ skills of the labour force.
- v. Working conditions.
- vi. Nature of work.

Ways of determining the rate of payment for the casual labour.

- i. Amount of task to be done.
- ii. Duration of work.
- iii. Market rate/ labour regulations.
- iv. Quality/Skill of labour.
- v. Nature of work.

Factors to consider when choosing a type of labor.

- i. Availability of labor.
- ii. Size of the enterprise.
- iii. Cost of labor/financial ability of the farmer.
- iv. Type of enterprise/type of work.

3. **Capital-** it refers to all man-made assets that is used in the production of desired goods e.g. machinery, tools, seeds, agrochemicals, farm structures, livestock, crop.

Qualities/ characteristics of capital.

- i. It is a basic factor of production.
- ii. It is manmade hence under the control of man.
- iii. It can depreciate/reduce in value.
- iv. Can be improved by technology.
- v. It is mobile.
- vi. It can vary in size.

Types of capital.

- i. Liquid capital/money.
- ii. Working/Circulating capital-raw materials used in production process e.g. fertilizers, pesticides, fuel and feedstuffs.
- iii. Fixed/durable capital- are not used completely in production e.g. farm structures, machinery etc.

Sources of capital.

- i. Personal savings.
- ii. Grants/donations from people or NGOs.
- iii. Agricultural credit/loans from friends, private moneylenders, commercial banks, co-operatives etc.

4. **Management**- this is the process of planning and decision making in the organization of other factors of production in the production process.

- Done by the farm manager.

Qualities of a good farm manager.

- Should have the knowledge about specific agricultural principles, marketing and accounting.
- Should be hard working and time conscious.
- Should have practical farming skills.
- Should be responsible, prudent, dynamic, competent and ambitious.
- Should be flexible in decision making.

Roles/ functions of a manager in the farm

1. Short term planning/ making quick decisions in order to carry on the activities in appropriate time and hence avoid a crisis (e.g. pest, parasite and disease control)
2. Long term planning/ making decisions which are linked to the future plans and operations in the farm (e.g. construction of zero grazing units, dips, stores, dams, fences, purchase of farm machinery).
3. Information gathering/ collecting and analyzing information related to the farm enterprise (e.g. marketing activities, production techniques).

4. Detecting weaknesses and constraints and finding ways and means of overcoming them.
5. Comparing standards of the farm/ enterprises with the set standards and making appropriate adjustments.
6. Keeping up to date farm records and accounts and using them in daily running of the farm.
7. Guiding and supervising the farm management.
8. Bearing consequences/ responsibility of plans/ decisions.
9. Making predictions of the likely outcome of possible alternative courses of action.
10. Hires and fires farm labor.

PRODUCTION FUNCTION (PF)

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- This is the physical relationship between inputs and outputs.
- **Inputs-** factors of production that are involved in production.
- **Outputs-** yield/outcome.

Types of inputs.

1. **Variable inputs-** they vary with the level of production.
- **Examples include:** casual labour, fertilizer, feeds, fuel, pesticides, livestock drugs. Their costs are called **variable costs**.

Characteristics of variable inputs.

- a) They change in quantity with the level of production in a given time.
- b) They are added to fixed inputs of production e.g. fertilizers added to land.
- c) Their costs depend on kind and quantity used.
- d) They are usually located to specific enterprises.
- e) Their cost is used to calculate **gross margin**.

Gross margin=total production-variable cost

2. **Fixed inputs**- They are inputs that do not vary with levels of production of an enterprise.

- **Examples include:** land, machinery, permanent labour.
- Their cost is called **fixed costs**.

Characteristics of fixed inputs.

- a) They do not vary with the level of production.
- b) Their costs are not allocated to specific enterprises of product.
- c) Their costs do not vary with costs of variable inputs.

Terms used in production function.

1. **Total product(TP)/total physical product (TPP)**- This is the quantity of the output produced by a given number of inputs over a period of time.
 - It is given in physical terms e.g. bags, kgs, litres etc and not monetary terms.
2. **Average product (AP)/average physical product (APP)**- This is the quantity of output per unit input.
 - It also refers to the ratio of output to the input unit.
3. **Marginal product (MP) /marginal physical product (MPP)**- It refers to the additional product resulting from addition of one unit of input.
 - It is obtained by subtracting the current output and the previous output.

Example 1. Input- output relationship.

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- Yield of maize at different rates of CAN fertilizer per hectare. **N/B- 20 kgs of CAN fertilizer represent 1 unit of input**

Fixed input (Hectare (ha))	Seed rate (kgs)	Variable Input (CAN fertilizer) (kgs)	Marginal input (kgs)	Output (TPP/TP) (90kg bags)	Marginal product(MP/M PP)	Average product (AP/APP)
1	25	0	0	6	6	6
1	25	20	20	12	6	12
1	25	40	20	19	7	9.5
1	25	60	20	29	10	9.7
1	25	80	20	36	7	9.0
1	25	100	20	42	6	8.4
1	25	120	20	48	6	8.0
1	25	140	20	53	5	7.6
1	25	160	20	57	4	7.1
1	25	180	20	59	2	6.6
1	25	200	20	60	1	6.0

TYPES OF PRODUCTION FUNCTION.

- They include:

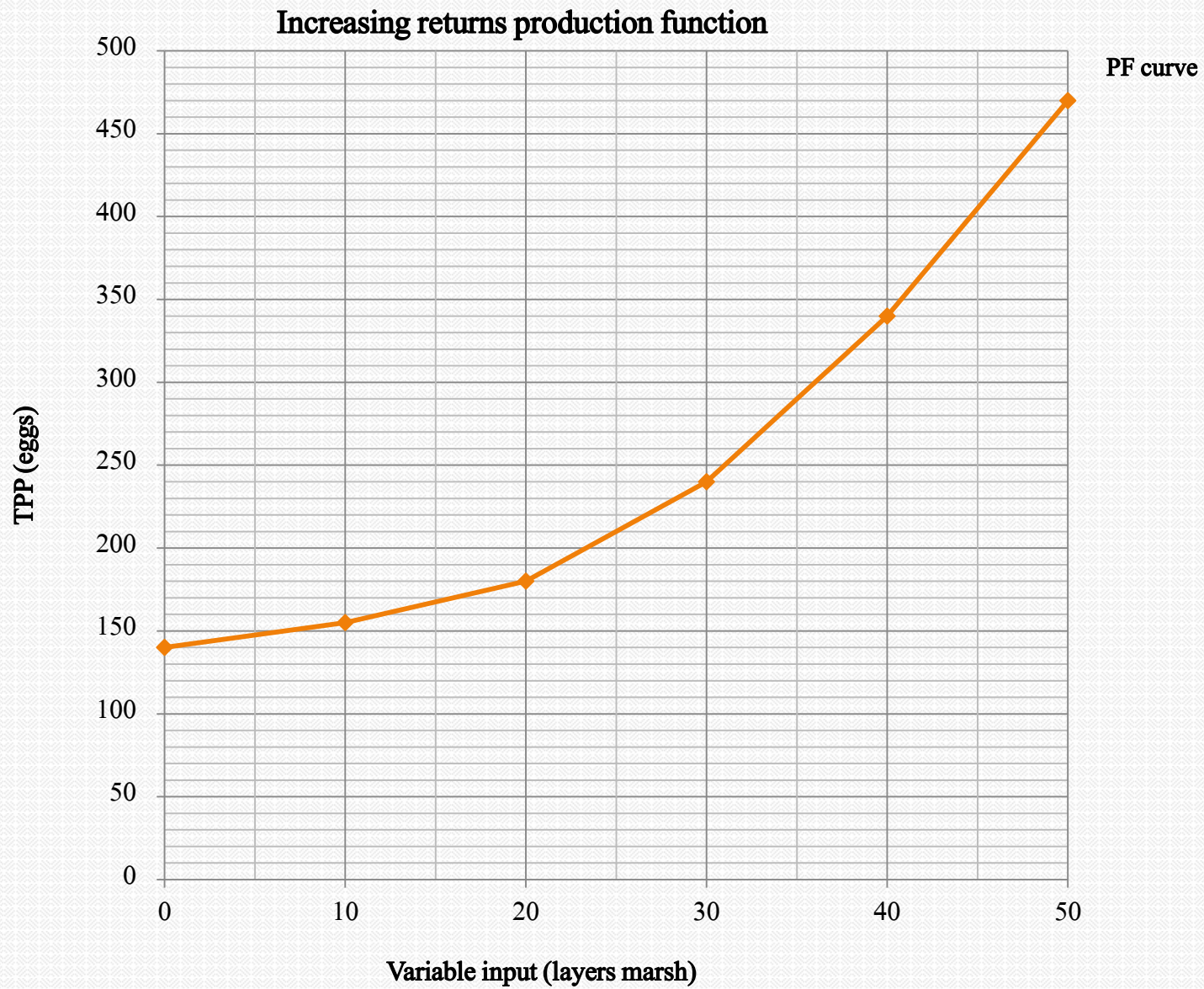
1. Increasing returns production function.
2. Constant returns production function.
3. Decreasing returns production function.

1. **Increasing returns production function-**
this is a type of production function in which each additional unit of input leads to greater/ larger increase in output than the previous/proceeding unit of input.
 - The resources are usually underutilized.
 - It is rare in agricultural production.

Example.

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Fixed Input(layers)	Variable input (layers marsh)	Total physical product (eggs)TP/TPP	Marginal product MP/MPP
100	0	140	0
100	10	155	15
100	20	180	25
100	30	240	60
100	40	340	100
100	50	470	130



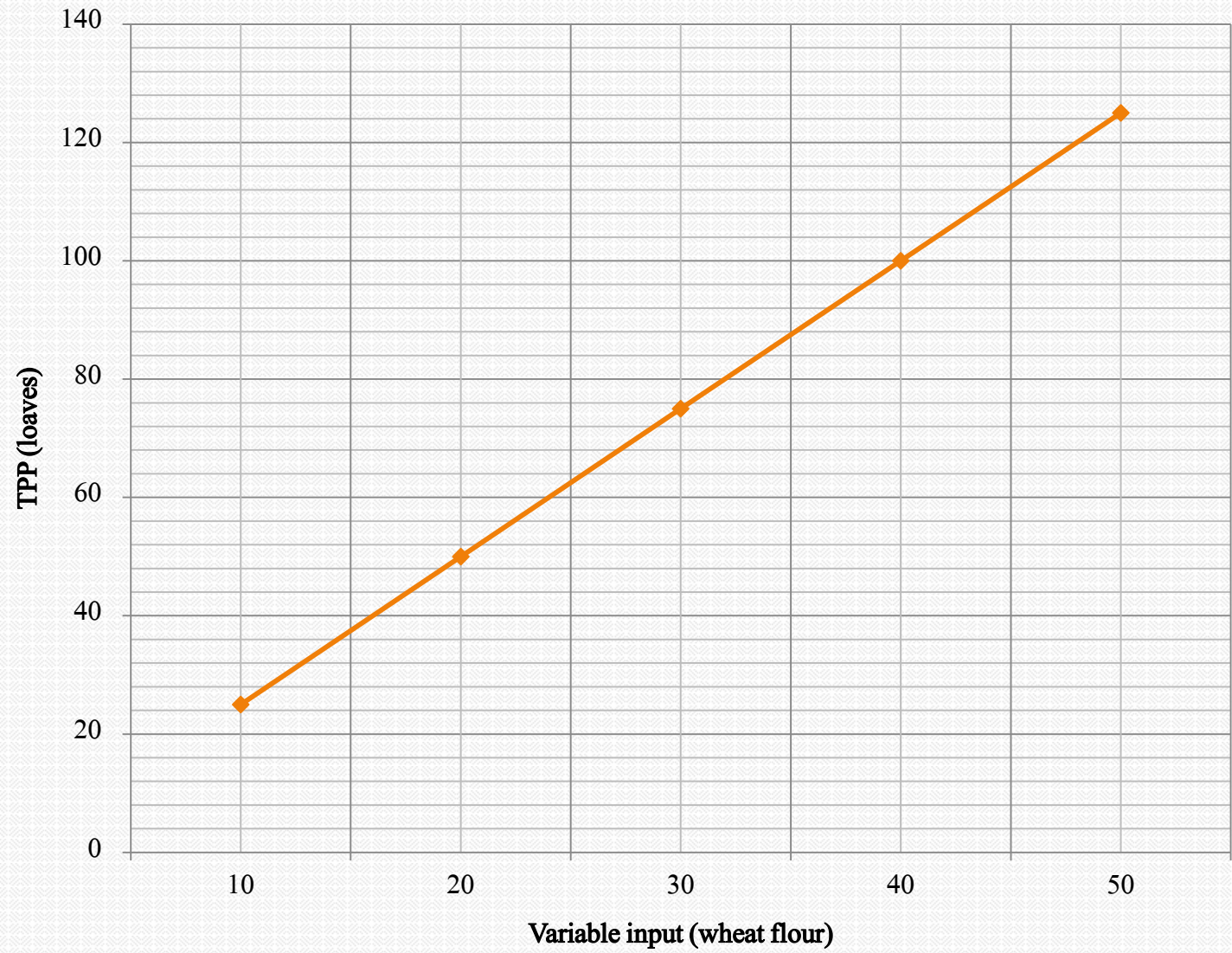
2. Constant returns production function-

- The amount of product increases at the same rate (amounts) for each additional unit of input i.e. the returns are constant to the input factor.
- It is rare in agriculture but it is common in industries.
- For example, in a bakery of a certain capacity where the labour is fixed at five man-days. If the amount of wheat were increased by equal successive units, the total and marginal production of loaves would be the same.

Example

Fixed input(Labour)	Variable input (wheat flour)	Output (loaves) (TP/TPP)	Marginal product (MP/MPP)
5	10	25	25
5	20	50	25
5	30	75	25
5	40	100	25
5	50	125	25

Constant returns production function



iii. Decreasing returns production function.

- In this type, an additional unit of input results in a smaller increase in output than the proceeding unit of input.
- It is common in agriculture.
- There is first an increase of output in an increasing rate which continues up to the point where the total output starts to increase at a decreasing rate.
- If there is continual addition of variable input the output will decrease.

Examples.

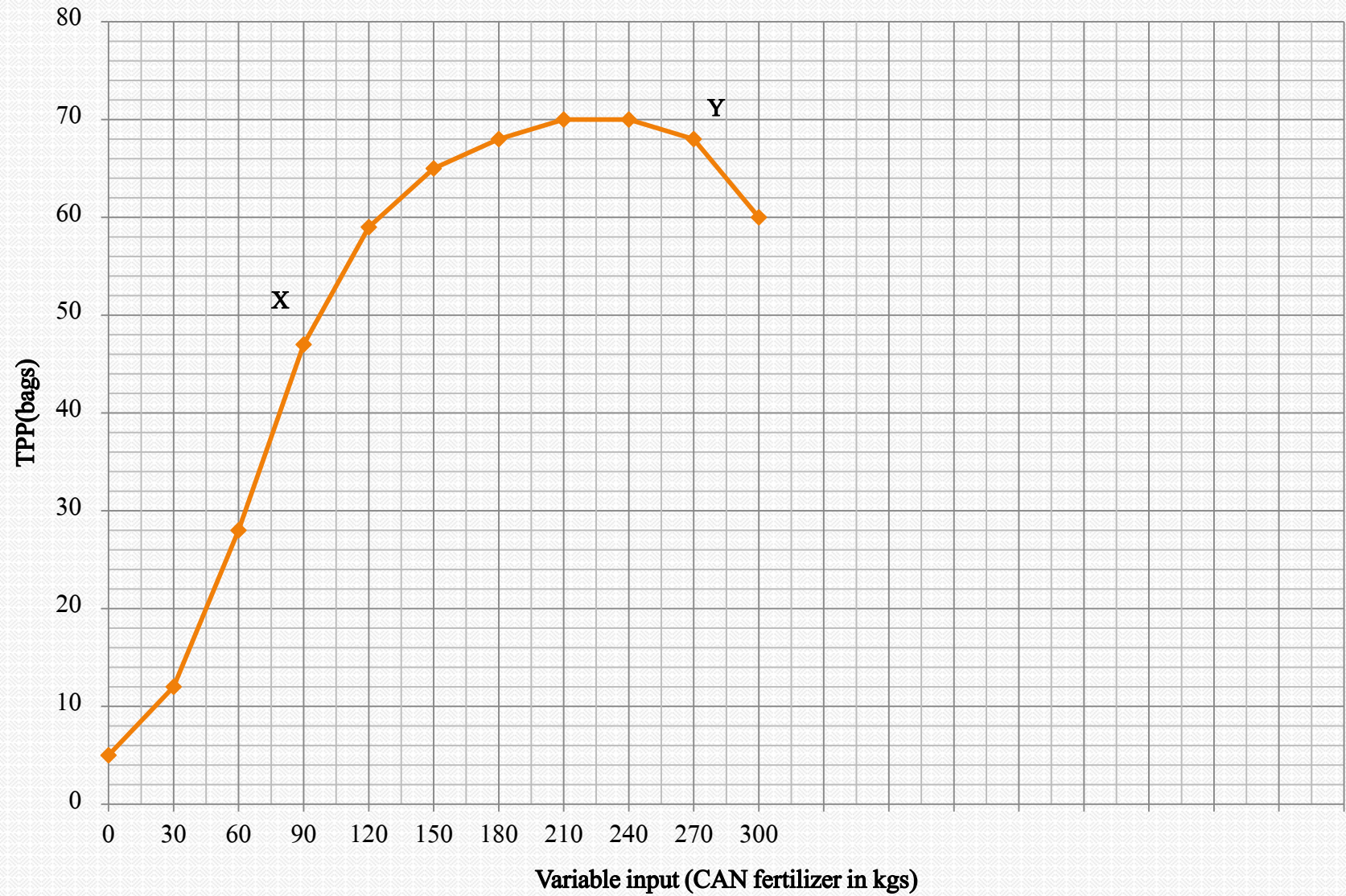
- a) When feeding concentrates to cows for milk production.
- b) Feeding pigs and broilers for weight gain.
- c) Feeding layers for egg production.
- d) Using fertilizers in crop production.

Example

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Fixed input (Hectare (ha))	Variable Input (CAN fertilizer) (kgs)	Output (TPP/TP)(bags)	Marginal product(MP/MPP)	Average product (AP/APP)
1	0	5	5	5
1	30	12	7	12
1	60	28	16	14
1	90	47	19	15.6
1	120	59	12	14.8
1	150	65	6	13.0
1	180	68	3	11.3
1	210	70	2	10.0
1	240	70	0	8.8
1	270	68	-2	7.6
1	300	60	-8	6.0

Decreasing returns production function



Points to note from the graph.

- There is first an increase in output at an increasing rate i.e. each additional unit of input leads to a larger increase in output than the preceding one.
- This continues up to the point when the total output starts to increase at a decreasing rate. This is where the law of diminishing returns starts to operate as represented by point X.
- Thereafter, the increase is at decreasing rate.
- At point Y, the maximum level of output is reached and there is no further increase in output at this point. Any further addition of fertilizer results in decline in output.

ECONOMIC LAWS AND PRINCIPLES.

- They include:
 - A. The law of diminishing returns.
 - B. The law of substitution.
 - C. The law of equimarginal returns.
 - D. The principle of profit maximization.

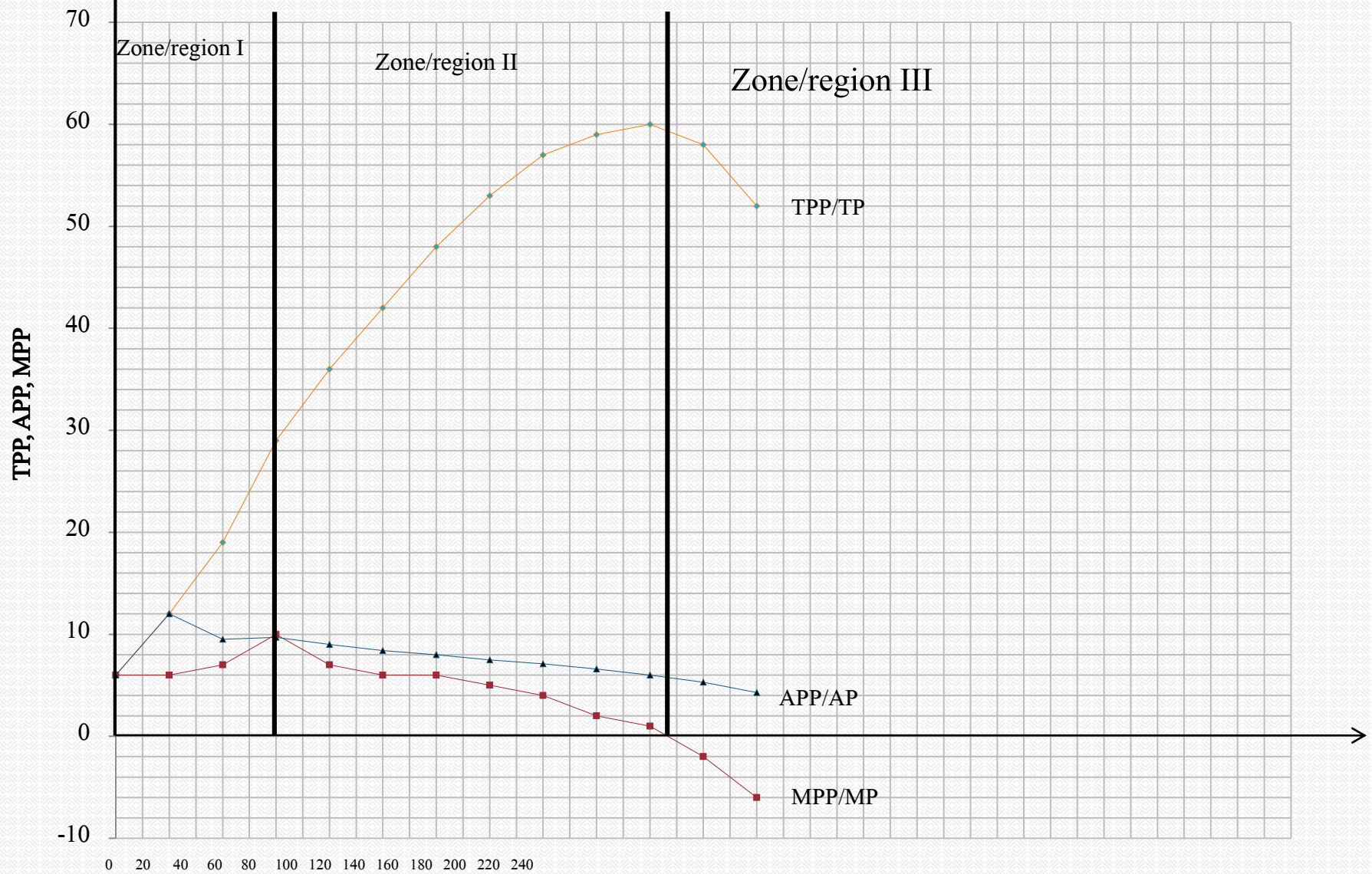
A. THE LAW OF DIMINISHING RETURNS

- It states that *if successive units of one input is added to fixed quantities of other inputs, a point is eventually reached when the additional product per additional unit of input will decline.*
- It is common in agriculture.

Fixed input (Hectare (ha))	Seed rate (kgs)	Variable Input (CAN fertilizer) (kgs)	Marginal input	Output (TPP/TP)	Marginal product(MP/MPP)	Average product (AP/APP)
1	25	0	0	6	6	6
1	25	20	20	12	6	12
1	25	40	20	19	7	9.5
1	25	60	20	29	10	9.7
1	25	80	20	36	7	9.0
1	25	100	20	42	6	8.4
1	25	120	20	48	6	8.0
1	25	140	20	53	5	7.5
1	25	160	20	57	4	7.1
1	25	180	20	59	2	6.6
1	25	200	20	60	1	6.0
1	25	220	20	58	-2	5.3
1	25	240	20	52	-6	4.3

Draw a graph of TPP, MPP and AP against the variable input (CAN fertilizer)

Graph of TPP, APP and MPP



Variable input (CAN fertilizer in kgs)

ZONES/ REGIONS OF A PRODUCTION FUNCTION CURVE.

- If perpendicular lines are drawn at the point where the MP and AP are equal and where the MP is equal to zero (where it cuts the horizontal line/axis) the graph will be divided into 3 regions, namely:

1. Region/zone I (increasing returns zone).
2. Region/zone II (decreasing returns zone).
3. Region/zone III (diminishing returns zone).

1. Region/zone I (increasing returns zone).
 - The total output (TPP/TP) increases at an increasing rate.
 - In this region the producer underutilizes the resources hence advised to produce more.
 - It is called **irrational zone** because available resources can yield more if put into more use.

2. Zone/region II/decreasing returns zone.

- The total output/TPP/TP increases at a decreasing rate.
- It starts at the point where $AP=MP$ up to the point where $MP=0$.
- It is economical to produce at this point because profit is maximized/ resources are used to maximum here therefore is called **rational zone**.

3. Zone/region III/diminishing returns zone.

- Starts at the point where the $MP=0$ (where it cuts X-axis)
- The total output decreases for more application of inputs.
- The resources are over utilized here hence it is uneconomical to produce at this region
- It is therefore called **irrational zone**.

Importance of law of diminishing returns.

1. Helps the farmer to identify the level of optimum fertilizer application.
2. Helps the farmer to determine the highest level of production.

B. THE LAW OF SUBSTITUTION.

- It states that *if the output is constant it is profitable to substitute one input factor for another so long as the second input factor is cheaper than the first one.*
- For example, milk production can be maintained by substituting dairy meal (which is cheaper) for dairy cubes (which is expensive).
- The law is based on the concept of input-input relationship and product-product relationship.

Importance of law of substitution.

- It enables the producer/ farmer to substitute a less profitable enterprise for a more profitable enterprise.

1. Input-input relationship.

- This is the way the inputs are combined in production process to maximize profits/ revenue.
- They are combined in the following ways:
 - a) Fixed proportion/no substitution –in this there is no substitution of inputs involved. For production to take place, both inputs must be present in the same proportion. It is not common in agriculture.

- b) Varying rate of substitution/imperfect substitution**- the inputs substitute one another at varying rates i.e. the amount of input factors can change so as to have the same amount of output e.g.
- a) Use of home made feed rations verses commercial feeds for livestock.
 - b) Use of hay and grain in feeding livestock.
 - c) Use of poultry manure and nitrogenous fertilizers.

- c) Constant rate of substitution (perfect substitution)** – the input factors substitute one another at a constant rate for each level of output regardless of the ratio of the two input factors used e.g. maize and sorghum as livestock feeds.

2. Product-product relationship

- This is the combination of enterprises with the aim of maximizing revenue.

Examples of product-product relationship.

- a) **Joint products relationship-** this is a situation whereby a farmer aims at producing one product, but automatically ends in getting another product.
- Example of joint products include:
 - a) Milk and butter.
 - b) Honey and wax.
 - c) Beef and hides.
 - d) Mutton and wool.
 - e) Cotton lint and cotton seed.

- b) **Competitive products relationship-** in this, the increase in production of one product will lead to a decrease of production of another. This occurs when the resources are limited.

- i. **Dairy and beef cattle-** if the number of dairy cattle is increased, there should be a reduction in the number off beef cattle.
- ii. **Wheat and maize-**if the wheat acreage is increased, then the maize acreage has to be reduced.

- c) **Complementary products relationship-** in this, an increase in the production of one product means simultaneous increase in the production of another e.g.
- i. A farmer can introduce a pig enterprise which will be maintained by the by-products of grain.
 - ii. Poultry production can lead to increase in vegetables because of use of manure.

- d) **Supplementary products relationship-** in this, increase in production of one product does not lead to a decrease in production of another e.g.
- i. A poultry enterprise can be introduced to supplement other enterprises.
 - ii. A farmer can grow an intercrop between the rows of main crop like beans/ cassava in coconut field or Irish potatoes between rows of citrus trees.

MARGINAL RATE OF SUBSTITUTION (MRS)

- This is how much one resource factor can be replaced by one unit of another factor but maintaining the same level of production.

$$\text{MRS} = \frac{\text{Change in input being replaced}}{\text{Change in input being added}}$$

Example

- A farmer using 60 kg of maize grain and 40 kg of wheat decides to change to 50 kg of maize grain and 47 kg of wheat to prepare a ration, calculate marginal rate of substitution.

Solution.

$$\begin{aligned} \text{MRS} &= \frac{\text{change in input replaced}}{\text{change in input added}} \\ &= \frac{\text{change in maize grain}}{\text{change in wheat grain}} \\ &= \frac{60-50}{47-40} \\ &= 1.42 \end{aligned}$$

Example 2

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- A farmer can combine dairy meal and home made feeds in order to obtain 40 litres of milk from a lactating cow as shown below.

a. Calculate the marginal rate of substitution.

Dairy meal (kg)	Home made feed (kg)	Marginal rate of substitution
1	48	-
2	39	9
3	32	7
4	27	5
5	23	4
6	21	2
7	20	1
8	19	1

Dairy meal (kg)	Home made feed (kg)	Marginal rate of substitution
1	48	-
2	39	9
3	32	7
4	27	5
5	23	4
6	21	2
7	20	1
8	19	1

- b. Given that the price of dairy meal is kshs. 800 per kg and that of home made feed is kshs. 200 per kg. Calculate the least cost combination (LCC).

Solution.

LCC occurs where,

$MRS = \text{price ratio}$

$= \text{price ratio} = \frac{800}{200}$

$= 4.0$

$= 4.0$

- *Therefore, LCC is where $MRS=4$ where 5 units of dairy meal are mixed with 23 units of home made feed*

C. THE LAW OF EQUI-MARGINAL RETURNS.

- It states that, if the amount of productive resources are limited, they should be allocated in such a way that the marginal return to those resources is the same in all alternative uses to which they are put.
- It is used together with opportunity cost in the process of maximizing the profit by use of scarce resources.
- For example, a farmer may choose to produce some maize. In this case, some resources used to produce wheat would be re-allocated for maize production.

D. THE PRINCIPLE OF PROFIT MAXIMIZATION.

- It states that ***profit is maximum where marginal cost is equal or almost equal to the marginal revenue.***
- Profit maximization is the point in a production process where the highest net return (net revenue) on invested capital is realized.
- When the difference between the total revenue (TR) and total cost (TC) is the highest/where profit is highest/where $MR=MC$.
- It aims at obtaining the highest profit/returns at a minimum cost.

Profit=total revenue- total cost of production.

□ The concepts of profit maximization are :

1. **Concept of cost.**
2. **Concept of revenue.**

1. THE CONCEPT OF COST.

- Cost is the price paid for goods and services rendered in production process.
- It is also called the **cost of production.**

Cost of production= quantity of input factor x price of input factor.

$$= Q \times P$$

Uses/ roles of cost in production.

- i. Used to calculate gross margin.
- ii. Used to calculate profit.
- iii. Used to determine the quantity of given product to be produced within a given period of time.
- iv. It helps to indicate the most profitable point of production (when converted into monetary value).

Types of costs.

1. **Fixed costs (FC)**- they are input costs which do not vary with the level of production e.g. rent, depreciation of farm machinery, cost of buildings, salaries of permanent labour.
2. **Variable costs (VC)**- these are input costs that vary with the level of production e.g. costs of feeds, cost of fertilizers, costs of fuel, wages of casual labour.
3. **Total cost (TC)**- this is the sum of all fixed and variable costs used in production of a given quantity of product. Total cost= FC + VC.
4. **Average cost (AV)**- total cost divided by the number of units of output.

$$AC = \frac{TC}{\text{Number of units of output}}$$

5. **Average variable costs (AVC)**- this is the total variable costs divided by total output.

$$\bullet \text{ } AVC = \frac{VC}{\text{Total output}}$$

6. **Average fixed costs (AFC)**- this is the total fixed costs divided by total output.

$$\bullet \text{ } AFC = \frac{FC}{\text{Total output}}$$

7. **Average total cost (ATC)**- this is the sum of average variable cost and average fixed costs. $ATC = AFC + AVC$.

8. **Marginal cost (MC)**- this is the extra cost incurred in the production in additional unit of output.

2. THE CONCEPT OF REVENUE.

- **Revenue** is the amount of money obtained from the sale of output/product.

Types of revenue.

1. **Total revenue (TR)**- this is the total physical product (TPP) multiplied by the unit price of the product e.g. 100 bags of Irish potatoes x ksh. 280 = 28, 000/-
2. **Net revenue (NR)/profit**- this is the difference between the total revenue and total cost of production.
3. **Marginal revenue (MR)**—this is the extra income from the sale of one additional unit of output.

WAYS OF INCREASING PROFIT.

1. Using modern methods of production e.g. mechanization.
2. Market research.
3. Applying the principle of Least Cost Combination.

Example.

In maize production carried over 8 seasons , a farmer uses one hectare of land each time and applies various of DSP fertilizer at the cost of ksh. 280 per 50 kg bag. The crop was harvested and sold at ksh. 200 per 90kg bag. Indicate the point where profit is maximized.

DSP fertilizer (50 kg bag)	Total yield (TPP) in 90 kg bags	Total revenue (kshs)	Total cost (Kshs)	Marginal revenue (MR)	Marginal cost (MC)	Net revenue/profit
0	15	3,000	0	0	0	3,000
1	35	7,000	280	4,000	280	6,720
2	52	10,400	560	3,400	280	9,840
3	68.5	13,700	840	3,300	280	12,860
4	70	14,000	1120	300	280	<u>12,880</u>
5	71	14,200	1400	200	280	12,800
6	70	14,000	1680	-200	280	12,320
7	68	13,600	1960	-400	280	11,640

Profit is maximized at the level of 4 bags of fertilizer where MR is almost equal to MC

Study question

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Explain ways through which the government can improve maize production.

1. Provision of training and extension services to advise farmers on modern method of production e.g. irrigation, use of certified seeds, disease and pest control.
2. Provision of subsidies on farm inputs e.g. fertilizers.
3. Provision of credit facilities/loan to finance farming operations.
4. Imposing high taxation on imported products to discourage importation and protect local farmers.
5. Quality control to ensure production of high quality products that can attract foreign markets.
6. Supporting research into new and improved varieties for high yields.
7. Supplying inputs to farmers.
8. Provision of market services.
9. Provision of drying and storage facilities.
10. Provision of tractor hire services.
11. Ensuring effective control of pests/diseases/weeds.
12. Ensuring effective soil and water conservation measures.

GROSS MARGIN.

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- It is calculated by subtracting total revenue/gross income and variable costs.
- **Gross margin= total revenue/gross income-variable cost.**
- Calculation of gross margin of different enterprises is called **gross margin analysis.**

Uses of gross margin.

1. Used to compare the performance of one farm and another.
2. Used to compare the performance of the farm between one season and another.
3. To compare the contribution of one enterprise and another in the same farm.
4. It helps to predict the out come of profit.

- **For example.**
- A farmer is considering undertaking the production of either maize or beans. Study the information below about the crops and answer the questions that follow.
 - a) Calculate the gross margin for each crop.
 - b) From your calculation which crop is most profitable to grow?

	Maize	Beans
Yield	5,500 kg	5,000 kg
Price	Kshs 15 per kg	Kshs 50 per kg
Cost of cultivation	Kshs 3,000	Ksh 3,600
Amount of seed	25 kg	20 kg
Cost of seed	Kshs 100 per kg	Kshs 80 per kg
Amount of DAP fertilizer	3 bags	2 bags
Cost of DAP fertilizer	Kshs 1,500 per bag	Kshs 1,500 per bag
Labour	50 man days	75 man days
Cost of labour	Kshs 150 per man day	Kshs 200 per man day
Amount of CAN fertilizer	3 bags	1 bag
Cost of CAN fertilizer	Ksh 1,000 per bag	Kshs 1,000 per bag
Cost of sprays	-	Kshs 1,000

Gross margin for maize.

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Item

Cost

• Revenue/gross income	$5,500 \times 15 = \underline{82,500}$
• Cost of labour	$150 \times 50 = 7,500$
• Cost of cultivation	3,000
• Cost of seed	$25 \times 100 = 2,500$
• Cost of DAP fertilizer	$3 \times 1,500 = 4,500$
• Cost of sprays	-
• Cost of CAN fertilizer	$3 \times 1,000 = \underline{3,000}$
• Total variable cost	20,500

Gross margin = total revenue/gross income – variable cost
= 82,500 – 20,500
= kshs 62,000

Gross margin for beans.

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Item

Cost

• Revenue/gross income	$5,000 \times 50 = 250,000$
• Cost of labour	$200 \times 75 = 15,000$
• Cost of cultivation	3,600
• Cost of seed	$80 \times 20 = 1,600$
• Cost of DAP fertilizer	$2 \times 1,500 = 3,000$
• Cost of sprays	-
• Cost of CAN fertilizer	$1 \times 1,000 = \underline{1,000}$
• Total variable cost	27,200

Gross margin = total revenue/gross income – variable cost

$$= 250,000 - 27,200$$

$$= \text{kshs } 222,800$$

It s more profitable to grow beans than maize

FARM PLANNING.

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- It involves drawing up a working programme for using the limited resources in the farm in order to achieve maximum profit.

ADVANTAGES/ REASONS FOR FARM PLANNING.

1. Enables the farmer to allocate resources effectively to various enterprises.
2. It guides a farmer to choose an enterprise which will maximize available resources.
3. Helps the farmer to carry out farming practices on time.
4. Enables the farmer to set targets on how much to produce.
5. Enables the farmer to assess the progress toward achieving the set objectives.
6. Helps the farmer to determine the most profitable combination of enterprises.
7. Helps the farmer to get/achieve maximum profit.
8. Helps to identify the weaknesses and strength of farm operations.
9. Assist in negotiating/ getting loan.
10. Helps in timely and careful decision making.

FACTORS TO CONSIDER DURING FARM PLANNING/SELECTING AN ENTERPRISE.

1. **Land size/Size of the farm-** a large number of enterprises can be established on a large scale farm compared to a small scale farm.
2. **Environmental factors-** climate, soil type and topography should be analyzed to determine crop and livestock enterprises that are suitable to local ecological conditions.
3. **Security-** enterprises which require more security should be near the farm house for security e.g. calf pens, poultry houses, and rabbit hutches.
4. **Risks and uncertainties-** enterprises should be analyzed to determine the risks involved.
5. **Current trend in labour market-** this will help to determine availability and cost of labour especially during the peak period.
6. **Farmer's objectives and preferences-** to ensure that the farmer who is the operator has a sense of ownership of the plan and to bring about motivation.
7. **Current market trends and prices of outputs-** to ensure consideration of enterprises with high profit returns.

8. **Availability and cost of farm inputs-** this will help to identify enterprises that are affordable and whose inputs are readily available.
9. **Government policy/ regulation-** this is to avoid enterprises and farming systems prohibited by the government.
10. **Communication and transport facilities-** this is to facilitate movement of output to the market and supply of input and also help in conveying improved methods of farming and market trends.
11. **Availability of capital-** to acquire farm inputs.
12. **Possible production enterprises-** they possible enterprises should be identified and analyzed so that suitable and profitable enterprises are selected.

STEPS FOLLOWED IN MAKING A FARM PLAN.

1. Determine the size of the farm by surveying and calculating the acreage.
2. Determine the environmental conditions of the area by collecting information on climate, soils and vegetation and analyzing the data to determine the all the feasible enterprises.
3. Determine the objectives and preferences of the farmer in order to eliminate the production possibilities that are undesirable.
4. Develop a tentative schedule by listing the selected enterprises and stating the types and costs of physical resources required so as to select one or combination of enterprises.
5. Determine the technical feasibility of the plan to make it realistic considering the external influences e.g. government policy.
6. Determine the expected yields and returns of various enterprises.
7. Determine the budget by translating the physical plan into monetary value.
8. Develop the financial flow in order to ensure that it is consistent, workable and desirable.
9. Implement the plan.
10. Observe and evaluate the plan in the course of implementation.

FARM BUDGETING.

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- Farm budgeting is the process of estimating the future results of a farm plan.
- A farm business budget is an estimate of the future income and expense of a proposed farm plan.

Importance of farm budgeting.

1. It helps the farmer in decision making to avoid over expenditure and impulse buying.
2. It enables the farmer to predict future returns. This helps the farmer to plan ahead.
3. It helps the farmer to avoid incurring losses by investing in less profitable enterprises.
4. It enables the farmer to secure/ get loan from financial institutions e.g. commercial banks and Agricultural finance Corporation.
5. It ensures a periodic analysis of the farm business.
6. It acts as a record which can be used for future reference.
7. It pinpoints efficiency or weakness in farm operations.

TYPES OF FARM BUDGETS.

1. **Incomplete budget/partial budget**- it is made when making minor changes in the farm business e.g. change in the size of the dairy herd, change in the size of the land under crop, change in production.

Questions asked in partial budgeting.

- a) What extra costs is the farmer going to incur as a result of proposed change? (-ve effect).
- b) What revenue is to be forgone as a result of proposed change? (-ve effect).
- c) What costs will be saved as a result of proposed change? (+ve effect).
- d) What extra revenue will be earned as a result of proposed change? (+ve effect).

- A farmer at wishes to change from arable farming to dairy goat production. In arable farming he has been spending ksh. 400 on weeding maize and ksh. 200 on weeding cabbages. He spends ksh. 500 and ksh. 300 on harvesting maize and cabbages respectively. He buys the following inputs: DAP fertilizer at ksh. 1,000, cabbage seeds at ksh. 400, maize seeds at ksh. 600, pesticides at ksh. 800 and ksh. 300 on shelling maize.
- The change in enterprise will have the following implications: he will buy 5 dairy goats at ksh. 2,000 each, pay milk man ksh. 3,000, control diseases at ksh. 1,500, fence at ksh. 1,500. The revenue he gets when growing maize is ksh. 10,000 and cabbages is ksh. 4,000. In dairy goat production, he will get ksh. 20,000 from sale of milk and ksh. 1,000 from sale of manure.
- Prepare a partial budget and advise the farmer whether the change is worthwhile or not.

Solution

Debit (DR-)			Credit (CR-)		
	Kshs	Cts		Kshs	Cts
<u>Extra costs</u>			<u>Extra revenue</u>		
Dairy goats(5x2000)	10,000	00	Milk	20,000	00
Milk man	3,000	00	Manure	<u>1,000</u>	00
Diseases	1,500	00	Sub total	<u>21,000</u>	
Fencing	<u>1,500</u>	00	<u>Costs saved</u>		
Sub total	<u>16,000</u>	00	Weeding (200+400)	600	00
<u>Revenue forgone</u>			Harvesting (500+300)	800	00
Maize	10,000	00	Fertilizer	1,000	00
Cabbages	<u>4,000</u>	00	Seeds (400+600)	1,000	00
Sub total	<u>14,000</u>	00	Pesticides	800	00
			Shelling	<u>300</u>	00
			Sub total	<u>4,500</u>	00
GRAND TOTAL	<u>30,000</u>	00	GRAND TOTAL	<u>25,500</u>	00
NET INCOME= 25,500- 30,000= -4,500 ; The change is not worth while					

2. **Complete budget-** it is made when the farmer wants to start a new business where both the variable and fixed costs are likely to be affected.

Guidelines followed when making a complete budget.

- a) **Formulation of farming goals-** the farmer states the reasons for setting up the farm business.
- b) **Taking the farm inventory-** includes listing farm building, land improvement e.g. irrigation, fencing, breeding stock, human labour, funds available, sources of power, machinery and farm equipment.
- c) **Planning the resources-** involves showing how these various resources like land, labour and capital are utilized.
- d) **Estimating production-** it involves finding out the gross production of the assets in the farm originating from crops, livestock and other activities.

- d) **Estimating income and expenditure-** involves preparing a statement of the income and expenditure based on existing prices and costs.
- e) **Analyzing the input-output relationships that exit on the farm.**
- f) **Analyzing existing production weaknesses in the farm-** this will enable the farmer to know what to eliminate first.
- g) **Making a number of alternative farm plans and choosing one for adoption.**
- h) **Putting the best chosen plan into operation and supervising its implementation.**

Example.

Enterprise	Hectarage	Gross margins (Kshs)
Maize	0.5	3,000
Irish potatoes	0.5	2,500
Beans	0.5	4,500
Onions	0.25	2,800
100 laying hens	0.25	15,000
	Total gross margin	27,800

Fixed cost	Kshs
Labour	4,800
Depreciation of farm structure.	150
Hand tools and equipment	1,000
Total fixed cost	5,950

Profit=total gross margin- total fixed cost.

$$=27,800 - 5,950$$

$$=21,850/-$$

AGRICULTURAL SUPPORT SERVICES AVAILABLE TO THE FARMER.

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1. **EXTENSION AND TRAINING-** it involves giving informal education to farmers on production techniques. This is mainly conducted by extension field officers of the Ministries of Agriculture and Livestock Development.
2. **BANKING SERVICES-** this enables the farmer to make payments and receipts made through the bank accounts.

Advantages of transacting business through the bank accounts.

- a) The money is safer in the bank.
- b) A bank cheque can be used as evidence of payments incase of a dispute.

- c) In transactions involving large sums of money, it is easier and quicker too write a cheque than count out large sums of money.
- d) The bank gives advice to farmers on how to use the loan/ credit given for maximum returns.
- e) A banker's statement can act as evidence of financial standing when the farmer wants to acquire/ lease some property.

3. **CREDIT/ LOAN-** this is borrowed credit. It can be **hard loan-** offered against substantial security e.g. machinery or **soft loan-** offered without or with little security.

Types of credit.

- a) **Short-term credit-** it is repayable within one year. It is given as working capital e.g. seeds, feeds and fertilizers.
- b) **Medium-term credit-** it is repayable within two years. It is used for farm development projects e.g. fencing, buying machinery, soil and water conservation and buying livestock.
- c) **Long-term credit-** it is payable within a period of 15 years. It is used for major developments.

Sources of agricultural credit.

- a) Co-operative societies
- b) Crop boards e.g. pyrethrum board of kenya, national irrigation board.
- c) Commercial banks.
- d) Agricultural finance corporation.
- e) Settlement fund trustees- offered to new settlers in settlement schemes.
- f) Individuals.
- g) Insurance companies.
- h) Traders.
- i) Hire purchase companies.

Problems associated with credits/ loans.

- a) Most farmers do not have enough security e.g. title deed.
- b) Loans are diverted to other uses for which they were not intended.
- c) The interest rates are usually high such that repayment becomes a problem.
- d) Non-payment of the loan may lead to the land or other assets used as security auctioned.
- e) Lack of knowledge and appropriate skills in management of credits may lead to misappropriation/ misuse of the funds.
- f) Lack of proper farm records may disqualify the farmers from getting credit/ loan.

4. ARTIFICIAL INSEMINATION (AI) SERVICES-

These help the farmers to produce high quality animals.

5. VETERINARY SERVICES- they help the farmer to treat and control livestock diseases and parasites.

6. FARM INPUT SUPPLIES- they supply inputs to farmers e.g. Kenya Farmers Association (KFA)

7. **AGRICULTURAL RESEARCH**- it has the following objectives:
- a) Improve crop and livestock production techniques.
 - b) Develop improved varieties and types of crops and livestock.
 - c) Improve pasture and fodder quality.
 - d) Develop techniques for controlling diseases and pests of various crops and livestock.
 - e) Determine suitable ecological zones for various crops.
 - f) Co-ordinate research work done to improve crops and livestock.

Examples of agricultural research stations in Kenya.

- a) Coffee Research station in Ruiru- it does research on coffee.
- b) Kenya Agricultural Research Institute (KARI) at Muguga- It does research on agronomy, plant pathology, entomology, forestry, livestock nutrition and diseases.
- c) National Plant Breeding at Njoro- it does research on wheat, barley, rye and oil crops e.g. sunflower, rape seed and simsim.
- d) National Agricultural Research Station in Kitale- it conducts research on maize and pasture crops.

- e) Pyrethrum research Station at Molo- carries research on pyrethrum.
- f) Horticultural Research Station at Thika- does research on fruits, flowers, pulses and grain legumes.
- g) National Sugar research Station at Kibos- it conducts research on sugar cane.
- h) Tea Research Foundation at Kericho- it conducts research on tea.
- i) Dryland Farming Research Station at Katumani in Machakos- carries research on soil moisture use, plant breeding, pest management, animal nutrition and farm systems economics.
- j) National Agricultural laboratories at Kabete- carries research on entomology and soil testing.
- k) Sunflower Research Station.
- l) International Centre for Insect Physiology and Ecology.
- k) Western Agricultural Research station at Kakamega- carries research on sweet potatoes, cassava and small ruminants.
- l) National Animal Husbandry research station at Naivasha- researches on livestock management and breeding.
- m) Kenya Veterinary vaccines production institute.
- n) Coast Agricultural research station at Mtwapa- conducts research on Maize and sugarcane.
- o) Cotton research station at Kibos in Kisumu.
- p) Embu Research Station- conducts research on medium altitude maize varieties.
- q) International Livestock Research institute in Nairobi.

8. **MARKETING SERVICES-** they help the farmers to sell their products. They include:

- a) National cereals and produce board- it sells cereals e.g. maize and wheat and pulses e.g. beans, pigeon peas, grams, groundnuts.
- b) Kenya co-operative creameries- which markets milk.
- c) Kenya planters co-operative union- which markets coffee.
- d) Pyrethrum board of Kenya- which markets pyrethrum.

9. **TRACTOR HIRE SERVICES-** involves hiring of tractors and implements by farmers who do not have them. It is possible to hire tractor service from the following:

- a) Government Tractor hire services.
- b) Private contractors.
- c) Individual farmers.
- d) Co-operative societies.

Advantages of Tractor Hire services-

1. Farmers who cannot afford to buy a tractor can get access to the tractor services.
2. Farmers do not incur the cost of servicing and maintenance of the tractor and its implements.
3. The services are more efficient than hand tool.

Disadvantages of tractor hire services.

1. They are not available to most farmers when they need it.
2. Some farmers may be overcharged especially by private and individual farmers.

RISKS AND UNCERTAINTIES.

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- A risk is the divergence between the expected and the actual outcome.
- Uncertainty is imperfect knowledge about future events/ state of not knowing about the future events.

Examples /types of risks and uncertainties.

1. **Fluctuation of commodity prices-** the farmer may not predict the future market prices.
2. **Physical yield uncertainty-** the farmer does not know how much to expect.
3. **Ownership uncertainty-** the farmer may lose part/ whole of the produce through theft, change in government policy, fire, death etc.
4. **Outbreak of pests and diseases-** this will affect the expected outcome.

5. **Sickness and injury uncertainty-** the farmer and member of the family or employee is affected and loses the ability to work due to sickness or injury.
6. **New production technique uncertainty-** the farmer may not be certain/ sure as to whether technology is as effective as the previous one.
7. **Obsolescence-** a farmer may invest in machines which may become outdated/ obsolete within a short time.
8. **Natural catastrophes-** they include floods, drought, earthquakes, storms, strong winds which may destroy the crops or kill animals.

WAYS OF ADJUSTING TO RISKS AND UNCERTAINTIES

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1. **Diversification of production**-involves setting up several and different enterprises on the farm so that if one fails, the farmer does not suffer total loss.
2. **Selecting more sure/certain enterprises**- it involves choosing an enterprise which earns a more steady income though less profitable than choosing a more profitable enterprise which has a high degree of income variation.
3. **Hedging/Contract production/ contracting**- farmers get into a contract with consumers to supply certain goods over a specified period of time at an agreed price. This guarantees a constant and fixed market for their produce.
4. **Insurance against losses/taking insurance policy**- this guarantees compensation in the event of a loss due to crop failure, death of livestock, theft, fire and accidents.
5. **Input rationing**- involves using less inputs than the optimum required for an enterprise so that in case of loss the farmers suffer less loss.
6. **Adopting modern methods of production**- they include spraying crops against diseases and pests, vaccinating livestock against diseases and irrigating crops so as to reduce risks.
7. **Flexibility in production methods**- the farmers may design the enterprises so that if there is need to change from one enterprise to another in response to change in demand, they may incur minimum/ less expenses.

5. AGRICULTURAL ECONOMICS 4.

(FARM ACCOUNTS)

Definitions.

- i. **Farm accounting-** analysis, classification and recording financial transactions.
- ii. **Farm accounts-** financial records of income and expenses of the farmer.

Importance of keeping farm accounts.

1. **They help the farmer to secure loans-** creditors assess financial records to determine whether a farmer is credit worthwhile or not.
2. **They help the farmer to make sound management decisions-** financial records provide valuable data that helps the farmer in planning and decision making.
3. **They help the farmer to determine whether the farm is making a profit or a loss-** financial records indicate whether the farm is running at a profit or at a loss.
4. **They help the farmer to prepare farm budgets-** financial records help the farmer to draw estimates for future income and expenditure and especially when preparing partial budget.
5. **They help the farmer to evaluate assets and liabilities-** financial records help in assessing the farmer's ability to pay what it owes others.
6. **They help the farmer to assess the tax chargeable-** financial records help the farmer to estimate the amount of tax that can be charged so that the farm is taxed fairly.

FINANCIAL DOCUMENTS.

1. INVOICE.

- A document issued by the seller to the buyer when goods are taken on credit.
- It informs the buyer of the goods delivered/amount he owes the seller.
- It is also used as evidence of debt and for accounting purposes.
- It is filled in duplicate. The original copy is given to the buyer and the duplicate is retained by the seller.

Information/ details contained in the invoice.

- i. Date of transaction.
- ii. Type of goods.
- iii. Quantity of goods.
- iv. Price of goods.
- v. Total amount of money involved.
- vi. Invoice/ serial number.
- vii. Terms of payment.
- viii. Name of the supplier.
- ix. Name of the buyer/ recipient.
- x. Signature of the supplier/ stamp of the supplier.

<i>Name and address of seller</i>	MKULIMA BORA P. O. Box MOMBASA	INVOICE Invoice no ... Date.....
<i>Name and address of buyer</i>	Dr. SAGALO FARM Box KWALE	

S.No	Particulars	Quantity	Price	Amount per unit
	Total			

Signature of the buyer.....

Fig. 5.1: Sample invoice.

SUNDI HARDWARE LTD.

P. O. Box _____

THIKA

RECEIPT

No. 517

Date 10/05/06

Received from

P. O. Box

Quantity	Particulars	Unit Price	Amount per unit (Kshs.)
10 bags	cement	695	6950
30Kg	nails	80	2400
380 feet	timber	50	19000

Cash/Cheque

Signature

Revenue
Stamp

Fig. 5.2: Sample receipt.

2. RECEIPT/ CASH SALE DOCUMENT.

- It is issued when cash payment for goods delivered or services rendered is made.
- It is a document showing that payment has been made.
- It also shows whether payment was made in form of cash, money order or cheque.
- It is prepared in duplicate/ triplicate.

Information/ details contained in the receipt.

- i. Date of transaction/ payment.
- ii. Name of the buyer.
- iii. Name of seller.
- iv. Amount of money involved.
- v. Signature and official stamp.
- vi. Revenue stamp.
- vii. Receipt serial number.
- viii. Quantity of goods sold.
- ix. Type of goods/particulars or services.

3. DELIVERY NOTE.

- It is given to the buyer by the seller and it accompanies the goods delivered.
- It acts as an evidence that goods have been physically delivered from the supplier to the buyer.
- It may be in duplicate/ triplicate depending on the nature of the transaction.

Information/ details contained in the delivery note.

- i. The date of delivery.
- ii. The quantity of goods delivered.
- iii. The type of goods delivered/ particulars.
- iv. The method of delivery.
- v. The person who receives the goods/ stamp of the receiver.
- vi. The condition in which the goods are delivered.
- vii. The serial number of the delivery note.
- viii. The person who delivers the goods/ from whom the goods were delivered.
- ix. Signature of the receiver.

DELIVERY NOTE

No

Date.....

From

P. O Box.....

To

P. O. Box

Please receive the following goods.

Serial No.	Quantity	Particulars
4770	108 bags	Maize
4771	65 bags	Rice

Delivered by Received by

(Name and signature)

(Name and signature)

4. PURCHASE ORDER.

- It is used to request for the supply of goods and services on credit.
- It may be accepted or rejected by the supplier.

Information/ details contained.

- i. Date of order.
- ii. Date within which the goods ordered are delivered.
- iii. Person who orders the goods.
- iv. Purchase order serial number.
- v. Quantity of goods being ordered.
- vi. Type of goods/particulars ordered/ requested.
- vii. Person who authorized the order.

5. STATEMENTS.

- They are of two types i.e. statements of account and bank statements.
- A statement of account is a document sent to the buyer by the seller as a reminder to inform him/ her what he/ she owes the seller. It is usually written at the end of the month.
- A bank statement is a financial document sent by the bank to the farmer every month to show the position of the bank account at the end of the month.

Information/ details contained in statement of account.

- i. Nature of goods supplied.
- ii. Price of goods.
- iii. Total amount of money paid.
- iv. Balance due.
- v. Statement serial number.

PURCHASE ORDER

From

No.

P. O. Box

Date.....

To:

P. O.

Please supply the following goods on or before:

Items no.	Units	Description	Unit price
1	2 bags	Dairy meal	1,800.00
2	1(25kg) bag	Maize seed (614)	2,500.00
3	1(50kg) bag	Diammonium Phosphates	1,700.00

Official rubber stamp

Ordered by.....

Signature.....

Authorised by

Signature.....

STATEMENT OF ACCOUNT

From

Date

P. O. Box

To:

of Acc. No.....

P. O. Box

Date	Particulars	Debit	Credit	Balance
01.07.86	Balance brought forward			
30.07.86	Amount due by 15.08.86			

Pay on or before..... Date

BOOKS OF ACCOUNT

1. LEDGER BOOK.

- This is the principal/ main book used in keeping financial records.
- It contains the details of all the things in the business owned by the farmer and contains all the accounts.

Details/ information

- i) Heading/account**-a page/ section in a ledger book given to a person/ firm/ enterprise and on which all the business transactions relating to that person/ firm are entered.
- ii) The left hand side of the account/ debit side (DR)**-where all the decreases (expenses and purchases) are recorded.
- iii) The right hand side of the account/ credit side (CR)**-where all the increases (income and sales) are recorded.
- iv) Date column**- the date is entered immediately the business transaction takes place.
- v) Particulars column**-type and description of the transaction is recorded here.

- vi) Folio column**- the number of the page on which the account is found is entered.

- ✓ A folio is a page where a particular account appears in the ledger.
- ✓ Each page in the ledger is numbered and each account should appear on its own page.

Uses of ledger.

- a)** It shows all that the farmer or the business firm owns.
- b)** It shows the amount the business owes other people (creditor) or firms and what the other firms owes the business (debitor).
- c)** It records all the expenses and purchase and sales and income.

EXAMPLE

DAIRY CATTLE									
DR					CR				
Date	Particulars	Folio	Amount		Date	Particulars	Folio	Amount	
			Kshs	cts				Kshs	Cts

STUDY QUESTION

- The table below represents an account in a financial book of a poultry farmer.

POULTRY									
DR					CR				
Date	Particulars	Folio	Amount		Date	Particulars	Folio	Amount	
			Kshs	cts				Kshs	Cts

- Identify the financial book.
✓ *Ledger / ledger book.*
- On 10th January 2011, the poultry farmer bought 5 bags of layers mash worth ksh. 10,000 and sold 100 trays of eggs for Ksh. 20, 000 on 10th January 2011. Enter these transactions in the account above.

POULTRY

DR

CR

Date	Particulars	Folio	Amount	Date	Particulars	Folio	Amount
10/1/11	Bought 5 bags of layers mash	1	10,000.00	10/1/11	Sold 100 trays of eggs	1	20,000.00

2. JOURNAL.

- This is a book where all unclassified non-cash transactions are entered.

Functions/ roles/ uses

- i. Recording non-cash transactions before posting them to their ledgers.
 - ii. Used to correct errors in other books of account.
- Information/ details recorded in a journal include:
 - a) The purchase on credit of a capital equipment.
 - b) The sale on credit of a capital equipment.
 - c) Rectification of errors i.e. if errors are made in other books, new entries are made correcting or cancelling the old entries.

- b) Transfer from one account to another in the ledger.
- c) Opening entries in new books/ balances from the old books are entered into new books. The balances are journalized first, the journal entries are entered as opening entries.

Details/ information contained in the journal.

- i. Date.
- ii. The name of the account to be debited.
- iii. Name of the account to be credited.
- iv. Amount of money involved.
- v. Brief description of the transaction.

Example

			DR		CR	
Date	Particulars	Folio	Amount		Amount	
			Shs	Cts	Shs	Cts
11/2/2003	Cash (cash A/C)	1			16,000	00
12/3/2003	Bought 2 heifers (Dairy cattle A/C)	2	10,000	00		
15/4/2003	Sold milk (Milk A/C)	3			2,000	00
16/4/2003	Bought 2 goats (Goats A/C)	4	8,000	00		

Subsidiary books of the journal

- They are books for recording other transactions to avoid filling the journal quickly.
- a) **Purchase book-** it is also known as creditor's account. A creditor is someone/ firm to whom the business/ farm owes money.
- It contains all the details of items bought on credit.
- b) **Sales book-** it is also known as debtor's account. A debtor is a person/ firm who/ which owes money to the farm business.
- It contains all sales of goods on credit.

3. INVENTORY.

- This is a book in which a record of all assets owned by a business or an individual is kept.

Function/ role

- It provides information which is used during valuation or drawing up a balance sheet and profit and loss account.

Types of inventory.

- i. **Consumable goods inventory-** it records goods which get used up in the production process, such as, fertilizers, fuel, animal feeds, drugs, construction materials, seeds, etc.
- ii. **Permanent goods inventory-** This shows a record of assets which do not get used up in the production process, e.g. tools, equipment, machines, buildings.

STUDY QUESTION

- The table below shows a farm record for Mwamuzi Farm.

RECEIPTS			ISSUES			
Date	Commodity/ item	Quantity	Date	Issued to	Quantity	Balance in stock

- Identify the farm record.
 - ✓ *Consumable goods inventory.*
- Give one reason for keeping the farm record illustrated above in farm accounting.
 - ✓ *It provides information used for drawing Profit and Loss Account and Balance sheet.*

- c) The farm received 20 bags of DAP fertilizer from Milimani Agrovat on 07/07/2018 and another 20 bags on 21/07/2018. on 28/07/2018, a gardener took eight bags of DAP and used it to plant potatoes on the farm. Enter this information in the farm record illustrated above.

RECEIPTS			ISSUES			
Date	Commodity / item	Quantity	Date	Issued to	Quantity	Balance in stock
7/7/18	DAP fertilizer	20 bags				20 bags
21/7/18	DAP fertilizer	20 bags				40 bags
			28/7/18	Gardener	8 bags DAP	32 bags

4. CASH BOOK.

- It is a financial book in which all transactions involving receiving and paying out of cash are recorded.
- It consists of sales and receipts side (left hand side) and purchases and expenditure side (right hand side)

Uses.

- i. It is used for recording all transactions involving receiving and paying out of cash.
- ii. It shows the net worth of the enterprise.

Details/ information contained in a cash book.

- a) Date.
- b) Receipt number.
- c) The person/ firm from whom money has been received.
- d) Total amount received.
- e) Particulars/ details of transaction.

CASH BOOK					
DR.			CR.		
SALES & RECEIPTS			PURCHASES AND EXPENSES		
Date 2003	Particulars	Amount	Date 2003	Details	Amount
		shs			shs cts
1st Jan	Balance brought forward	2000 00	Jan 3	Bought 3 bags layers mash	1800 00
Jan 10	Sold 2 bags of Irish potatoes	3000 00	Jan 14	Bought 2 bags of fertiliser	2400 00
Jan 15	Sold 20 trays of eggs	2400 00	Jan 20	Bought pangas and slashers	1200 00
Jan 22	Sold carrots	600 00	Jan 21	Bought broiler finisher mash	3000 00
Jan 25	Sold broilers	5000 00	Jan 22	Bought acaricide	2000 00
Jan 28	Sold green maize	4000 00	Jan 29	Bought poultry vaccine	1000 00
Jan 31	Sold dairy culls	6000 00	Jan 30	Bought dewormer	1500 00
		23000 00	Jan 31	Balance carried down	10100 00
					23000 00
Jan 31	Balance carried down	10100 00			

FINANCIAL STATEMENTS.

- They are used to summarize the financial information obtained from financial books.
- They include:
 - A. Balance sheet
 - B. Profit and loss account.
 - C. Cash analysis.

A. BALANCE SHEET.

- This is a financial statement drawn to show the financial position of the farm business at a particular period of the year.

Uses/ benefits of a balance sheet.

- i. It shows the financial status of the enterprise/whether it is solvent or insolvent.
- ii. It enables a farmer to know whether he is making profit or loss/value of assets and liabilities.
- iii. Used for fair taxation.
- iv. For obtaining loan/credits.
- v. To show the value of land.

FORMAT

- The balance sheet has two sides, namely:
 - a) Left hand side (LHS)- where liabilities are recorded.
 - b) Right hand side (RHS)- where assets are recorded.

A. Liabilities

- These are all the debts that the farmer or business owes other business firms e.g. loans and mortgages, bank overdrafts, debts payable for goods and services received and services for which other people have paid to the farm business in advance.

Types of liabilities.

- i) **Long term liabilities-** they are debts whose repayable period is more than one year e.g. long term loans from banks or Agricultural Finance Corporation (AFC)
- ii) **Current/ short term liabilities-** they are debts which are payable/ repaid within a period of one year e.g. rent, electricity bill, water bills, telephone bills, wages, bank overdrafts debts payable.

B. ASSETS-

- They are properties owned by the business and have a monetary value attached to them e.g. cash at hand and in the bank, stored produce and farm inputs, livestock, growing crops, farm building, machinery, equipment, debts receivable for goods delivered and services offered.

Types of assets

- Fixed assets:** They are used by the business for a long period of time, e.g. land, buildings, machinery and motor vehicle, tools and equipment, perennial crops, breeding stock.
- Current/ short term assets:** These are property or items which are likely to be held for a short period of time/ less than one year i.e. they can be converted to money within a short period/ one year e.g. cash at hand, cash in bank, debts receivable, livestock, farm produce to be sold, inputs in store.

1. **Heading:** it shows the name of the person or business firm, title of the statement (balance sheet) and date on which it is being prepared preceded by the words "as at".
 2. **Asset side:** This is usually put on the right hand side and shows current assets and fixed assets.
 3. **Liability side:** It is usually put on the left hand side and shows current liabilities long-term liabilities and capital/ net worth/ owner's equity
- If the value of liabilities exceeds the value of assets, the firm business is said to be **insolvent** hence it can not be able to pay all its debts.
 - If the value of assets exceeds the value of liabilities, the business is said to be **solvent** hence the business is able to meet all its debts and still have a balance.
 - When drawing a balance sheet the following orders are use:
 - a) Order of permanency i.e. fixed assets followed by current assets/ long term liabilities followed by short term/ current liabilities.
 - b) Order of liquidity- the reverse order.

EXAMPLE

- The inventory for Jashi Farm as at 31st December 2011 was as shown below. Use the information to prepare a balance sheet

Cash in hand	7,000
Layers	25,000
Dairy cattle	1 30,000
Cassava in store	5,000
Calves	19,000
Buildings	80,000
Land	180,000
Machinery	85,000

- On the same day, the following information was obtained from the farmer's records:

Bank loan	210,000
Eggs sale on credit	10,000
Milk sale on credit	15,000
Vegetable sales on credit	5,000
Farm inputs purchased on credit	20,000
Cash in bank	25,000
Interest payable	2,500
Taxes payable	800

JASHI FARM
BALANCE SHEET AS AT 31ST DECEMBER 2011

LIABILITIES	Kshs	Cts	ASSETS	Kshs	Cts
<u>Long liabilities</u>			<u>Fixed assets</u>		
Bank loan	210,000	00	Land	180,000	00
			Buildings	80,000	00
			Machinery	85,000	00
			Dairy cattle	130,000	00
			Calves	19,000	00
			Layers	25,000	00
			<u>Sub total</u>	<u>519,000</u>	<u>00</u>
<u>Current liabilities</u>			<u>Current assets</u>		
Interest payable	2,500		Cassava in store	5,000	00
Tax payable	800	00	Eggs sale on credit	10,000	00
Wages payable	6,200	00	Milk sale on credit	15,000	00
Farm inputs	20,000	00	Vegetables sale on credit	5,000	00
<u>Sub total</u>	<u>29,500</u>	00	Cash in bank	25,000	00
<u>Total liabilities</u>	<u>239,000</u>	<u>00</u>	Cash in hand	7,000	00
Net worth/ capital/ owner's equity		<u>00</u>	<u>Sub total</u>	<u>67,000</u>	<u>00</u>
	<u>346,000</u>	<u>00</u>	<u>Total assets</u>	<u>586,000</u>	<u>00</u>
<u>Grand total</u>	<u>586,000</u>	<u>00</u>	<u>Grand total</u>	<u>586,000</u>	00

B. PROFIT AND LOSS ACCOUNT.

- This is a financial statement showing whether the business made a profit or loss.

- It has:

a) Heading- it shows the name of the person or business firm, title of the statement (profit and loss account) and date on which it is being prepared preceded by the words "for the year ended".

b) The income side.

- It is placed on the right hand side and contains:

i) Closing valuation- includes all the assets in the farm/ business by the end of financial year.

ii) Sales and receipts- include all the produce items sold from the farm business and any debts receivable by the end of financial year.

b) Expenditure side

- It is placed on the left hand side and contains:

i) Opening valuation- includes all assets in the farm/ business at the beginning of the financial year.

ii) Purchases and expenses- they include items bought and debts payable to the firm business for the financial year.

- If the sum of expenditure side is more than the sum on income side then the business has made a loss.
- If the sum of income side is more than the sum on expenditure side then the business has made a profit.

EXAMPLE

- The following information was obtained from farm records of Hekima farm for the year ended 31st December 2014.
- a) Using the information provided below prepare a profit and loss account for Hekima farm.
- Opening valuation- Ksh. 6,000/=
- Paid wages- Ksh. 5000/=
- Bought equipment worth- Ksh. 8,000/=
- Bought pig feeds worth- Ksh. 4,000/=
- Sold mature pigs worth- Ksh. 7,000/=
- Bought drugs worth- Ksh. 3,200/=
- Sold maize worth- Ksh. 3,000/=
- Closing valuation- Ksh. 40,000/=

HEKIMA FARM

PROFIT AND LOSS ACCOUNT FOR THE YEAR ENDED 31st DECEMBER 2014

EXPENDITURE	Kshs	Cts	INCOME	Kshs	Cts
Opening valuation	6,000	00	Sold pigs	7,000	00
Wages paid	5,000	00	Sold maize	3,000	00
Bought equipment	8,000	00	Closing valuation	40,000	00
Bought pig feeds	4,000	00			
Bought drugs	3,200	00			
<u>Sub total</u>	<u>26,200</u>	<u>00</u>	<u>Sub total</u>	<u>50,000</u>	<u>00</u>
Net profit	23,800	00			
<u>Grand total</u>	<u>50,000</u>	<u>00</u>	<u>Grand total</u>	<u>50,000</u>	<u>00</u>

Calculate percentage profit

$$\begin{aligned}\% \text{ profit} &= \frac{\text{Profit}}{\text{opening valuation}} \times 100 \\ &= \frac{11,000}{8,000} \times 100 \\ &= 137.5 \%\end{aligned}$$

C. CASH ANALYSIS.

- ✓ This is a financial statement showing all cash transactions within a financial year.
- ✓ It records income (sales and receipts) and expenditure (purchases and expenses).

- ✓ Each entry appears twice; once on the total column and once on the column of the specific enterprise onto which the transaction was made.
- ✓ This statement indicates how much each enterprise has earned or spent.

Example

- On 1st January 2009, Kaburu Farm started farm operations with Ksh 30,000 cash.
- During the month, the farm made the following transactions. Study the transactions and prepare a cash analysis for Kaburu Farm for the month of January.

Date	Transaction	Amount (kshs)
05/01/09	Livestock sales	80,000
08/01/09	Crop sales	50,000
15/01/09	Bought seeds for planting	7,500
20/01/09	Paid K.F.A. for fertilizer	16,400
25/01/09	Bought livestock feeds	50,000
30/01/09	Paid wages for planting and weeding	56,000
31/01/09	Received cash from KCC for milk delivery	120,000
31/01/09	Paid transport charges for delivery of milk	9,000

Cash analysis for Kaburu farm

Purchases and expenses								Sales and receipts					
Date	Description	Amount	Seeds	Fertilizer	Feeds	Wages	Transport	Date	Description	Amount	Livestock	Crop	Milk
15/1/9	Bought seeds	7,500	7,500					1/1/9	Opening balance	30,000			
20/1/9	Paid fertilizer	16,400		16,400				8/1/9	Sale of livestock	80,000	80,000		
25/1/9	Bought feeds	50,000			50,000			8/1/9	Sale of crop	50,000		50,000	
30/1/9	Paid wages	56,000				56,000		31/1/9	Cash for milk	120,000			120,000
31/1/9	Transport charges	9,000					9,000						
Sub-Total		138,900	7,500	16,400	50,000	56,000	9,000			280,000	80,000	50,000	120,000
Closing balance		141,100											
Grand total		280,000								280,000			

6. AGRICULTURAL ECONOMICS 5 **(AGRICULTURAL MARKETING AND ORGANISATIONS)**

Definition of terms.

1. A **market-** is a place where buyers and sellers meet to buy and sell goods and services/ carry out business transactions. OR
 - ✓ A group of buyers and sellers in sufficiently close contact with one another for exchange to take place between them.
2. **Marketing** -refers to the flow of goods and services from the producer to the consumer.
 - ✓ It includes all the activities involved in the transfer of goods and services from the producer to the consumer.

Types of markets.

- A. **Perfect/ competitive market-** sellers and buyers are unable to determine the market price.
 - ✓ The market price is acceptable/ equilibrium and any buyer can purchase from any seller and vice versa.
- B. **Imperfect /non-competitive market-** sellers and buyers are able to determine the prices in the market.
 - ✓ Buyers and sellers are not aware of the prices offered by others.
 - ✓ Examples of non-competitive markets are **monopoly, oligopoly and monopsony.**

Types of market situations/ structures.

1. Monopoly / Monopolistic

competition- monopoly consists of one seller. Monopolistic competition consists of large number of small firms/sellers competing against each other. Examples include clothing stores, fast food restaurants.

✓ It has the following characteristics/ features:

- i. Existence of few sellers of the commodity.
- ii. The product quality varies from one firm to another.
- iii. The products are close substitutes.
- iv. There is a perfect product differentiation between the firms.
- v. A single seller/ firm may dominate the market due to technological superiority.

2. Monopsony- market situation consisting of one buyer. The individual buyer determines market price.

3. Oligopoly- it consists of few firms/ sellers such that mutual interdependence/ collaboration exists among them.

✓ When each firm/ seller makes its policies it takes into consideration the effect on other firms.

4. Pure/ perfect competitive market- it consists of many sellers and buyers.

✓ The buyers and sellers do not have any control over the prices.

Conditions for purely competitive market

- i) The number of sellers and buyers must be large.
- ii) The product must be homogenous/ similar/ identical.

DEMAND.

- **Demand** refers to the quantity of goods and services buyers/ consumers are willing and able to buy at a specific price in a given market at a given time.

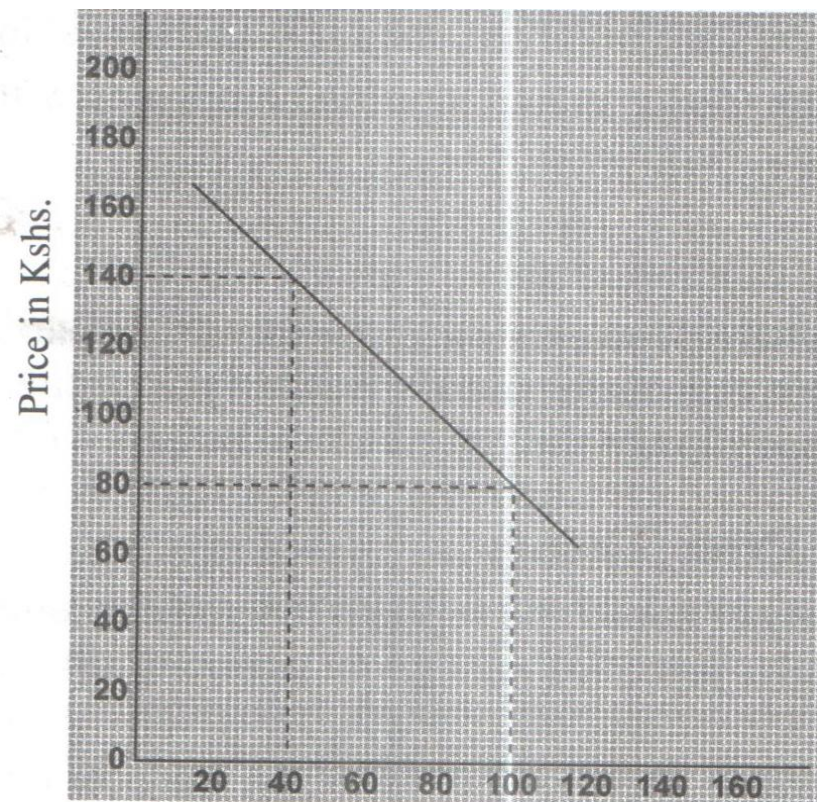
The law of demand.

- This law states that *the lower the price of a product, the higher the quantity demanded and the higher the price of a product, the lower the quantity demanded of it.*

OR

- *The quantity of products demanded varies inversely with the price.*
- The relationship between demand and price is known as the demand schedule e.g.

Price (kshs)	160	140	120	100	80
Quantity (kgs)	20	40	60	80	100



Quantity of meat demanded (Kgs)

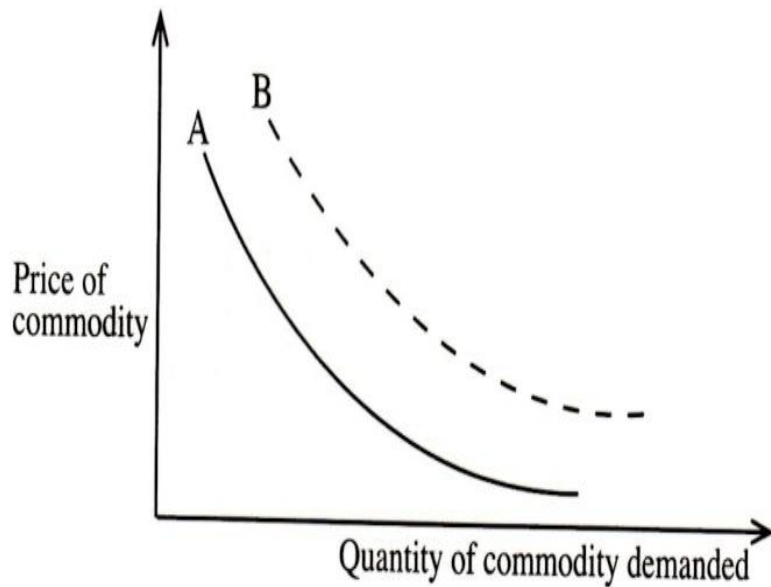
Fig.6.1: Demand curve for meat.

FACTORS WHICH INFLUENCE DEMAND OF A COMMODITY.

1. **Price of related goods or services/substitutes-** When the price of a substitute increases, the demand of the commodity increases, e.g. if the price of butter increases the demand of margarine (which is a substitute of butter) increases.
2. **Population size of buyers/ consumers.-** When population size increases, the demand for goods and services will also increase.
3. **Income level of consumers/ buyers-** An increase in the income level of consumers will cause an increase in demand for commodities.
4. **Beliefs, customs and taboos-** they will determine the total demand of a given good or service e.g. religious beliefs of Muslims forbid consumption of pork, thus decreasing demand.
5. **Taste and preferences of the consumer-** A change in consumers' tastes and preferences in favor of a commodity increases the demand of particular commodities.
6. **Price expectations-** If in future the prices of certain commodities are likely to go up, then the demand of such goods may increase now and vice versa.
7. **Level of taxation-** if the government increases tax on certain goods, this will increase their prices hence decreasing the demand.
8. **Advertisement-** it promotes the sale of a given good or service hence increasing its demand.
9. **Perishability/change in quality-** when agricultural goods e.g. meat, milk, tomatoes and fruits deteriorate in quality, their demand decreases.
10. **Future expectation/ uncertainty-** when there is fear of shortage of a given commodity in future, buyers/ consumers tend to buy more for stocking thus increasing demand.

STUDY QUESTIONS

1. Explain the factors that:
 - a) Decrease the demand of tea in the market
 - b) Increase the demand of tea in the market.
2. The diagram below illustrates the law of demand in agricultural marketing. Study it and answer the questions that follow.



- a) Give a reason for the shape of the curve labeled **A**.
 - *As the price of the commodity increases the quantity demanded decreases and vice versa.*
- b) If the price of the commodity remains constant, explain **three** factors that can cause the curve to shift from **A** to **B**.
 - *If there is an increase in the income of consumers.*
 - *Effective advertisement / sales promotion.*
 - *Increase in the price of a related goods / substitute.*
 - *If there is an increase in population.*
 - *Change in taste and preference in favor.*
 - *If the quality of the commodity goes up.*
 - *Fear of shortage in future.*

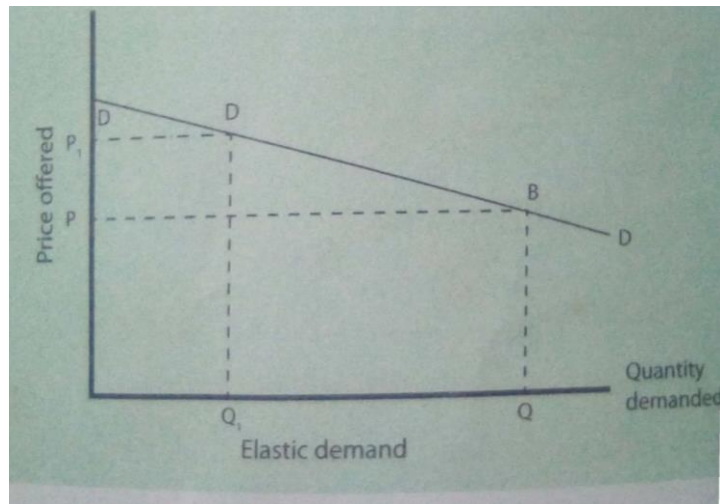
ELASTICITY OF DEMAND (Ed).

- This is the degree of responsiveness of demand for a product when there is a change in price.

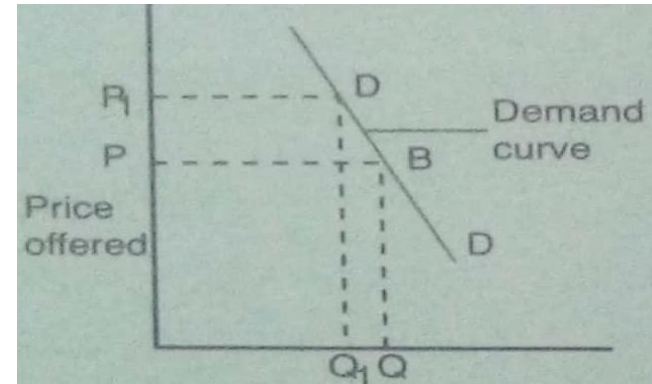
$$Ed = \frac{\% \text{ change } (\Delta) \text{ in quantity demanded}}{\% \text{ change } (\Delta) \text{ in price}}$$

Types of elasticity of demand

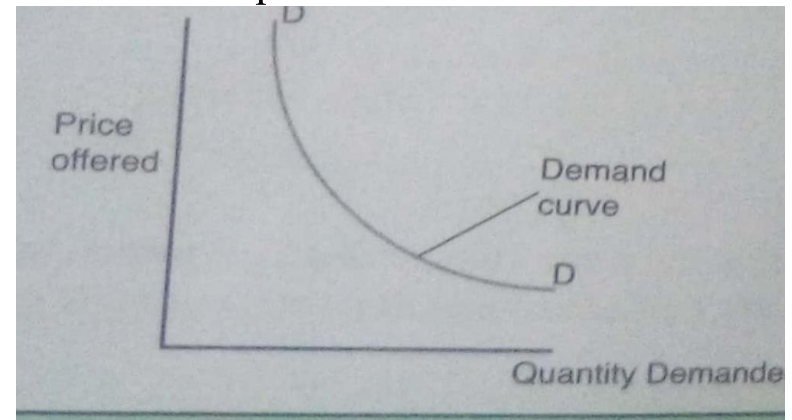
1. **Elastic Ed**- There is more than proportional (higher) increase in demand with change/ decrease in price. Ed is more than 1.



2. **Inelastic Ed**- There is less than proportional (lower) increase in demand with change/ decline in price. Ed is less than 1.



3. **Unitary Ed**- This is when demand for a product responds by the same proportion as the change in price. Price elasticity of demand is equal to one.



Example.

- ✓ Calculate the elasticity of demand when 1000 loaves of bread are demanded at a price of Kshs. 20 per loaf while only 600 loaves were demanded at Kshs. 23 per loaf.

$$Ed = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}}$$

$$\begin{aligned}\% \Delta \text{ in Quantity} &= \frac{\text{change } (\Delta) \text{ in quantity}}{\text{Initial quantity}} \times 100 \\ &= \frac{1000 - 600}{600} \times 100 \\ &= \frac{400}{600} \times 100 \\ &= 66.6\%\end{aligned}$$

$$\begin{aligned}\% \Delta \text{ in price} &= \frac{\text{change } (\Delta) \text{ in price}}{\text{Initial price}} \times 100 \\ &= \frac{20 - 23}{23} \times 100 \\ &= \frac{-3}{23} \times 100 \\ &= -13.04\%\end{aligned}$$

$$\begin{aligned}Ed &= \frac{\% \Delta \text{ quantity demanded}}{\% \Delta \text{ in price}} \\ &= \frac{66.6\%}{-13.04\%} \\ &= 5.11 \text{ (Negative sign is ignored)}\end{aligned}$$

Ed is therefore elastic.

FACTORS THAT DETERMINE / AFFECT ELASTICITY OF DEMAND.

1. **Availability of substitutes-** commodities with many substitutes usually have elastic demand.
✓ If a commodity has no substitute e.g. salt its demand is inelastic and has to be bought at any price .
2. **Degree of necessity-** a commodity e.g. salt is of great necessity hence has to be bought at any price hence has inelastic demand.
3. **The number of uses the product can be put to-** the commodity that can be put into many uses has elastic demand.
4. **Time lag-** if the commodity can be postponed to another day, then it has elastic demand e.g. cement for construction.
5. **Time span-** elasticity of demand is usually greater in the long run because some adjustments can easily be done while it is small in the short run.
6. **Proportion-** commodities that form the large proportion of the total expenditure have elastic demand.

SUPPLY

- Supply is the quantity of goods and services that sellers/ suppliers/ producers **are willing and able** to sell/ supply at a specific price in a given market and period of time.

The law of supply.

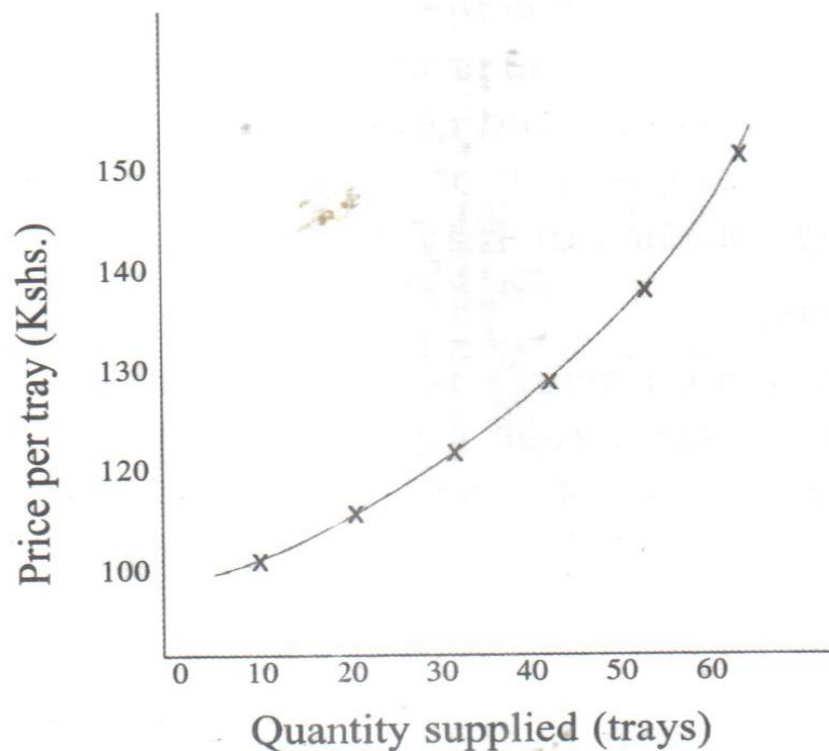
- The law states that *as the price of the goods and services increases the quantity of goods and services supplied increases and vice versa.*

OR

- The quantity of goods and services supplied is directly proportional to the price of the commodity.*
- The relationship between supply and price is referred to as **the supply schedule**.
- When the supply schedule is represented graphically, a **supply curve** is obtained.

Example

Price (kshs)	100	105	113	120	130	150
Quantity Supplied (Kg)	10	20	30	40	50	60



FACTORS WHICH INFLUENCE / DETERMINE SUPPLY.

1. **Number of sellers/suppliers-**
When the number of sellers dealing with the same product increases the supply of the same will increase.
2. **Price of related goods /substitutes-** When the price of a substitute of a commodity increases, the supply of the commodity decreases e.g. if the price of margarine increases in comparison with that of butter, suppliers will increase the supply of margarine and reduce that of *butter*.
3. **Technology-** modern techniques of production will increase production hence increase supply.
4. **Weather-** Favorable weather conditions lead to increased production thus increasing supply.
5. **Change in prices-** When the price increases, the farmers will have incentive to produce more hence increasing supply.
6. **Price expectation-** if the future price of the a commodity is expected to rise the supply will decrease now i.e. the suppliers will hoard the commodity to release it when the price has increased and vice versa.
7. **Government policy-** Government imposition of tax on a commodity raises its cost of production, thus, fewer farmers are willing to produce it. This lowers the supply of that particular commodity and vice versa.
8. **Cost of production-** If the cost of inputs such as fertilizers and seeds is low/ decreases, then it is easy for farmers to buy them and apply on crops hence increasing yield/ supply.
9. **Change in the supply of associated /related goods-** Increase in the supply of associated commodity increases the supply of the other e.g. supply of meat will automatically lead to increase in supply of hides and skins.
10. **Transport system-** improved and efficient transport will facilitate the delivery of farm produce to the market and hence increase supply.

ELASTICITY OF SUPPLY (E_s).

- ✓ This is the degree of responsiveness of the quantity supplied of a product to change in price.

$$\text{Elasticity of supply} = \frac{\% \text{ change in quantity supplied}}{\% \text{ change in price}}$$

Types of elasticity of supply.

1. **Inelastic supply**- This is when the in price leads to a slight change in supply. Elasticity of supply (E_s) is less than one.
2. **Unitary elastic supply**- This occurs when a change in price brings a change in supply in equal proportions. The elasticity of supply (E_s) is equal to 1.
3. **Elastic supply**- This is when price increases, the quantity supplied increases at a higher proportion. The elasticity of supply (E_s) is more than one.

Example

If the price of millet changes from ksh. 10 to ksh. 12 results in a change of supply from 400 kg to 600 kg. Calculate the elasticity of supply.

$$\begin{aligned}\% \Delta S &= \frac{(600 - 400) \times 100}{400} \\ &= \frac{200 \times 100}{400} \\ &= \frac{20000}{400} \\ &= 50\%\end{aligned}$$

$$\begin{aligned}\% \Delta P &= \frac{(12 - 10) \times 100}{10} \\ &= \frac{2 \times 100}{10} = 20\%\end{aligned}$$

$$E_s = \frac{\% \Delta S}{\% \Delta P} = \frac{50}{20} = 2.5$$

PRICETHEORY

- This is the mechanism by which the price is determined in the market.
- **Price** is the amount of money paid in exchange for a good or service.

Determination of market prices.

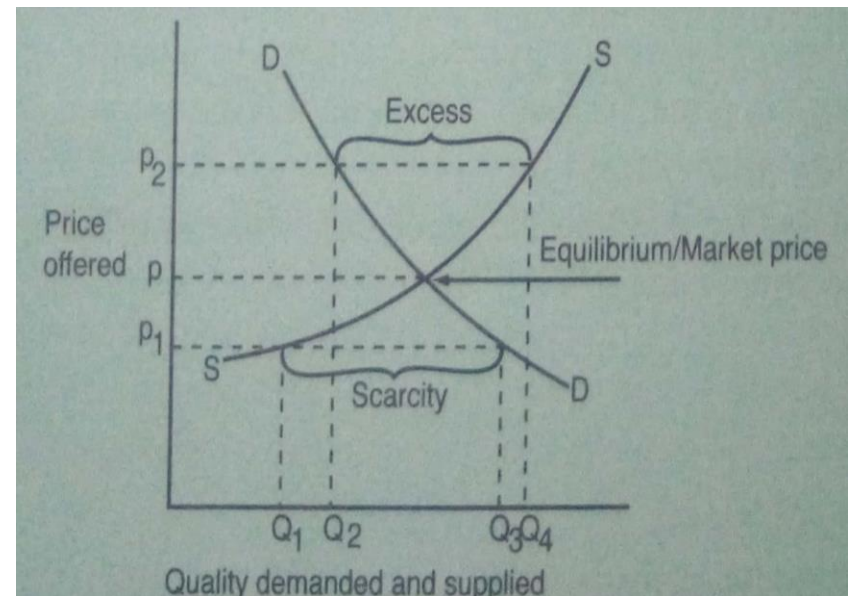
- The market/ equilibrium price is attained when supply and demand are equal.

Price control

- The government takes the following measures to ensure and maintain production:
 1. Gives subsidy by reducing the costs of production inputs.
 2. Fixes prices of the related products.

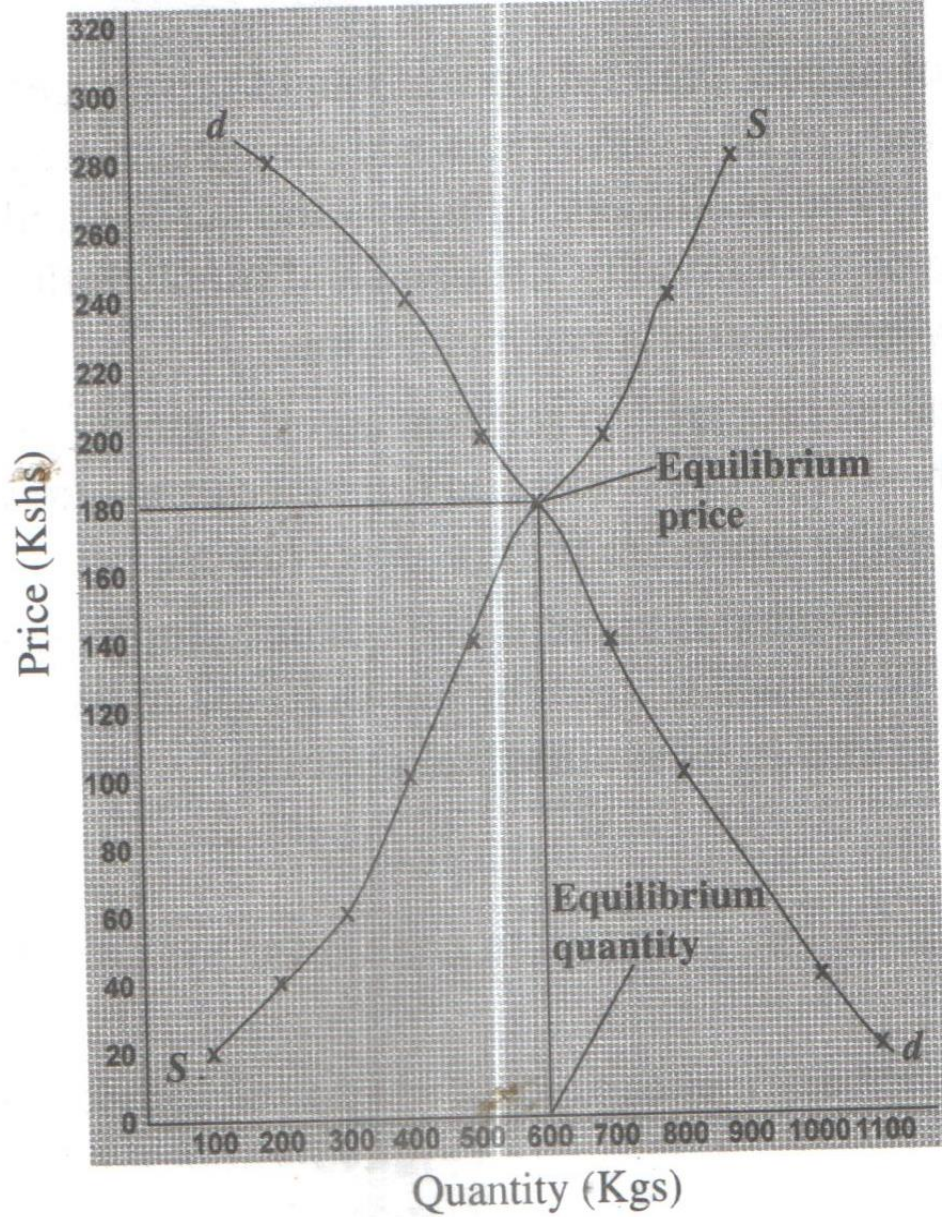
FACTORS WHICH AFFECT THE PRICE OF COMMODITY

1. Price of substitutes/related goods.
2. Price expectations in the future.
3. Quality of commodity/ product.
4. Tastes and preference of the buyer.
5. Number of sellers/supply
6. Number of buyers/demand.
7. Change in production technique.



Example

Price per kg (Shs)	Quantity demanded (kg)	Quantity supplied (kg)
320	200	1000
280	300	900
240	400	800
200	500	700
180	600	600
140	700	500
100	800	400
60	900	300
40	1000	200
20	1100	100



Explanation.

- The market price/equilibrium price is Shs.180 while the quantity demanded and supplied (equilibrium quantity) is 600 kg.
- A rise in price would change both the quantity supplied and demanded.

MARKETING FUNCTIONS.

- **Marketing** refers to all the processes involved in the transformation and flow of goods and services from the farm to the consumer.
- **Marketing functions** are the processes involved in marketing. They include:
 1. **Buying and assembling-** buying involves acquisition of goods from farmers on payment of cash. Assembling means collecting of produce from individual farms and concentrating them at convenient points.
 2. **Storage of farm produce-** to make commodities and minimize losses.

3. **Selling on behalf of farmers-** presenting the produce to consumers in the most attractive manner.
4. **Advertisement-** this increases demand of the produce.
5. **Transporting and distributing-** done to areas of consumption to ensure consumers /buyers get the product at the right time and at the right place.
6. **Processing-** it involves changing a product from a raw state to a finished state.
 - ✓ It improves quality and reduces bulkiness hence facilitating transportation.
6. **Packing-** refers to wrapping the product into a case, container, sack.
 - ✓ It is done to:
 - i) Ease transport and reduce storage space.
 - ii) Avoid damage, theft and adulteration on its way to the market.
 - iii) Facilitate the measurement of the commodities into quantities that can easily handled.

7. Grading and standardization-

- ✓ **Grading** is the sorting of the produce into different lots each with same characteristics e.g. size, shape, flavor, degree of ripeness.
- ✓ **Standardization** is the establishing of uniformity in quality and quantity of the product.
- ✓ Grading is done to provide uniform standard / cater for various consumers, allowing the buyers to purchase what they want.

8. **Packaging-** refers to the process of designing and creating a cover for the product.

Reasons/aims/objectives of packaging.

- i. Ease transport and reduce transport space.
- ii. Enables the farm produce to be handled correctly.
- iii. It makes marketing more convenient.
- iv. Prevents physical deterioration, adulteration and substitution.
- v. Offers easy labelling, measurement and attachment of sales instructions and descriptions.
- vi. It promotes sales.

9. **Collection and analysis of market information/ market research-** done to determine the appropriate market price of a product, where and when to sell.
10. **Financing-** the finance is obtained from public and commercial banks which is used to finance all the activities from the original buying of the raw produce to the final sale of the finished goods.
11. **Bearing of risks-** the risks include price fluctuations, change in government policy, theft of the produce, change in consumers preferences, e.t.c.
 - ✓ It is done by use of chemicals/ insurance to protect the farm produce.

PROBLEMS OF MARKETING AGRICULTURAL PRODUCE AND POSSIBLE SOLUTIONS.

1. **Perishability-** most agricultural products are perishable leading to losses due to spoilage and farmers incur extra costs in transportation/ processing/ storage.
 - ✓ To prevent losses due to perishability, the products must be stored in a refrigerator (refrigeration), sold immediately after production/ processed, canned or dehydrated.
2. **Seasonality-** most agricultural products are seasonal in nature hence creating storage problems/ causes oversupply at times leading to price fluctuation.
 - ✓ The problem can be solved by storing surplus production in form of buffer stocks to be releasing during scarcity.

3. **Bulkiness of most agricultural produce**-this increases storage and transport costs/ making it difficult and expensive to transport.
 - ✓ The problem is solved by processing the products.
4. **Poor storage**- leading to heavy losses of the produce.
 - ✓ The problem can be solved by pooling their resources together in co-operatives in order to purchase expensive storage facilities.
5. **Poor transport system/ lack of vehicles**- this leads to losses due to spoilage.
 - ✓ The problem can be solved by improving the road network and formation of co-operatives.
6. **Changes in market demand**- this leads to price fluctuation.
 - ✓ The problem is solved by contracting/ contract production.
7. **Inadequate capital**-this leads to poor financing of various marketing functions.
 - ✓ The problem can be solved by formation of co-operatives
8. **Poor quality of produce**-this leads to price fluctuation.
 - ✓ The problem can be solved by processing farm produce and using modern technology of production.

9. **Lack of perfect market information-**this makes selling difficult/ exploitation by middlemen
 - ✓The problem can be solved by formation of co-operatives, proper market research and training of farmers.
10. **Changes of supply/ competition from cheap imports-** this leads to price fluctuations.
 - ✓The problem can be solved by using modern technology, contract production and surplus to be stored to regulate supply.
11. **Government interference-**through its agents leading to price fluctuations.
12. **Lack of knowledge in marketing-** this leads to heavy losses.
13. **Delayed payments-**this leads to lack of capital for farm operations.

MARKETING AGENTS/ AGENCIES

- ✓ Are bodies which facilitate the marketing process.
- ✓ They include:
 1. **Wholesalers-** they buy the produce in large amounts/ bulk from farmers and resell to others/ process them.
 2. **Retailers-**they are traders who buy goods in bulk from the wholesalers and sell to consumers in small quantities.
 - ✓ They also offer special services e.g. restaurants/ hotels.
 3. **Itinerant traders/ middle men-**they buy products from individual farmers and resell them.
 4. **Brokers/ commission agents-** they act on behalf of other business people for a fee/ commission.

5. **Co-operative societies/ unions-** they buy the produce from farmers and help to sell them e.g. Kenya Co-operative Creameries limited for selling milk.
6. **Marketing boards-** they promote production and marketing of agricultural produce and buy produce from farmers.
7. **Processors/ manufacturers-** they buy and process produce from farmers.

AGRICULTURAL ORGANIZATIONS

✓ These are bodies that promote agricultural activities.

✓ They include

- A. Marketing/ statutory boards.
- B. Co-operative societies.
- C. Farmers associations and unions.

A. MARKETING (STATUTORY) BOARDS.

- A marketing board is a statutory organ established by an Act of Parliament.
- Marketing boards carry out one or more of the marketing functions.

FUNCTIONS OF MARKETING (STATUTORY) BOARDS.

1. They buy farmers' produce or delegate the responsibility to an approved agent.
2. They fix prices of farm produce in consultation with the government.
3. They collect farm produce from areas of production and deliver it to the stores or factories.
4. They inspect the production process to ensure high quality of produce.
5. They arrange for the supply of farm inputs such as seeds and fertilizers.
6. They sell the products on behalf of farmers.
7. They provide storage facilities for farmers' produce.
8. They provide finance/ credit/ loan facilities to farmers.

8. They provide technical advice / advisory services on production/ extension services where applicable.
10. Some process and package farm produce. An example is the Kenya Tea Development Agency (KTDA) which processes tea.
11. They undertake research services on techniques of production.
12. They regulate production to prevent under supply of the produce.
13. They package/ pack the farm produce.
14. They invest the accrued profit.
15. They advertise/ promote sales of the produce.

EXAMPLES OF MARKETING/ STATUTORY BOARDS.

1. **Coffee Board of Kenya**-has the following roles/ functions:
 - a) Licensing coffee producers and producers.
 - b) Carrying out research on all aspects of coffee production and processing.
 - c) Acts as a government agent on all matters related to international coffee agreement e.g. prices.
 - d) Marketing of coffee.
2. **Kenya meat commission**- it has the following roles/ functions:
 - a) Buying cattle from beef farms.
 - b) Slaughtering beef animals.
 - c) Grading carcasses.
 - d) Marketing beef locally and overseas.

3. **Pyrethrum Board of Kenya-** it has the following functions/ roles:

- a) Providing advisory services to the pyrethrum farmers.
- b) Managing pyrethrum nurseries which provide planting materials to farmers.
- c) Processing pyrethrum from farmers.
- d) Marketing processed products.
- e) Buying pyrethrum from farmers.
- f) Carrying out research to obtain the best cultivars through selection and breeding.

4. **National Irrigation Board-** developing and improving irrigation projects.

5. **Cotton Board of Kenya-** it has the following functions/ roles:

- a) Planning, monitoring and regulation of cotton growing and ginning (separation of seeds from raw cotton fibres).
- b) Licensing and control of cotton ginners.
- c) Regulation and control of quality of raw cotton.
- d) Regulation of the export or import of cotton lint or cotton seed/fibres.
- e) Regulation and control of quality and supply of seed through ginneries.
- f) Carrying out and promotion of research and development in cotton production and processing technology.
- g) Providing or co-ordinating training for any sector of the cotton industry.

6. **Kenya Sisal Board-** has the following roles/ functions:

- a) Promoting production of sisal.
- b) Regulation of production, grading and marketing of sisal.
- c) Registration of sisal products.
- d) Licensing of sisal factories.
- e) It examines all sisal for export to ensure and maintain high quality.
- f) Rebaling sisal which may be spoilt on transit from sisal estate to the port.

7. **Horticultural crops development authority-** it has the following roles/ functions:

- a) Offering advisory services to farmers (through extension officers)
- b) Collecting produce from farmers.
- c) Sorting and grading the produce.
- d) Marketing the products locally and internationally.

8. **Kenya sugar authority-** it has the following functions/ roles:

- a) Advise on development of sugarcane production (for the manu.facture of sugar)
- b) Advise on rules and regulations necessary for development of sugar industry.
- c) Formulate and advise on the price of sugarcane.
- d) Develop and implement a sugarcane testing service and quality control.
- e) Advise on all aspects of sugar processing.
- f) Register all sugarcane producers within sugar factory zones.
- g) Ensure availability of adequate statistics relating to sugarcane industry.
- h) Advise on utilization of sugar by-products.

9. National cereals and produce

board- it has the following roles/ functions:

- a) Regulation and control of production and storage of cereals (e.g. maize, wheat)
- b) Buying and storing cereals (which are then sold to millers and consumers)
- c) Advising the minister on production, importation or exportation of cereals.

10. Agricultural finance

corporation- it has the following roles/ functions:

- a) Providing agricultural credit/ loans to farmers at reasonable rate.
- b) Providing technical services to the farmers to ensure the best use of the borrowed loan/ credit.
- c) Ensure repayment of loan.

11. Agricultural development

corporation- it has the following roles/ functions:

- a) Running and operation of state farms.
- b) Raising high quality livestock which are sold to farmers as breeding stock.
- c) Bulking planting materials e.g. maize, Irish potatoes and Napier grass.
- d) Promotion of agricultural production through field demonstrations.

B. CO-OPERATIVE SOCIETIES.

- ✓ A co-operative is an organization of people who have joined together voluntarily with a common purpose for a mutual economic benefit.

Types of farmers' Co-operative societies.

- i) **Farmer/Producer co-operatives-** are formed by groups of individual farmers who pool resources together, buy large pieces of land and operate as a group.
 - ✓ At the end of the year each farmer is paid dividends according to members' shares.
- ii) **Consumer co-operatives-** they buy inputs in bulk and sell them to the members at lower prices.
- iii) **Marketing co-operatives-** they are formed to help small scale farmers market their produce collectively.
 - ✓ Each farmer produces a certain product, pooled together and transported to the market thus reducing production costs.

PROCESS OF FORMATION OF CO-OPERATIVES.

- ✓ A minimum number of **ten adults** are required for the formation of co-operative.
- ✓ Members interested elect interim committee.
- ✓ The committee decides on the name of the co-operative and draws/ writes up its constitution.
- ✓ The committee works with local leaders to register the co-operative with the commissioner for co-operatives.

Factors necessary for proper functioning of farmers co-operative societies

1. Availability of adequate funds or capital/ inputs.
2. Training of personnel or availability of advisory services on managerial skills.
3. Loyalty on the part of all farmers, co-operators and officials to support their organization.

4. Proper and accurate record keeping and accountability for all operations.
5. Efficiency in production and marketing of produce.
6. Honesty on the part of personnel regarding finances.
7. Timely payment of farmers dues.

PRINCIPLES OF CO-OPERATIVES.

1. **Open membership-** anybody can join regardless of race, religion, sex, education and political inclinations after payment of required amount.
2. **Equal rights-** co-operatives are run democratically where the principle of one-man one-vote operates.
3. **Principle of share limit-** A member may buy shares in the co-operative up to a specific maximum limit to avoid domination by one member or a group of members.
4. **Interest on shares-** profit is shared to members as dividends on the basis of share contributions.
5. **Education-** Members should be continually educated to ensure that they have relevant skills of the co-operative society.
6. **Withdrawal from membership-** members are free to join or withdraw voluntarily.
 - ✓ Upon withdrawal, members get back their share contribution.
7. **Loyalty-** Members are expected to be faithful and loyal to their cooperative society.
 - ✓ Members of a producer co-operative must sell their produce through their co-operative only.
8. **Co-operative principle-** Co-operatives are supposed to join the co-operative movement at the primary, district, national and international levels.
9. **Non-profit motive-** Co-operatives are essentially non-profit making organizations but their main objective is to improve the living standards of their members.

FUNCTIONS OF AGRICULTURAL CO-OPERATIVE SOCIETIES.

1. They pool resources together to buy expensive machinery e.g. tractors and land.
2. They provide education/ technical information to members / farmers.
3. They provide credits/ loans to farmers.
4. They negotiate for higher prices for members/ farmers.
5. They reduce overhead costs e.g. transportation, storage and use of machinery.
6. They bargain with suppliers to give discount on seed, fertilizer and other inputs/ provide inputs at a lower prices.
7. They provide employment for members/ farmers and other people.
8. They benefit the farmers from lower taxes charged.
9. They help to market the farmers produce.
10. They provide strong bargaining power for members on policy issues.
11. They invest and pay dividends to members.
12. They help to negotiate loans for their members without security e.g. title deed.
13. They provide banking services to its members.

C. AGRICULTURAL ASSOCIATIONS AND UNIONS.

1. Agricultural Society of Kenya (ASK).

Roles/ functions of ASK

- a) Holds competitive agricultural shows/ exhibitions.
- b) Encourages breeding and importation of pure breeds of livestock.
- c) Encourages and assists in official milk recording scheme.
- d) Organizing the running of Young Farmers Club.
- e) Organizing the National Ploughing Contests.
- f) Organizing tours for its members.
- g) Publishing the Kenya Stud book.
- h) Publishing a monthly journal called “The Kenya Farmers”.
- i) Awarding bursaries for local and overseas students.
- j) Organizing local and international exchange programmes.
- k) Organizing national tree planting.

2. Youth organizations- they include:

A. 4-K clubs (Kuungana, Kufanya, Kusaidia Kenya).

- ✓ Youth organization whose members are drawn from primary schools.
- ✓ The clubs are run by the Ministry of Agriculture but at school level teachers and other local leaders assist members.
- ✓ The motto is “Learn to do by doing”.

Roles/ functions of 4 k-clubs.

- i) Teaching by carrying out practical projects to show that agriculture can be a profitable project.
- ii) Exposing the youth to existing and improved Agricultural technologies.
- iii) Developing and enhancing leadership qualities among the youth.
- iv) Using the youth as agents of change by taking part in competitive shows.
- v) Involvement in field trips to places of Agricultural interest.

b) Young Farmers Club of Kenya (YFC)

- This is a youth organization under the Agricultural Society of Kenya (ASK).
- Members are from secondary schools and middle level colleges.

Roles/ functions.

- 1) Participating in exhibitions and competitions at ASK shows.
- 2) Involvement in agricultural projects at club level.
- 3) Participating in Young Farmers Club (YFC) rallies.
- 4) Involvement in workshops and seminars related to agriculture.
- 5) Participation in national tree planting activities.
- 6) Participation in exchange programs both locally and abroad.
- 7) Participating in national ploughing contests.

3. Kenya National Federation of Agricultural Producers (KENFAP)

- Its members are individual farmers as well as from various commodity associations e.g.
 - i. Kenya Horticultural Producers Association.
 - ii. Cereal Growers Association.
 - iii. Cotton Growers Association.
 - iv. Coffee Growers Association, etc.

Roles/ functions.

- i) Bargaining for better prices of farm produce.
- ii) Ensuring adequate and timely supply of farm inputs.
- iii) Bargaining for reasonable and affordable prices of farm inputs.
- iv) Provision of better infrastructure including roads, electricity, telephone services to facilitate quick delivery of farm produce to the market.

- v) Provision of loan facilities.
- vi) Adequate control of crop and livestock pests and diseases.
- vii) Looking for markets of farmers produce both locally and overseas.
- viii) Offering technical services to farmers.
- ix) Representing Kenyan farmers in the International Federation of Agricultural Producers.
- x) Publishing a monthly magazine known as the "*Farmers' Voice*".

4. Agricultural-based Women Groups.

- They are women groups mainly dealing with agricultural activities, a majority of which are located in rural areas.
- These women groups engage in agricultural activities such as growing vegetables or particular cash crops, rearing dairy goats, poultry production, rearing of dairy cattle, fish rearing.

Objectives of women based groups.

- i. To alleviate poverty among the rural women.
- ii. To empower women financially in order to improve their living standards.

Role of agricultural based women groups

- i. Loaning members to finance their farming activities.
- ii. Enlightening members on improved/modern farming techniques/emerging issues
- iii. Establish income generating activities for members
- iv. Assist in marketing agricultural produce for the members.
- v. Buy farm inputs in bulk and sell to members at a low price
- vi. Collectively assist members in their farm operations
- vii. Guarantees members for loans.
- viii. Gathering information on intended projects/feasibility study.
- ix. Acts as agencies of change in a community.

Problems that Agricultural based women groups face.

1. Lack of managerial skills e.g. in accounting and book keeping.
2. Lack of technological knowledge to improve ways of carrying out certain agricultural activities they are involved in.
3. Lack of funding.
4. Limited access to credit facilities.
5. Low level of education of women especially those in rural areas.
6. Women's multiple roles and responsibilities. This hinders them from devoting more time to their groups, which means the groups do not exploit their full potential.

7. AGROFORESTRY.

1

- Agroforestry is the practice of growing trees/ shrubs, crops, forage (fodder or pasture) and keeping livestock on the same piece of land with the aim of improving productivity of soil.

Forms of agroforestry.

1. **Agrosilviculture-** this is a combination of growing trees/ shrubs and crops in agricultural production.
2. **Silvopastoral-** this as a combination of trees/ shrubs and keeping of livestock/ growing of forage.
3. **Agrosilvopastoral-** this is a combination of growing trees/ shrubs, crops and keeping livestock/ growing of forage.

Importance of agroforestry.

1. **Source of firewood/ remedy for deforestation-** provides a sustainable source of fuel wood. This saves forest trees hence reducing deforestation.
2. **Source of income-** products of trees and shrubs e.g. fruits, poles, timber and forage can be sold for cash.
 - Legumes fodder provide feed for livestock hence saving money that could have been used for expensive feeds.
3. **Environmental benefits-** trees protect the soil from strong rains, sun and wind thus preventing soil erosion, improves water holding capacity and improves water catchment of in a given area.
 - Legumes fix atmospheric nitrogen thus improving soil fertility.
4. **Labour saving-** it saves labour on fetching firewood from distant places. This labour could be used in productive work in the farm.
5. **Aesthetic value-** trees and shrubs help to add beauty to the environment and make the place more pleasant to live in.

Study questions.

1. State four benefits of agroforestry to a maize crop.

- ✓ Leguminous trees fix nitrogen into the soil.
- ✓ Trees act as windbreaks.
- ✓ Trees stabilize soil against soil erosion.
- ✓ Leaf litter decompose to form humus/ recycle nutrients.
- ✓ Trees improve and act as water catchment areas.

2. State four ways in which trees improve soil productivity

- ✓ They conserve moisture.
- ✓ They improve soil structure.
- ✓ They control soil erosion.
- ✓ They contribute plant nutrients/ organic matter

Characteristics of trees and shrubs suitable for agroforestry.

1. **Fast growth rates-** this reduces the time taken for the farmer to get the end products of trees and shrubs . Examples include *Eucalyptus spp*, *calliandra spp*, *pnes etc.*
2. **Deep rooted-** this ensures minimum competition for mineral nutrients and moisture with the crop plant. It also helps to resist drought. Examples include *Eucalyptus spp*, *Grevillea spp*
3. **Nitrogen fixing/ leguminous-**to improve soil fertility. Examples include *Calliandra spp*, *Leucaena spp*, *Cajanus cajan*, *Sesbania spp*.
4. **Highly leafy/ highly branched/ Good in by-product production e.g. timber, fruits and poles-**the trees and shrubs should produce the by-products without affecting the crop plant. Examples include *Grevillia*, *Markamia spp*, *Croton spp*.
5. Should not compete for sunlight.
6. Should be hardy/ should have high survival rate

Forms of agri-silviculture.

1. **Wide row planting:** It involves planting trees at a wider spacing of 4 m between the rows and 8 m to 12 m within the rows.
2. **Intensive hedgerow:** This involves planting of trees and shrubs between rows of crops.

Trees and shrubs to avoid planting at certain sites.

1. Eucalyptus spp should not be planted near the water sources because they absorb a lot of water and the river may end up drying if they are many.
2. Bushy trees should not be planted near the homestead since they harbour dangerous animals.
3. Cypress and eucalyptus should not be planted in arable land because their leaves have allelopathic effects on crops.
4. Trees should be planted in a multi-storey structure to avoid tall plants shading other crops.
5. Fast growing crops should be planted on soil conservation structures e.g. soil bunds.
6. Leguminous trees and shrubs can be planted in a land that is left fallow because they improve soil fertility

Tree nursery

6

1. Types of tree nurseries.

a) Direct nurseries- seeds are planted directly on the ground without any containers.

✓ They are easy to establish and maintain but seedlings have low survival rate after transplanting.

b) Containerised nurseries- seeds are shown directly in containers e.g. pots, polythene bags, plastics or tins which are filled with soil mixtures. The containers are then arranged under a nursery shed.

2. Seed collection and preparation

a) Acquiring seeds- high quality seeds should be used with the following qualities:

i) Good size.

ii) Healthy/ Free from pests and diseases.

iii) Should be fresh to ensure high viability.

▪ Seeds can be obtained from government departments or collected from mature trees, threshed and used or stored for future use.

b) Seed preparation and treatment.

- After collection of seeds preparation practices include:
 1. Extraction to remove the seeds from pods/ fruits.
 2. Drying to reduce moisture content.
 3. Testing to verify the seed quality.
 4. Seed treatment to:
 - ✓ To break seed dormancy and ensure rapid germination.
 - ✓ To improve vegetative growth associated with nitrogen fixation.
 - ✓ To control pests and diseases.

Methods of seed treatment

1. Seed inoculation- coating the leguminous seeds with inoculant to improve nitrogen fixation.
2. Seed dressing- coating of seeds with appropriate chemicals to control pests and diseases.
3. Mechanical breaking- done by nicking or filing the seed coat with a knife to facilitate water entry e.g. seeds of croton.

4. **Soaking in water-** to soften the seed coat and make it permeable to water.
5. **Chemical treatment-** seeds are put in dilute sulphuric acid to soften the seed coat and make it permeable to water.
6. **Light burning-** trash is spread over the seeds already covered with a thin layer of soil then burnt.
- ✓ This is done to soften the seed coat and make it more permeable to water.
7. **Hot water treatment-** used to soften the seed coat and make it more permeable to water in Leucaena, calliandra and acacia.
 - Hot water (about 80°C) is poured on the seeds in a container and allowed to soak for 3-5 minutes.
 - The hot water is drained and replaced with cold water and allowed to soak for 24 hours until they swell.

3. Nursery management practices.

- a) **Mulching-** a light mulch should be applied on the nursery bed to prevent excessive evaporation and also moderates soil temperatures.
 - ✓ The mulch should be removed immediately the seedlings start to emerge to prevent smothering the seedlings.
- b) **Watering-** it is done in the mornings and evenings using a watering can to supply and maintain appropriate amount of moisture.
- c) **Weed control-** it is done through uprooting to avoid competition for nutrients and moisture.

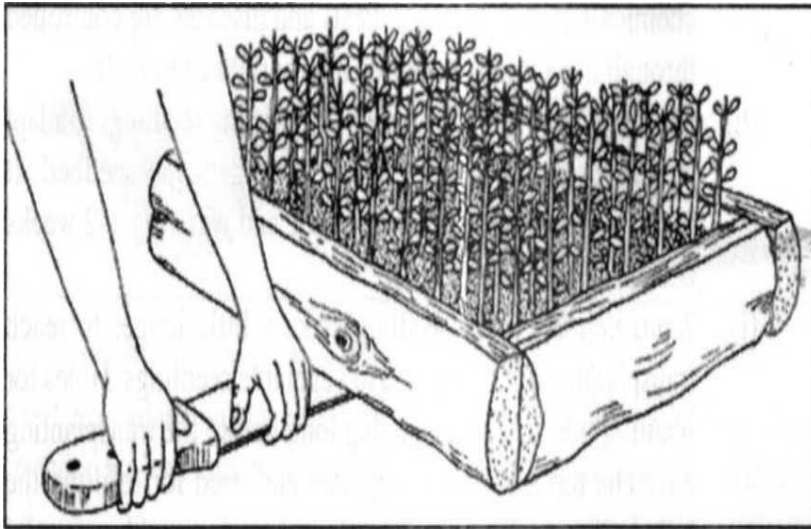
- d) **Pricking out-** removal of some seedlings from overcrowded nursery and planted in another nursery (seedling bed).
 - ✓ This is done to allow seedlings to grow strong and healthy without much competition.
- e) **Shading-** A shade should be erected over the nursery but dark conditions should be avoided.
 - ✓ It is done to protect the seedlings against harsh environmental conditions.

- f) **Hardening off-** It is the practice of preparing seedlings to adapt to the ecological conditions prevailing in the seedbed.
 - It involves gradual reduction of shade and watering 1-2 weeks before transplanting.
- g) **Pest and disease control-** it is done by sterilizing the soil through heat treatment or application of chemicals (e.g. furadan).
 - It can also be done by application of appropriate chemicals.

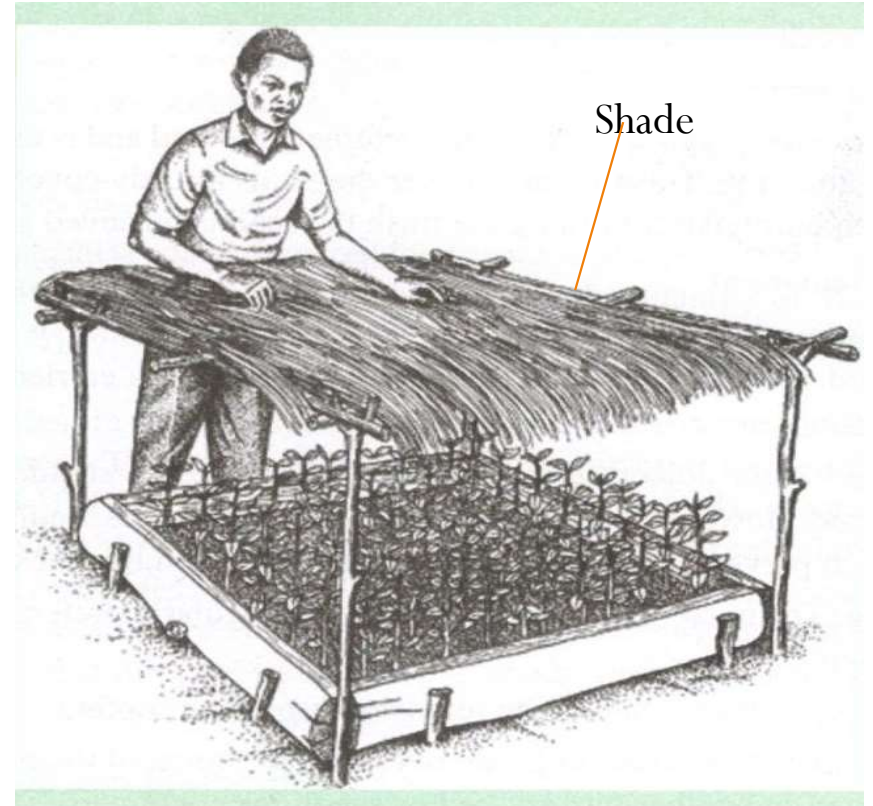
- h) **Root pruning-** cutting of roots when they become longer than the depth of the polythene sleeve by use of a knife, razor blade or scissors.

Reasons/importance/advantages of root pruning.

- i) Encourages the development of short, dense and strong rooting system.
- ii) Enables easier lifting of the seedlings.
- iii) To minimize root damage during transplanting.



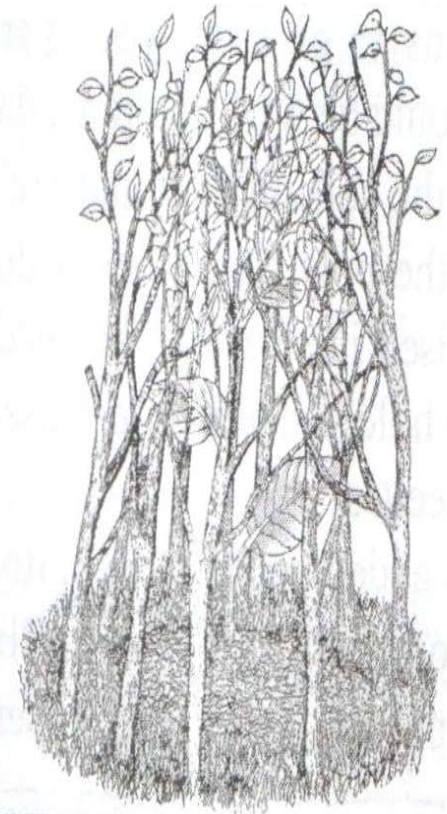
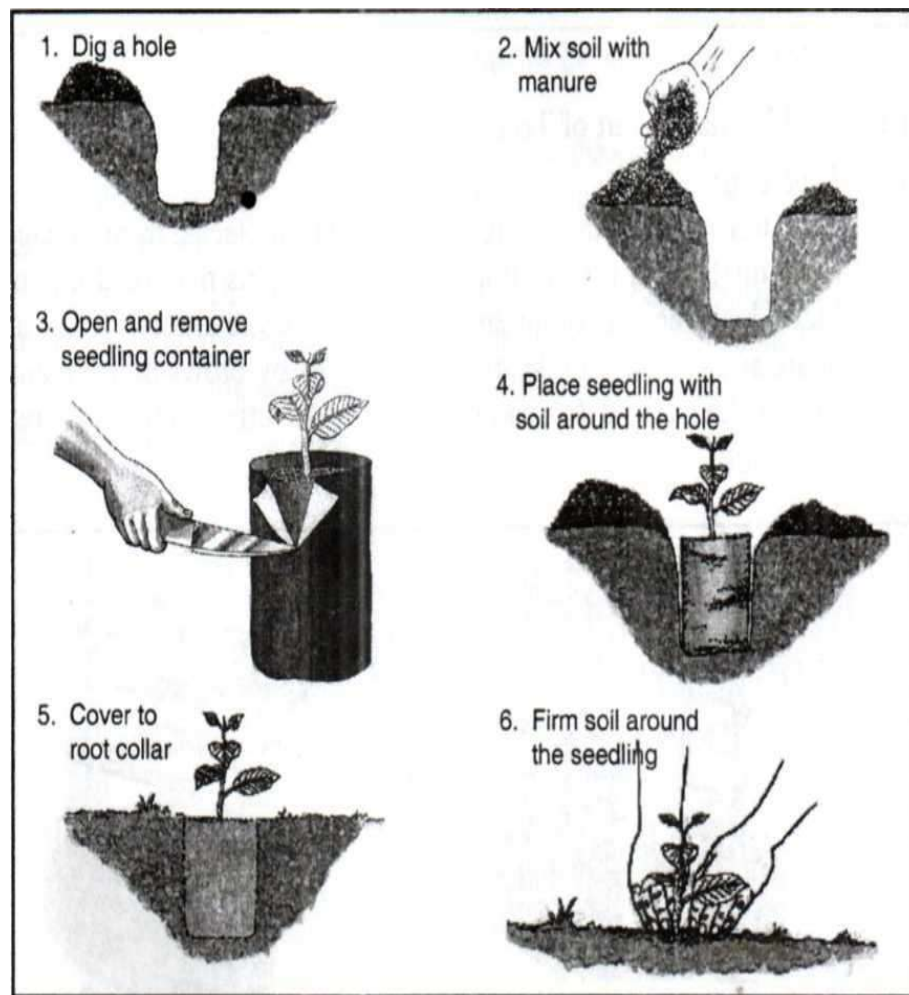
Root pruning



Procedure of transplanting of tree seedlings

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- ✓ Transplant at the onset of long rains to give young seedlings a good start.
- ✓ Dig planting holes few days before transplanting day.
- ✓ Keep top soil and sub-soil separate to be used to fill the planting hole halfway.
- ✓ Water the seedlings one day before transplanting to make soil stick around the roots and for ease removal of polythene sleeves.
- ✓ Mix top soil with well rotten manure/ phosphatic fertilizer
- ✓ Place the seedling at the centre of the planting hole.
- ✓ Use a sharp knife/ razor blade to cut and remove the polythene sleeve.
- ✓ Add more soil and firm gently around the plant until the hole is completely full.
- ✓ Plant the seedling at the same depth as it was in the nursery.
- ✓ Water and mulch the seedlings then provide a temporary shade to conserve moisture.

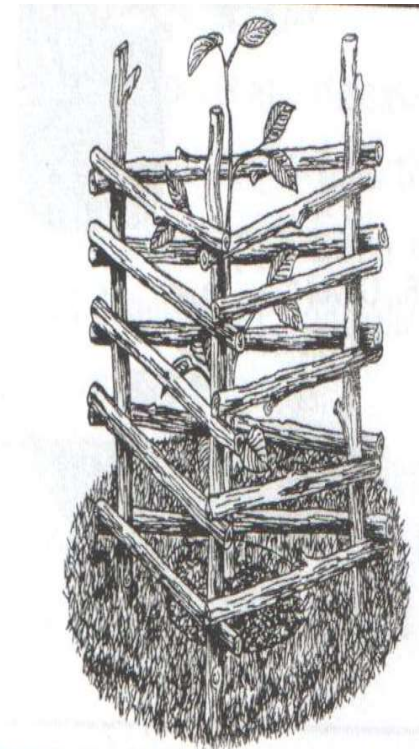
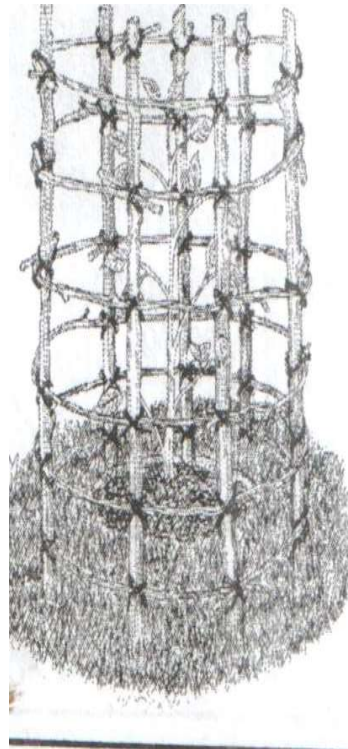


Temporary shade for the transplanted tree seedlings

Care and management of trees

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1. **Protection-** It is done on individual seedlings or the entire field to protect them against destruction by animals.



2. **Pruning and training-** involves the removal of extra or unwanted parts of a tree.

- The unwanted parts may be due to breakage, overcrowding, pest and disease attack or low productivity

Reasons for pruning trees.

- i. To reduce shading the crops.
- ii. To provide fuel wood or fodder.

3. **Grafting Old Trees-** refers to the practice of uniting two separate woody stems/ scion and root stock to grow as one.

Reasons/ advantages for grafting agroforestry trees.

- i. To allow propagation /growth /perpetuate trees that cannot be reproduced through other asexual means.
- ii. To get trees with desirable characteristics.
- iii. To change the varieties of established fruit trees.

4. **Weeding/weed control-** done through use of appropriate chemicals, tillage, slashing to reduce competition for nutrients and moisture with weeds.
5. **Disease and pest control-** done keep the trees healthy and ensure vigorous growth using appropriate chemicals.
6. **Thinning/selective harvesting-** some trees are removed from overcrowded field to reduce competition for moisture/water and nutrients.
7. **Irrigation /watering –** it is done during the dry season to supply enough moisture.
8. **Shading/mulching-** involves erecting a shade of application of mulch around the trees to reduce evaporation /moisture loss/ conserve moisture.
9. **Fertilizer/ manure application-** involves application of well rotten manure/ inorganic fertilizers to supply required nutrients.
10. **Gapping-** it involves filling the spaces/ gaps where seedlings failed to establish to obtain optimum plant population.

Agroforestry practices

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A. Alley Cropping/ intensive hedgerow/ intercropping.

- This is the growing trees/ shrubs between the rows of crops.
- Common species used include *Leucaena* and *Calliandra*

Benefits/Advantages of alley cropping/hedgerow /Intercropping.

1. Acts as a source of wood fuel, fruits, medicine and fodder.
2. Legumes improve soil fertility through nitrogen fixation/ nutrients cycling.
3. Helps to control soil erosion/ Facilitates soil and water conservation when roots bind soil particles.
4. Leaves and branches are used as mulch.
5. It provides green manure.
6. It maximizes utilization of available land.

- B. Border planting:** It involves planting trees/ shrubs on the edges/on borders of the farm.

Importance of border planting.

- 1) It acts as wind breakers.
- 2) Marks boundaries in the farm
- 3) It acts as a live fence.
- 4) It provides wood, fuel, timber and poles.
- 5) Has aesthetic value/add beauty.

C. Woodlots in the farm.

- These are plots of land set aside for trees only.
- Woodlots are usually established on parts of the farm that are not productive or where other forms of agricultural production would encourage soil erosion.
- Trees for woodlot should be fast growing e.g. *Acacia sp*, *Eucalyptus sp*.

D. Multo-storey cropping.

- This is the growing of trees and crops forming different levels like storeys.
- Common species used include Whistling pine, African blackwood etc.

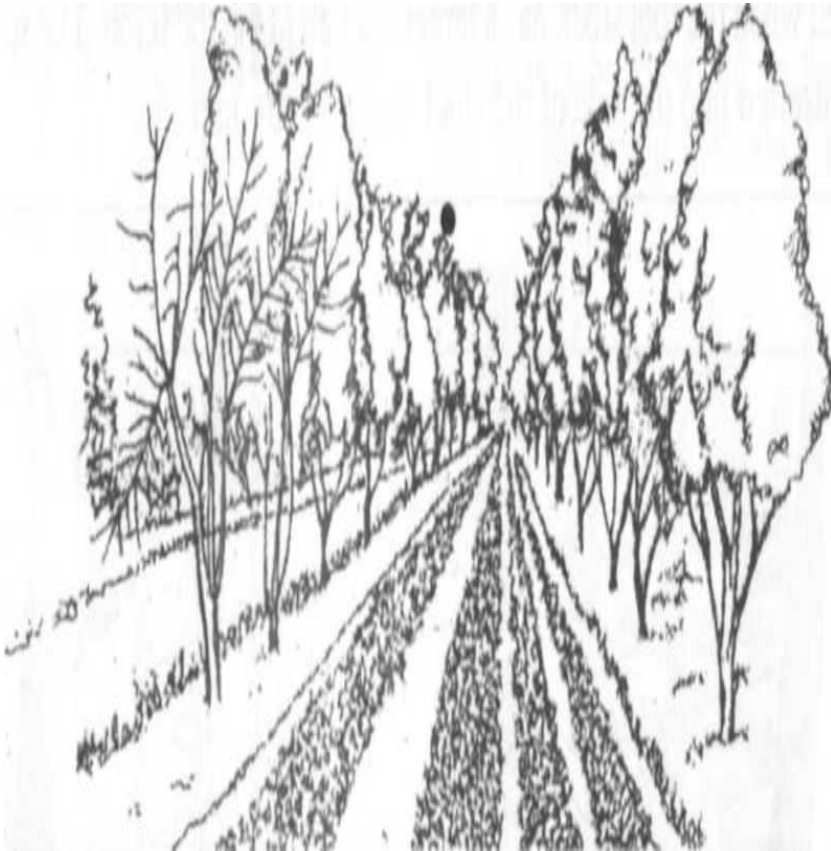
Importance/advantages of multi-storey cropping.

- 1) The trees act as windbreaks for crops.
- 2) Trees create suitable micro-climate for the crops.

- 3) Trees act as a source of timber and firewood.
- 4) Shrubs/trees provide fruits.

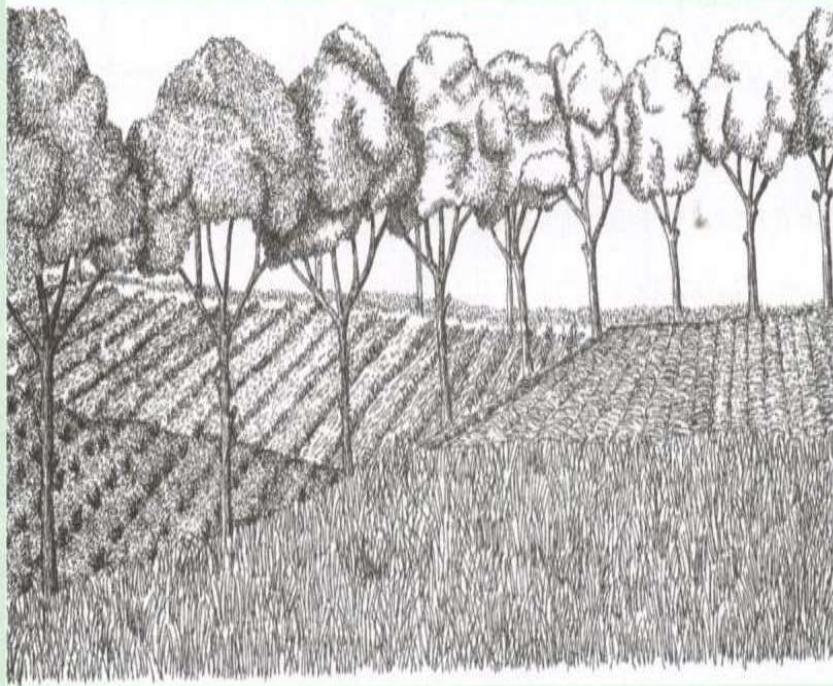
a) Alley Cropping/ intensive hedgerow/ intercropping.

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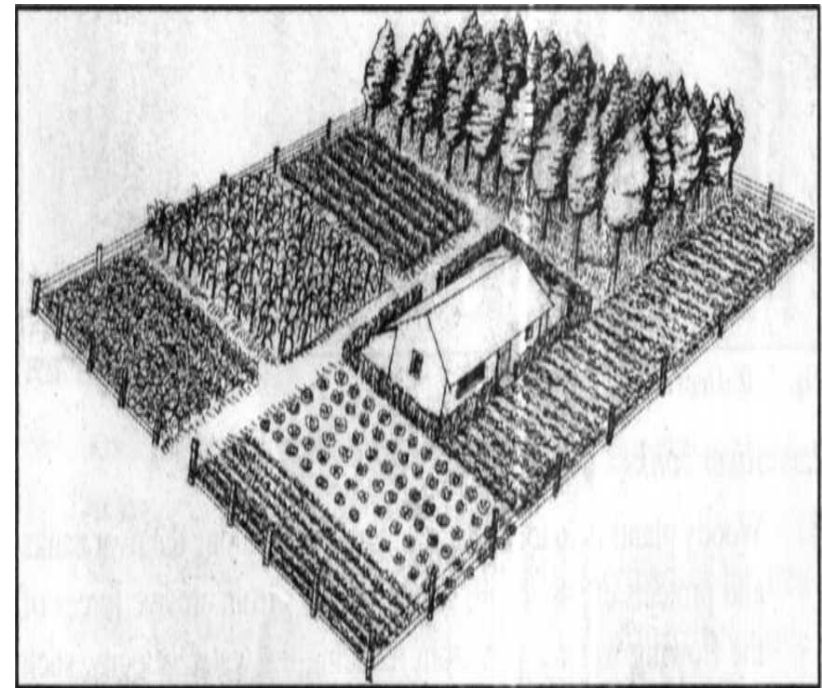


Calliandra

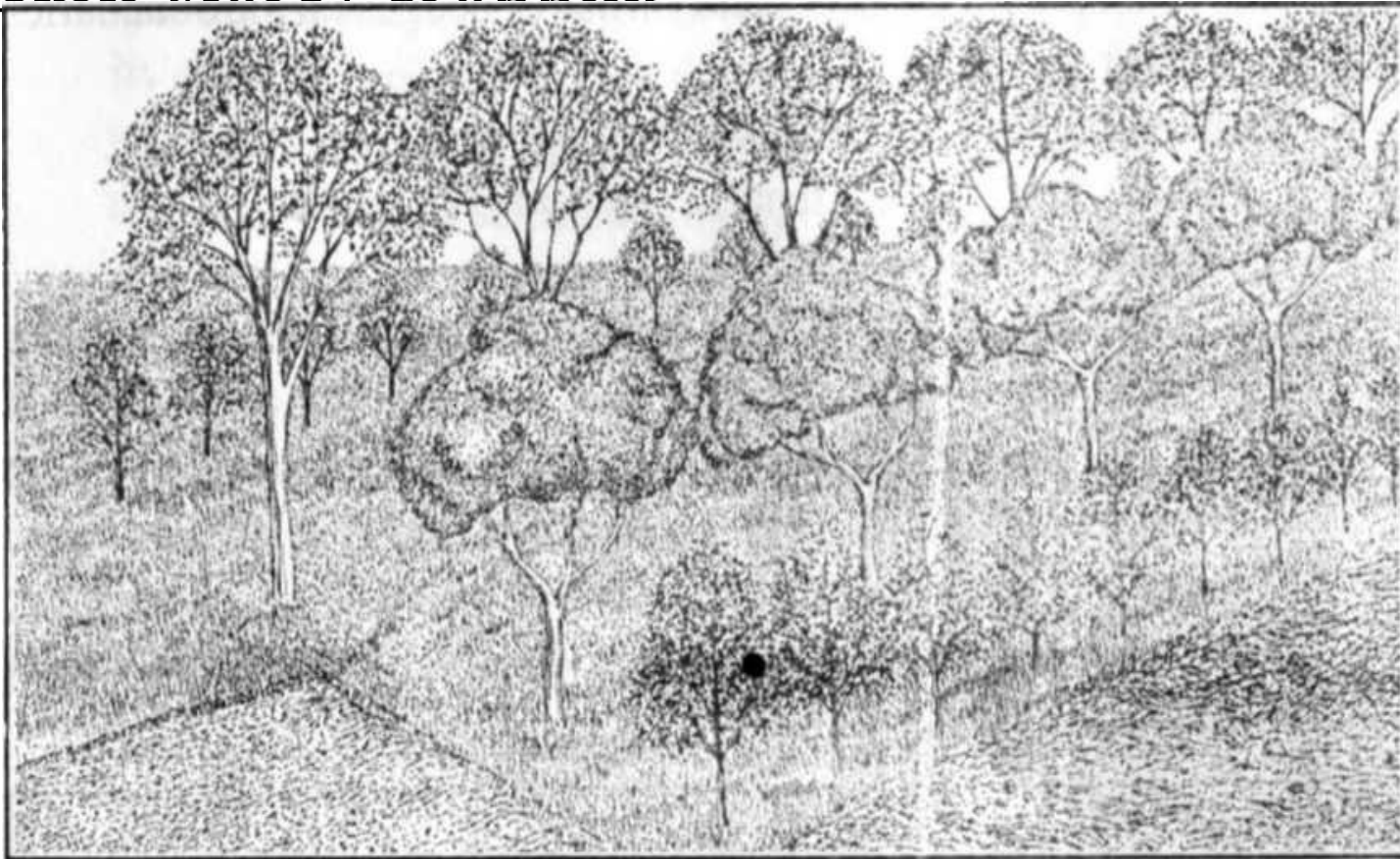
b) Border planting



c) Woodlots



Multi-storey cropping



Sites for Agroforestry Trees.

23

1. **Boundaries**- trees/ shrubs/ hedges are planted along the boundaries e.g. Eucalyptus sp, Jacaranda sp, Grevillea sp, Markhamia sp.

✓ They help to:

- a) Protect the farm as a fence.
- b) Mark the boundary.
- c) Form wind breaks/ act as wind breakers.
- d) Provide wood, fodder, fruits and timber for fuel and building as well as for sale.

2. **River banks**- trees and shrubs are planted along the river banks to:

- a) Control river bank erosion.
- b) Slow down speed of surface run off.
- c) Provide fuel wood, fodder, fruits and poles

3. **Terraces-** trees and shrubs (e.g. Grevillea, Sesbania, Calliandra, Mango, Avocado and Paw paw) are planted on terraces:

- a) To reduce the speed of surface runoff.
- b) To trap soil.
- c) For roots of trees to hold soil particles together reducing its erosivity.
- d) To provide shade which protect the soil against rain drops thus controlling splash erosion.

4. **Slopes-** woodlots are established on sloppy areas to:

- a) Stabilize the soil /maintain soil structure/ reduce soil erosion.
- b) Reduce the speed of surface runoff.
- c) Trap soil in erosive water.

5. **Homestead-**trees and shrubs are planted around the house to:

- a) Provide shade.
- b) Provide beauty/ aesthetic value.
- c) Provide firewood, fruits and timber.

Tree harvesting methods.

25

1. **Pruning-** this is the removal of branches from the lower part of the tree crown. It is done to:
 - i. Reduce shade.
 - ii. Improve the quality of the trunk.
 - iii. Provide fodder and fuel.

2. **Lopping--**this is the removal of branches from trees in an haphazard manner.

2. **Pollarding-** this is the cutting of all the branches and top part of the tree. It is done to provide fodder, poles, timber and wood fuel. It is practiced in Croton sp, Casuarina, Grevillea, Jacaranda etc.

4. **Coppicing-** this is the cutting of the whole tree about 30 cm above the ground.
- It is done to provide fodder, wood and mulching materials.
 - It is practiced in Calliandra, Cassia, Eucalyptus, Leucaena, Markhamia, Grevillea, Sesbania.

5. **Thinning-** this is the cutting down of trees to avoid overcrowding. The thinned trees are used as a source of fuel wood, poles and fodder.
- It is common where trees have been established by direct seedling or planted with little space.

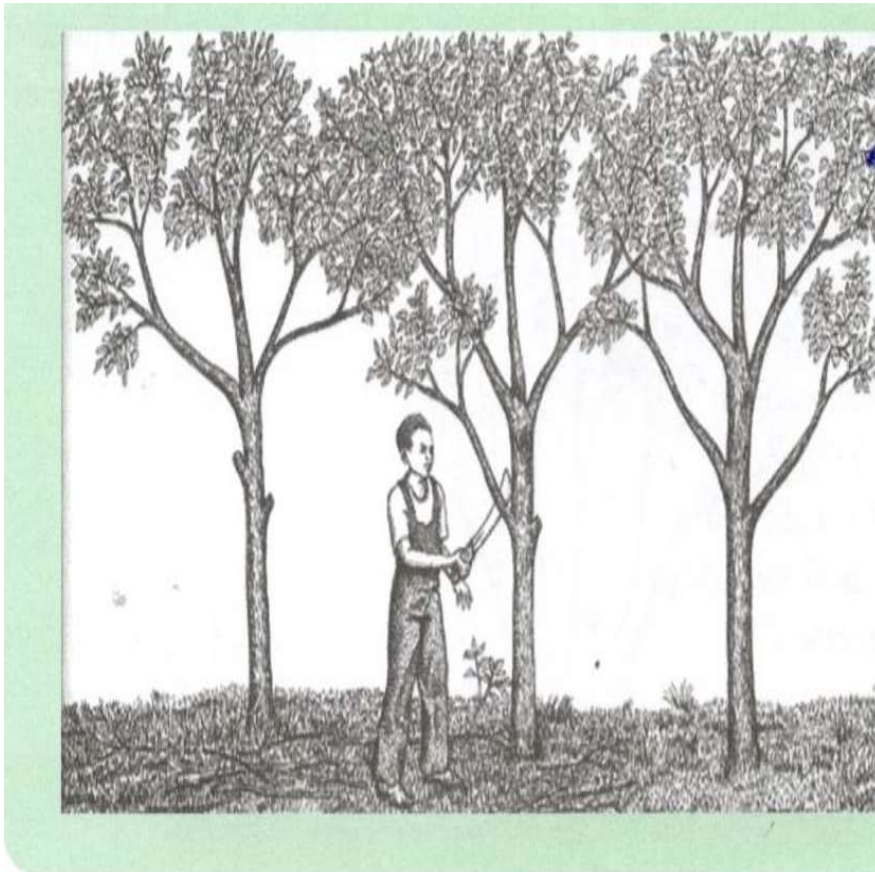


Fig. 7.11: Pruning trees

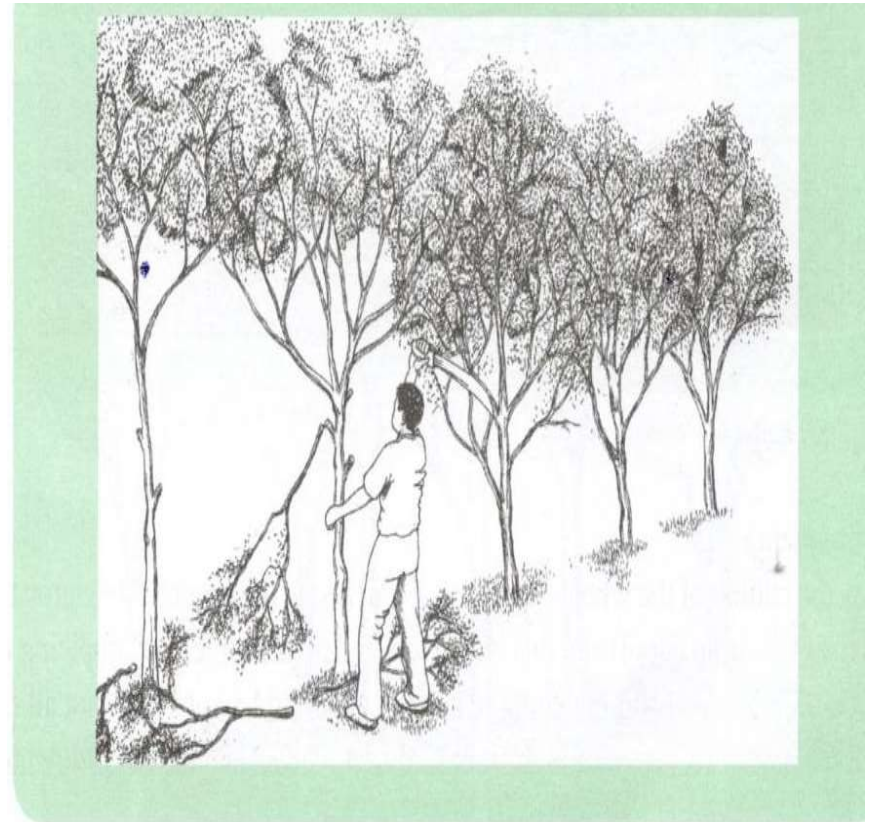


Fig. 7.12: Lopping trees

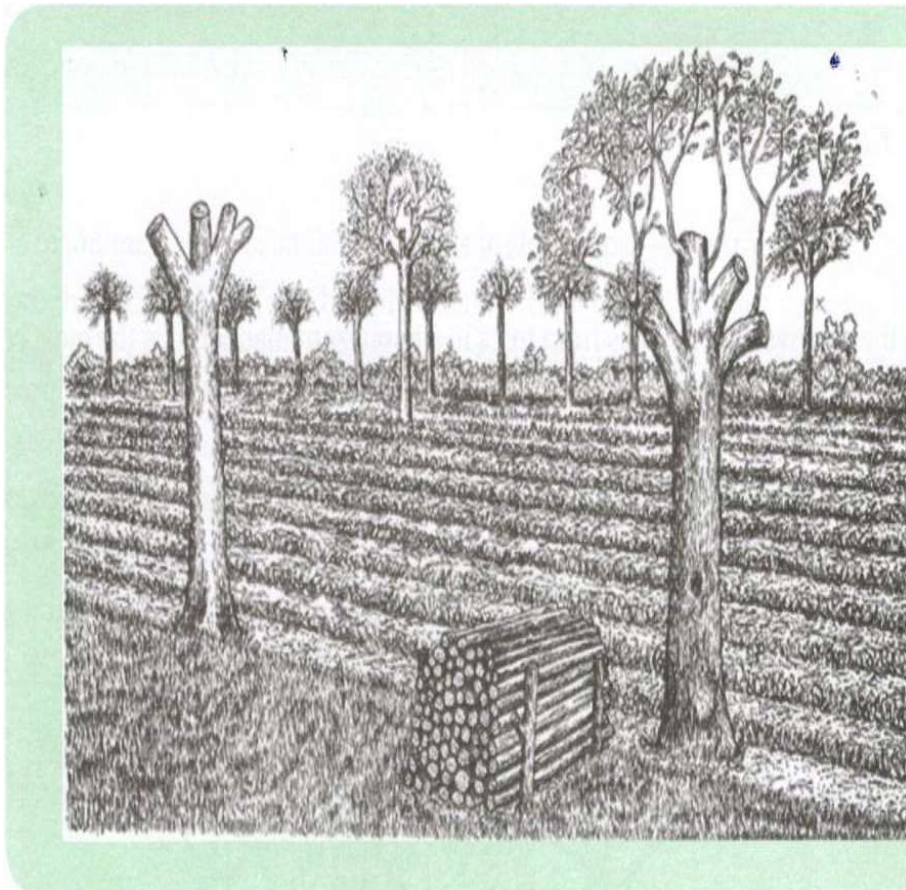


Fig. 7.13: Pollarding technique

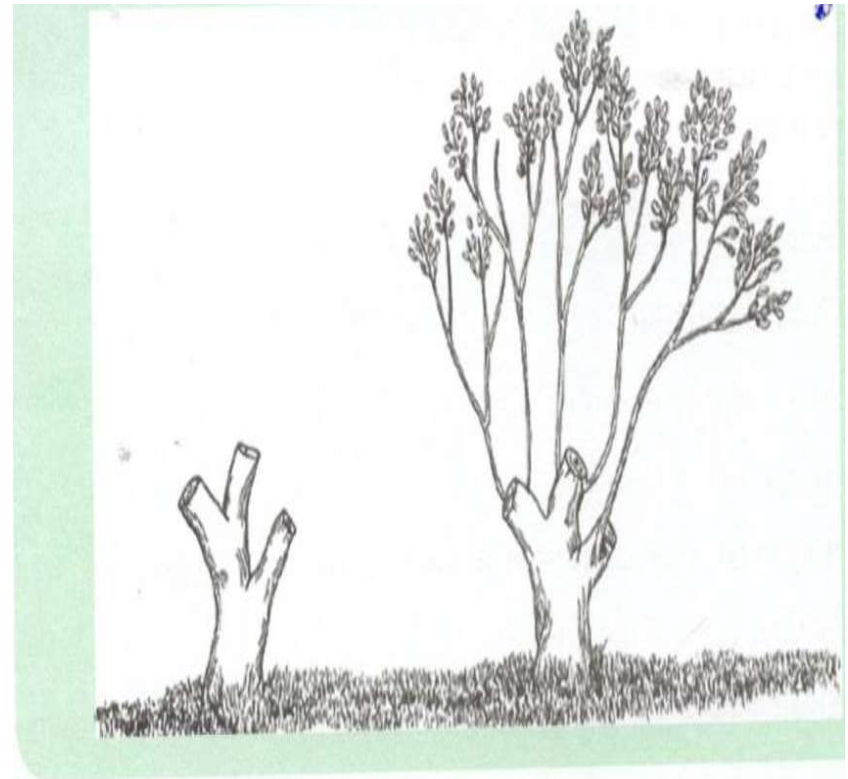


Fig. 7.14: Coppiced trees

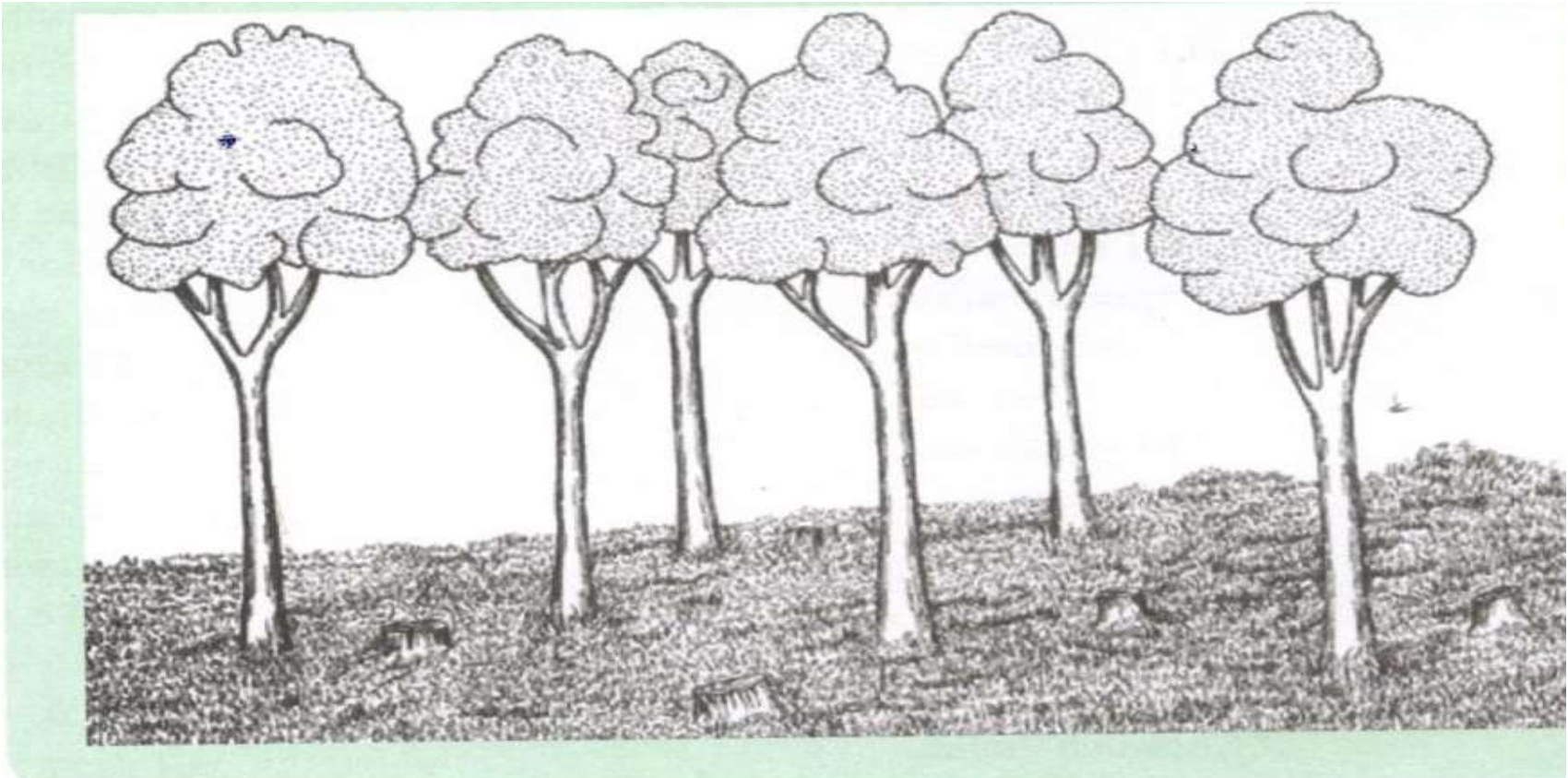


Fig. 7.15: Thinning trees